

Beginning —with— Tableau Cloud

A beginner's guide to accessing, sharing, and
managing content on Tableau Cloud



Urmisha Patel



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About the Author

Urmisha Patel is a seasoned data analytics professional and Tableau expert based in Pennsylvania, USA, with over eight years of experience in transforming raw data into actionable insights. Passionate about empowering individuals and organizations to harness the power of data visualization, she has worked across diverse industries, helping teams unlock their data's potential through effective storytelling and impactful dashboards.

With a background in data science and business intelligence, she combines technical expertise with a strong commitment to teaching. As a certified Tableau professional, she has conducted numerous workshops, webinars, and training sessions, mentoring professionals of all skill levels to become confident Tableau users.

In addition to her work as a practitioner, she regularly contributes to the data analytics community, sharing insights, tutorials, and solutions through blogs, forums, and YouTube channels. This book reflects her dedication to making Tableau accessible to everyone, whether a complete beginner or a seasoned user looking to deepen their skills.

Outside of work, she enjoys creating educational videos on YouTube channel, always seeking inspiration for the next visualization project. She firmly believes in the transformative power of data and is excited to help readers embark on their Tableau journey.

She pursued her post-graduation in business analytics and business intelligence, specializing in artificial intelligence and machine learning from Great Lakes Institute of Management, and a bachelor's degree in computer engineering from Gujarat Technological University in India.

About the Reviewer

Anshu Bantra is a senior data analyst and emerging analytics engineer/scientist based in Melbourne, Australia, with a robust background in statistical analysis, data visualization, and insight generation. He is currently advancing his expertise in data science, machine learning, and statistical modeling, with a focus on deriving actionable insights from complex data sets through predictive analytics, A/B testing, sophisticated statistical methods, and building better data flows. Committed to continuous learning, he is deepening his knowledge of algorithmic techniques and statistical approaches, with a strong emphasis on workflow automation and optimization for enhanced data processing efficiency. Beyond his primary role, he is a dedicated educator who actively shares his expertise through workshops, classes, and publications.

Acknowledgement

I would like to extend my deepest gratitude to the people and organizations who made this book possible. First and foremost, I would like to thank the Tableau team for their continuous innovation in cloud-based analytics and for providing an incredible platform that serves as the foundation of this book. Their dedication to empowering users to turn data into actionable insights is what inspired me to write this guide. A special thank you to the Tableau Product Team for providing insights and resources that helped refine the content and ensure accuracy.

To my colleagues at **Tata Consultancy Services**, I am grateful for your collaboration, feedback, and inspiration. You pushed me to think critically and provided valuable insights into the practical application of Tableau Cloud in real-world scenarios.

A heartfelt thank you to **BPB Publications**, whose feedback ensured the content was both clear and technically sound. Your attention to detail and thoughtful suggestions helped me create a book that is accessible and valuable to readers at all levels of experience with Tableau Cloud.

I would also like to acknowledge the **mentors/instructors** at the **Great Lakes Institute of Management** for their guidance throughout my career, particularly in the realm of data analytics. Your mentorship has been invaluable in shaping both my understanding of data tools like Tableau and my approach to communicating complex ideas clearly.

To my family, especially **my husband**, **my mother**, and **my brother**, thank you for your unwavering support, patience, and encouragement throughout this project. Your belief in me motivated me, even during the most challenging days.

Finally, to the readers—thank you for picking up this book. I hope it provides you with the tools and insights you need to unlock the full potential of Tableau Cloud and to make data-driven decisions that will transform the way you work.

Thank you to everyone who has played a part in making this book a reality.

Preface

Data can drive decisions, but only if it is accessible, understandable, and actionable. As businesses continue to move towards cloud technologies, Tableau Cloud has emerged as a leader in making access not only easier but more dynamic. This book is for anyone looking to navigate the ever-expanding world of Tableau Cloud, whether you're new to data analytics or an experienced Tableau user who wants to fully embrace the flexibility and collaborative power of the cloud.

When I first started using Tableau Cloud, I quickly realized that its potential went beyond just being a tool for visualization. It allowed teams to collaborate, make decisions in real time, and democratize data in ways never seen before. However, I also noticed that many people struggled to make the most of the cloud environment, facing challenges with setup, data connection, and sharing insights efficiently. That is why I decided to write this book: to give you a comprehensive, step-by-step guide on how to leverage Tableau Cloud effectively.

This book is divided into chapters that guide you through every stage of using Tableau Cloud—from setting up your account and connecting data sources to building advanced dashboards and creating meaningful visualizations. Each chapter is designed to provide practical advice that you can apply right away, along with tips and tricks I have learned while working with the platform over the years.

Thank you for picking up this book. I am excited to guide you through the world of Tableau Cloud and to help you master the platform to transform your data analysis and decision-making process.

This book begins with an introduction that highlights the importance of data visualization in today's data-driven world and introduces Tableau as a leading tool for creating impactful visualizations. It discusses why Tableau

stands out among other tools, emphasizing its ease of use, flexibility, and robust capabilities. The introduction also outlines the book's structure and explains what readers will gain from each chapter.

Chapter 1: Getting Started with Tableau - This chapter provides an overview of Tableau's history and its suite of products, including Tableau Desktop, Tableau Public, Tableau Server, and Tableau Prep. It walks readers through the installation process and offers a guided tour of Tableau Cloud's interface, helping them familiarize themselves with the workspace, menus, and tools. It further covers tableau cloud architecture, security in cloud, and types of licenses.

Chapter 2: Managing Data Connections - In the second chapter, readers will learn about the variety of data sources Tableau supports, such as flat files, databases, and cloud services. This chapter covers the basics of establishing data connections over the web and choosing between live and extract connections. It also introduces creating or upgrading the extracts over the web and configuring the connection with an analytics extension.

Chapter 3: Data Visualization - The third chapter, focuses on creating various types of charts and graphs, such as bar charts, line charts, scatter plots, and heat maps. Readers will learn how to choose the right visualization for different types of data and analysis. The chapter also introduces Tableau's powerful Show Me feature and walks through best practices for crafting visually appealing and meaningful charts. Readers will gain hands-on experience with sorting, filtering, grouping, and creating hierarchies to organize data. The chapter also introduces basic aggregation techniques and table calculations and explores calculated fields for applying custom formulas and performing basic data analysis.

Chapter 4: Advanced Data Visualization - The fourth chapter, is dedicated to combining multiple visualizations into cohesive dashboards and narrative-driven stories. It covers dashboard objects, animation, and accelerator. It further covers principles of dashboard design and interactivity with parameters, filters, and actions.

Chapter 5: Content Management and Interaction Tools - This chapter explores how Tableau Cloud enables users to organize, manage, and maintain their published content effectively. It starts by introducing the

concept of projects as the primary organizational structure for content, including workbooks, data sources, and flows. Readers will learn about Tableau file types and organize content hierarchically to align with team workflows.

The chapter continues with subscription features, enabling users to receive regular email updates on specific views or dashboards. Alerts and notifications based on **key performance indicators (KPIs)** are explained, helping users stay informed about critical data changes in real time.

Chapter 6: Site Management - This chapter shifts focus to administering Tableau Cloud environments at the site level. It begins with an overview of Tableau Cloud's site and planning, detailing how multiple sites can be used to separate and manage different business units or projects. Readers will learn about user and group management, including creating groups, assigning roles, and setting permissions to balance accessibility with security. This chapter also discusses monitoring and maintaining site health using Tableau's administrative views, which provide insights into usage, performance, and content engagement.

Chapter 7: Publishing and Scheduling - The seventh chapter, explores publishing workflows, explaining how users can publish workbooks and data sources to Tableau Cloud while maintaining data integrity and security. Advanced topics included in this chapter are Tableau Bridge, scheduling extract refreshes, and autosaving the workbooks.

Chapter 8: Troubleshooting Tableau Challenges - The eighth chapter, provides a comprehensive guide to identifying and resolving common issues encountered while using Tableau Cloud. It begins by emphasizing the importance of troubleshooting skills and outlines the typical challenges users may face, from data connection errors to performance bottlenecks.

Chapter 9: Miscellaneous Projects - In the final chapter readers will be provided with hands-on experience on the case studies that cover connecting the data source from desktop and web, building innovative and creative graphs, advanced analytics, dynamic dashboards, and stories. The chapter covers scheduling refreshes and extracting data refreshes.

This structured approach ensures that readers, whether beginners or those with limited exposure to Tableau, will build a solid foundation and gain confidence in leveraging Tableau for data visualization and analysis.

I hope this book serves as a reliable companion on your journey to mastering Tableau Cloud. Let us unlock the power of data together!"

Tableau Files and Coloured Images

Please follow the link to download the
Tableau Files and the *Coloured Images* of the book:

<https://rebrand.ly/57f435>

The Tableau files for the book are hosted on GitHub at <https://github.com/bpbpublications/Beginning-with-Tableau-Cloud>. In case there's an update to the files, it will be updated on the existing GitHub repository.

We have code bundles from our rich catalogue of books and videos available at <https://github.com/bpbpublications>. Check them out!

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CHAPTER 1

Getting Started with Tableau

A picture speaks a thousand words -Anonymous

Introduction

In this data-driven world, it is important to present the data in such a manner that people and organizations can explore and manage data faster to discover and share insights that can change the business and the world. Number crunching is not the way to make everyone understand; visuals play a vital role for any novice to understand and make their business decisions.

In this chapter, we will explore the powerful analytics platform Tableau Cloud, which is the market favorite for modern business intelligence on the web.

Structure

In this chapter, we will cover the following topics:

- Understanding data visualization
- Tableau products
- Understanding Tableau Cloud
- Data and analytics in Cloud
- Capabilities and scope of Tableau Cloud
- Overview of Tableau architecture
- Security in Tableau Cloud

- Licensing
- Downloading Tableau Online
- Sign-in to Tableau Online
- Touring the Tableau Online User Interface

Objectives

By the end of this chapter, readers will understand the power of Tableau in Cloud, after an in-depth introduction to data visualization and its importance.

Readers will also gain insights into the architecture of Tableau Cloud. Further, they will gain a detailed understanding of Tableau Online workspace, focusing on how Cloud manages published content. They will also understand the types of licenses and site roles in this product. After reading this chapter, readers will be one step closer to understanding the robust business intelligence solution Tableau Cloud, popularly known as Tableau Online.

Understanding data visualization

Data visualization is the art of presenting the data in such a way that a layperson can also understand it. It could be the perfect blend of charts, graphs, colors, dimensions, and labels, which can create the visuals as a masterpiece, hence revealing interesting facts or insights, which in turn helps businesses to make informed decisions. In other words, data visualization can be used to explore the dataset, extract the information, and express and identify the vital relationships in charts and plots that are more helpful to the business. The art and practice of visualizing data are becoming even more important to analyzing and finding hidden patterns. Visuals can help us find patterns in data that numbers fail to show.

Data visualization is an inevitable part of the world of analytics. Hence, to make a quick decision, business managers or decision makers embrace visualization software at all levels as more data sources are used. Currently, there are a lot of visualization tools available in the market, and amongst them, Tableau is the most popular and fastest evolving **business intelligence (BI)** and data visualization tool.

Importance of data visualization

Before starting our journey with Tableau, we must emphasize the importance of data visualization and how it helps businesses make informed decisions. Let us look at some advantages in detail:

- **Faster decision-making:** Humans process visuals better than the tedious tabular forms of reports. If data is communicated well, then decision-makers can quickly make decisions.
- **Analyzing data in a better way:** Visual representation of data communicates in better ways than just number crunching. The visuals help the analyst understand the key points needed for their business. *Data Viz.* strengthens the impact of messaging on your audiences and presents the data analysis in the most persuasive way.
- **Making sense of complicated data:** Data Viz. allows you to glance at large volumes of data and communicates insights visually to internal and external audiences in a better way. It benefits them by allowing them to recognize the new patterns and errors in the data.

Data visualization has a credible impact on businesses. In the present big data era, data is growing faster than ever, and business intelligence is taking a huge set of data and turning it into insightful data points. Data Viz. is doing its part by quickly presenting the data to transfer the information to a human brain with a clear message.

Understanding Tableau

There are numerous business intelligence tools available in the market to perform data visualization; however, now we will explore why we choose Tableau. *Figure 1.1* shows the logo of Tableau:



Figure 1.1: The logo of Tableau Let us look at some factors that make Tableau a suitable choice for data exploration:

- **Quick and interactive visualizations:** Tableau is a lot more creative than you think. With simple drag-and-drop features, you can easily create an interesting and dynamic dashboard that helps us analyze the data and share crucial insights easily.
- **Easy to use:** Compared to other BI tools, Tableau allows you create rich visualizations in just a few seconds with simple drag-and-drop functionalities to perform complex tasks.
- **Handles large amounts of data:** Tableau can handle huge amounts of data (millions of rows of data) without impacting the performance. You can easily

connect with any data sources e.g., Excel, **Statistical Package for Social Sciences (SPSS)**, Cloudera Hadoop, **Statistical Analytical System (SAS)**, Microsoft SQL Server, PostgreSQL, Oracle, Teradata, and so on. You can even connect with live data sources to provide companies with real-time results on key performance indications.

- **Mobile-friendly dashboard:** Tableau dashboards can be viewed and operated on several devices, such as mobile phones, desktops, tablets, and so on. By default, Tableau automatically adjusts the report viewing based on the device. All views and dashboards in Tableau software are compatible with mobile and tablet devices.
- **Integrates with scripting languages:** This BI tool can easily integrate with Python or R to help understand data with visual analytics. Users can perform basic and complex calculations in Tableau.
- **Real-time analytics:** Tableau can have a direct connection to data servers, which can provide real-time analysis. The data should be at your fingertips so you can use it for everyday decision-making. To be most effective, the data should be as up-to-date as possible.
- **Easy to integrate and connect a variety of data sources with the existing technology:** Tableau offers an enormous number of options for connecting to the data sources. This software integrates easily with the existing technology and can grow with you as your technology changes over time.
- **Intuitive dashboard and user interface:** Tableau's drag-and-drop functionality, intuitive interface, and effortless creation of visualizations speed up the work of data analysts or novices alike and can explore the data. Based on their permission and access level, you can easily see all the possible data sources in Tableau and combine them to create new views and uncover new insights.
- **Simple sharing and collaboration:** This software is capable of sharing the developed report on the Tableau server or online to make sharing and collaboration within the organization effortless so that you can consistently start discussions and infuse decisions with data without wasting time on creating or disseminating reports. Creators can merely share and publish the workbook or dashboard to Tableau Server or Online and can also edit it once published.
- **User-level security:** Tableau software gives you ultimate control. To manage unintended alterations to your data or to ensure security, user-level security

filters out records based on row-level/role-based permissions.

- **Community support:** Tableau has an expansive and supportive community that helps in learning across the organizations and equips employees with extensive training and tutorial options, forums, and support.

Visit this community support link: <https://www.tableau.com/community>

As per *Gartner's Magic Quadrant 2022*, Tableau is one of the top three market leaders across all BI products. The magic quadrant for analytics and business intelligence platforms can be seen in *Figure 1.2*:



Figure 1.2: Magic Quadrant for analytics and business intelligence platforms *Tableau products*

Tableau offers various services according to business needs and types of work. The following table lists Tableau products:

Product	Purpose	Free tool?	Download Link
Tableau Desktop	Self-service analytics are made for individual or professional use through simple drag-and-drop actions.	14-day free trial version	https://www.tableau.com/products/desktop/download
Tableau Prep	An ETL tool to get ready data for analysis. It visually combines data from multiple sources.	14-day free trial version	https://www.tableau.com/products/prep/download
Tableau Public	For anyone to publish interactive data online.	Free tool	https://public.tableau.com/en-us/s/download
Tableau Cloud (Online)	For individuals or organizations, business intelligence is in the Cloud.	14-day free trial version	https://www.tableau.com/products/online/thanks-dl
Tableau Server	Share and collaborate with the organization.	14-day free trial version	https://www.tableau.com/support/releases/server
Tableau Reader	It allows you to read the files saved on the Tableau desktop.	Free tool	https://www.tableau.com/products/reader/download
Tableau Mobile	A free mobile app for accessing Tableau Server or Online on your mobile devices. It can also provide offline viewing by automatically caching favorite views. Available for both IOS and Android users.	Free tool	https://www.tableau.com/products/mobile

Table 1.1 : List of Tableau products This chapter aims to give a basic understanding of Tableau Cloud (Online). We will also touch upon the architecture and licenses of Tableau Cloud, which affect accessibilities and privileges when using this product. We will also explore the user interface of Tableau Online after downloading its free trial version (14 days).

Understanding Tableau Cloud

Tableau Cloud is a secure, fully hosted, cloud-based solution for sharing, authoring, distributing, and collaborating on content created in Tableau. It empowers you to make smarter decisions with fast, flexible, and easy analytics. Tableau Cloud enables more people and teams to uncover insights and become faster and more confident decision-makers, ultimately leading to better, data-driven outcomes. In other words, Tableau Online is the power of Tableau in Cloud.

Tableau Cloud is popularly also known as **Tableau Online**. Hence, these terms might be used interchangeably in this book.

Data and analytics in Cloud

Cloud technologies have revolutionized every aspect of the business pattern. Due to this data and analytics on Cloud, organizations can run fast and scale seamlessly as per their needs to grow. The modern and central way to perform data analytics in Cloud is to reduce maintenance costs, increase performance and security, and make data more accessible anywhere. Thanks to *Tableau Cloud*, individual organizations can do visualization and spark intuitive insights with any data, anywhere, at any time. All are easily accessible from a browser or on the go with your mobile apps. [Figure 1.3](#), gives a brief understanding of Tableau Cloud:

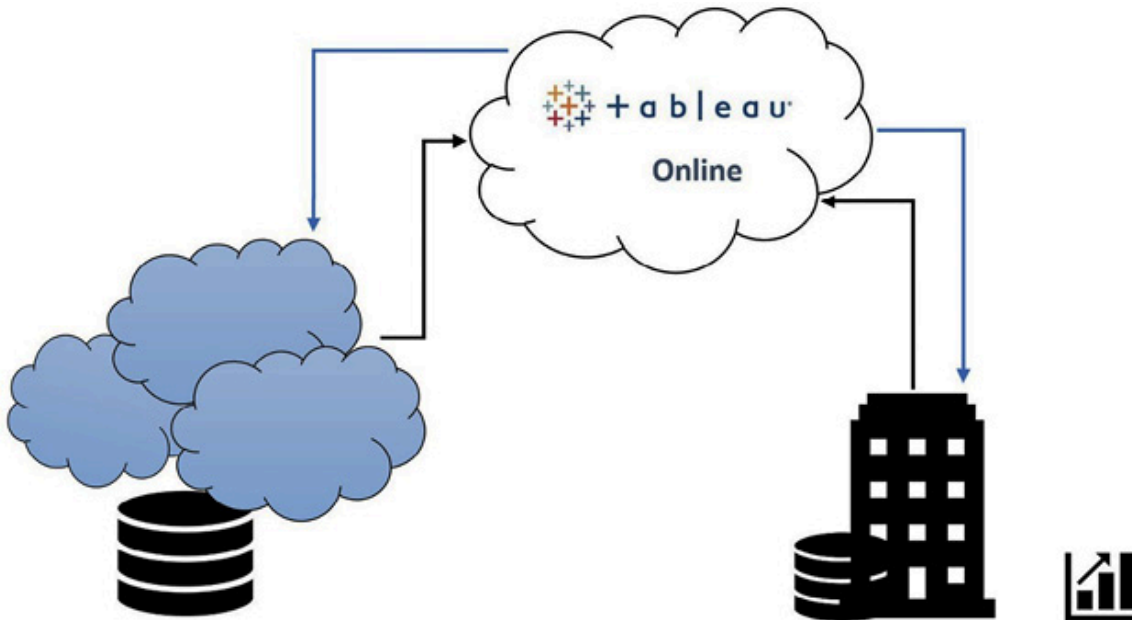


Figure 1.3: Understanding Tableau Cloud *Capabilities and scope of Tableau Cloud*

So far, we have covered a basic understanding of Tableau Cloud. Now, we will unleash its capabilities. We will understand and look at the capabilities plus scope of Tableau Cloud:

- **Skip the setup time and hardware cost:** Tableau online is a fully hosted solution, so the user has no need to configure the server, manage software upgrades, or scale hardware capacity.
- **Interact, edit, and author on the web:** Gives people the power to explore the interactive dashboards from anywhere. You can drill down, highlight, filter, and even edit on the web, and the authors can publish new workbooks.
- **Connect to the data anywhere:** Automatically refresh data from the web apps. Live query on-premises databases or schedule the extract refreshes.
- **Dashboard starters:** In just a few clicks, you will be able to connect to any web applications and create pre-built dashboards. This starter is designed to help run your data in a few minutes.
- **Empowering organization with data:** To make data-driven decisions from the dashboards, users can create and share their Viz data sources with the organization.
- **Stay secure, by design and by default:** Empower site admins to easily manage security, permissions, and authentication for the users, contents, and data.
- **Access anywhere, embed everywhere:** External users can securely access analytics, and no virtual private network is required. Easily embed interactive dashboards on the web.
- **Always up to date:** There is no downtime. It is no longer needed to manage the updates and patches deployment.

Overview of Tableau architecture

Tableau Cloud is a large-scale service with a worldwide global reach. It has 16,000+ customer sites with over 3,00,000 users. It is important to note that Tableau Online is fully backed by **Amazon Web Services (AWS)** and its applications.

Many users still wonder whether Tableau Cloud is secured. They wonder if the data is protected or how the data is transferred on the tool. Let us have a close look at the security perspective.

Security in Tableau Cloud

Inbound and Outbound communication are encrypted in transit, with inbound communication encrypted by modern Hypertext Transfer Protocol Secure (HTTPS) and outbound communication depending on customers' data source capabilities. You still must be wondering whether your data (including Viz., data, etc.) are protected or not. Yes, all permanent data stores and **Elastic Block Storage (EBS)** volumes are encrypted at rest. Most importantly, Tableau Online is backed by AWS. Hence, the infrastructure security leverages **Virtual Private Cloud (VPC)** security groups and access controls.

There is continuous ongoing compliance and analysis going on at the backend, which includes security assessments at regular intervals (internal and external), security reviews, static analysis, and complaints with *EU-US* privacy shield certifications, SOX, and SOC2.

Now, you might think of what will happen in case the server is down or fails. There might be confusion regarding the loss of data. To address all such questions, let us understand Tableau Online architecture as shown in *Figure 1.4*:

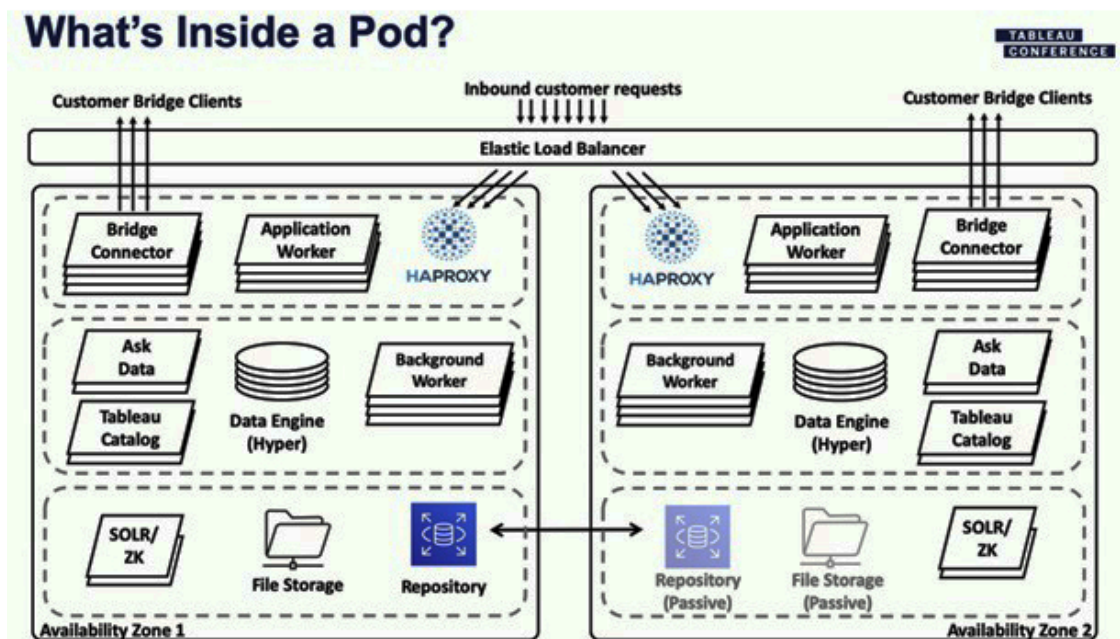


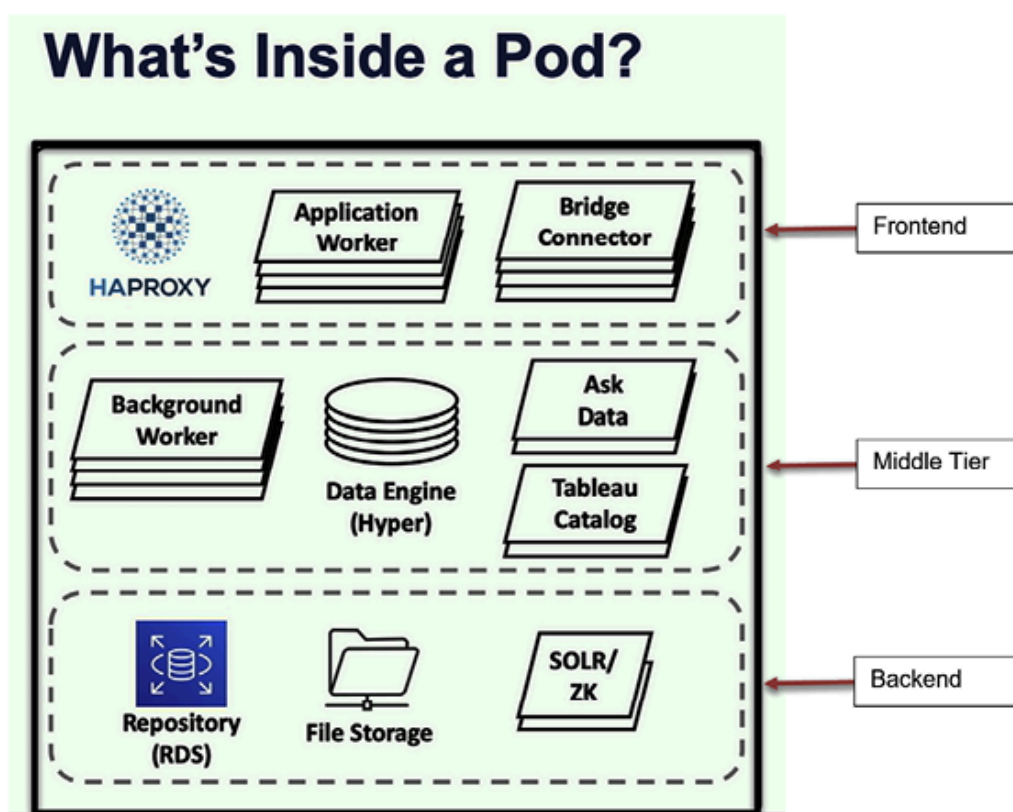
Figure 1.4: The detailed Tableau Online architecture Source: <http://www.tableau.com>

Every customer request comes from the **Elastic Load Balancer (ELB)**. This ELB automatically distributes the traffic across the availability zones. The great thing is high availability & redundancy.

The availability zones are mirror images of each other. They have separate data centers and power. Even if one fails, the other is always ready to fulfill the customer's request. Only one file storage and repository run actively at a given time. The replication process (i.e., streaming happens all the time in the background) is continuously set up between two **availability zones (AZs)**. If any AZs fail, then backup is set, and things continue to run smoothly.

Let us zoom in on one of the AZs and explore what is inside the **Point of Delivery (POD)**.

This AZ is divided into three tiers with various application services, as shown in the following figure:



Source: www.tableau.com

Figure 1.5: Three tier architecture Source: <http://www.tableau.com>

- Front-end services
- Middle services
- Back-end services

Let us look at these tiers in detail:

Front end tier

Front end tier comprises of:

- Ingress/Egress for POD
- Routes to middle tier and backend applications

This tier includes HAproxy, application worker, and bridge connector.

- **HAproxy:** This load balances traffic to application workers within the availability zone and leverages the caching experience with Tableau online. Hence, it maintains the sticky sessions.
- **Application worker:** It handles all the user/traffic. It leverages AWS Elastic Cache in-memory and can search, edit, explore, and publish the Viz.
- **Bridge connector:** It can connect to data sources on premises. It serves as a bridge connection for live or extracted queries from on-premises data sources.

Middle tier

Middle tier performs the following functions:

- It takes the request to the backend data server.
- It collects a lot of information about data

This tier includes background workers, data engine, ask data, and Tableau catalog:

- **Background workers:** This helps optimize workloads and scale optimization. The background worker works in five different roles (Extract refreshes, alert, subscription, system jobs, and flows). These five roles are independent of each other. Nobody waits until the other completes its role.

For example, if you schedule an extract refresh and, in the meantime, the user does any subscription, then the background worker will not wait for the extract refresh to complete. It will do its own job.

- **Data engine:** The Data Engine, popularly known as **Hyper**, is a fast analytical query processing engine. It can be deployed on standalone nodes and optimizes query workloads.
- **Ask data:** This is a newly popularized application. It has the power to ask data capabilities. It integrates with managed Elasticsearch.

- **Tableau catalog:** It has a role in pulling in metadata form from workbooks, flows, data sources, and so on. It queries role to service queries from workers.

Backend tier

Backend tier performs the following functions:

- This database tier is set up in such a manner that data can be backed up every 24 hours.
- It has an automatic failover/replication, which makes the Tableau Online product robust.

This tier includes Repository, File storage, and SOLR/Zookeeper:

- **Repository:** It runs on AWS RDS PostgreSQL. It dynamically scales up the instance size to accommodate the growth.
- **File storage:** It is a network storage file that is easily scalable and replicable.
- **SOLR/Zookeeper:** It is known as a search service and coordination service. It has three instances across the POD for a quorum.

In the next section, we will examine different types of licenses.

Licensing

Tableau's powerful analytics platform empowers everyone across the organization with the data. Depending on the user's needs, they can select the deployment options that give them accessibility and permissions.

Let us explore those licensing options from the following table:

License	Purpose	Includes	Cost
Tableau Creator	This role gives privileges to perform end-to-end analytics workflow to discover insights with powerful suites of products.	This license includes Tableau Desktop, Tableau Prep Builder, and one creator license on Tableau Cloud.	\$70/per month
Tableau Explorer	They can explore the data with full self-service analytics; they have access to the subset of creator capabilities. Can add new content based on the existing published data sources and workbooks.	This license includes one explorer license of Tableau Cloud.	\$42/per month

Tableau Viewer	User can view and interact with dashboards and visualization in a secure platform but cannot edit the workbook.	This license includes one viewer license of Tableau Cloud.	\$15/per month
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Table 1.2 : Licensing in Tableau Cloud

Note: Every deployment requires at least one Creator.

It is important to select the deployment option wisely to meet the organization's needs and the mix of user types.

After gaining knowledge on licensing, follow these steps to download the Tableau Cloud.

Downloading Tableau Online

Let us download the trial version (14 days) of Tableau Cloud to get hands-on experience with this amazing tool. In this section, you will learn how to install/setup the Tableau Online.

Follow these steps to get started:

1. Go to the official Tableau (Link: <https://www.tableau.com/products/cloud-bi>).
2. On the top right page, you will find a **TRY NOW** button, as seen in the following figure, and click on it.

Or,

You can click on the **Start A Free Trial** button as shown in the following figure:

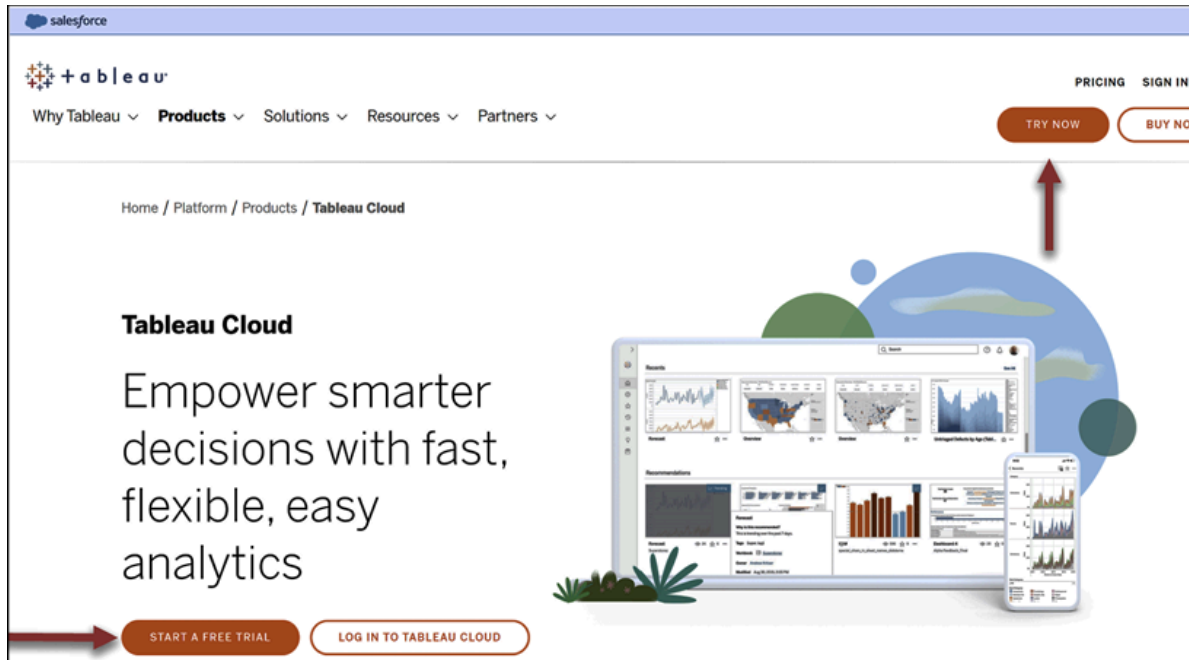


Figure 1.6: Download free trial

3. You will be redirected to a new page. Click on **Start A Free Trial** button, and you will be asked to enter the details and email as shown in the following screenshot:

The image shows the Tableau Cloud sign-up form. The form is titled 'Let's sign you up (or sign in to Tableau Cloud)'. It includes a 'CLOSE' button in the top right corner. The form fields are: 'First Name' (with a red error message 'First Name field is required.'), 'Last Name', 'Business E-mail', 'Organization', '- Company Size -', and '- Department -'. There are also 'TRY NOW' and 'BUY NOW' buttons in the top right corner.

Figure 1.7: Tableau Cloud sign in form

4. After entering the details, click on **Request A Free Trail** option. This can be seen in *Figure 1.8*:

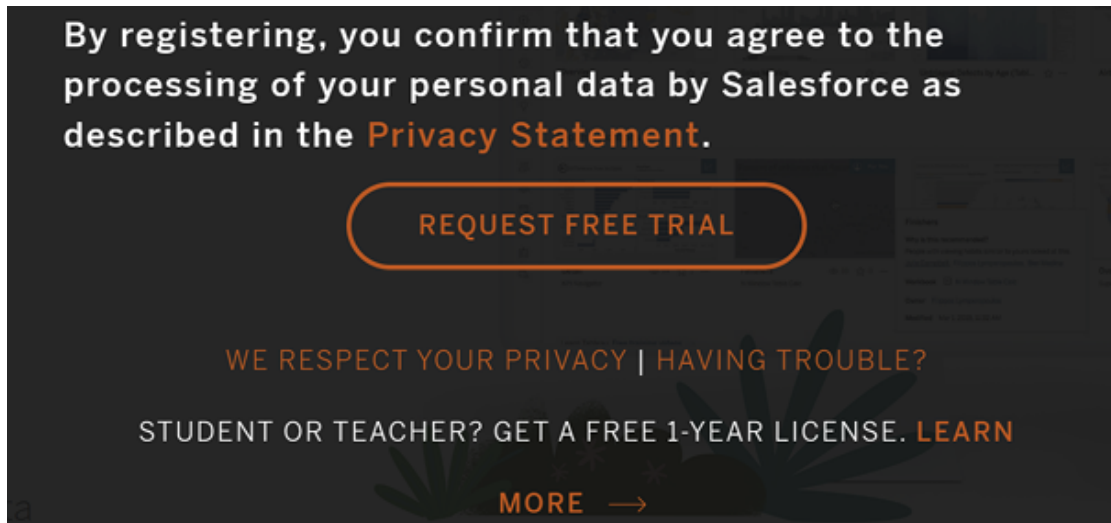


Figure 1.8: Request Free Trial

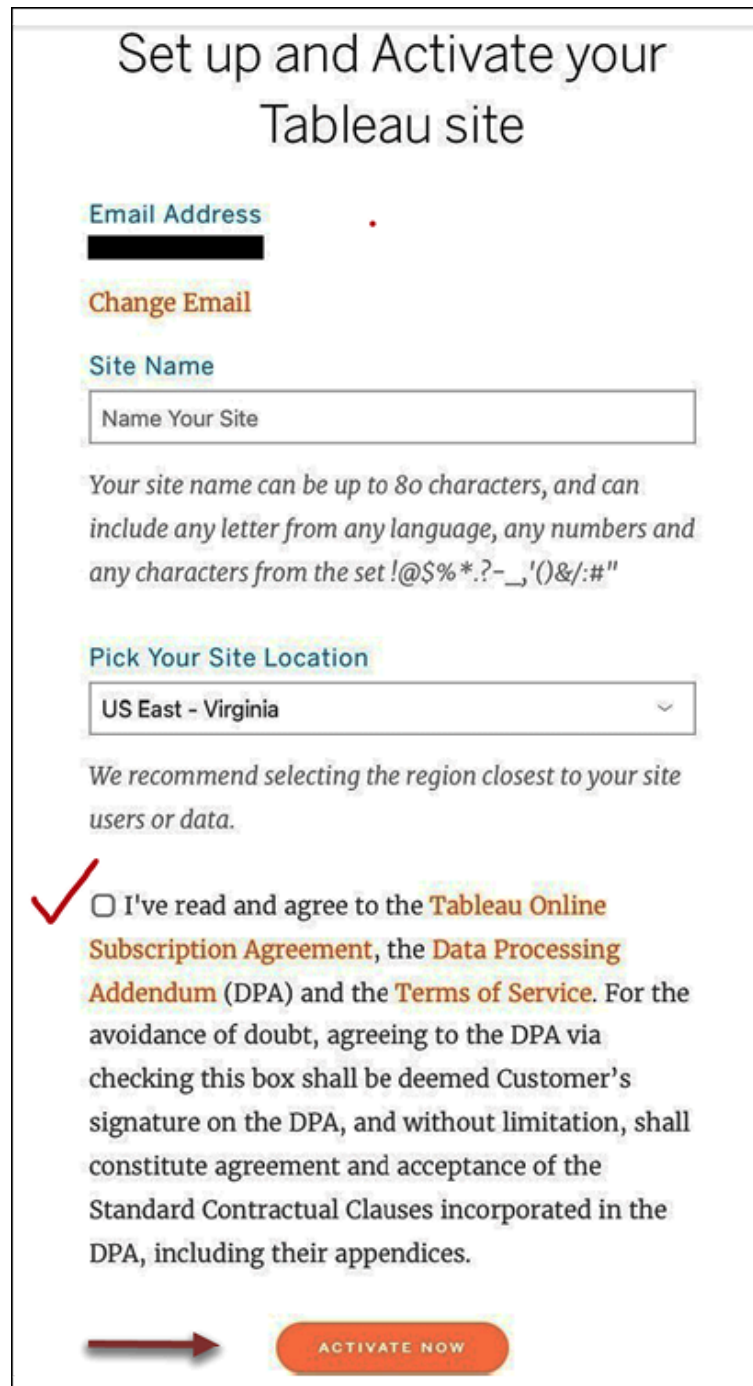
5. Next, you will receive an email in your inbox within five minutes for verification and activation of the Tableau trial version.
6. Verify your address as shown in the following figure:
7. Click on **Activate My Site**:



Figure 1.9: Activate My Site You will get 48 hours to activate the trial.

8. A browser window will open on a new page.

9. You need to enter a few more details in the fields provided on the page, as shown in the following figure. Once you enter all the details, check the terms and conditions box.
10. Finally, click on the **Activate My Site** button:



The screenshot shows the 'Set up and Activate your Tableau site' page. It includes an 'Email Address' field with a red error dot, a 'Change Email' link, a 'Site Name' field with the placeholder 'Name Your Site', and a 'Pick Your Site Location' dropdown menu set to 'US East - Virginia'. A red checkmark is placed to the left of the terms and conditions checkbox. The 'ACTIVATE NOW' button is highlighted with a red arrow.

Set up and Activate your Tableau site

Email Address .

[Change Email](#)

Site Name

Name Your Site

Your site name can be up to 80 characters, and can include any letter from any language, any numbers and any characters from the set !@\$%*.?~_'()&/:#"

Pick Your Site Location

US East - Virginia

We recommend selecting the region closest to your site users or data.

☒ I've read and agree to the [Tableau Online Subscription Agreement](#), the [Data Processing Addendum \(DPA\)](#) and the [Terms of Service](#). For the avoidance of doubt, agreeing to the DPA via checking this box shall be deemed Customer's signature on the DPA, and without limitation, shall constitute agreement and acceptance of the Standard Contractual Clauses incorporated in the DPA, including their appendices.

 [ACTIVATE NOW](#)

Figure 1.10: Set up and Site Activation

11. Congratulations! You have successfully setup the **Tableau Online** platform:

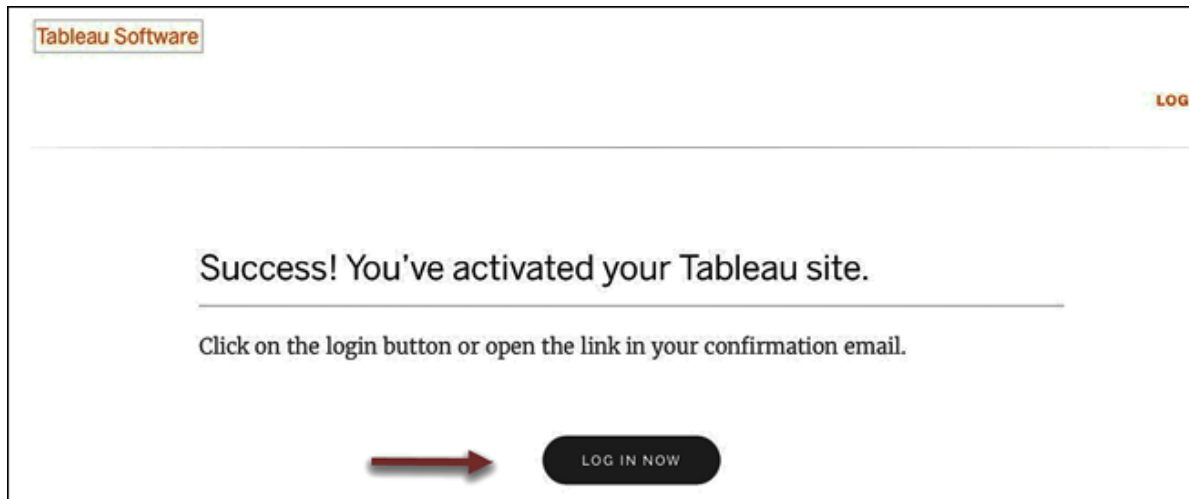


Figure 1.11: Site Log In

12. The following figure shows the home page of Tableau Online:

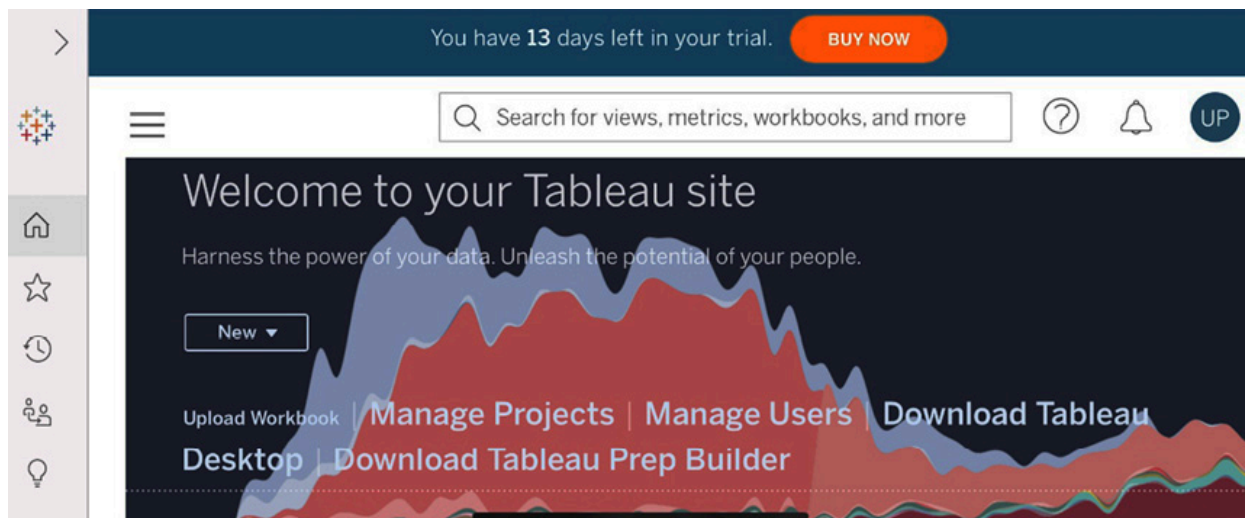


Figure 1.12: Home Page of Tableau Site Please note that this will only be a 14-day trial version.

You will receive an email on the successful setup, as seen in the following figure:

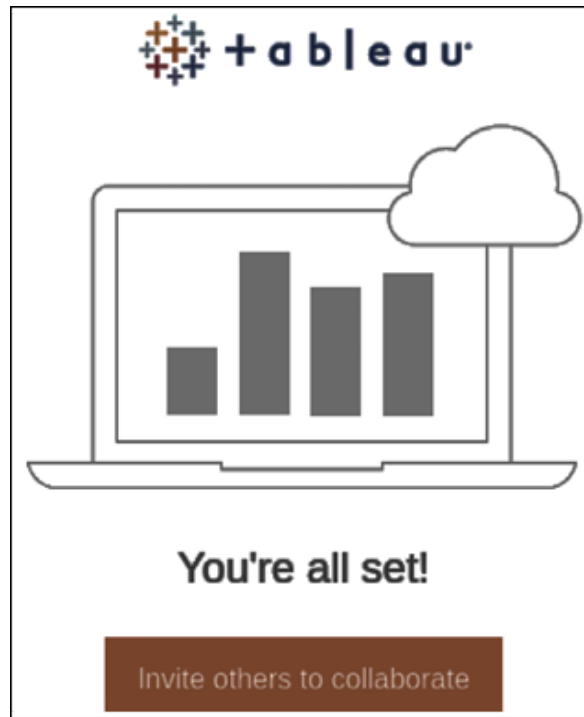


Figure 1.13: Successful setup You can connect 40+ data sources and load the data to explore, analyze, and get findings from the data.

Note: It is mandatory to install Tableau Desktop or Tableau Public in your laptop.

Sign-in Tableau Online

To access the content on Tableau Cloud, you can sign in via web browser, Tableau desktop, Tableau mobile app, or Tableau Bridge.

However, you must set up your Tableau Online site in one of the following ways to authenticate:

- **Single sign-on (SSO)** : SSO means your admin must have set up Tableau Online so that you can enter the same username (email address) and password for the other applications in your company or for your client.
- **Multi-factor authentication (MFA)**: MFA is made up of a username (email address), password, and verification methods to access Tableau Online. This is one of the most secured ways to authenticate.

For this, you must register for multi-factor authentication and choose the verification method: Register MFA:

URL:

https://help.salesforce.com/s/articleView?id=sf.mfa_supported_verification_methods.htm&type=5

Let us explore *four* different ways to sign in:

- **Sign-in from Tableau Desktop**

Signing in to Tableau Cloud via Tableau Desktop is one of the easiest and most widely used by developers, BI analysts, or users.

Follow these steps to sign in successfully:

1. In the Tableau Desktop, select **Server|Sign in**.
2. Click on **Tableau Online** in the bottom left corner or enter the Tableau Online URL: <http://online.tableau.com>
3. Enter the username (email address) and password/your credentials you have used for Tableau Online for authentication.
4. Click on Sign-in.
5. This method of sign-in from Tableau Desktop is used to access or publish any content on Tableau Online.

Note: Tableau Desktop will keep you sign-in from session to session if you do not sign-out.

- **Sign in from web browser**

Follow these steps to sign in from web browser:

1. Enter the Tableau Online URL in the address bar.
2. URL: <http://online.tableau.com>.
3. Enter your credentials/username (email address) and password for site authentication.
4. Click on **Sign-In**.
5. This method is used to sign in directly to the web browser.

- **Sign-In via mobile app**

Follow these steps:

1. Enter your credentials/username (email address) and password.
2. Click on **Sign-In**, then tap to connect to Tableau Online via Tableau Mobile app.

- **Sign-In via Tableau Bridge**

Follow these steps:

1. In Tableau Desktop, select **Server -| Start** Tableau Bridge Client.
2. This Tableau Bridge will help to connect to Tableau Online.

Touring the Tableau Online UI

When you sign into the Tableau Online site, the first page looks like the following figure:

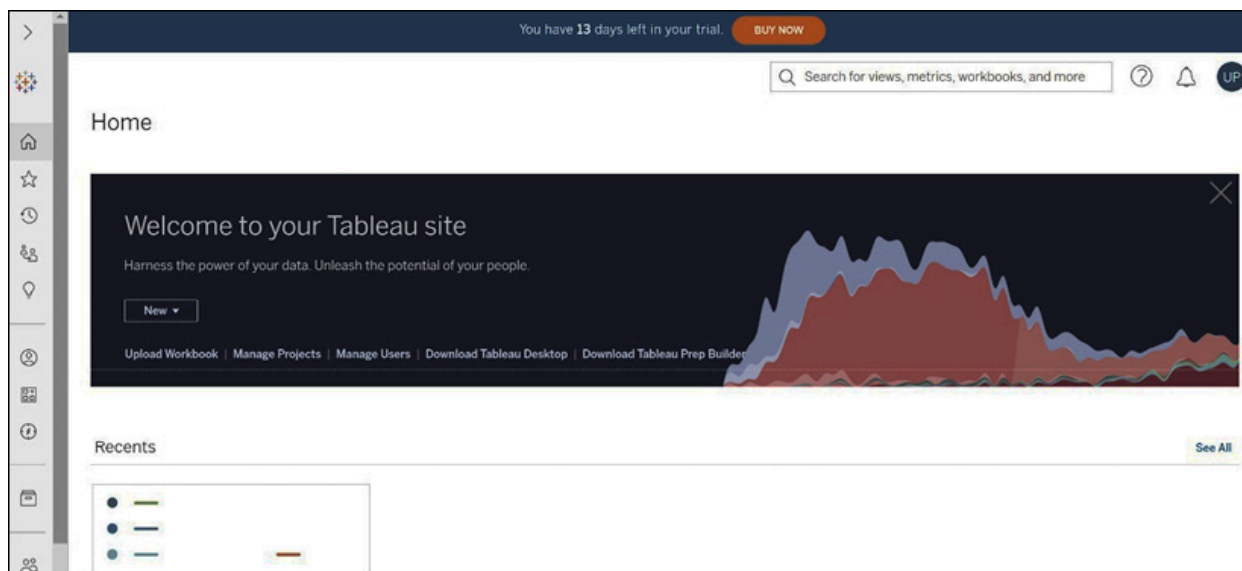


Figure 1.14: Touring Tableau UI

Note: You might see few additional features/functionalities if your site is licensed for data management, virtual connections, etc.

Let us start exploring the Tableau site.

Follow these steps to browse and find the content:

1. In the left-most navigation menu, click **Explore** as shown in *Figure 1.15*:

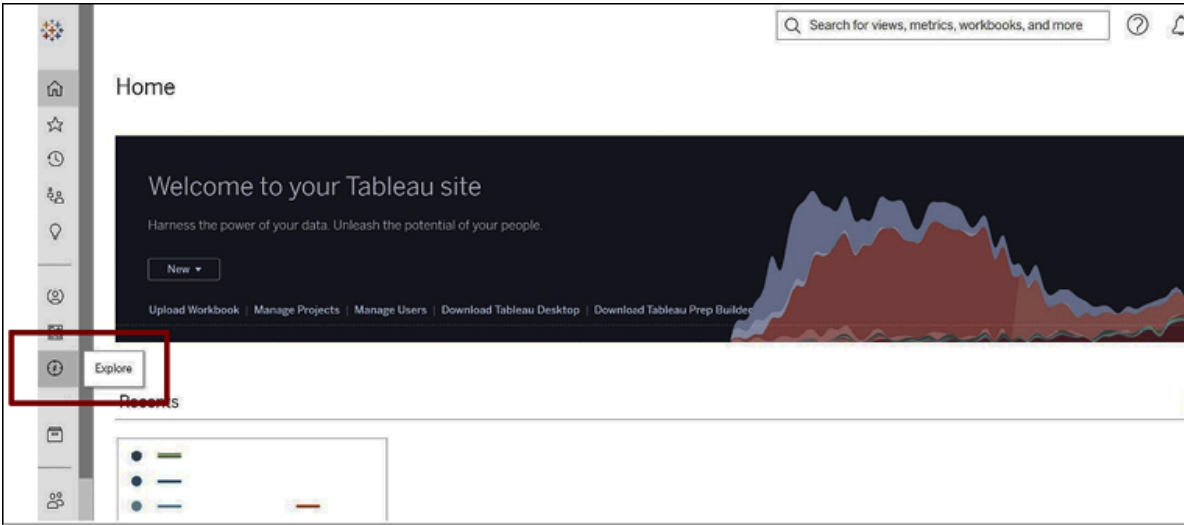


Figure 1.15: Explore

2. Once you click on **Explore**, you will see all the contents you have the access of across the Tableau site as per your site role and permissions:

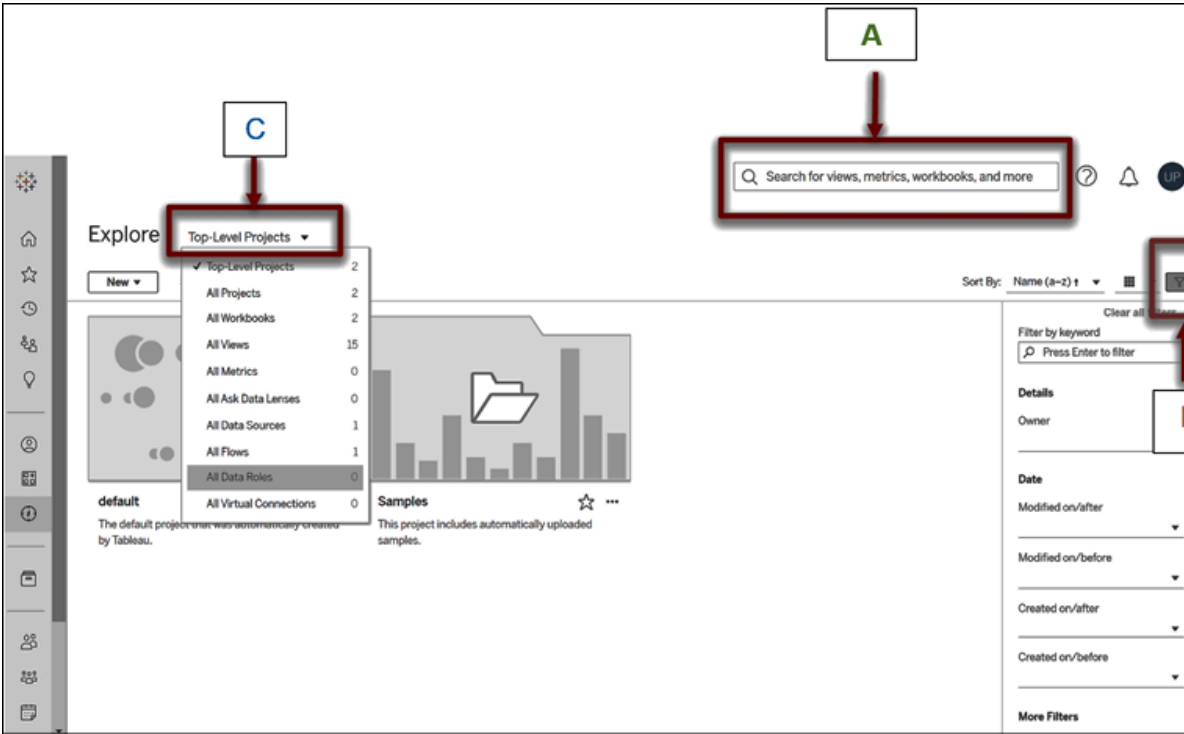


Figure 1.16: Explore page Highlighted in the figure are the following details:

- **A** - Quick search
 - **B** - Your content settings and quick filtering
 - **C** - Content type menu, at what level content to be viewed
3. By default, explore shows **Top-Level Projects**; select the type of content from the drop-down menu to view the contents at the specific level.

Let us look at them in detail:

- **Projects:** When you select **All Projects** from the drop-down menu, you will see the list of projects in Grid View. If you select the specific project, then contents within the selected project, including nested projects, will be seen.

In the following **Samples** is one of the projects. By selecting this project, you will see the contents (including nested projects, workbooks, etc.) within it:

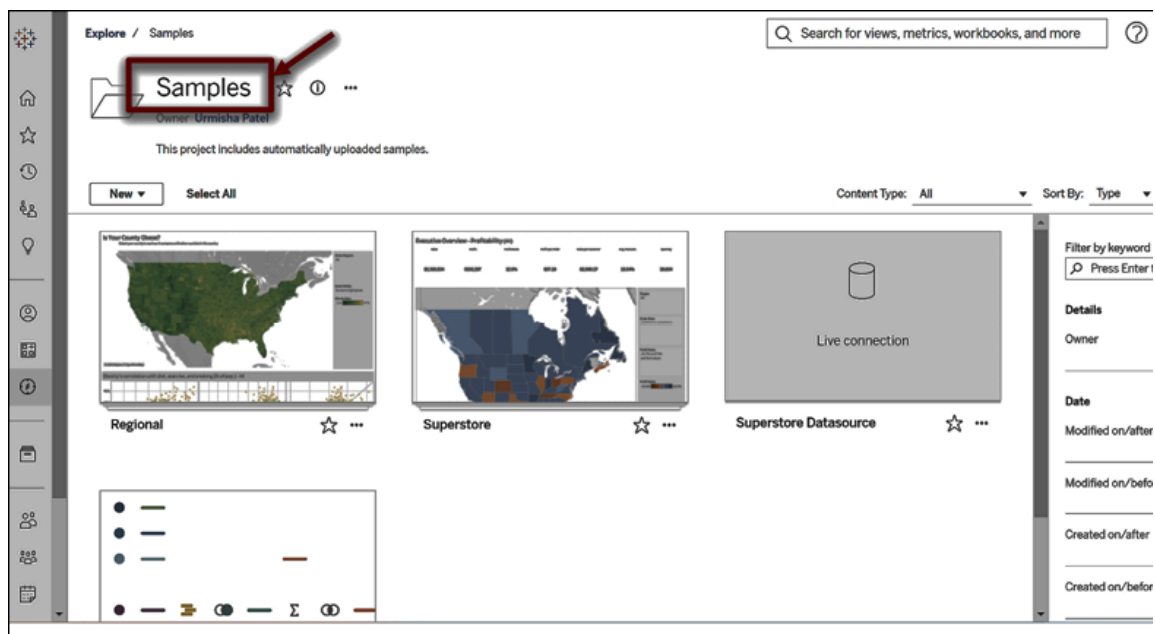


Figure 1.17: Sample page

- **Workbooks:** When you select **All Workbooks** from the drop-down menu, you will browse to the workbook page. This page includes views, connected metrics, user subscriptions, data sources, lineage, and custom views in the workbook.

The following figure gives a glimpse of the workbook page:

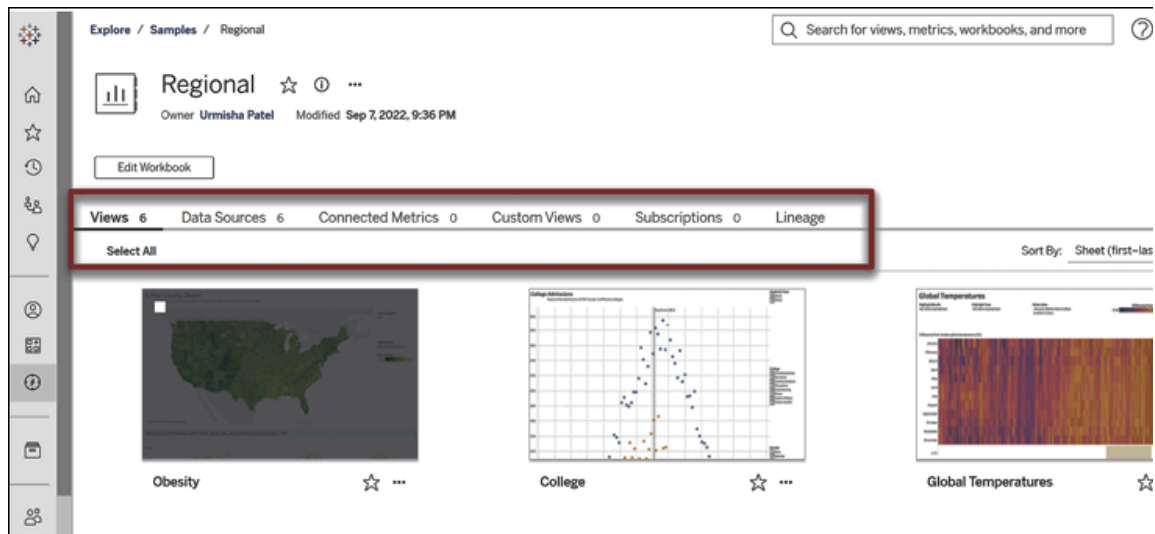


Figure 1.18: Workbook page

- **Views:** Select **All** Views from the drop-down menu. The View page allows us to interact with the view, including sharing, downloading, subscription, alerts, editing, etc.

The Options available in the toolbar vary depending on the site configuration and permissions, as seen in the following screenshot:

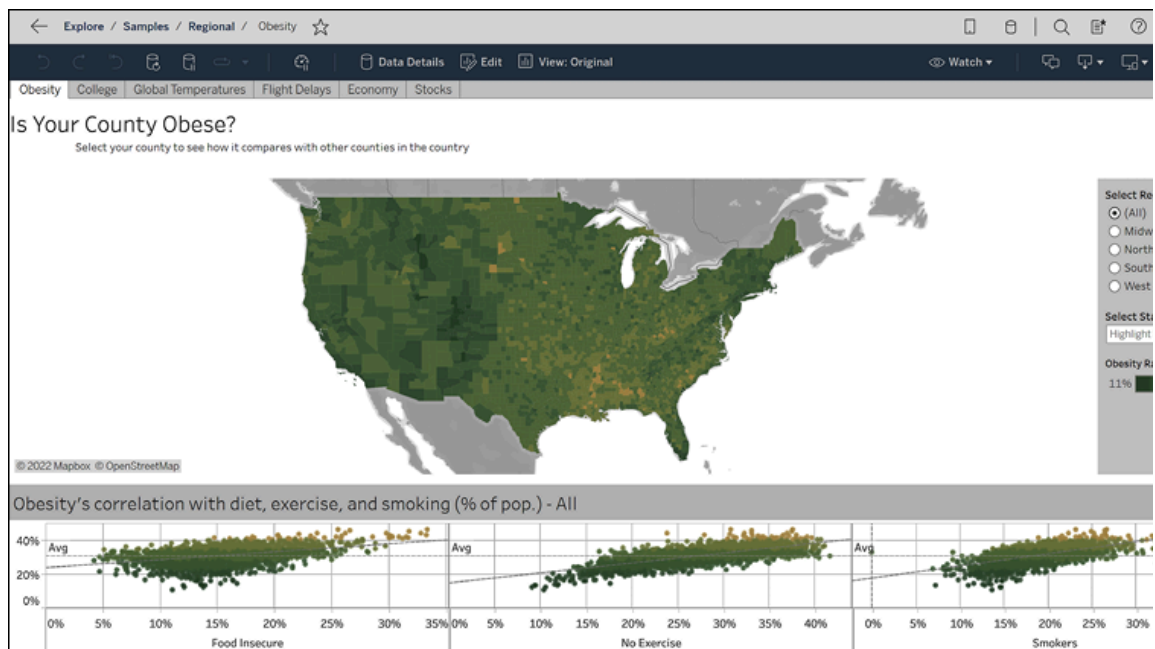


Figure 1.19: Views page

- **Metrics:** Select **All Metrics** from the drop-down menu. The metrics page will display the historical values for the metric; this includes the time the

metric last refreshed, and details about the metric such as description, owner, project, definition, description, last modified, and so on.

The metrics page is available once you start connecting with the data source or scheduling the data extracts.

- **Data sources:** Choose **All data sources** from the drop-down menu. The data source page shows the connection to database or file-based data, as well as the workbooks connected to the data source. If the data source is an extract, it will also show the scheduled extract refreshes. Refer to the following figure:

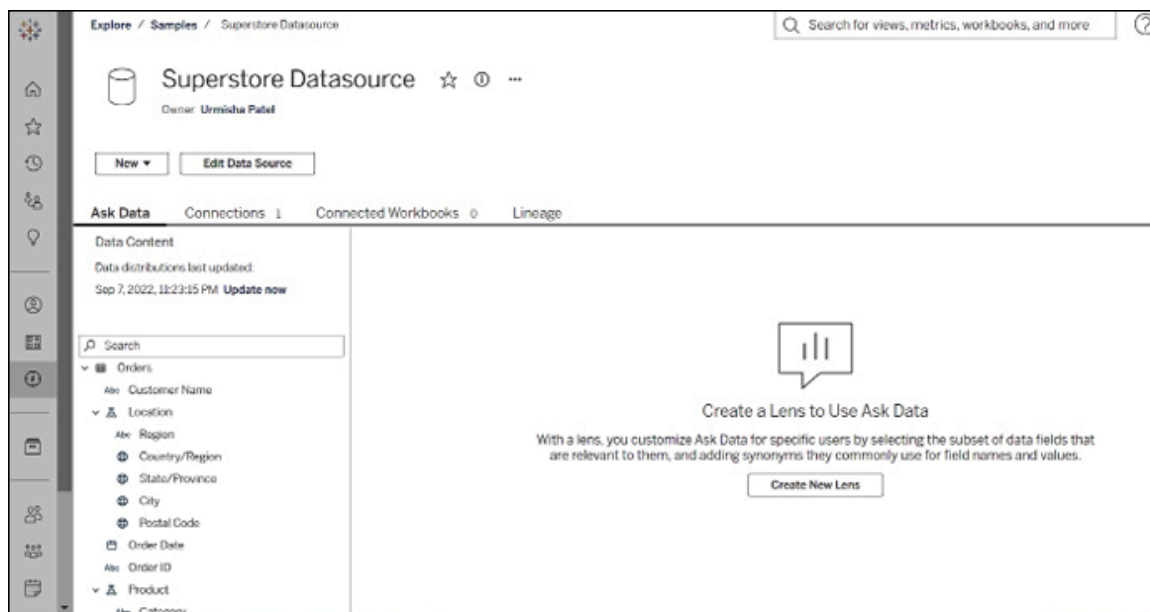


Figure 1.20: Data source page

- **Ask Data lenses:** Choose **All Ask Data lenses** from the drop-down menu, this page will give privilege to easily and quickly go through the data from the data source.

If **Ask Data** lenses are not available, we can create Ask Data lenses from the Data Source page as well. This page can be seen in the following figure:

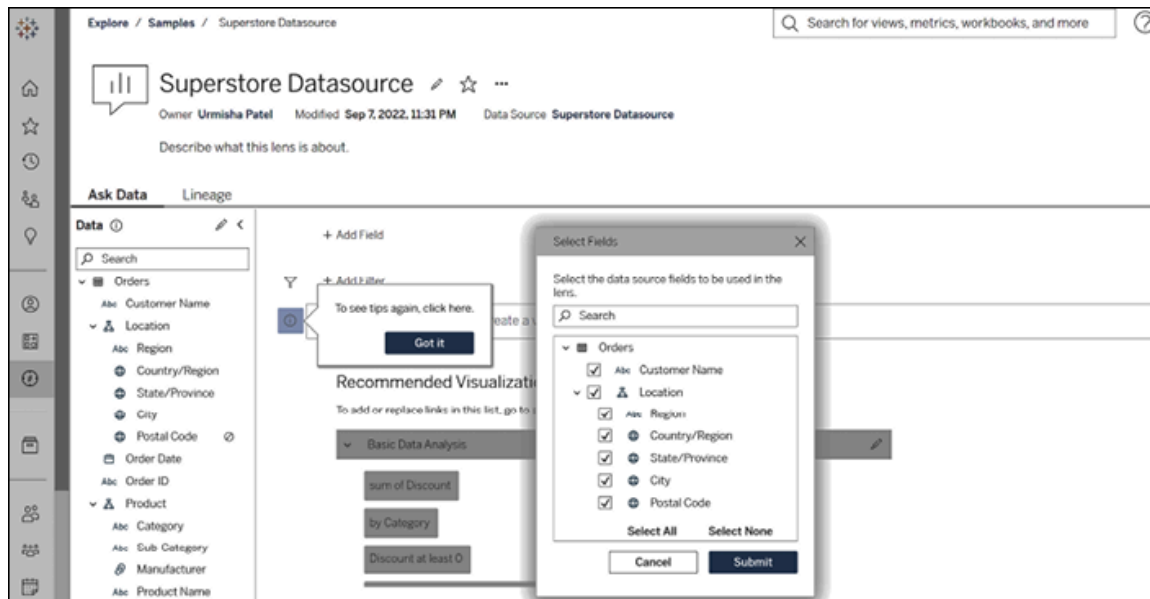


Figure 1.21: Ask Data lenses page

- **Data roles:** Choose **All data roles** from the drop-down menu. This page will help the user validate the field values when cleaning the data. You can create custom roles from the Tableau Prep-builder using the field values in your dataset and publish them on Tableau Online.
- **Flows:** Select **All Flows** from the drop-down menu. This page enables you to edit, view, monitor, and maintain the flow. This is linked with data source, data connection and data management.
- **Virtual connection:** Select all virtual connections from the drop-down menu. The virtual connection will work at row-level security and provide the central access point to data. You can use a virtual connection to create a data source, workbook, or flow.

View content as a list or grid

You can display and view the content by selecting either the thumbnails or a list view by clicking on the View As icon, as shown in the following figure:

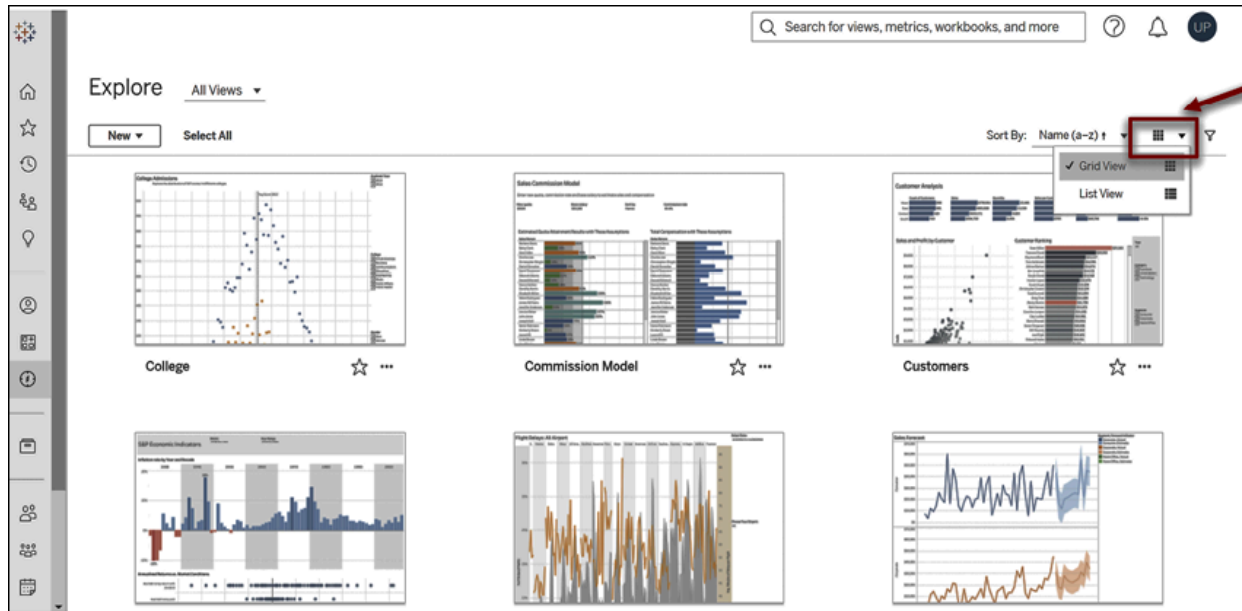


Figure 1.22: View content

- **Sort content:** To sort items, click the **Sort By** drop-down menu. You can sort the content based on name, viewed by users, workbooks, sheet, owner, modified, locations in ascending or descending order. You can see this illustrated in [Figure 1.23](#):

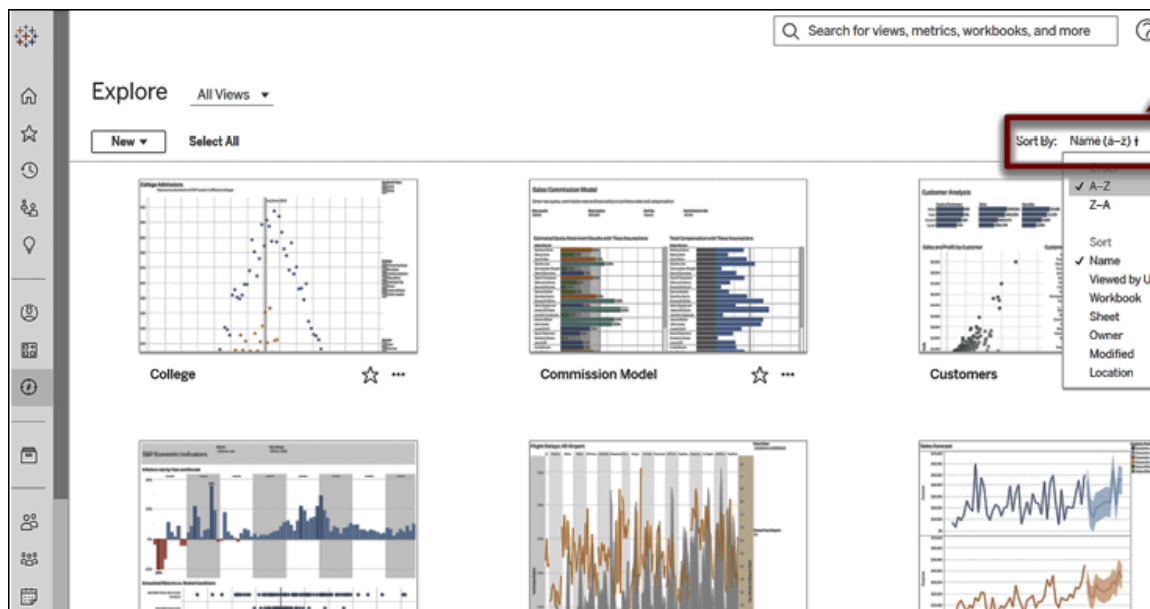


Figure 1.23: Sort content

- **Access important content quickly:** Tableau is smart enough to keep a track of the items that are important to you. Hence, in addition to browsing and

exploring the site, you can also take advantage of quick content access. This can be seen in [Figure 1.24](#):

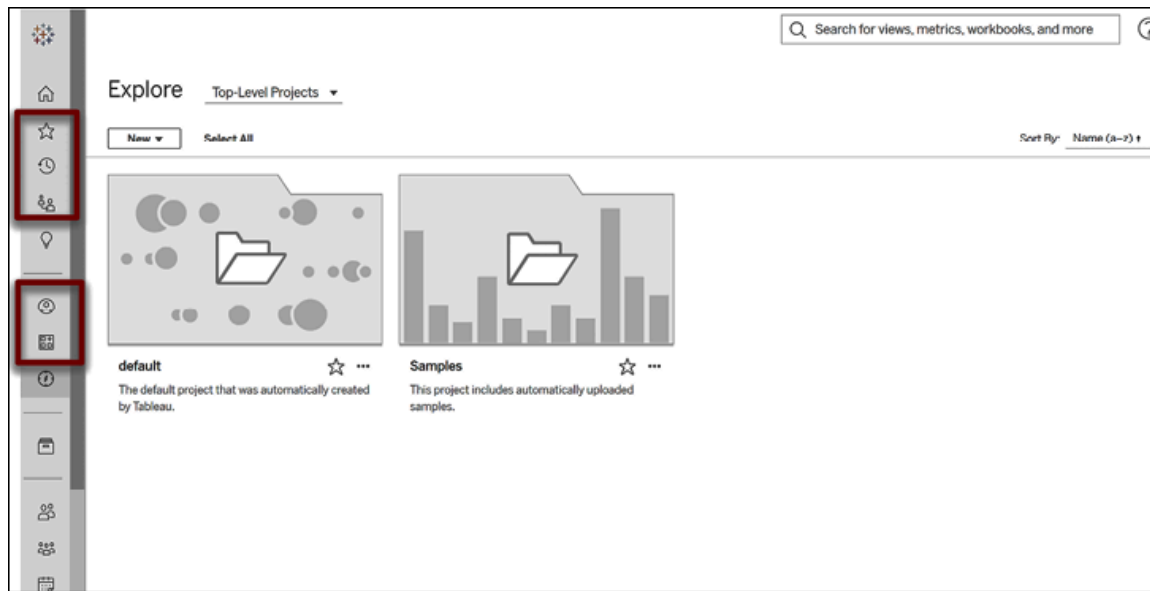


Figure 1.24: Quick access

The highlighted five icons on the leftmost side of the Tableau site are the important quick content access icons:

- **Favourites:** You can add any type of content to your favorite.
- **Recent:** You can see the recently accessed files, the views, metrics, flows, and workbooks.
- **Shared with me:** You can find the contents/items shared with you by other users.
- **Collections:** You can create custom collections of related items.
- **Personal space:** A private location where creators and explorers can edit or save their work.

Conclusion

In this chapter, we explored in detail Tableau Cloud after a high-level introduction to data visualization and its importance in a data-driven business world. It is vital to explore the dataset, extract the information, to express and identify the vital relationships in charts and plots that are more helpful for businesses. Tableau Cloud is one of the many Tableau products on the Cloud, which is a secured, fully

hosted, cloud-based solution for sharing, authoring, distributing, and collaborating content created in Tableau.

In the next chapter, you will learn how to connect your data on the web in live or extract data connections. We will understand how to manage the data connections.

Multiple choice questions

1. Select the appropriate answer: What are the different types of Tableau Products?

- a. Tableau Desktop
- b. Tableau Mobile
- c. Tableau Online
- d. All of the above

2. Select the appropriate answers: Into which tiers the availability zones are divided?

- a. Middle tier
- b. Application tier
- c. Front tier
- d. All of the above

3. Select the appropriate answers: What does the art of presenting data in a graphical format, including graphs, plots, colors, dimensions, charts, and labels, called?

- a. Data art
- b. Data visualization
- c. Data analysis
- d. All of the above

4. Select the appropriate answers: What is the fully hosted, Cloud-based, enterprise-grade solution in the analytics platform called?

- a. Tableau Desktop
- b. Tableau Online
- c. Tableau Prep
- d. Tableau Mobile

Answers

1.	d
2.	a,c
3.	b
4.	b

Questions

1. How is inbound and outbound communication encrypted in transit?
2. What are the different ways to sign in to Tableau Online?
3. How can important content be accessed quickly in Tableau Online?
4. What are the features of Tableau Online?
5. What is the difference between Tableau Creator and Tableau Viewer roles?

Points to remember

Here are the key takeaways from this chapter:

- Data visualization is the art of presenting the data in such a way that a layperson can also understand it. It could be the perfect blend of charts, graphs, colors, dimensions, and labels, which can create the visuals as a

masterpiece, hence revealing interesting facts or insights, which in turn helps businesses to make informed decisions.

- At present, there are a lot of visualization tools available in the market, and among them, Tableau is the most popular and fastest-evolving **BI** and data visualization tool.
- Tableau offers various services according to business needs and type of work. There are seven various types of services. They are **Tableau Desktop, Tableau Cloud, Tableau Public, Tableau Reader, Tableau Server, Tableau Prep, and Tableau Mobile**.
- **Tableau Cloud** is a secure, fully hosted, Cloud-based solution for sharing, authoring, distributing, and collaborating on content created in Tableau and empowers you to make smarter decisions with fast, flexible, and easy analytics.
- Capabilities and scope of Tableau Cloud
 - Interact, edit, and author on the web
 - Connect to the data anywhere
 - Dashboard starters
 - Empowering organizations with data
 - Stay secure, by design and by default
 - Access anywhere, embed everywhere
 - Always up to date
 - Skip the setup time and hardware cost
- To protect the data in Tableau Online, the Inbound and Outbound communications are encrypted in transit, where inbound communication is encrypted by modern HTTPS, and outbound communication depends on customers' data source capabilities.
- Depending on the needs, the user can select the deployment options which give them accessibility and permissions. There are three types of licenses. They are:
 - Tableau Creator
 - Tableau Explorer
 - Tableau Viewer

- You can download the 14-day trial version of Tableau online and start unleashing the power of Tableau Cloud.
- You can set up your Tableau Online with one of the authentication methods: Single sign-on or multi-factor authentication.
- There are multiple ways to sign in Tableau Online. They are:
 - Sign-in from Tableau Desktop
 - Sign-in from web browser
 - Sign-in from the Tableau Mobile App
 - Sign in via Tableau Bridge
- You might see few additional features/functionalities if your site is licensed for data management, virtual connections, etc. in the Tableau Online UI.

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<https://discord.bpbonline.com>



CHAPTER 2

Managing Data Connections

*Errors using inadequate data are much less than those using no data at all -
Charles Babbage [Introduction](#)*

In the previous chapter, we configured the Tableau Online site with the logo and explored its user interface. Now, you can start organizing the content (including data, reports, and more) for you and your users who want to share and publish Tableau data. Depending on the type of license, users can connect and publish the contents (including data, reports, and more) from the Tableau Desktop or Tableau Online via a web editing environment.

Your data is at the center of Tableau. Here, we will focus on connecting your data, managing the data connection, and unleashing Tableau Online capabilities.

Structure

In this chapter, we will cover the following topics:

- Connecting data to the web
- Connection: live versus. extract
- Run Initial SQL
- Extract upgrade to .hyper format
- Creating extracts on the web
- Direct connection to Cloud data
- Configure connections with analytics extension
- Editing connections on Tableau Online

Objectives

This chapter will familiarize readers with managing the data connection (including Cloud base data sources, on-premises data, and so on). They will learn to connect your data, share and publish it with you and your business users, and so on.

After reading this chapter, as readers, you will be able to start managing their data connection on the Cloud with various data sources, either live or extracted. Using accelerators, you can quickly start analyzing your data with pre-built-in dashboards.

Connecting data to the web

Before we create a workbook and start building different Viz to analyze your data on the web, first, you need to connect to your data. Tableau supports connecting the data sources on the web and publishing through Tableau Desktop, or directly connecting the data through Tableau online, server, or public.

Here, we will explore how to connect data to the web directly through Tableau Online.

Follow these steps:

1. Once you sign in the Tableau Online, you can connect the data directly on the web in two ways:

- a. **Home | New | Workbook** Or

- b. **Explore | New | Workbook**

This can be seen in the following screenshot:

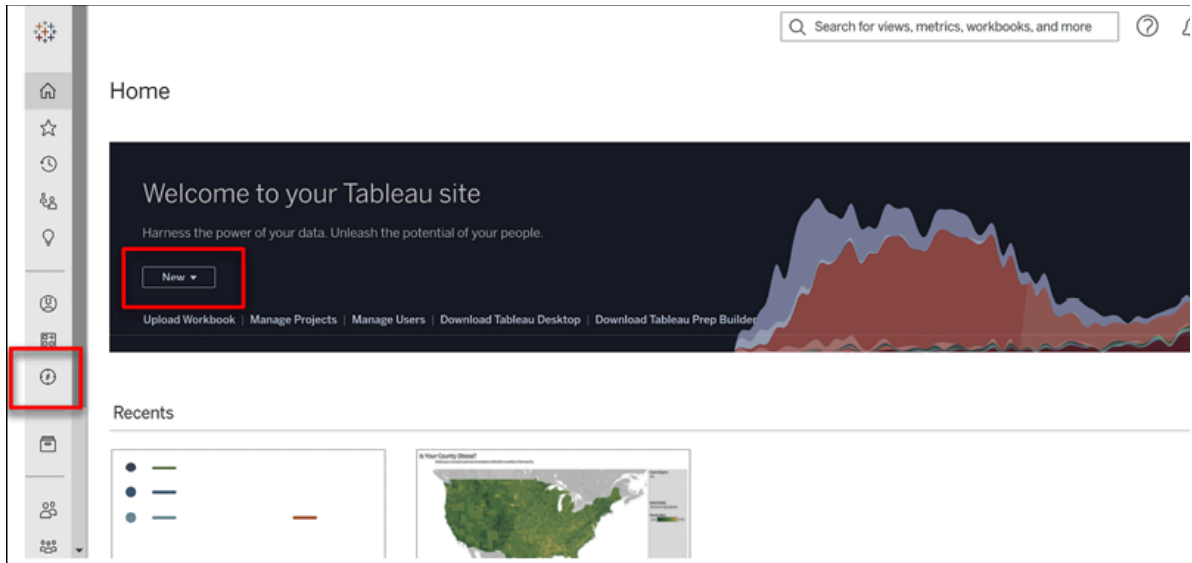


Figure 2.1: Home page of Tableau site

2. To connect your data, select the **On The Site** tab, as shown in the following figure:

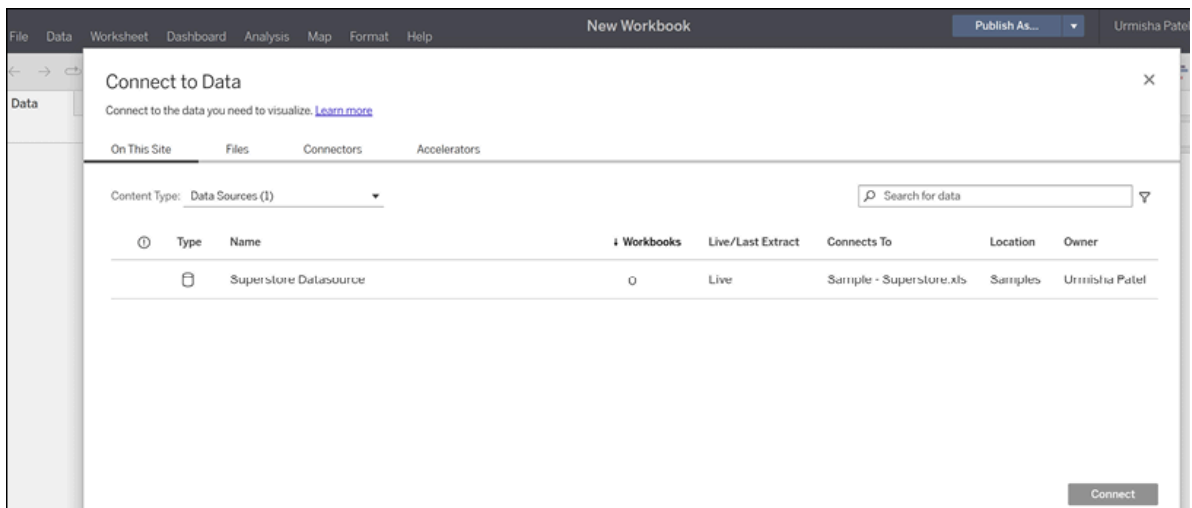


Figure 2.2: Connect to data page

- a. **On This Site**, as shown in [Figure 2.2](#), you can connect or search your data from the existing data source. You can select the data source and click on **Connect** button.

However, you have multiple options to connect to your data.

- b. Click on **Content Type** dropdown menu, and you will find the following options as shown in the following screenshot:

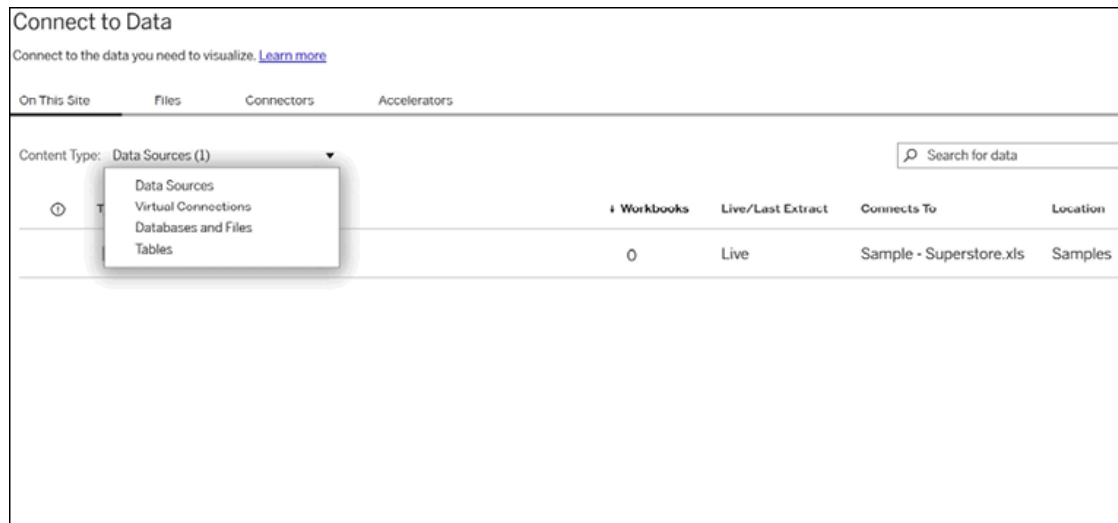


Figure 2.3: On This Site page

- c. If you select **Databases and Files** or **Tables**, you can filter the assets by **Data Quality Warning** (**All Assets**, **Assets without Warnings**, and **Assets with Warnings**).
3. **Files:** Tableau supports uploading data files (**.xlsx**, **.csv**, **.txt**) directly from your device to the browser. In **Files** tab of the **Connect to Data** page, connect excel or text files by simply dragging and dropping into the field or by clicking on **Upload from computer** as shown in *Figure 2.4*:

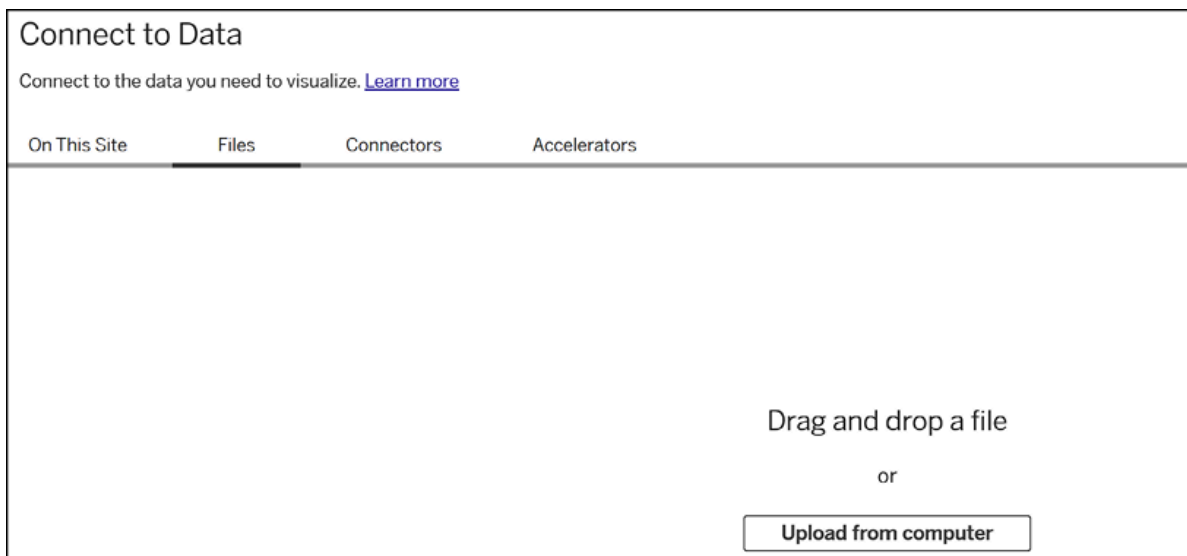


Figure 2.4: Files page

4. **Connectors:** Click on the **Connectors** tab, and you will find the list of Tableau-supported connectors. So, you can be able to connect to the data hosted in the

cloud database or server in your enterprise. Whenever you make any data connection, you just simply need to add the database connection information.

Connect to Data			
Connect to the data you need to visualize. Learn more			
On This Site	Files	Connectors	Accelerators
Alibaba AnalyticDB for MySQL		Azure Data Lake Storage Gen2	Denodo
Alibaba Data Lake Analytics		Azure SQL Database	Dremio
Amazon Athena		Azure Synapse Analytics	Dropbox
Amazon Aurora for MySQL		Box	Esri
Amazon EMR Hadoop Hive		Cloudera Hadoop	Exasol
Amazon Redshift		Databricks	Google BigQuery
Apache Drill		Datorama	Google Cloud SQL
			Google Drive
			Impala
			Kyvos
			Microsoft SQL S
			MongoDB BI Cor
			MySQL
			OData

Figure 2.5: Connectors page For example, if your data belongs to Impala SQL then click on **Impala** as shown in the following figure:

Connect to Data			
Connect to the data you need to visualize. Learn more			
On This Site	Files	Connectors	Accelerators
Alibaba AnalyticDB for MySQL		Azure Data Lake Storage Gen2	Denodo
Alibaba Data Lake Analytics		Azure SQL Database	Dremio
Amazon Athena		Azure Synapse Analytics	Dropbox
Amazon Aurora for MySQL		Box	Esri
Amazon EMR Hadoop Hive		Cloudera Hadoop	Exasol
Amazon Redshift		Databricks	Google BigQuery
Apache Drill		Datorama	Google Cloud SQL
			Google Drive
			Impala
			Kyvos
			Microsoft SQL Server
			MongoDB BI Connector
			MySQL
			OData

Figure 2.6: Connect to data Refer to the following figure:

Figure 2.7: Connectors page Here, you need to enter the server name and sign-in information.

Similarly, for any other database server, you mostly need to provide the sign-in information, and you are ready to connect the data from that server.

- 5. Accelerators: Tableau Accelerators** are pre-built-in dashboards designed to jumpstart the data analysis. Accelerators are built on sample data, so you can replace that with your data, and find your insights with minimal setup. Refer to the following figure:

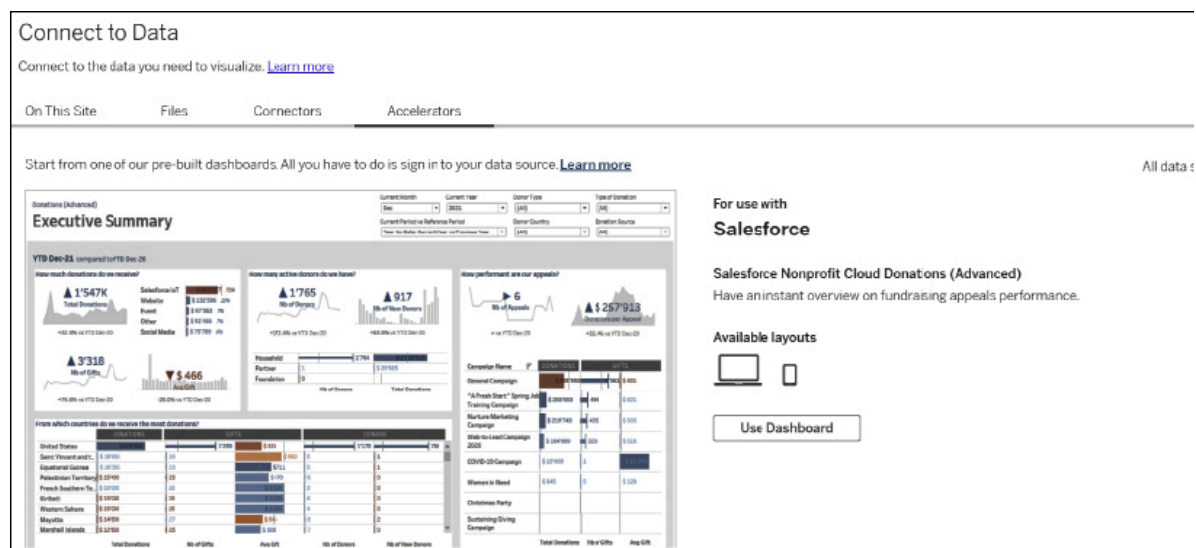


Figure 2.8: Accelerators page

There are a limited number of accelerators available to use when you create the workbook in Tableau Online. Refer to the following figure for a better understanding:

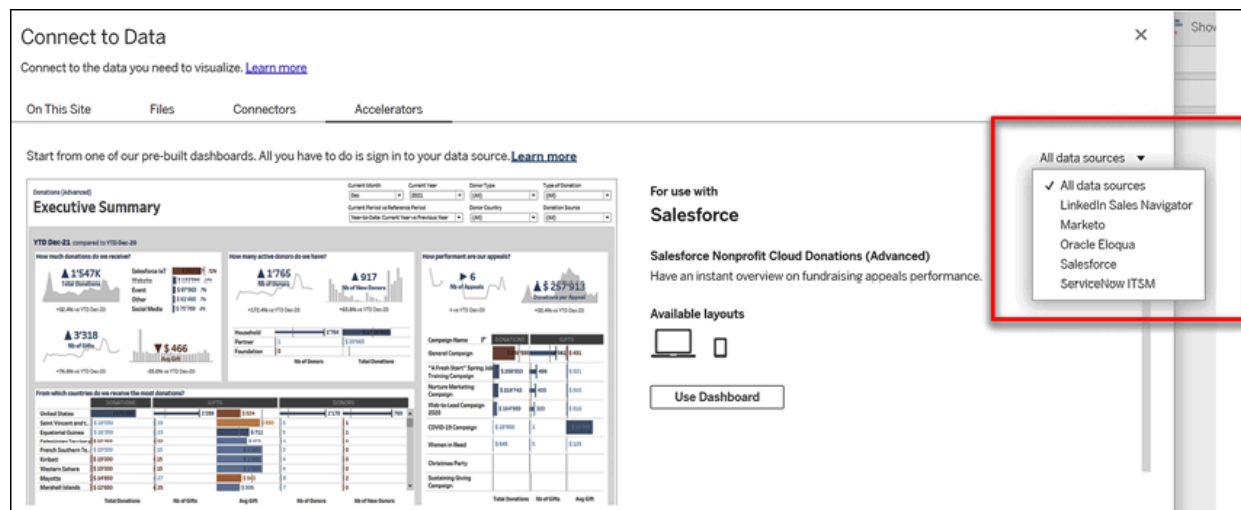


Figure 2.9: Data sources in accelerators page Follow these steps to get started with pre-existing designs:
Sample: Salesforce

1. Click **Use Dashboard**, from the list of pre-existing designs, select your data source, business metrics, **key performance indicators (KPIs)** you need:

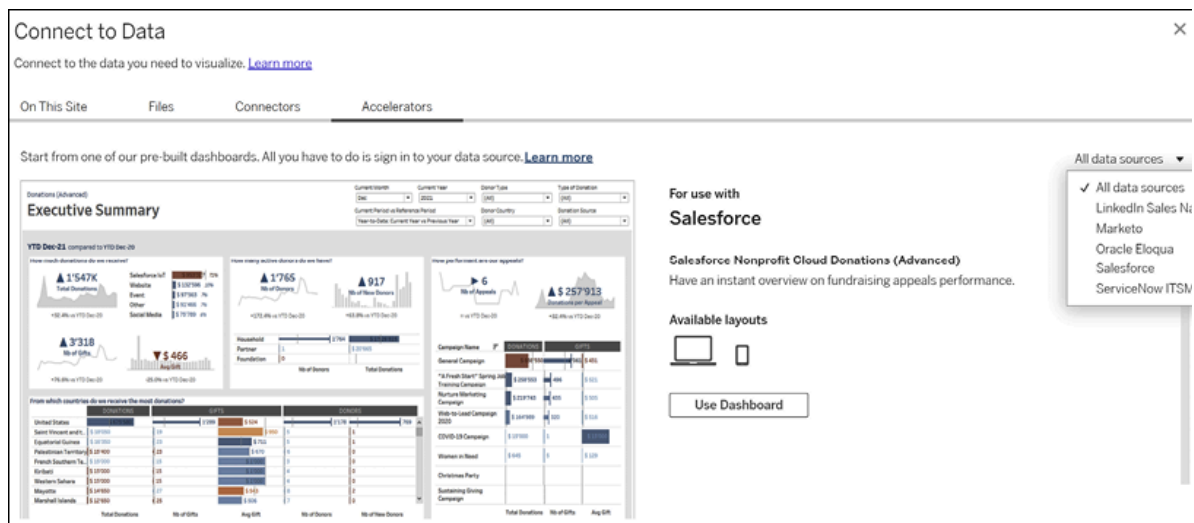


Figure 2.10: Dashboard usage in accelerator page

2. Select the project and click **Create workbook**:

Create workbook

Workbook name

Salesforce Nonprofit Cloud Donations (Advanced)

Select project

▼ Projects

- default
- Samples

Cancel Create Workbook

Figure 2.11: Workbook creation

3. Here, **Samples** | Click **Create Workbook**.
4. If you quickly want to see how the workbook looks with the existing sample data, click **Continue without signing in** or **Continue** to create the workbook with your data:

Connect to Salesforce

Before you connect

Sign in to Salesforce to see your data in the dashboard. Or continue without signing in to see sample data.

Cancel Continue without signing in Continue

Figure 2.12: Connecting to Salesforce

5. If you choose to create workbook with your data, then enter the salesforce credentials in the data source connection.

Note: If a salesforce account is not set up, then sign in to a new account (30-day free trial version).

Refer to the following figure for a representation of the Salesforce homepage:

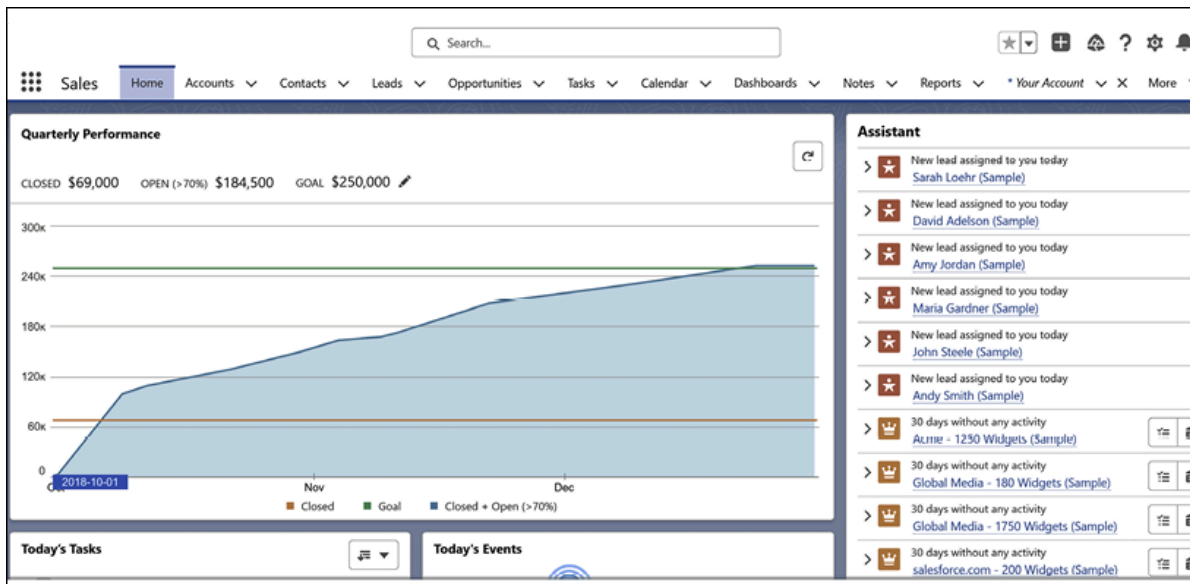


Figure 2.13: Salesforce homepage

6. Select **Account** from the top-menu as shown in the following figure:

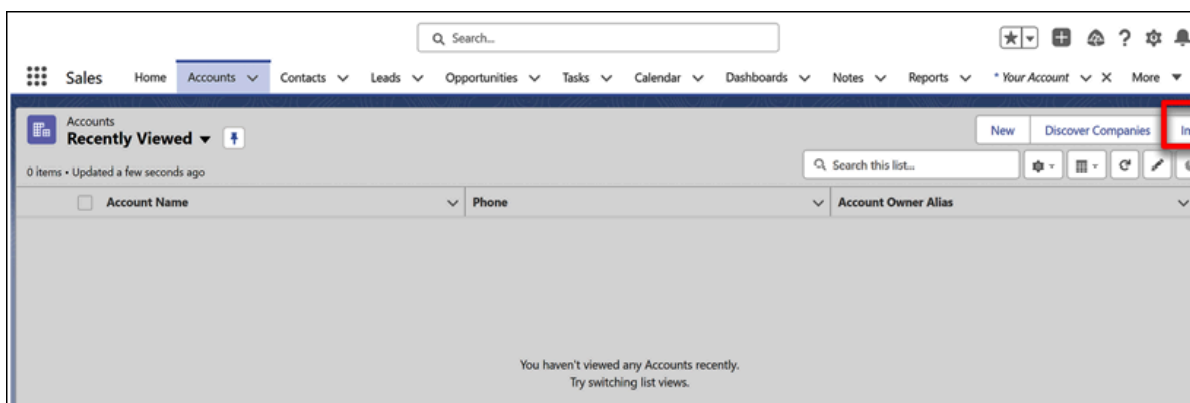


Figure 2.14: Salesforce account page

7. Click on **Import** from the **Account** page.
8. Select the type of data file you want to insert as shown in the following figure, example, CSV file:

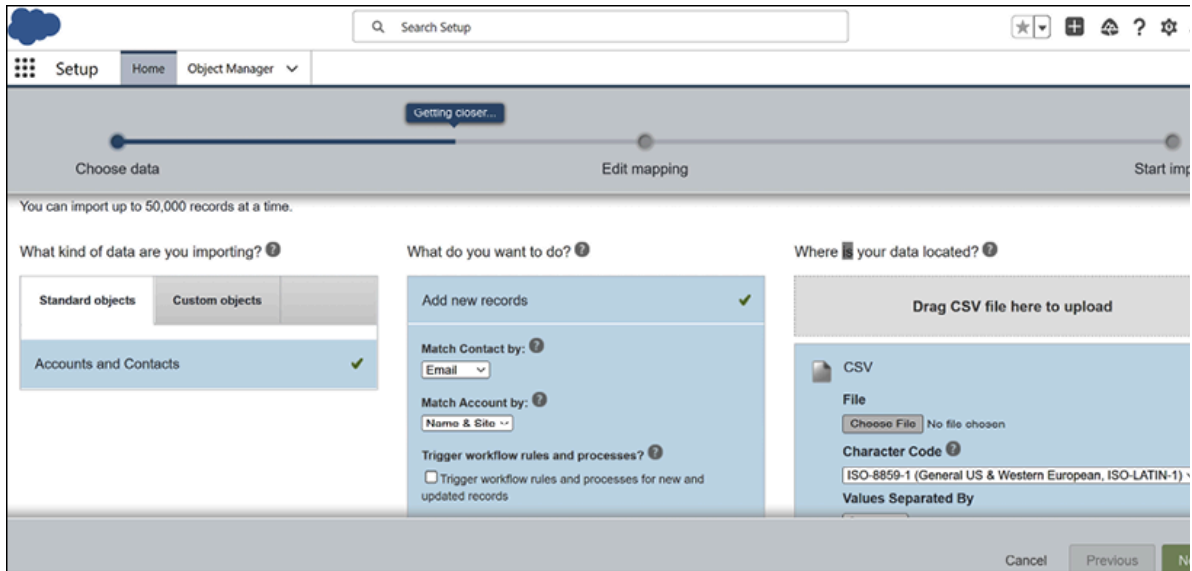


Figure 2.15: Import the data in Salesforce Refer to the following figure:

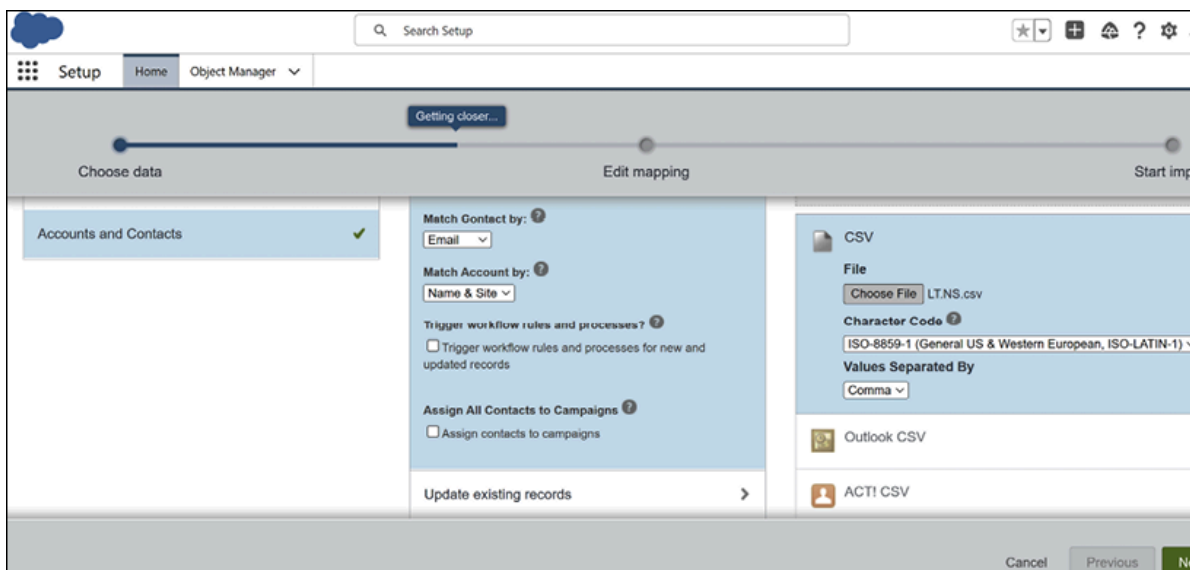


Figure 2.16: Importing data in Salesforce

9. Click on **Next**
10. Click on **Ok**.

Connection: Live vs. extract

In the previous section, we explored diverse ways to connect your data. But the question comes whether it is in live connection or data extracts? You are going to hear these terms often about live vs. extract data connections in Tableau.

Let us drill down into this and understand the distinction between data extracts and live connections, and when and where to use them.

Tableau data extracts are the snapshots of data loaded into the system's in-memory data engine to quickly recall for visualization. Extracts tend to be faster than live connections, especially when complex visualization, filters, parameters, and calculations are applied to the data. Data extracts created from a local file (**.xlsx**, **.csv**, **.txt**, and so on) or on-premises database can speed up the workbook optimization. To build the visualization, there is no need for a database; instead, Tableau can use the in-memory data engine queries and extracts directly. The extracts will need to refresh to get the most updated data from the original data source, whether it is from a local file or on-premises database at a given point in time, as extracts are snapshots of the data. Whereas Live connection offers real-time data, with any change in the data source reflected in Tableau.

Live connection depends on the database for all queries. So, with live connections, your data queries are only as fast as the database itself. For fast performance, databases need to optimize. And unlike extracts, databases are not optimized for fast performance. Workbook speeds are affected by various other factors, including any custom SQL, network speed, traffic on that network.

Extract or Live connection

Both types of connections have their place. The choice depends on the situation's necessity. If real-time decisions need to be made, then the data must be kept in a live connection. There must also be visualizations that monitor daily or weekly trends. For these analytics, using an extract of the data source helps build a faster workbook.

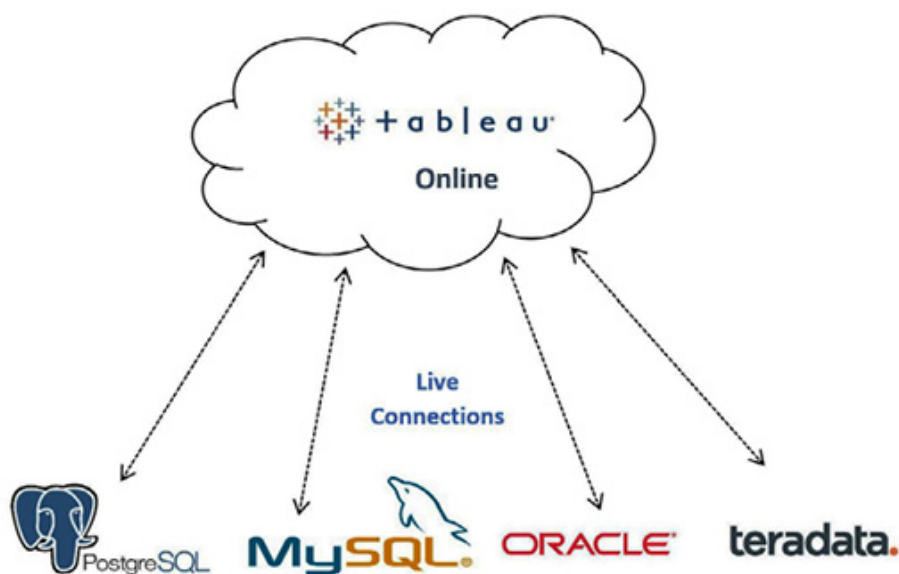


Figure 2.17: Live connections Tableau Online currently supports live connections to the following cloud-hosted data sources. Few of them are as follows:

- SAP HANA
- Microsoft Azure SQL Database
- Amazon Redshift
- Amazon Aurora
- Hive and Cloudera Impala
- Google Cloud SQL
- Google BigQuery
- Snowflake
- Spark SQL
- MySQL
- Microsoft SQL Server
- Oracle
- Teradata
- PostgreSQL and many more.

Refer to the following figure for an illustrated example:

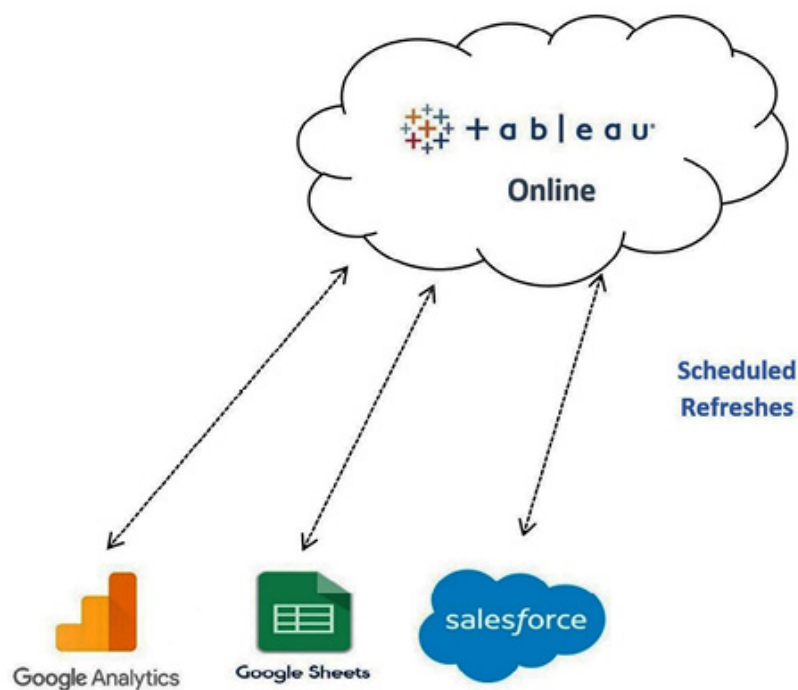


Figure 2.18: Extract data connections With scheduled data extract refreshing from web application, workbooks can connect to data from the following cloud applications.

- Google Analytics
- Salesforce
- QuickBooks Online
- Google Sheets and many more

All other connections will use the extracts.

You can also use Tableau Bridge to keep the data fresh and updated, but we will cover that topic in [Chapter 7, Publishing and Scheduling](#).

Run Initial SQL

Tableau 9.3 brought a wonderful new feature called **Initial SQL**, which allows you to declare a set of SQL commands to run before any queries within a particular Tableau Server user's session.

When connecting to the databases, you can write an initial SQL command that will run whenever the connection is setup with the database.

For example, when you open the Tableau website, the Impala database looks like the following figure:

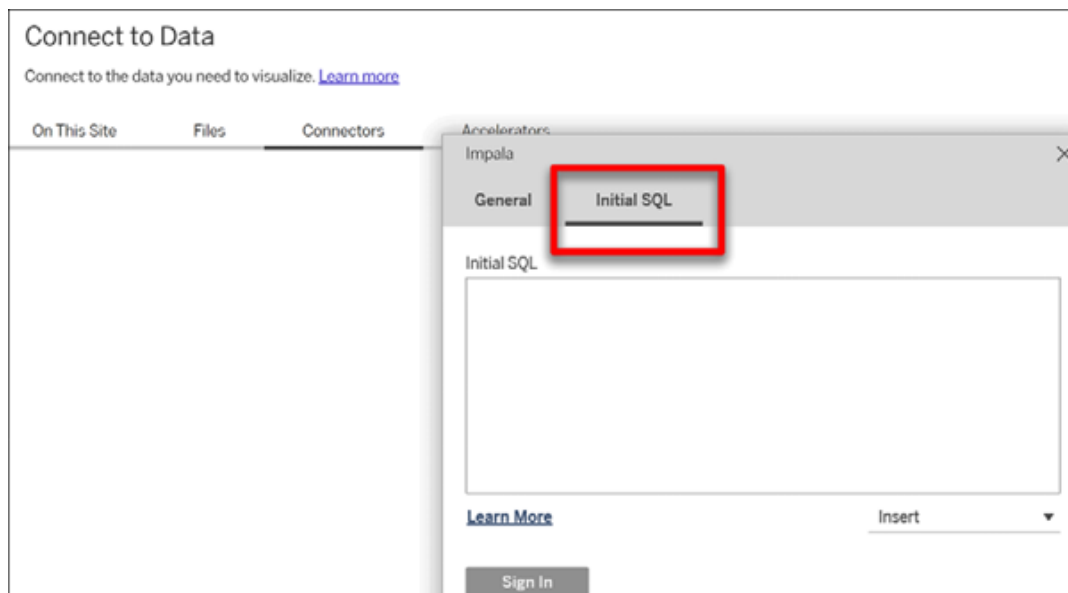


Figure 2.19: Initial SQL

Follow these steps:

1. Click on **Connectors** | select **Impala**.
2. Select **Initial SQL**.
3. You can add an initial SQL command in the **Server Connection** dialog box or on the **Data Source** page.
4. You can use this command:
 - To set a custom data environment.
 - During the session, create the temporary tables to use.

Note: Adding the initial SQL command is not mandatory. However, it is advisable to do so.

Do not get confused with the Initial SQL and the Custom SQL connection. A custom SQL connection is defined as the relation (or table) to issue against the queries. When you refresh the extracts, open a workbook, sign-in to Tableau Server, or publish to the Tableau Server/Online, the Initial SQL will run and not when you refresh your views.

Use the initial SQL by following these steps:

1. Enter the SQL command into the **Initial SQL** dialog box. The following statement is the sample query used to set the environment. This will set the pool in which query must run:



Figure 2.20: Usage1—Initial SQL

2. You can insert the parameter in the **Initial SQL** statement by simply using the **Insert** drop-down menu to pass parameters to your data source. Refer to the

following figure for a better understanding:

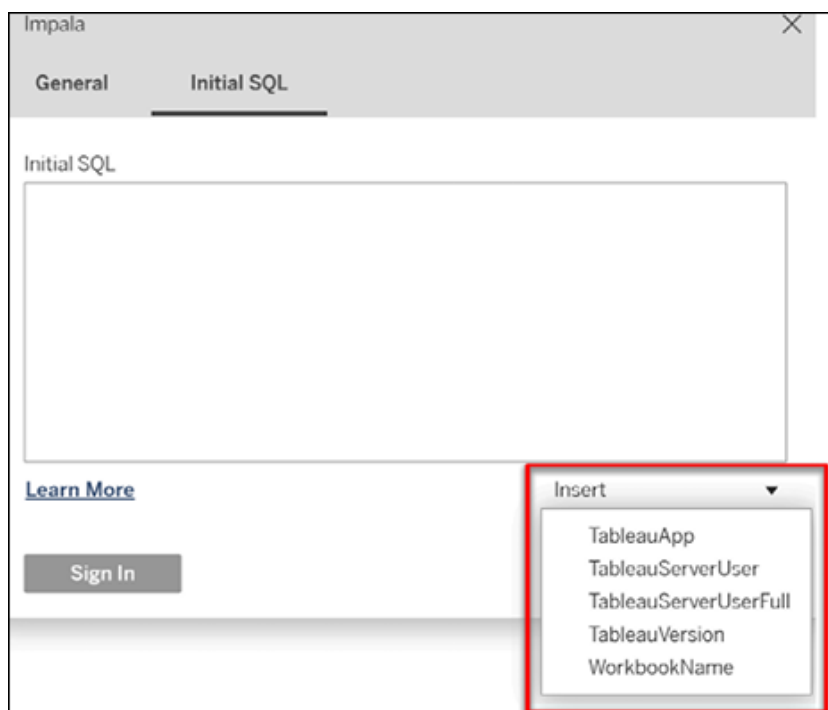


Figure 2.21: Insert parameters in Initial SQL

There are a few reasons why passing parameters are useful. The following parameters are supported in an initial SQL statement:

Parameter	Description	Example
Tableau Server User	Used for impersonating the username of the current server user. Return an empty string if the user is not signed into the Tableau Server.	urmishapatel
Tableau Server User Full	Used for impersonating the username and domain of the current server user. Return empty string if the user not signed into Tableau Server.	domain\urmishapatel
Tableau App	The name of the tableau application.	Tableau Desktop Professional Tableau Server
Workbook Name	The name of the tableau workbook. Use it when the workbooks with the embedded data source.	Financial Analysis
TableauVersion	The version of the tableau application.	10.1

Table 2.1 : Parameters in Initial SQL

For example, if you want to pass **TableauServerUserFull** parameter in the Initial SQL then, write the command as written in the following figure or, Write, set [space] [Choose TableauServerUserFull from Insert drop-down menu]:

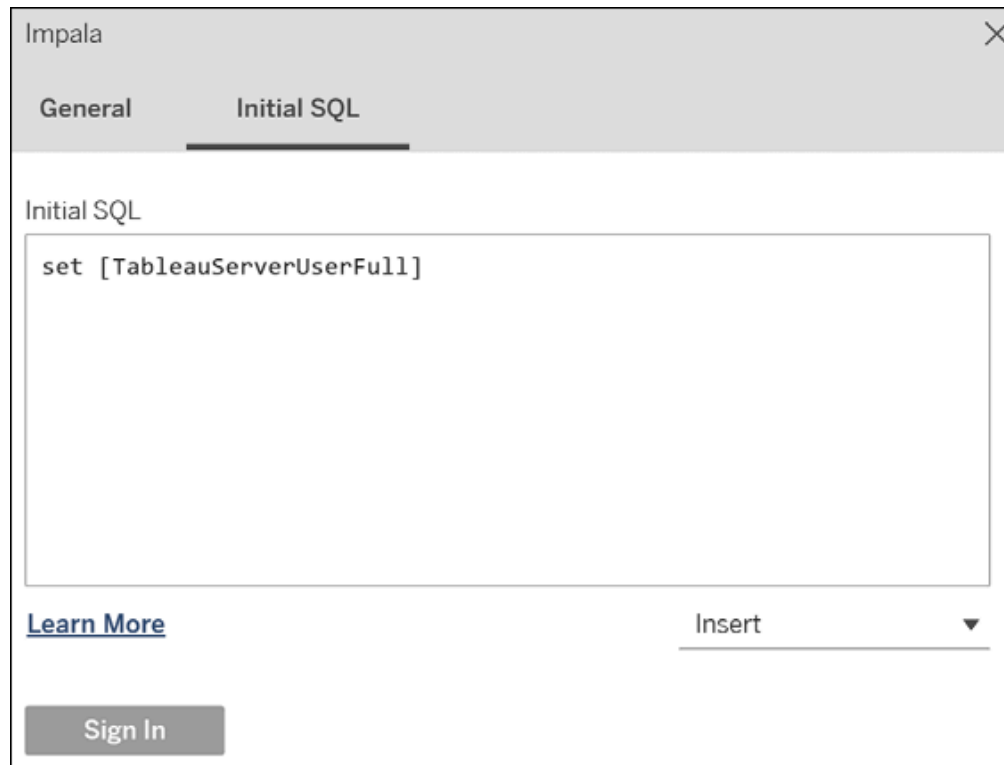


Figure 2.22: Usage2—Initial SQL

Extarct upgrade to .hyper format

At the beginning of Tableau version 10.5, the new extracts are in the .hyper format instead of the **.tde** format. Tableau upgraded its extract format with an improved data engine and larger extracts, which supports the same query performance and fast analysis as the data engine before it.

There are various benefits of using **.hyper** formats. They are as follows:

- **Create larger extracts:** The **.hyper** format can support more data that you can consolidate. The extracts in the **.hyper** format contain billions of rows of data. Previously, **.tde** extract had to create the extracts separately into a single.
- **Faster extract creation and refreshes:** This release supports fast extract creation and refreshes the larger data sets.

Better performance when interacting with views for the larger extract data sources. The views that use extract data sources can perform more efficiently using **.hyper** format than previously using **.tde** format.

So far, you have understood the extract upgrade benefits. However, you still have questions about what causes the extract to upgrade. Let us now look at the limitations in **.tde** format or what to do if your extract is in **.tde**.

Extract upgrade causes

Nevertheless, you can continue to open and use **.tde** extracts; when you perform an extract in **.tde** format, then Tableau upgrades it to **.hyper** format. After the extract is upgraded, you must know their backward limitations as well:

- You cannot convert the upgraded **.hyper** format extract to **.tde** format.
- You cannot open the upgraded extract (**.hyper** format) in the earlier version.
- You cannot use export as the version in Tableau Desktop to downgrade a workbook that contains **.hyper** format.
- If a workbook contains **.hyper** format, then you cannot open the workbook in Tableau Desktop prior to 10.4 version or if you have downloaded earlier version from Tableau Online using the **Download Tableau Workbook**.

Upgrade the extracts

There are three ways to upgrade the extract from **.tde** to **.hyper**:

- When extract is updated manually in Tableau Desktop 2022.2 or older versions.
- During the extract refresh (full or incremental).
- When appending data to an extract.

We will explore these methods in detail while scheduling the extract refreshes in [Chapter 6, Publishing and Scheduling](#).

Creating extracts on the web

To ameliorate the data source performance and additional analytical functions, you can choose to extract the data on the web, where Tableau will copy the data from the data store and send it to Tableau Online. In the web, you can extract data while in web authoring or in content server.

Let us understand both ways to extract data.

- Create **Extracts** in web authoring.
- You can extract data directly in web authoring with default extract settings.

Follow these steps to extract the embedded data source in web authoring:

1. On the home page, select **Explore | Data Sources**
2. Select the data source by clicking on the **Data Source Name**.
3. Click on **Edit Data Source**:

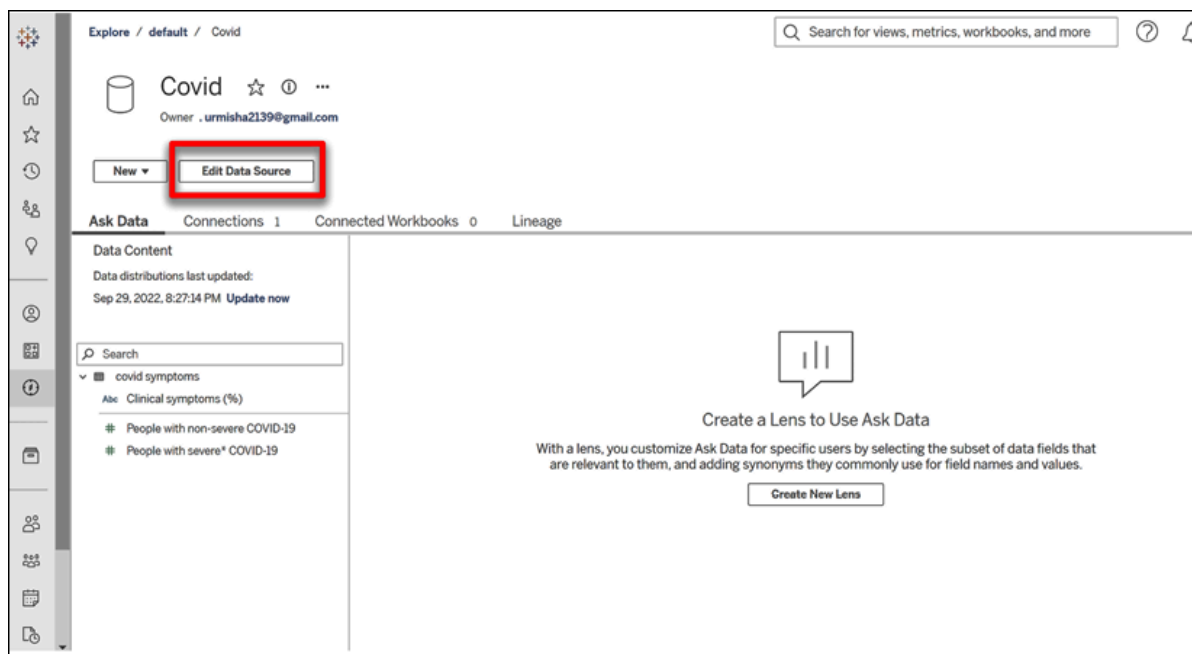


Figure 2.23: Edit source

4. Change the connection type from **Live** to **Extract** in the top-right corner of **Edit Data Source** page, as shown in the following figure:

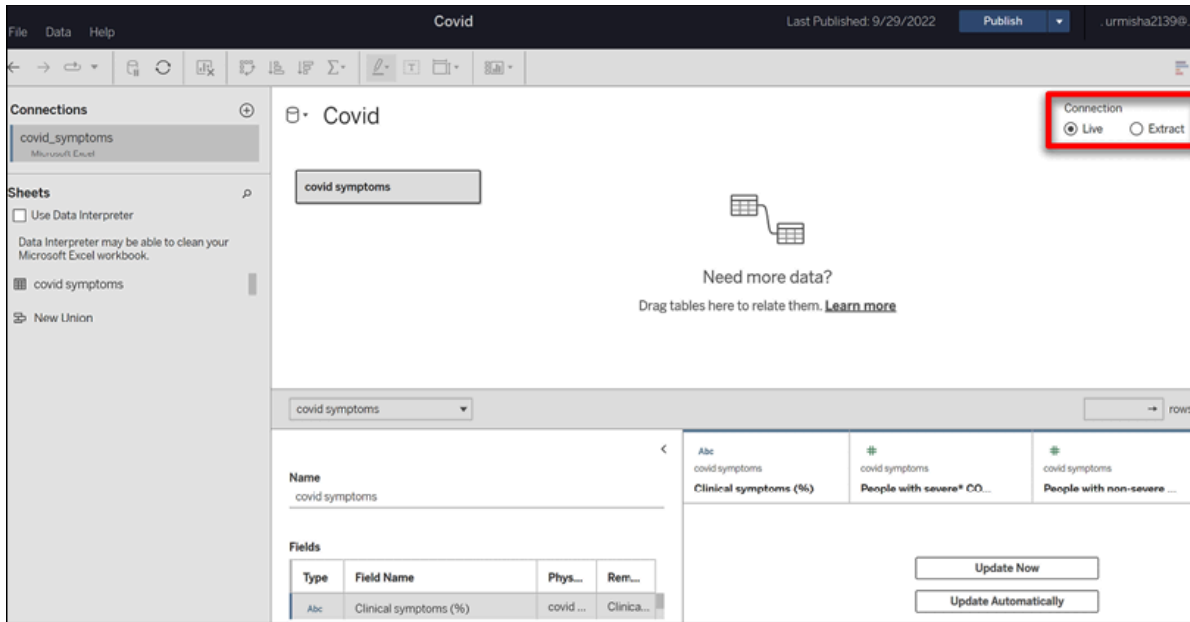


Figure 2.24: Edit connection type

5. Once you change the connection type to **Extract**. This will enable **Edit** and **Create Extract**.
6. Now, click on **Edit**. You will see the **Creating Extract** dialog box.
7. Click on **Create Extract**:

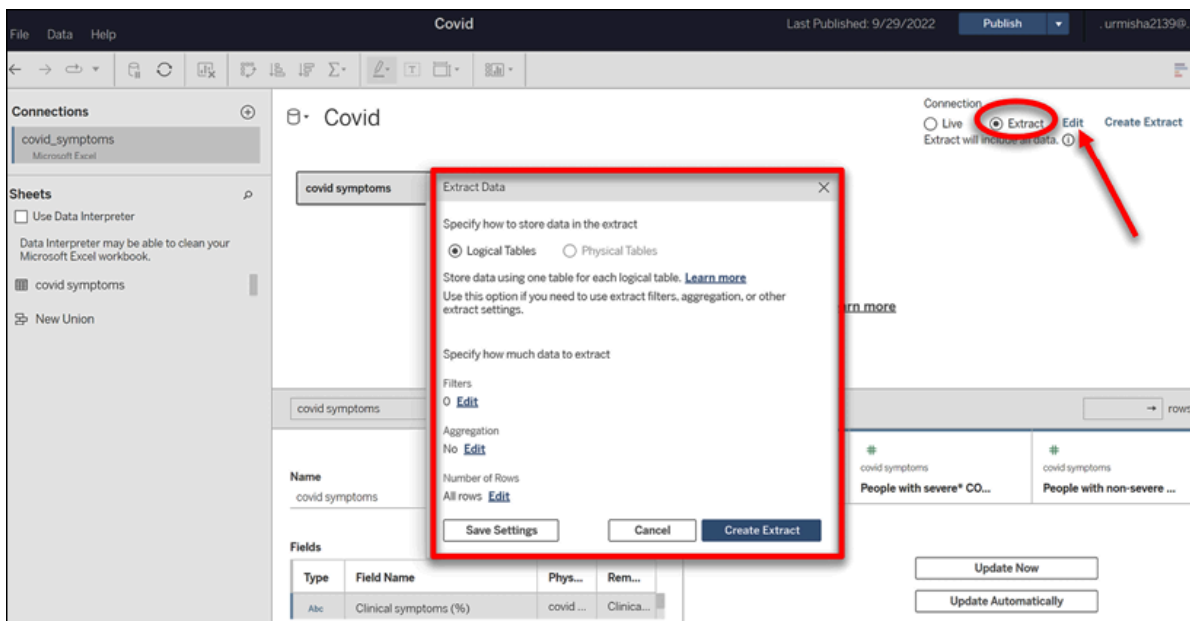


Figure 2.25: Create Extract

Before you click on **Create Extract**, understand the **Extract Data** dialog box.

Let us configure it to tell Tableau how to store the extract based on which fields you want to filter the data and limit the amount of data in your extract.

1. Specify how to store data in the extract. You can choose one of the two provided structures (schema) to store the data. You must choose the option depending on what you need.

- a. **Logical tables**

- i. Physical tables that define logical tables are merged and stored with that logical table.
- ii. Select logical tables when you want to limit the records in your extract and with additional extract properties, including filters, aggregation, Top N, or other features that require denormalized data.

- b. **Physical tables:** For each physical table in the data source, store data using one extract table.

This option will reduce the extract file size and improve the performance certainly and cost. When the extract is stored as physical tables, you cannot append data to it.

Conditions for using the **physical tables** option:

- All joins between the physical tables (=) are equality joins.
- No incremental refreshes configured.
- No additional extract properties (Filters, Top N).
- No RAWSQL used.
- The data types of columns used for relationships or joins data types must have to be identical.

2. Specify how much data to extract:

- a. Filter

- i. Select **Edit**.

- ii. Click on **Add filters** to add one or more filters/fields to limit the data based on fields and their values.

- iii. You can **Edit**, and **Delete** the filters anytime and change the extraction.

b. Aggregation

- i. You can select **Aggregate data for visible dimensions** to aggregate the measures using their default aggregation.

- ii. You can also select **Roll up dates** to the specified data level such as **Year**, **Month**, and so on to aggregate the data form the drop-down menu.

c. Number of rows

- i. Select the number of rows you want to extract.

- ii. You can extract **All rows** or the **Top N** rows.

Note: Not all data sources support Sampling options in the Extract dialog box.

For sampling, you must first select filter aggregation and then extract the number of rows.

Extracting creation may take a long time, depending on the data source, connection, and records. To ensure your extraction is not lost, click **Notify Me When Complete** in the dialog box to specify the location for the extracted workbook to be saved. If the extract creation fails, you will be notified that the extract cannot be created, and you can restore the unsaved changes by reopening the original workbooks.

Creating extract on the content server

Follow these steps to create the extracts from the embedded data sources:

1. Sign-in as **Site Admin** or as the owner of the data source.
2. Click **Explore**.
3. Select **All Data Sources** from the drop-down menu in the **Explore** page.
4. Click on the **Data Source Name** to select the data source.
5. Click **Actions**, as shown in the following figure.
6. Click **Extract**.

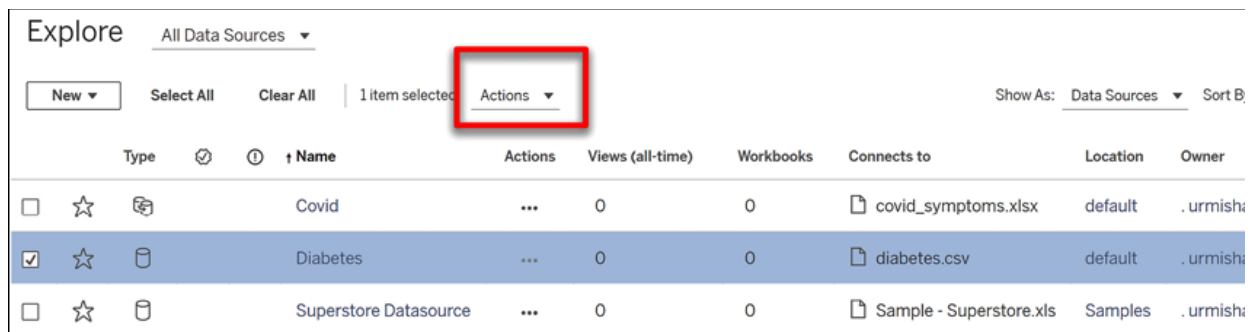


Figure 2.26: Create extract on content server Follow these steps to create the extracts from the published data sources:

1. Sign-in as **Site Admin** or as the owner of the data source.
2. Click **Explore**.
3. Select **All Data Sources** from the drop-down menu in the **Explore** page.
4. Select the data source by clicking on the **Data Source Name**.
5. At the top of the screen, under the **Data Source Name**, select the drop-down menu says **Live**.
6. Click on **Connection type** to change **Live** to **Extract**.
7. Click **Save**.

Direct connection to Cloud data

In the earlier topic, *Connect Data to the Web*, we saw various Tableau-supported connectors used to connect your data with different platforms. Surprisingly! Tableau also allows you to set a direct connection to the maintained SQL-based data hosted on the public cloud platform when you publish workbooks or data sources. In such a scenario, there is no need to publish the static extract of the data. Depending on the data type and business requirement, you can keep the data fresh either by live connection or by scheduling the extract refreshes.

Connecting directly to the data cloud increases the risk to data security. Addressing this issue, Tableau came up with a solution by **encrypting connections using SSL**. You can enable SSL encryption for the connections to your hosted SQL data when you create the connection in Tableau Desktop.

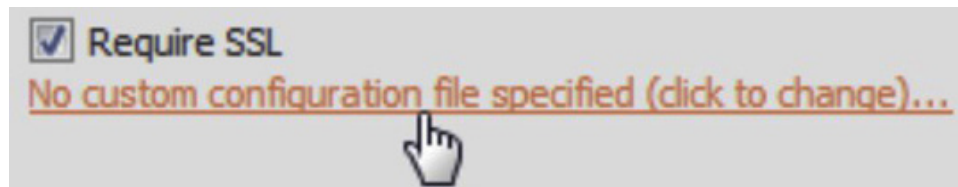
Follow the steps to enable the encrypted connection:

1. In the Tableau Desktop, connect to the data.

2. In the **Server Connection** dialog box, select the **Require SSL** check box, as shown in the following figure.

Or,

There is also an option to self-sign or use another custom certificate to connect to the data over SSL.



*Figure 2.27: Encrypted SSL connection*¹

3. Click **Ok**, to finish.

Configure connections with analytics extension

Tableau can perform dynamic calculations in a workbook with languages like R and Python, Einstein Discovery, and other tools and platforms using the *Analytics Extension*. Let us configure this analytics extension at your site in Tableau Online.

Follow these steps to configure analytics extension settings:

1. As a site administrator, sign in to your Tableau Online site.
2. Click on **Settings**.
3. Click the **Extensions** tab then scroll to **Analytics Extensions** in the **Settings** page.
4. Select **Enable analytics extension** for the site by clicking the checkbox.
5. Click **Create a new connection**. Refer to the following figure:

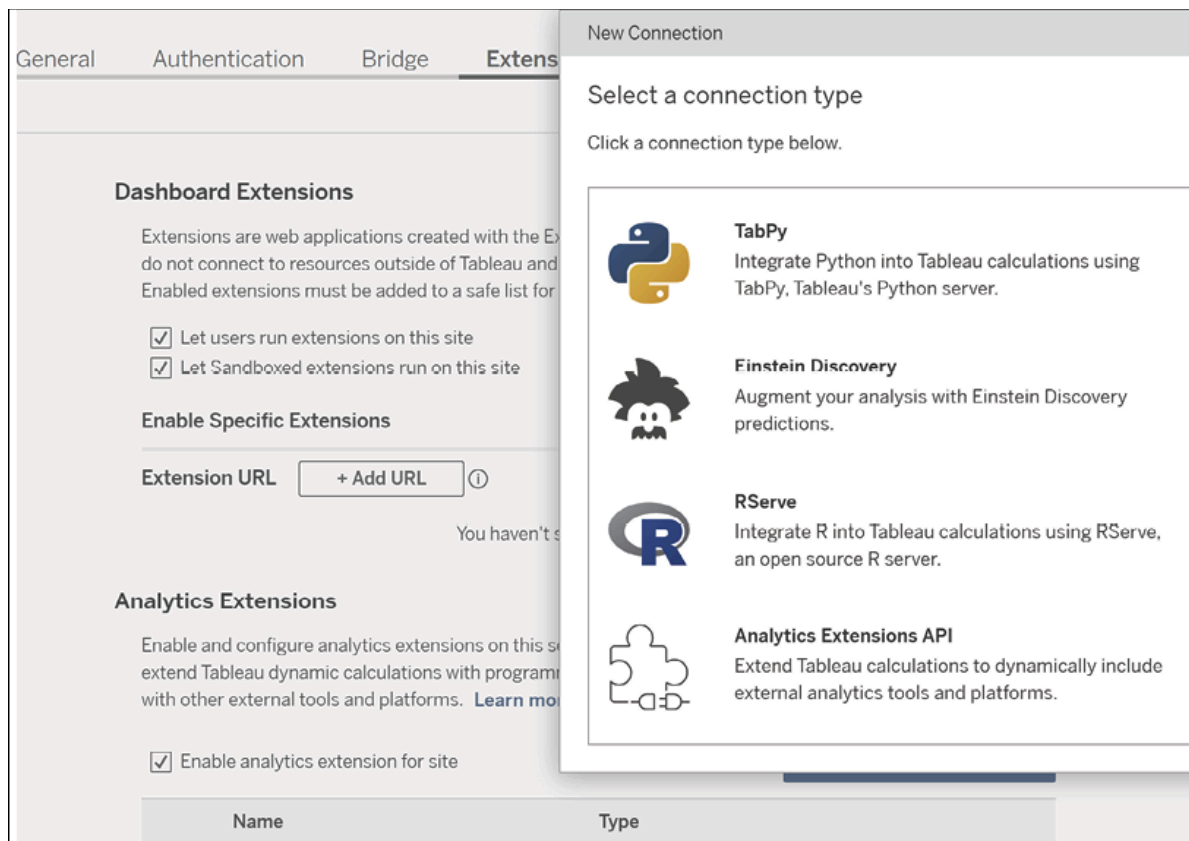


Figure 2.28: Connection type

6. Select the connection type you want to add, enter the configuration details, and connection information for your analytics service.

Example, Python integrates with Tableau using *TabPy*:

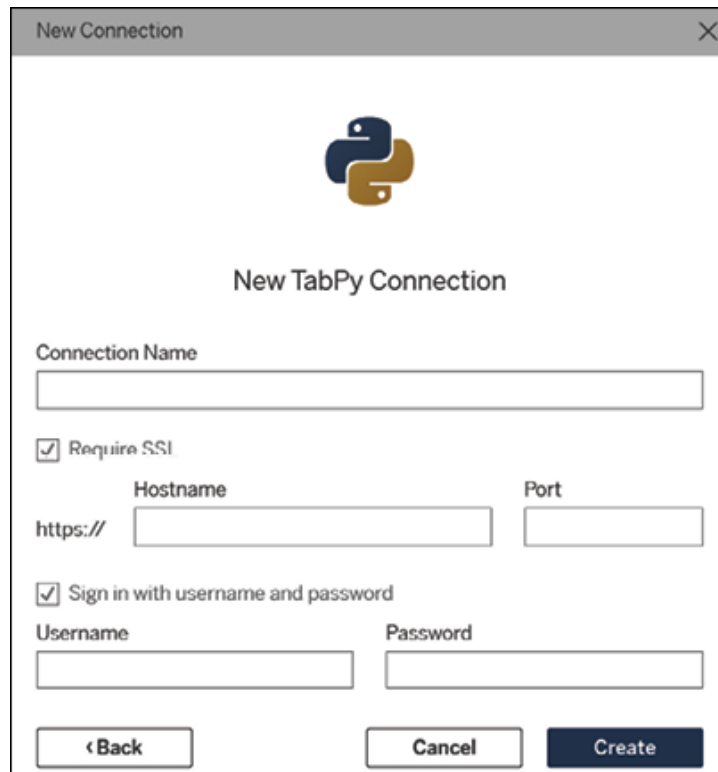
A screenshot of a 'New Connection' dialog box. At the top is a Python logo. Below it is the title 'New TabPy Connection'. There is a text input field for 'Connection Name'. Below that is a checked checkbox labeled 'Require SSI'. Under this checkbox are two input fields: 'Hostname' (with 'https://' pre-filled) and 'Port'. Below these is another checked checkbox labeled 'Sign in with username and password'. Under this checkbox are two input fields: 'Username' and 'Password'. At the bottom are three buttons: 'Back', 'Cancel', and 'Create'.

Figure 2.29: TabPy connection

7. Click **Save**.

Edit or delete the analytics extension connection by following these steps:

1. Scroll to **Analytics Extension**.
2. To change the configuration, select edit or delete icons:

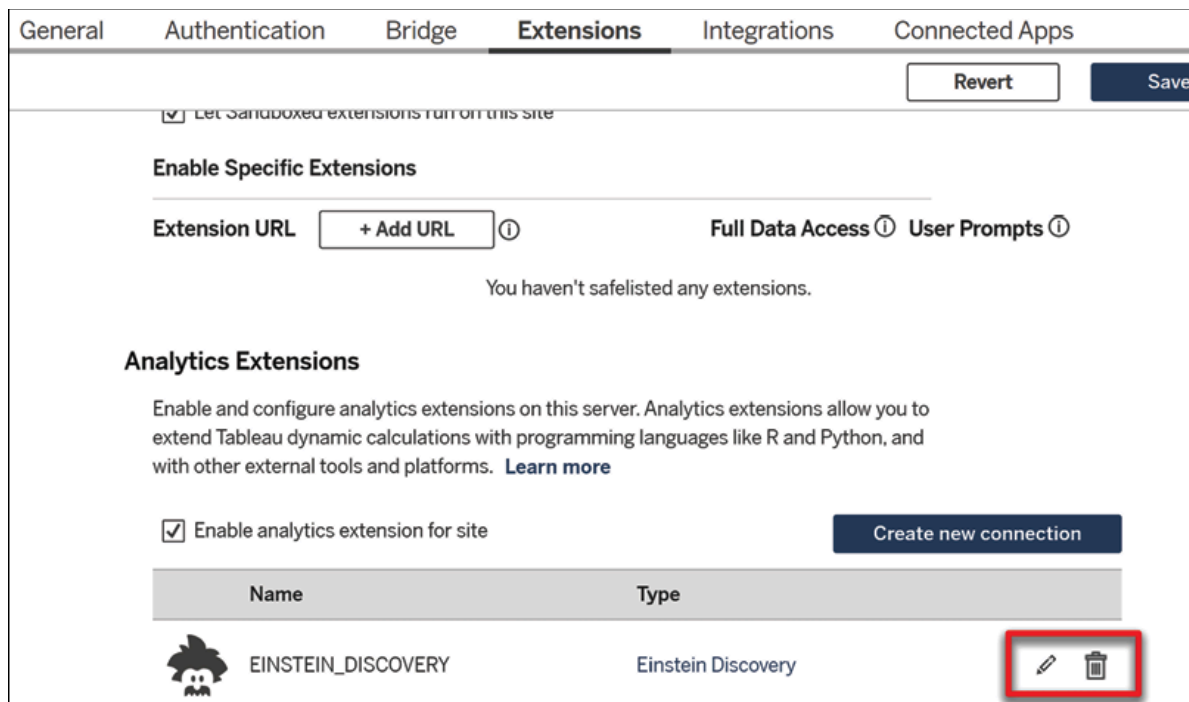


Figure 2.30: Edit or delete analytics extension

Editing connections on Tableau Online

The site administrators and data source owners can manage the published data source connection information directly on the site. The connection information describes the database owner, data source, and how to access the data. You can embed database credentials in the connection to allow users to access the data directly or enable scheduled refreshes to extract.

You can set the connection so that users are prompted to sign in. You can also update the tag and change the data source owner without having to republish the data source and change the server name.

Follow the steps to edit the connection:

1. Open the **Explore** page.
2. From the drop-down menu, select **All Data Sources**.
3. If there are a large number of data sources, use filters to narrow down the list.
4. Select the check box of the data source you want to update, and on the **Actions** menu, select **Edit the Connection**.
5. Update the connection information and click **Save**.

6. Refresh the data connection page for your changes to take effect. Refer to the following figure:

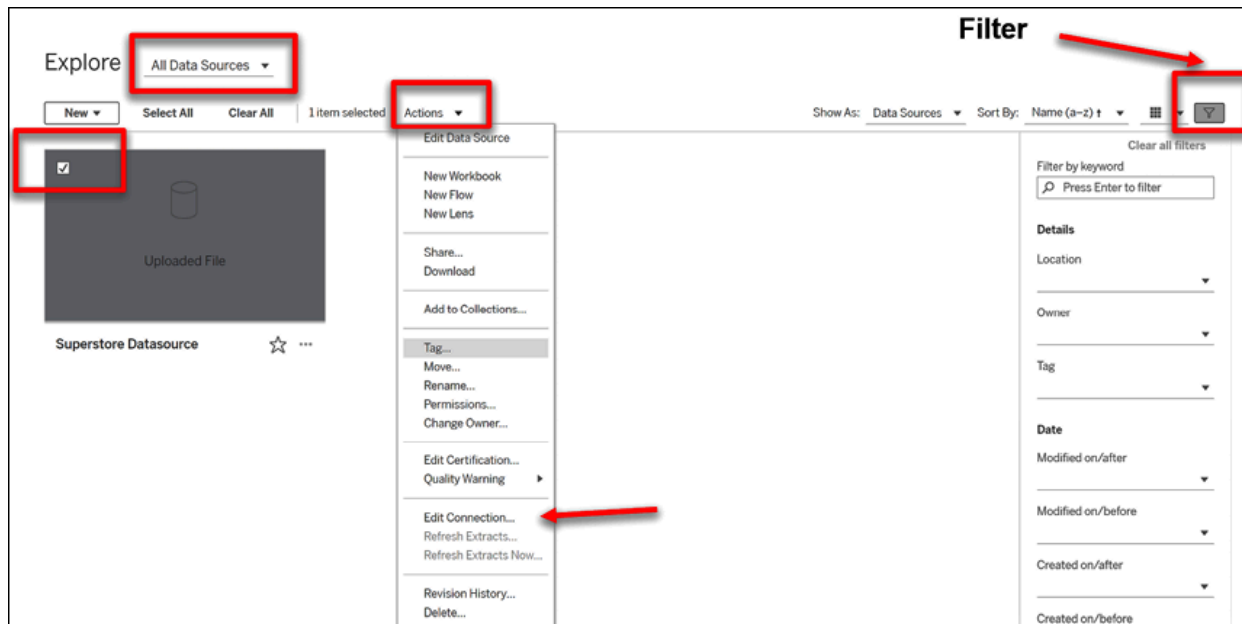


Figure 2.31: Edit the connections *Conclusion*

In this chapter, we unleashed the power of Tableau Online on managing data connection.

In Tableau Desktop, you can upload your workbook or data sources to Tableau Online and publish it via the server menu. However, the same thing can be easily achieved directly on Tableau Online by connecting the data sources on the web via a number of connectors, various file types, accelerators, or directly on the site. The publishing steps require you to make decisions on how you and others in your organization can access your data source or workbook. You can publish workbooks and data sources using live or extract connections. Tableau supports various connector types that support live and extract connections. Tableau improved its storage capacity. However, a site has approx. 100 GB storage limit for extracts and workbooks.

In the next chapter, you will see the creative side of Tableau Online. You will create the data visualization worksheet, including graphs, plots, filters, calculated fields, LODs, etc., in Tableau Online directly on the web.

Fill in the blanks

1. _____ used to extend the dynamic calculation in workbooks.

2. _____ to declare a set of SQL commands to run before any queries within a particular Tableau Server user's session.
3. _____are pre-built dashboards designed to quickly jumpstart the data analysis.
4. _____ format data extract supports more data than _____ format?

Multiple choice questions

1. Select the appropriate answers: To connect the data on the web, we will use:

- a. On this site
- b. Connectors
- c. Accelerators
- d. All of the above

2. Select the appropriate answers: The type of connections in Tableau Online are:

- a. Live
- b. Extract data
- c. Analytics extensions
- d. Scheduled extracts
- e. All of the above

3. Tableau also allows you to set a direct connection to the maintained SQL based data hosted in the cloud platform. Which type of cloud platform/s?

- a. Private
- b. Hybrid
- c. Public
- d. All the above

4. Select the appropriate answers: Options to extract the data are:

- a. Filters, aggregations, number of rows
- b. Logical or physical tables
- c. Live
- d. Incremental, full extract

Questions

1. What is the distinction between live connection versus extract data?
2. What are the benefits of .hyper over .tde extract format?
3. How to connect data on the web?

Answers

1.	d
2.	a,b
3.	c
4.	a,b

Points to remember

Here are the key takeaways from this chapter:

- Tableau supports connecting the data sources on its web environment via N number of connectors, various file types, and accelerators directly on the site.
- There are two types of connections: live and extract.
- Live connection used when real-time decisions need to be made. Extract connections are used when monitoring daily or weekly trends and to build a faster workbook.

- Tableau 9.3 brought a wonderful new feature called Initial SQL which allows you to declare set of SQL commands to run before any queries within a particular Tableau Server user's session.
- Tableau upgraded its extract format and improved the data engine for larger extracts, which supports the same query performance and fast analytical support as the data engine before it. The extracts were upgraded from .tde to .hyper format.
- Tableau allows site administrators or data owners to create extracts on the web and can change the connection type for the published and embedded data sources.
- Direct connection to cloud data is possible with Tableau Online simply with an encrypted connection using SSL.
- Tableau increased its horizon and gives the flexibility to perform dynamic calculations in a workbook with languages like R and Python, *Einstein* Discovery, and other tools and platforms using analytics extension.

1. Source: www.tableau.com

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CHAPTER 3

Data Visualization

When you visualize, then you materialize - Anonymous

In the previous chapter, you learned how to organize and manage your data and connections. To make inferred business decisions, it becomes vital to understand your data and represent it in a graphical manner. In this chapter, you will uncover the hidden wealth of information in your data through graphical representations (including plots, charts, maps etc.,). Let us build the visuals and analyze your data.

Structure

In this chapter, we will cover the following topics:

- Tour of Tableau workspace
- Workbooks and sheets
- Add, rename, duplicate, reorder, and delete worksheets
- Types of charts
- Formatting
- Filters
- Calculated fields
- Table calculations
- Dashboard
- Story

Objectives

This chapter will elevate the creative side of Tableau Online. Readers will learn various charts, formatting, table calculations, filters, and so on. They will also understand where and when to use the various types of charts, along with building the dashboard and story in Tableau Online.

After reading this chapter, readers will be able to start building charts by simply dragging and dropping and creating dynamic dashboards. This will help readers examine and present meaningful insights from your data to higher authorities, stakeholders, reporting managers, etc., and make better business decisions.

Tour of Tableau workspace

Tableau workspace comprises of data pane, analytics pane, menu bar, status bar, shelves, toolbar, sidebars, and so on.

Let us get on the tour of Tableau workspace by having a look at the following figure:

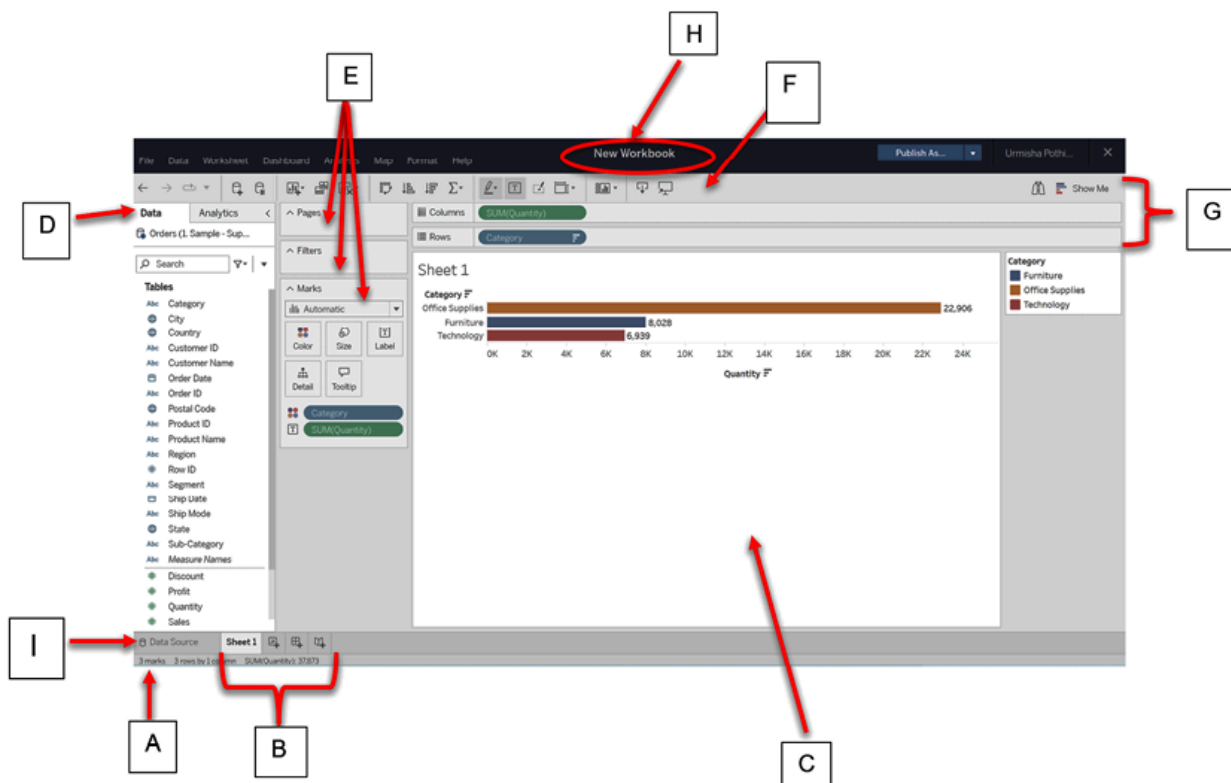


Figure 3.1: Tableau workspace Various elements in Figure 3.1 are explained as follows:

- A. **Status bar:** This bar displays the information related to the view or graph on the workspace. It provides the number of rows and columns information, measured fields, and number of marks used.
- B. **Sheet tabs:** The “Sheet1” label indicates the name of the worksheet. The icons next to it typically allow users to add a new worksheet, a new dashboard, or a new story.
- C. **View:** The area/canvas where you build your visualization (popularly known as Viz.) is called view.
- D. **Sidebar:** This bar consists of a Data pane and an Analytics pane in a worksheet.
- E. **Cards:** To enhance the data visuals, you can drag the fields in this zone. This element of the workspace also has the following functions:
- **Pages shelf:** This is used to control the display of data over time or to break a view into a series of pages.
 - **Filters shelf:** This section is used to add filters to the view, allowing you to control which data is displayed.
 - **Marks card:** This area allows you to control how data is displayed in the visualization. One can change the mark type (e.g., bar, line, circle), and you can drag fields to the various property buttons (Color, Size, Label, Detail, Tooltip) to control how your data is represented visually. We will discuss all these topics in detail further.
- F. **Toolbar:** This bar comprises navigation tools, analysis, access commands, and many more. Refer to [Figure 3.2](#) for more details.
- G. **Shelves:** Drag the fields to rows and columns shelves to add data to your view.
- H. **Workbook name:** A workbook consists of sheets. These sheets can be worksheets, dashboards, or stories.
- I. **Data source page:** Click on this tab, as it will take you to the data source page and you can view your data.

Tableau workspace toolbar reference

Now, let us understand how to use the toolbar to perform the common actions when creating or editing the view, as shown in the following screenshot:

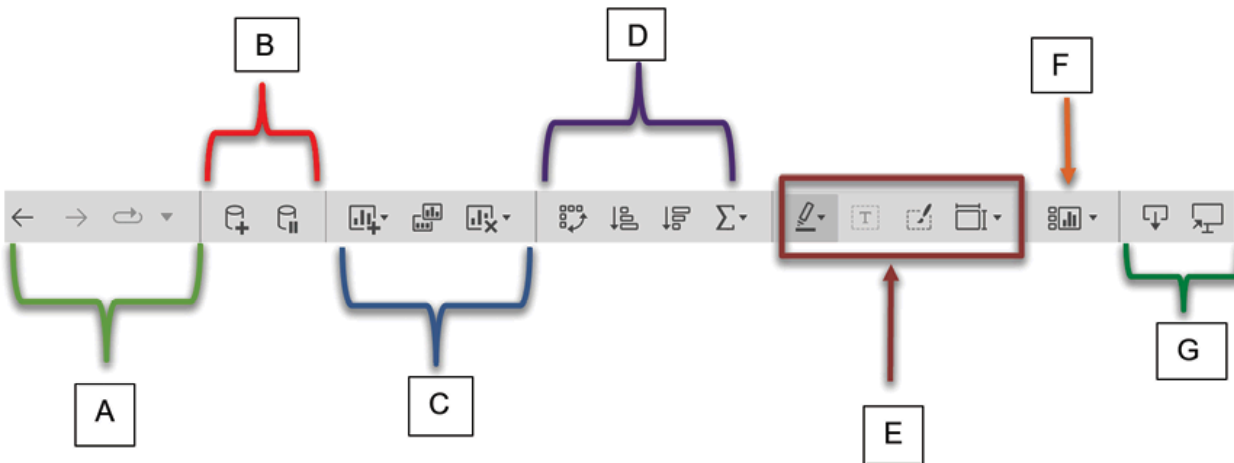


Figure 3.2: Tableau workspace toolbar

A. Undo, Redo and Replay Animations

-  **Undo:** Clicking on this icon will reverse the most recent action in the workbook you have performed.
-  **Redo:** Clicking on this icon will repeat the action you reversed with undo. This icon will be enabled after the undo button is clicked.
-  **Replay animations:** This button will enable any field in Page shelf and performs animations.

B. New Data Source, Pause Auto Updates

-  **New Data Source:** This button will take you to Connect to Data Page, where you can connect to any published data source or can upload a new data file.
-  **Pause Auto Updates:** This button will control the updates when any changes are made on Tableau.

C. New Worksheet, Duplicate, Clear

-  **New Worksheet:** You can add a new blank worksheet (worksheet, dashboard, or story) from the drop-down menu in your active workbook.

-  **Duplicate:** This button will create a duplicate copy of the current worksheet.
-  **Clear:** Clear can be used to wipe clear everything from the selected sheet by selecting Clear Sheet. Otherwise, specifics like applied formatting or applied sorting can be cleared from the sheet using Clear Formatting / Clear Sorts, respectively.

D. Swap, Sort Ascending, Sort Descending, Totals

-  **Swap:** This button will move the fields from column shelves to row shelves and vice-versa.
-  **Sort Ascending:** To arrange the selected fields in ascending order based on the measured or dimension fields in the view.
-  **Sort Descending:** To rearrange the selected fields in descending order based on the measured or dimension fields in the view.
-  **Totals:** This button is the shortcut key to compute the grand totals or sub-totals for the data in the view. Following options are available:
 - **Show Row Grand Totals:** This will add a column to the extreme right of the table/visual, showing row wise totals for all measured in the view.
 - **Show Column Grand Totals:** This will add a row to the bottom of table, showing column wise totals for all categories in the view.
 - **Row Totals to Left:** This will move totals column to the left.
 - **Column Totals to Top:** This will move totals to the top.
 - **Add All Subtotals:** This will add subtotal rows and columns in the view depending on dimensions in the column or row.
 - **Remove All Subtotals:** This will remove the subtotal rows or columns.

E. Highlight, show Mark Labels, Format Workbook, Fit

-  **Highlight:** You can highlight the values for the selected sheet using drop-down menu options. Having this option marked, highlights relevant categories in the table/visual, when selected from legends etc.
-  **Show Mark Labels:** This will show the mark labels for the active sheet.
-  **Format Workbook:** This will open the format workbook pane to change the size, font, tooltips, titles, lines, etc.,
-  **Fit:** You can select (Standard, Fit-Width, With-Height, or Entire View) from the drop-down. This fits the table visual on the current sheet in accordance with the selection.

F. Show/Hide Cards

-  **Show/Hide Cards:** You can easily show or hide the cards (Title, Caption, Parameters, Highlighter, and Filters) in the worksheet.

G. Download, edit Workbook in Tableau Desktop

-  **Download:** In Tableau Online, use this button to download the Images, Data, Crosstab, and PDF.
 - **Images:** Downloads the selected sheet, dashboard or story, as a PNG image.
 - **Data:** This will display the data in a new browser window with two tabs: Summary (aggregated data for the fields) and Full Data (complete data for selected fields in the visualization). Also provides option to download full data as a CSV file.
 - **Crosstab:** You can download the data used for the selected sheet in form of a Cross Tab, either as an Excel or CSV file.
 - **PDF:** You can save the current view or select specific sheets as PDF file.
 - **PowerPoint:** Download the current view or select specific sheets to be downloaded as PPT file.

-  **Edit Workbook in Tableau Desktop:** This button will allow you to download the workbook after publishing it on Tableau Online and then you can further edit in Tableau Desktop

Workbooks and sheets

Before we head start towards building the charts, it is important to understand the basics of Tableau Online.

Like Microsoft Excel, Tableau Online/Desktop also uses a workbook and sheet file-like structure. A single workbook contains N number of sheets. This sheet could be either a worksheet, a dashboard, or a story.

Refer to the following figure for a better understanding:

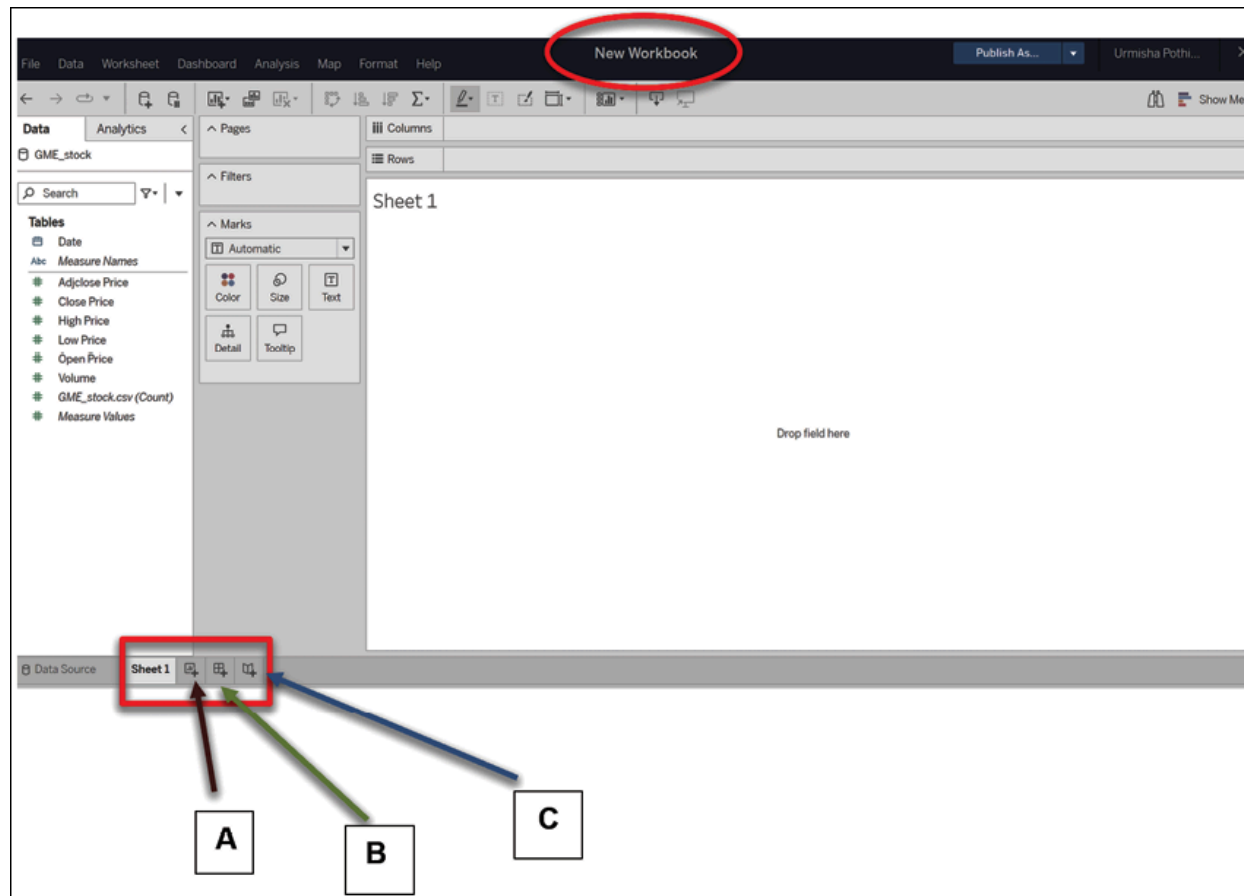


Figure 3.3: Workbook and sheets The following points have been highlighted in the preceding figure:

- A. **Worksheet:** In [Figure 3.3](#), you will see **Sheet 1**. This is called a **worksheet**, which contains the marks, data pane, analytics pane, filters, shelves, and so

on. You can also build the views (also known as worksheets) on your data by simply drag n drop of fields on the shelves.

B. **Dashboard** is the combination of one or more worksheets and/or their components to represent one or more insights from the data, which can also be used to find further details.

C. **Story** is the sequence of dashboards/worksheets that collectively present the information from your data in an intuitive manner.

In simple terms, the combination of **Worksheets** | **Dashboards** | **Story**. In other words, the group of worksheets creates a dashboard, and that group and sequence of dashboards make a story.

Add, rename, duplicate, reorder, and delete worksheets

After understanding the basics, let us play around and understand more about worksheets.

Adding a new worksheet

You can add a new worksheet by:

- Clicking on **New Worksheet**

Or,

Pressing *Ctrl + Alt + T*:

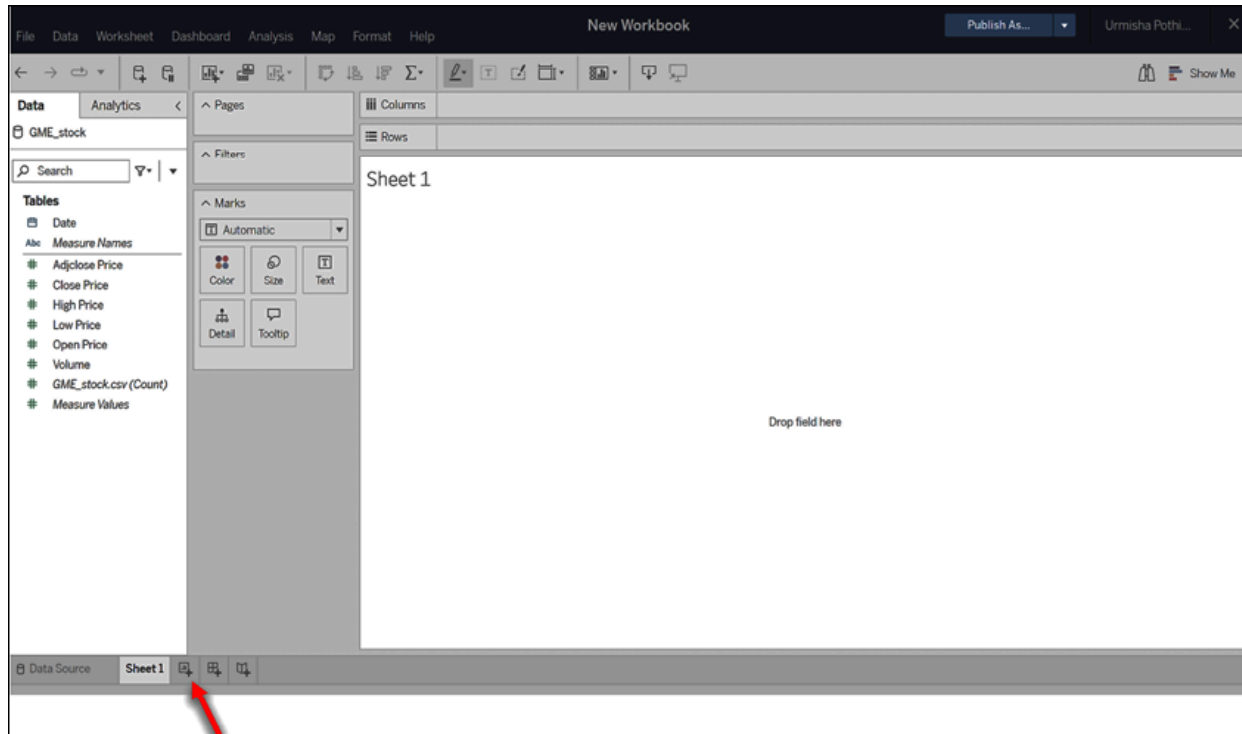


Figure 3.4: Add a new worksheet *Renaming a worksheet*

A worksheet can be renamed by:

- Right-click on **Sheet 1**, then select **Rename** from the drop-down menu, as shown in the following figure: Or,
- You can rename the active sheet Sheet1 by double clicking on **Sheet 1** in the sheet tab, typing the new name, and then click *Enter*. Refer to the following figure:

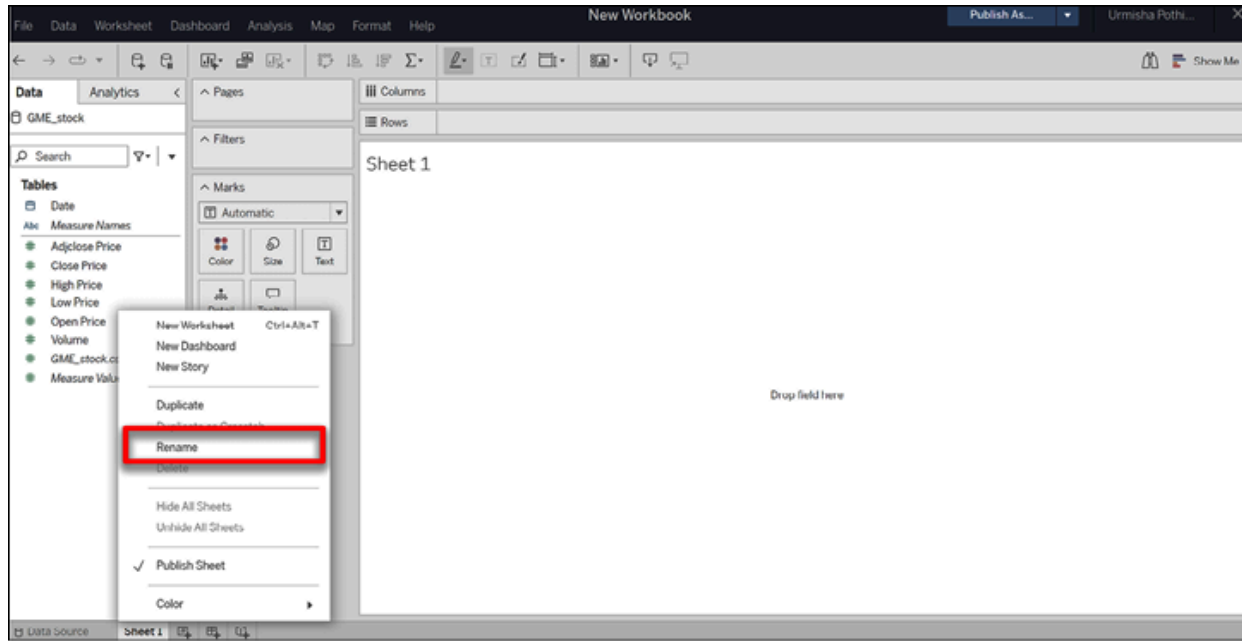


Figure 3.5: Rename a worksheet *Duplicating a worksheet*

Creating a duplicate sheet creates a carbon copy of the desired worksheet, however there is no connection between the two.

Note: When you create the duplicate dashboard, the new version of the dashboard will still refer to the same sheets used for the original dashboard, and any filter actions/selections will reflect on both the original and duplicated dashboards.

There are two ways to duplicate your worksheet. Let us understand them one by one:

- To duplicate a worksheet: Right click on the sheet tab you wish to duplicate, then select **Duplicate** from the drop-down menu, as shown in the following figure:

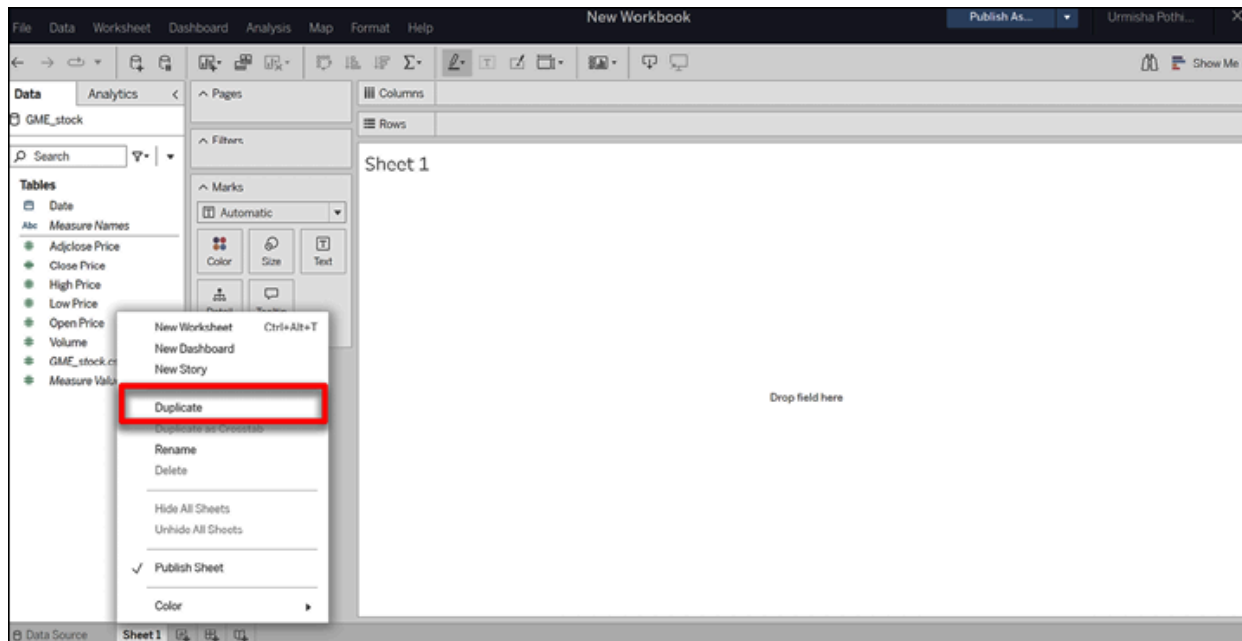


Figure 3.6: Duplicate a worksheet ***Duplicate as crosstab:** To present your graphical data as a table summarizing the information in rows and columns, you can choose the Duplicate as Crosstab option by right-clicking the sheet tab to create a new worksheet with the numbers in a Crosstab format.*

Crosstab is also referred as Pivot-Table. Follow these steps:

1. Right-click on **Sheet 1** on the sheet-tab.
2. Select the **Duplicate as Crosstab** from the option as shown in the following figure:

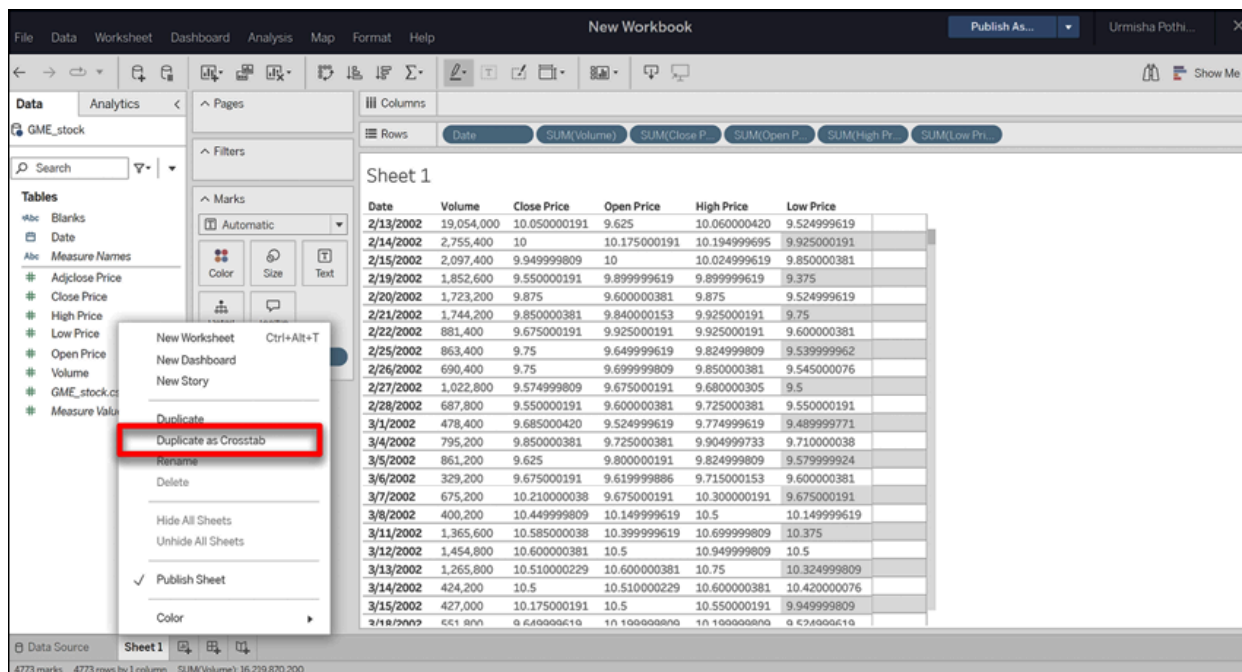


Figure 3.7: Duplicate as Crosstab a worksheet

Note: Dashboards and Story cannot be duplicated as crosstabs.

Reordering worksheets: Worksheets in a Tableau Workbook can be re-arranged into any desired sequence simply by using a mouse to click on Sheet Tab hold and drag to the desired location.

Deleting a worksheet

You can permanently remove a worksheet from the workbook. However, Tableau will not allow a worksheet to be deleted if it is the only worksheet in your workbook.

To delete a workbook, right-click the sheet tab and select **Delete** from the options, as shown in the following figure:

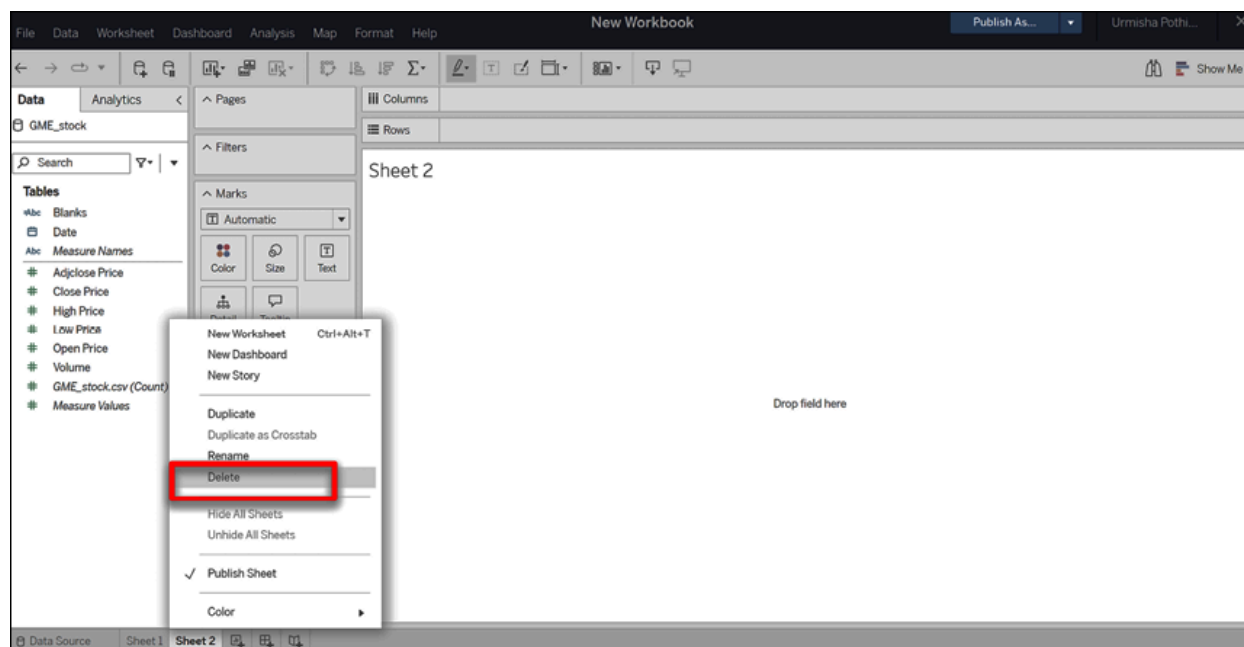


Figure 3.8: Delete a worksheet

Note: You can also perform above all the operations for Dashboards and Stories as well.

Types of charts

Tableau supports more than twenty-one charts. Here, we will learn how to build a few of those most popular charts.

Bar chart

Bar charts are the most common and widely used data visualization to compare data across multiple categories, find the historical highs and lows, show the frequency of each data category, and highlight the differences.

Bar charts are pictorial representations of different categories in the data, where each bar represents a category, and the length/height of the bar corresponds to the value of that category.

There are a variety of bar charts:

- Horizontal bar chart (i.e. bar chart)
- Vertical bar chart (i.e. column chart)
- Stacked bar chart
- Grouped bar chart (i.e. clustered bar chart)

Examples of bar charts:

- Sales by customer age
- Average transactions on various payment types
- Plotting pageviews counts over the period of six months

Horizontal and vertical bar chart

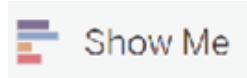
Let us get hands-on with horizontal and vertical bar charts: **Pre-requisites:** To create a horizontal or vertical bar chart, you need at least one dimension and one measured field.

Steps to build a horizontal or vertical bar chart:

1. Connect to the data **Sample-Superstore** data source.
2. Drag the dimension field **Category** from **Orders sheet** to row shelves.
3. Drag the measured field **Quantity** to column Shelves.
4. By default, Tableau will aggregate the measured field as SUM. However, you can change the aggregation option from the drop-down menu by selecting the measured field in column shelves.
5. To select the chart type, go to **Marks** and select **Bar Chart** from the drop-down menu.

Or, on the **Marks** cards, choose **Bar** from the drop-down menu to change the type of chart.

Or, click **Show Me** on the Toolbar:



6. Then, in the **Show Me** pane, Select the **Horizontal Bar Chart**.

You will also find the Tableau suggested charts based on the dimensions and measures in column and row shelves.

Adding various attributes to the bar chart: To add labels to the graphs, you can select:

- **Show the Label Marks** icon on the toolbar. 

Refer to the following figure:

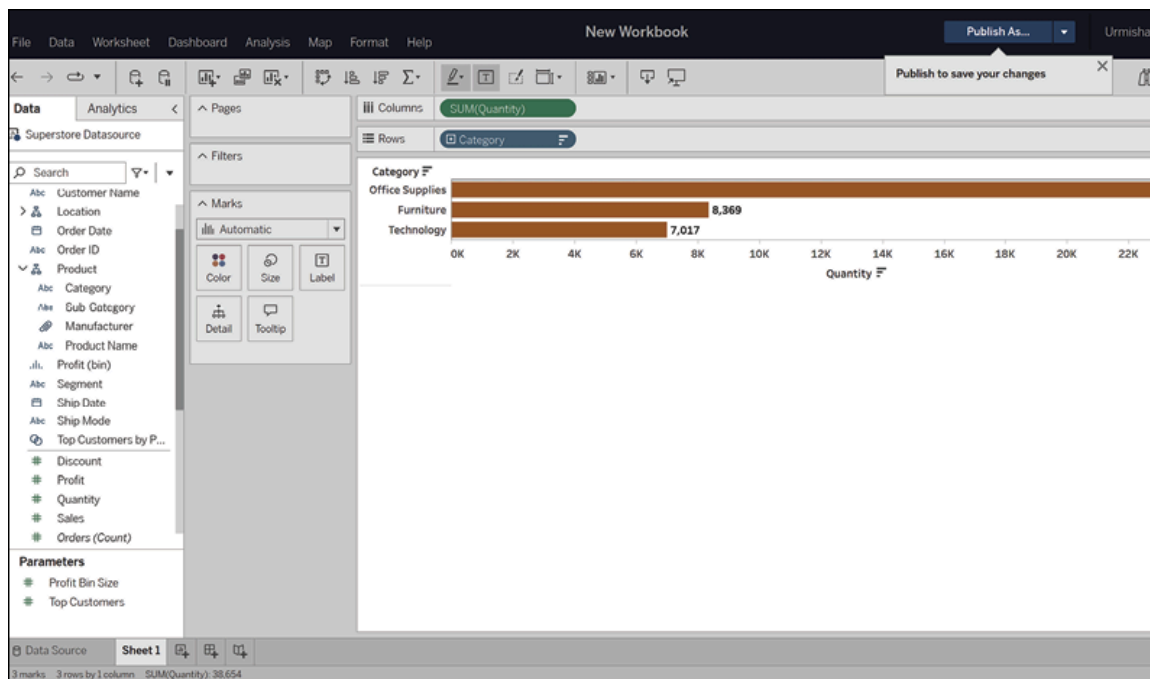


Figure 3.9: Horizontal bar chart

- To build the vertical bar chart, select the swap rows and columns icon on the toolbar. 
- To change the color of the bars, simply select **Color** from **Marks Card**. Refer to the following figure for a better understanding:

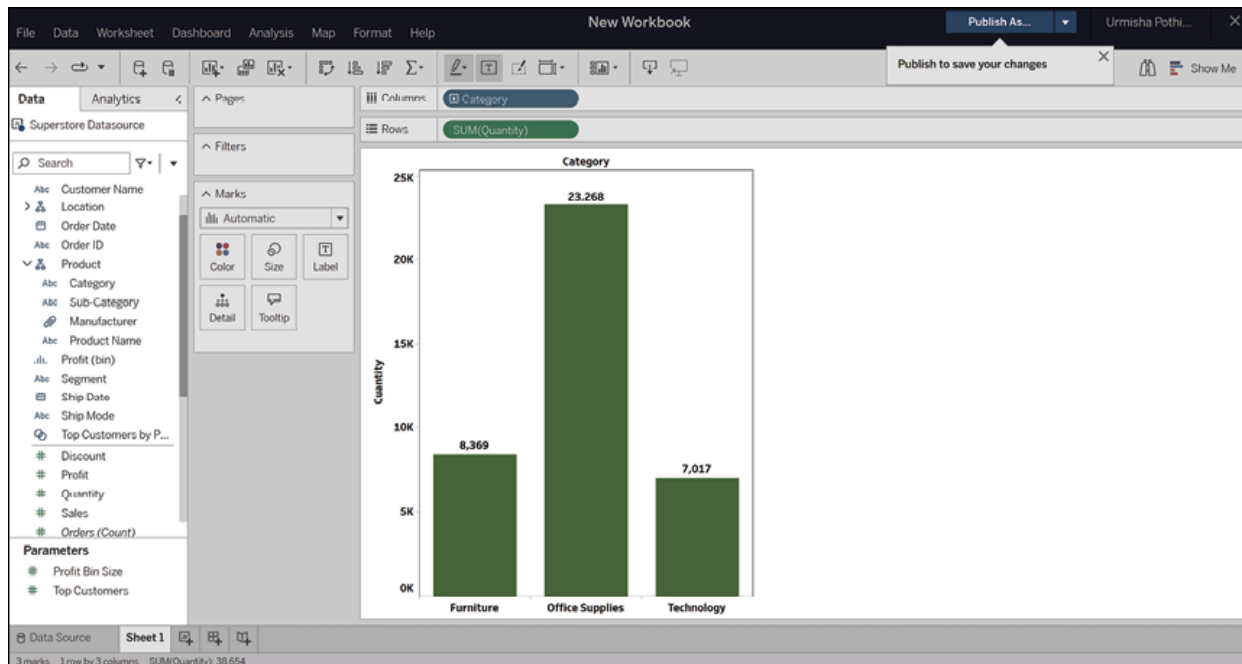


Figure 3.10: Vertical Bar Chart The vertical and horizontal bar charts mainly work for single dimension/categorical fields, while the stacked and grouped bar charts work for multiple categorical fields.

Stacked bar chart

Now, let us build the stacked bar chart.

Pre-requisite: To create a stacked bar chart, you need at least two dimensions and one measured field. Follow these steps to build a stacked bar chart:

1. Connect to the data **Sample-Superstore** data source.
2. Drag the dimension field **Category** from **Orders sheet** to row shelves.
3. Drag the measured field **Quantity** to column Shelves.

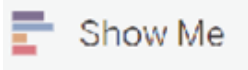
By default, Tableau will aggregate the measured field as SUM. However, you can change the aggregation option from the drop-down menu by selecting the measured field in column shelves.

4. Drag the dimension field **Ship-Mode** to **Color Marks Card**.
5. To select the chart type, go to **Marks** and select the Bar Chart from the drop-down menu.

Or,

On the **Marks** cards, choose **Bar** from the drop-down menu to change the type of chart.

Or,

Click **Show Me** on the Toolbar 

Then in the **Show Me** pane, select **Stacked Bar Chart**.

6. To add labels to the graphs, you can select **Show Label Marks** icon on the toolbar.



7. To change the horizontal stacked bar to vertical stacked bar charts and vice-versa, Select **Swap Rows and Columns** icon on the toolbar.



Refer to the following figure for a better understanding:

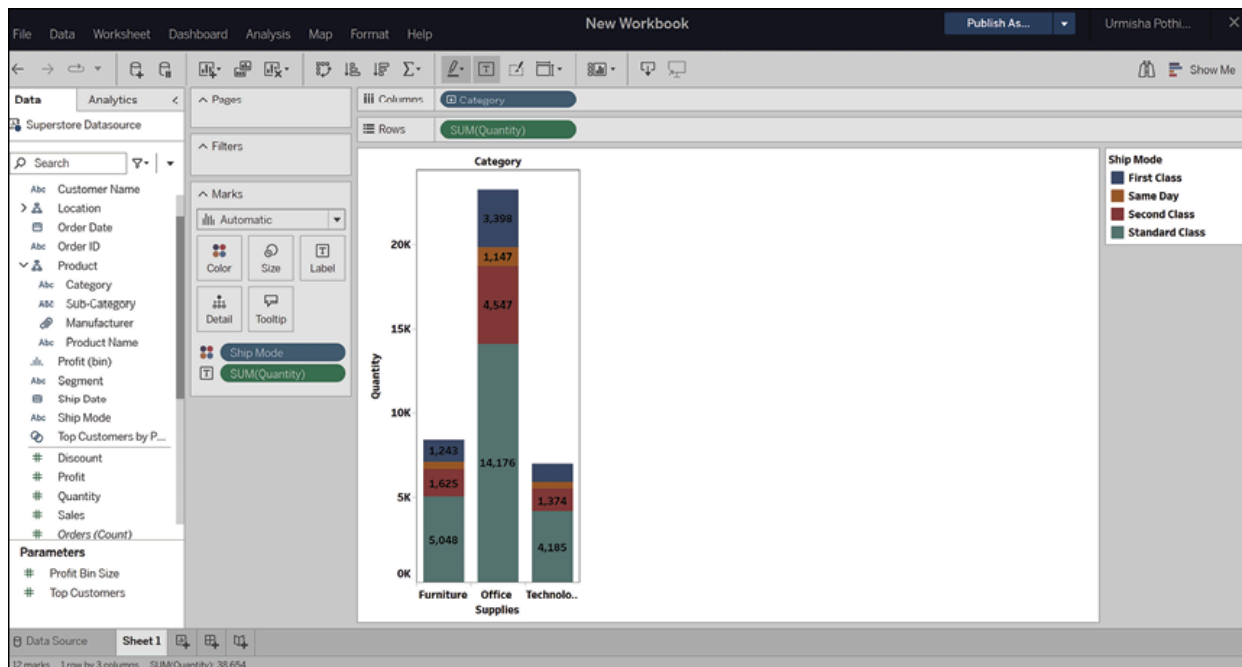


Figure 3.11: Stacked bar chart *Grouped bar chart*

Now, let us get hands-on with grouped bar chart.

Pre-requisite: To create a grouped bar chart, you need at least two dimensions and one measured field. Follow these steps:

1. Connect to the data **Sample-Superstore** data source.

2. Drag the dimension field **Category** from Orders sheet to row shelves.
3. Drag the measured field **Quantity** to column shelves.

By default, Tableau will aggregate the measured field as SUM. However, you can change the aggregation option from the drop-down menu by selecting the measured field in column shelves.

4. Add dimension field **Sub-Category** next to **Category** field in Column shelves.
5. To select the chart type, go to **Marks** and select the **Bar chart** from the drop-down menu.
6. To add labels to the graphs, you can select **Show Label Marks** icon on the toolbar.



7. To change the horizontal grouped bar to vertical grouped bar charts and vice-versa, Select **Swap Rows and Columns** icon on the toolbar.



Refer to the following figure for a better understanding:

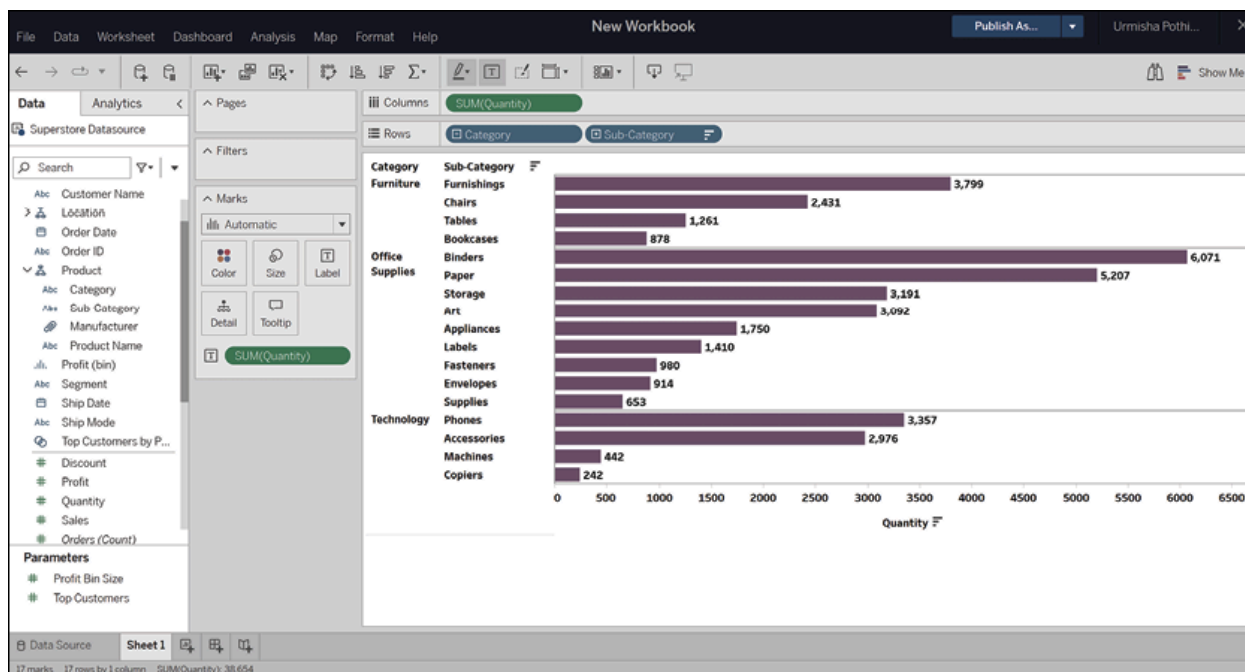


Figure 3.12: Grouped bar chart *Pie chart*

A pie chart is a circular graphical representation of data where the circle represents the entire data, and the slices represent the proportional of the quantity represented by each category in the data. The data world has a lot of ongoing debate about the use of pie charts. In simple words, pie charts are a good ratio tool for ratio analysis (i.e. Yes/No, or North/South, etc.), but we should refrain from using them if data has more than two categories.

Examples of pie charts include the following:

- Favorite type of movies
- Recommended diet
- Job classification

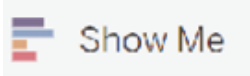
Building a pie chart

Pre-requisite: To generate a pie chart, you need at least one or more dimensions and one or two measured fields.

Note: This chart is not ideal for the dimension field, which has a large number of categories.

Follow these steps to build a pie chart:

1. Connect to the data **Sample-Superstore** data source.
2. Drag the dimension **Segment** to row shelves and drag the measured field **Orders(Count)** to column shelves.
3. By default, Tableau will aggregate the measured field as SUM and display the horizontal bar chart.

4. Now, Click on **Show Me** on the toolbar 

Select the **Pie Chart** from the type of charts.

5. To add the labels to the chart, drag the dimension **Segment** to **Label** in **Marks Card** and the measured field **CNT(Orders)** to **Label** in **Marks Card** from the Data pane.
6. To change the measured field **CNT(Orders)** to percentage format, Right-Click the field **CNT(Orders)**, and now select **Add Table Calculation Percent of Total**.

Refer to the following figure for a better understanding:

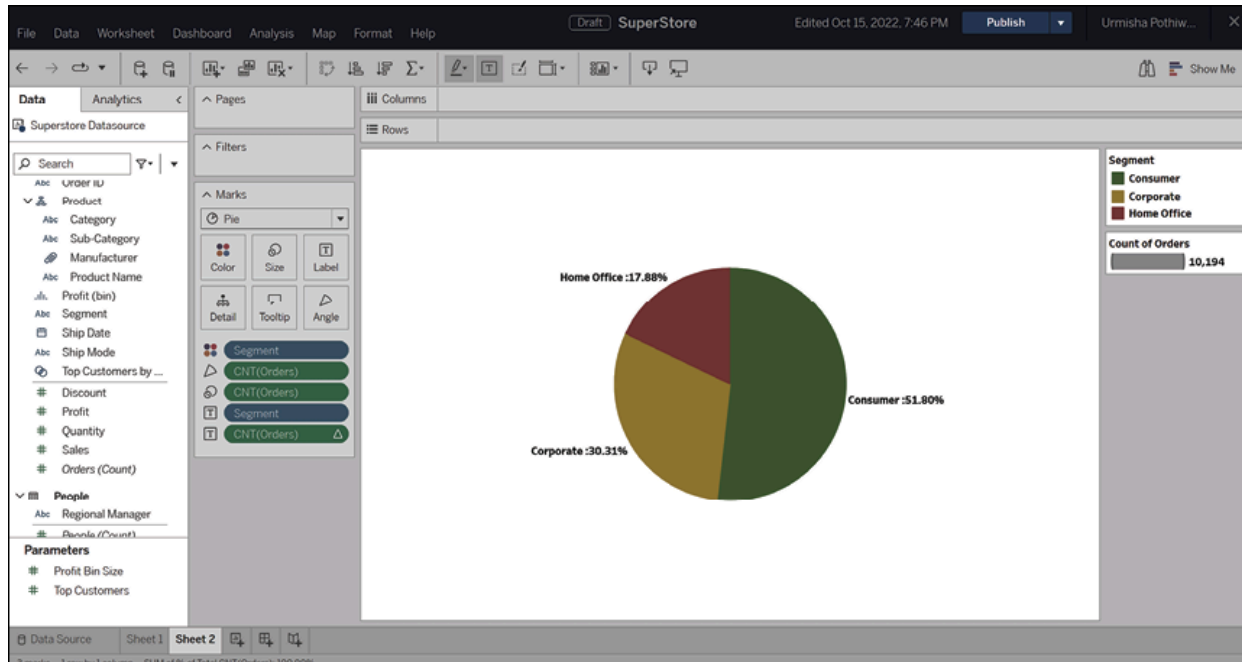


Figure 3.13: Pie chart *Line chart*

Line chart or line graph is used to view the changes (trends) over time. This graph is created by connecting several data points. Line charts are used to track the changes over short or long periods of time.

Examples of line graphs:

- Stock analysis
- Average monthly rainfall
- Yearly earnings
- Annual sales trend
- Daily mobile usage and so on

Building a line chart

Pre-requisite: To generate a line chart, you need at least one date field and one or more measured fields.

Follow these steps to build a line chart:

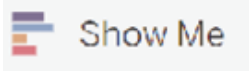
1. Connect to the data **GME Stock** data source.
2. Drag the **Date** to **Column** shelves.

By default, Tableau will aggregate the date by year and create the column headers.

3. You can also change the granularity of the aggregation level by selecting the date options from the drop-down menu and selecting the date field on column shelves.
4. Drag **Volume** to **Row** shelves.

By default, tableau will aggregate the volume as SUM and display the simple chart (Automatic in Marks Card).

5. On the **Marks** cards, choose **line** from the drop-down menu to change the type of chart.

or click **Show Me** on the Toolbar 

Then, in the **Show Me** pane, Select the **Line Chart**.

You will also find tableau suggested charts based on the dimensions and measures in column and row shelves.

6. To add labels to the graphs, you can select **Show Label Marks** icon on the toolbar.



7. [Optional] To enhance your graph, you can also change the **Title**, Format the **font style, size, gridlines, borders, edit axis, colors**, and so on.

Refer to the following figure for a better understanding:

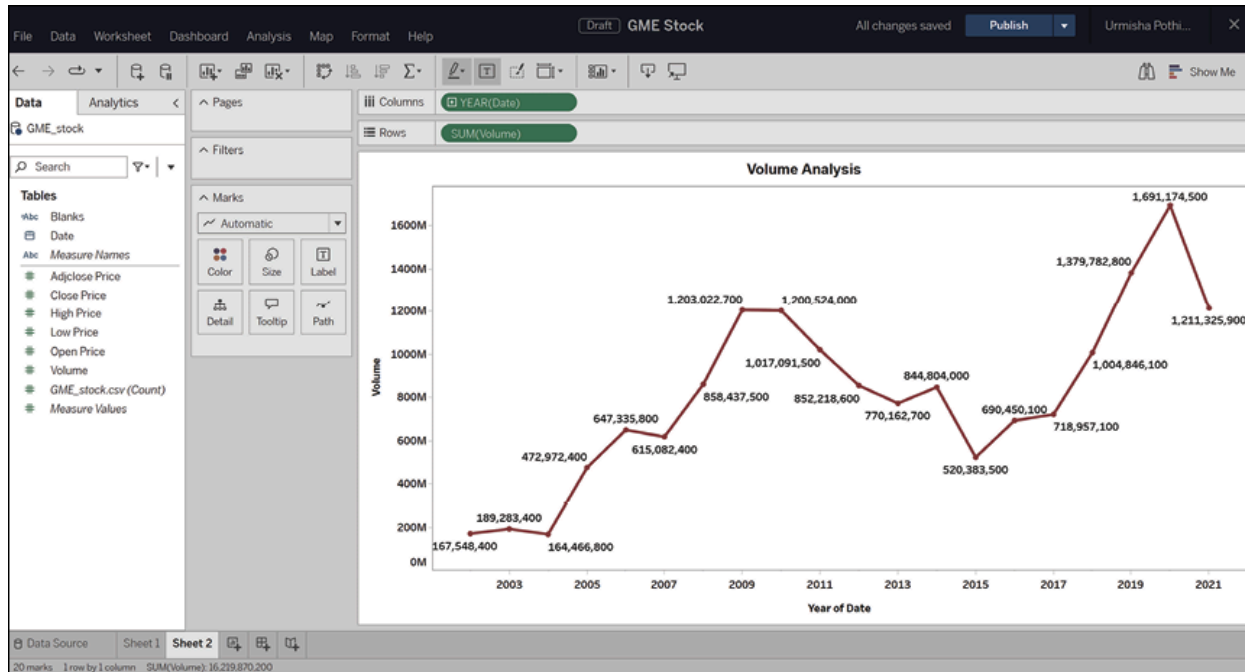


Figure 3.14: Line chart *Text table chart*

Text charts, also known as crosstabs, are a type of visualization in Tableau that displays data in a table format, with rows and columns. Each cell in the table shows the value of a measure at the intersection of the row and column headers. The text table charts use a text mark type. The Text table chart in Tableau is the same as the pivot table in Excel.

When users want to work on numbers instead of graphical representation then crosstabs are used. This does not mean that your table has to be completely boring just to display numbers, you can make it interesting and appealing.

Examples of text table chart:

- Expense Text Table
- Order – Sales Text Table

Building a text table chart

Pre-requisite: To create a text table, you need at least one measured field and at least one dimension field.

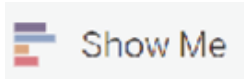
For example, if you want to create a text table on Total Sales, Number of Locations for Food Chains, then follow these instructions:

1. Connect to the data **FoodChain** data source.

2. Change the **Mark** type to **Text**.
3. Drag the measured fields **Sales** and **#Locations** to **Column** shelf.
4. Drag the dimension field **Chains** to **Row** shelf.
5. This will automatically create cross tabulation.

Or

click **Show Me** on the Toolbar



6. Then in the **Show Me** pane, select the **Text Table**.
7. When finished, click **OK**.

Refer to the following figure for a better understanding:

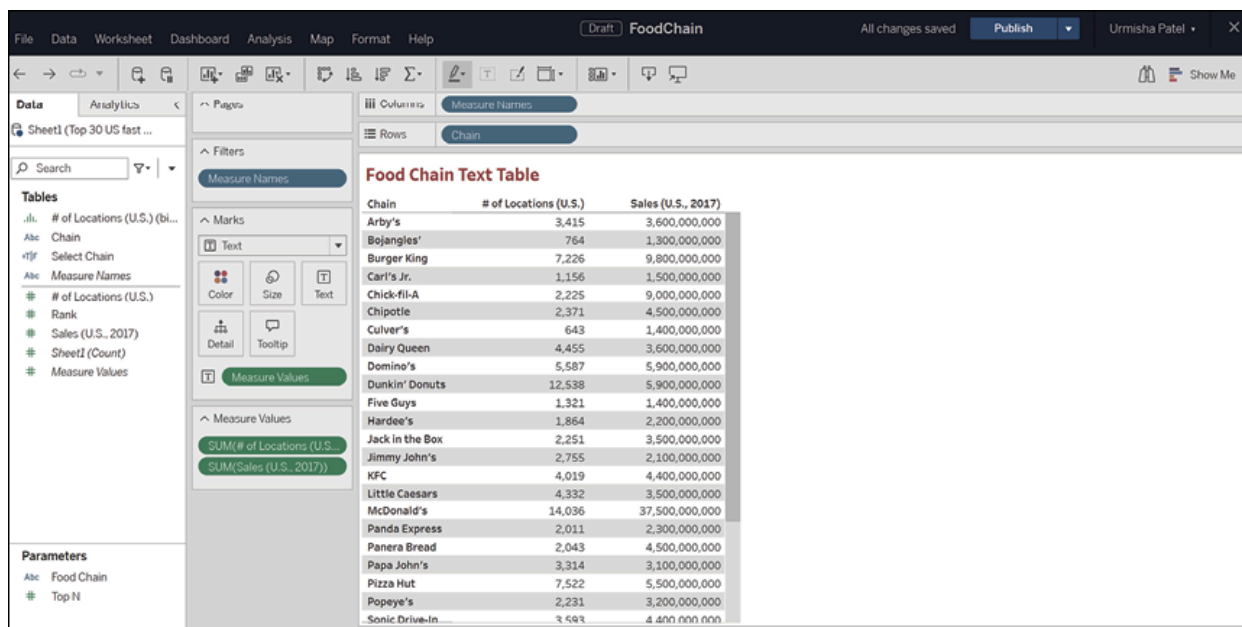


Figure 3.15: Text-table chart *Scatter plot*

Scatter plots are the most effective way of finding relationships between different numerical fields. This plot gets enhanced with analytics like trend lines or cluster analysis.

The two-dimension graph where the two measured values decide its position, i.e., one on the X-axis and another on the Y-axis.

Examples of scatter plots are as follows:

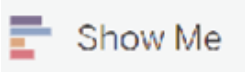
- Finding the relationship between Electricity Bill vs. Temperature.
- Scatter plot between Car Price vs. Car Age.
- Finding a trend between Online Hotel Revenue vs. the Number of Travel Agents.

Building a scatter plot

Pre-requisite: To create a scatter plot, you need at least two measured fields and zero or one-dimension fields.

Follow these steps to build a scatter plot:

1. Connect to the data **FoodChain** data source.
2. Drag the measured fields **Sales** to the **Column shelf** and **#Locations** to the **Row** shelf.
3. Go to **Analysis**, then disable the **Aggregate Measures**.

Or click **Show Me** on the Toolbar 

Then, in the **Show Me** pane, Select the **Scatter Plot**.

4. [Optional] Change the Shape of value points by clicking **Shape** Card.

By default, the scatter plot represents the value points with hollow circle shapes.

For instance, change it to a filled circle shape.

5. [Optional] To find the trends and patterns between the two fields and their relationship, then select **Analysis Pane** next to Data Pane and drag the **Trend Line**. Then, select the appropriate trend line type, which gives accurate results based on R-Square value.

To edit or format the trend line, select the trend line and perform the respective Edit or Formatting on the trend lines.

6. [Optional] Change the color of the shapes by clicking **Color** Card. Select the color.
7. [Optional] Change the size of the shapes by clicking **Size** Card. Then, adjust the size of the shape.
8. When finished, click on **OK**.

Refer to the following figure for a better understanding:

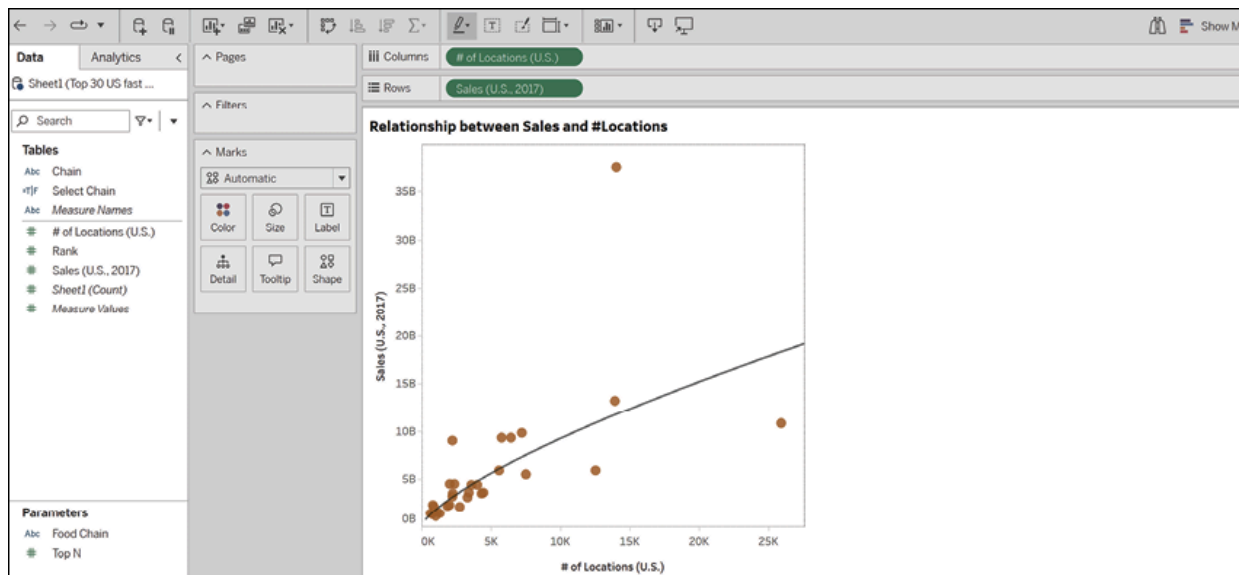


Figure 3.16: Scatter Plot with trend lines *Filled map*

Filled maps impart location information in the visuals. If you have any geographic information (includes country, state, area, province, postal code, etc.,) associated with the data, then filled maps are the best choice to understand how location correlates with trends in the data.

Examples of filled maps:

- Data that help find out information regarding obesity in a county.
- Data that tell us which *Indian* cities produce Ethanol.

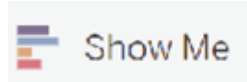
Building a filled map

Pre-requisite: To create a filled map, you need at least one geographic dimension field, at most two measured fields, and zero or more dimension fields.

For instance, you want to create a filled map on State/Province field based on the total number of orders from the Sample SuperStore, then follow these instructions:

1. Connect to the data **Sample SuperStore** data source.
2. Open the **Location** folder and double-click on **State/Province** geographical dimension field.
3. This will automatically generate a basic map.
4. Go to the **Marks** section, change the graph to **Map**.

Or, click **Show Me** on the **Toolbar**



Then, in the **Show Me** pane, Select the **Filled Map**.

5. Drag **Country/Region** to **Color Marks Card**. To change the colors to define unique value with different colors.

To display the total number of sales in each **State/Province**, drag **Orders** from **Measure** section to **Label**. Then click on the label to configure the font style, color, and size of the label. Refer to the following figure for a better understanding:

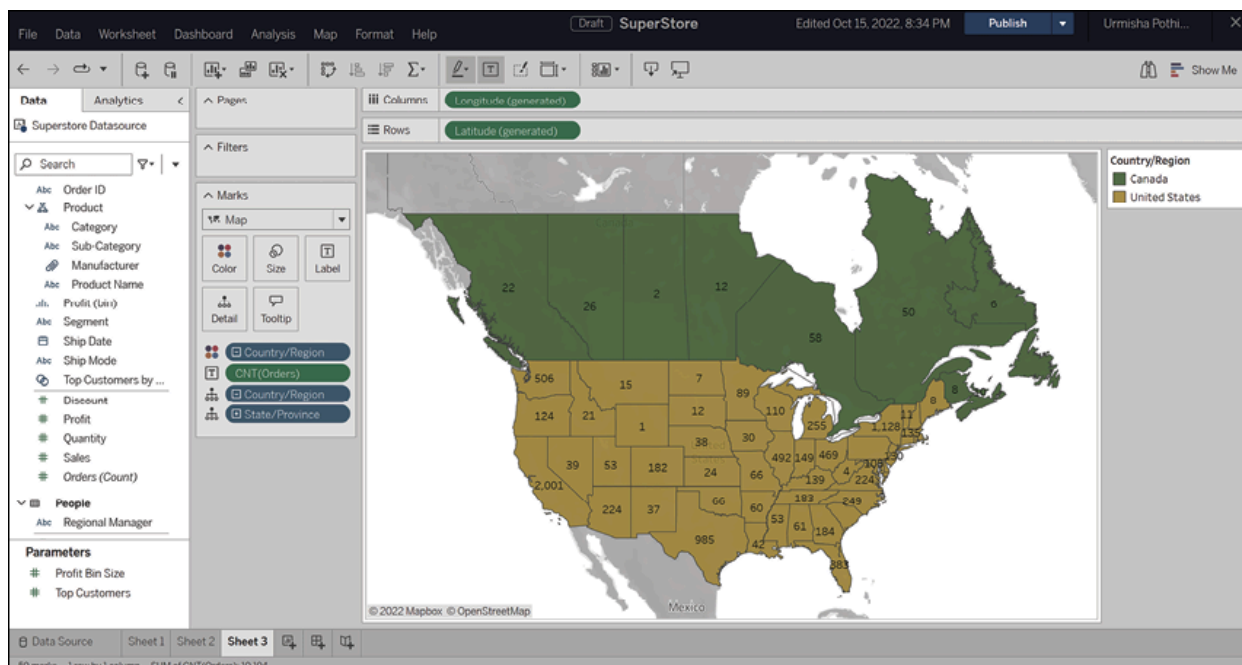


Figure 3.17: Filled map [Histogram chart](#)

A histogram chart is a graphical representation of the frequency distribution of the continuously measured field. This chart plays a vital role in understanding the statistics of data and the distribution of data and its shape.

A histogram chart looks like a bar chart. However, a bar chart compares the discrete measured field. While a histogram chart depicts the distribution of the continuous measured field.

Examples of histogram charts:

- Histogram of flight arrivals per minute
- Distribution of property sales between Jan'2021 to Dec'2022

- Histogram on pharmacy drug dispensing turnaround time

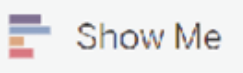
Building a histogram

Pre-requisite: To build a histogram, you need one continuous measured field and no dimensions.

For Example, you want to build a histogram on the distribution of locations.

Follow these steps to build a histogram:

1. Connect the data **Sample-SuperStore** data source.
2. Drag the continuous measure field **#Locations** in the row shelf or double-click the continuous measured field **#Locations**. Then, Click **Show Me** on the

Toolbar 

In the **Show Me** pane, Select the **Histogram**.

This will create a new field named as **#Locations (bins)**.

[Optional] To **Edit** this **Bin**, right-click the bin and select **Edit**.

Refer to the following figure for a better understanding:

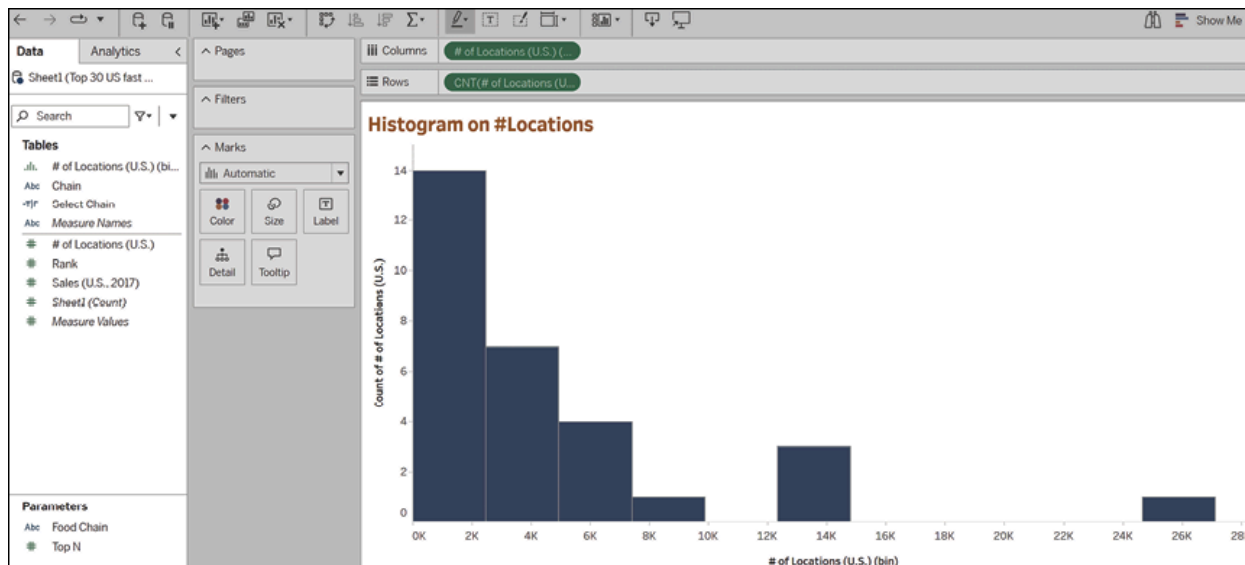


Figure 3.18: Histogram chart The above figure depicts that there are 14 locations belong to 1st bar and seven locations belong to 2nd bar, and so on.

Formatting

Formatting becomes crucial in choosing the right format when it comes to your presentation and analysis. Formatting allows you to make changes in your worksheet, workbook including font size, style, alignment, colors, header, borders, gridlines, graph lines, axis, legends, highlighters, filters, parameters, borders, and dividers, etc.

Note: There are fewer formatting options available in Tableau Online on the web-edit environment than Tableau Desktop.

These formatting can be done at two different levels.

- Formatting at Workbook level
- Formatting at Worksheet level

We will explore each of these levels in detail.

Formatting at workbook level

Formatting at the workbook level refers to quickly changing the fonts, including worksheet titles, tool tips, worksheet font size, colors, dashboard titles, story titles, and lines, including grid lines, trend lines, zero lines, axis ruler lines, axis ticks by configuring the format settings at the workbook level instead of at the sheet level.

Follow these steps to format at workbook levels:

1. Go to the **Tableau Menu bar**, then Click on the **Format** menu.
2. Select **Workbook**.
3. You will get the Format Workbook pane like the following [Figure 3.19](#), which consists of a series of drop-down lists where you can make the changes in the format settings in a workbook.

Click on **More** drop-down and make the changes in a workbook setting:

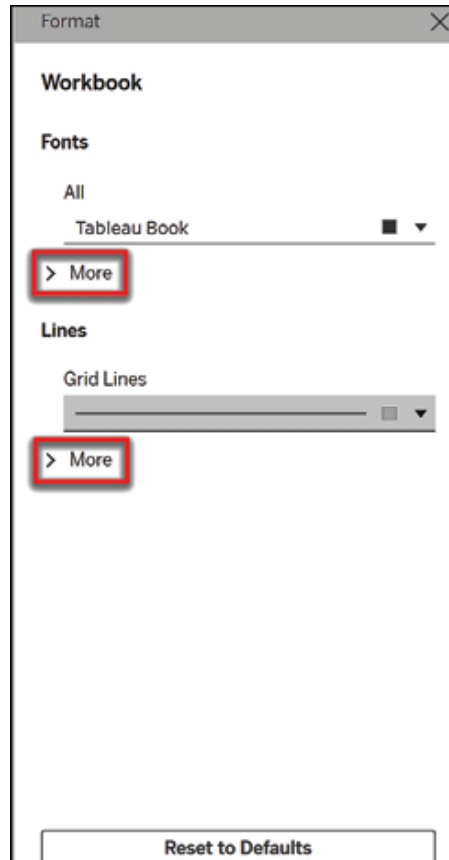


Figure 3.19: Format settings at a workbook Let us understand further the options available within the drop-down list.

- **Font:** The font formatting drop-down list includes tool tips, worksheet titles, dashboard titles, story titles where you can change the font size, style, color palette as shown in the following figure:

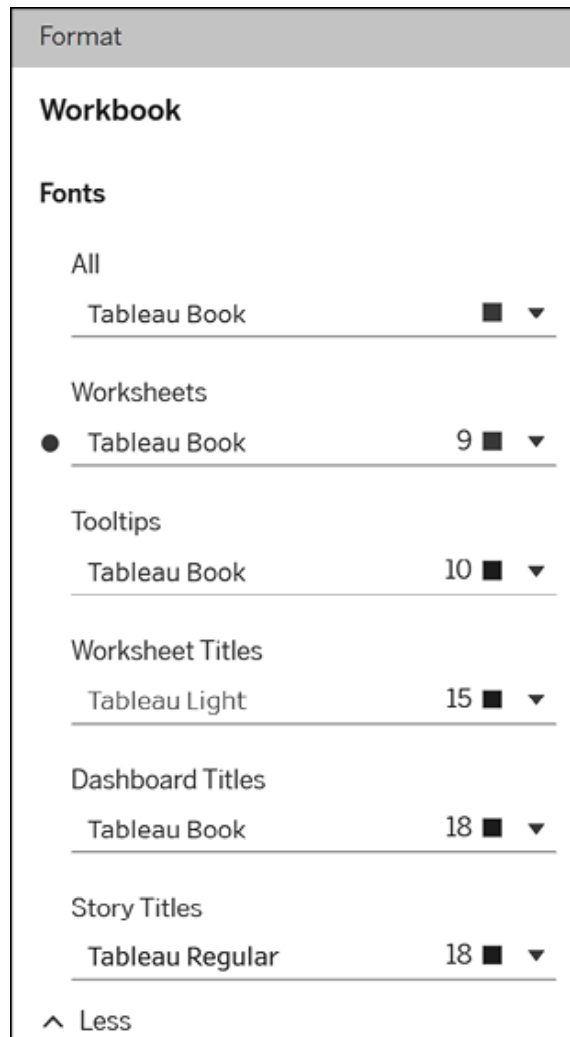


Figure 3.20: Font settings at a workbook

- **Lines:** The lines formatting drop-down list includes grid lines, zero lines, trend lines, reference lines, axis rulers, and axis ticks, where you can change the line types, line size, color palette, and line opacity.

Note: If you change the line opacity and set different color settings for the line type, then the change in line opacity reflects at the worksheet level instead of the workbook level.

Refer to the following figure for a better understanding:

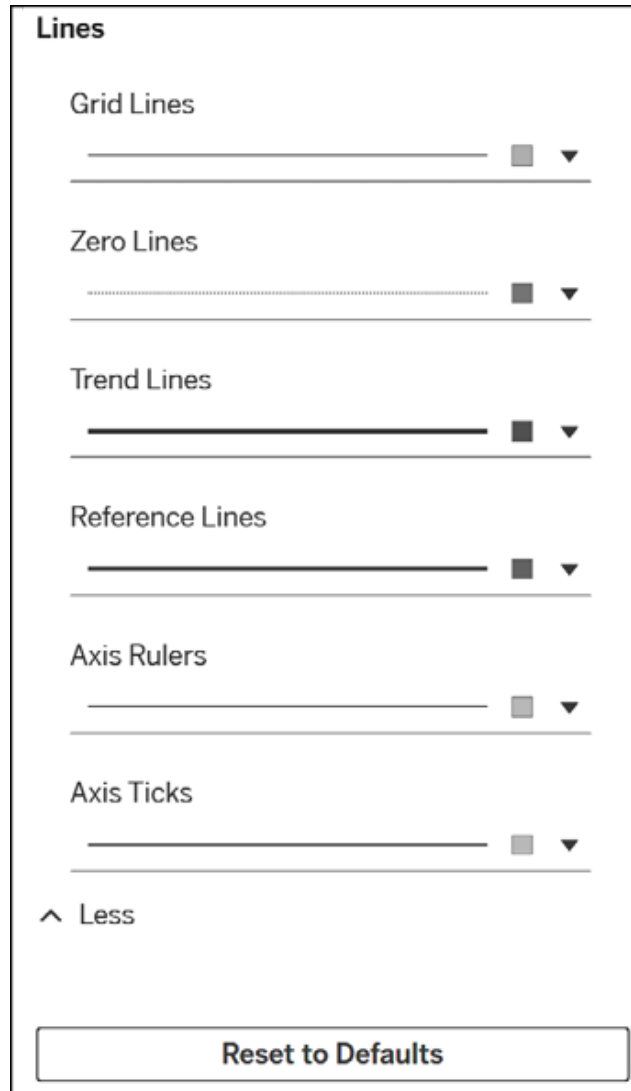


Figure 3.21: Line settings at a workbook *Formatting at worksheet level*

You can configure the format settings at the worksheet level, where you can change the fonts, styles, lines, interactive controls, borders, and dividers. This formatting will apply only to the current view.

Follow these steps to format at worksheet levels:

1. Go to the **Tableau Menu bar**, then click on the **Format** menu.
2. Select **Worksheet**.
3. You will get the **Format** worksheet pane like the following *Figure 3.22*, which consists of a series of drop-down lists where you can make the changes in the format settings in a worksheet:

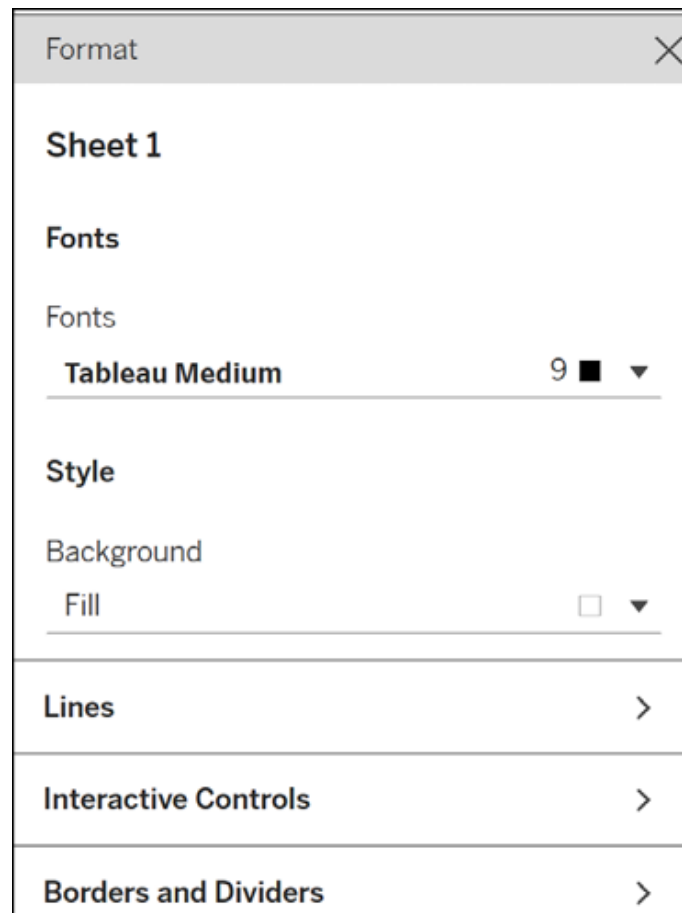


Figure 3.22: *Format Settings at a Worksheet Let us explore the drop-down list and the options available in detail:*

- **Fonts:** Under Fonts settings, you can change the color, font style, and size, color palette, and so on, as shown in the following figure:

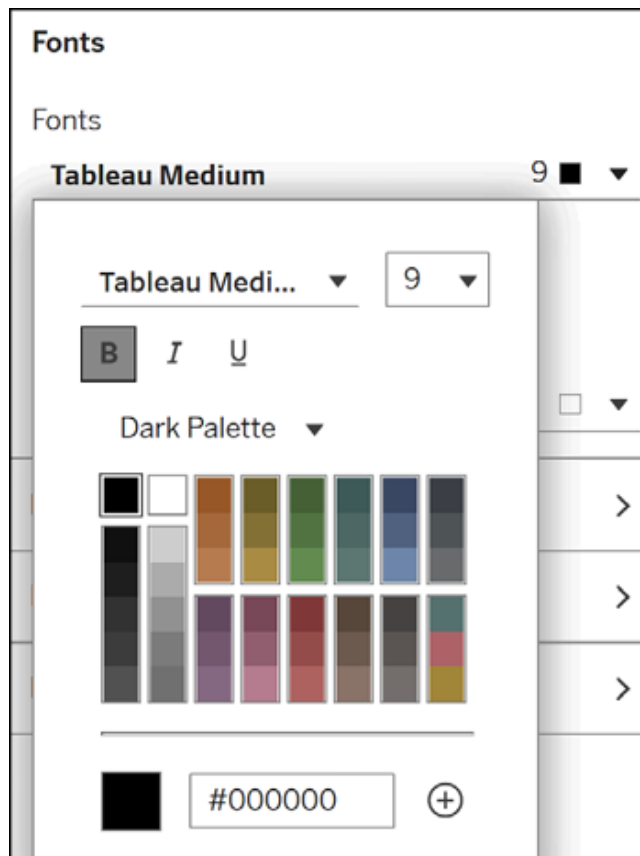


Figure 3.23: Font Settings at a Worksheet

- **Style:** Under Style settings, you can change the background color palette, color, and so on, as shown in the following figure:

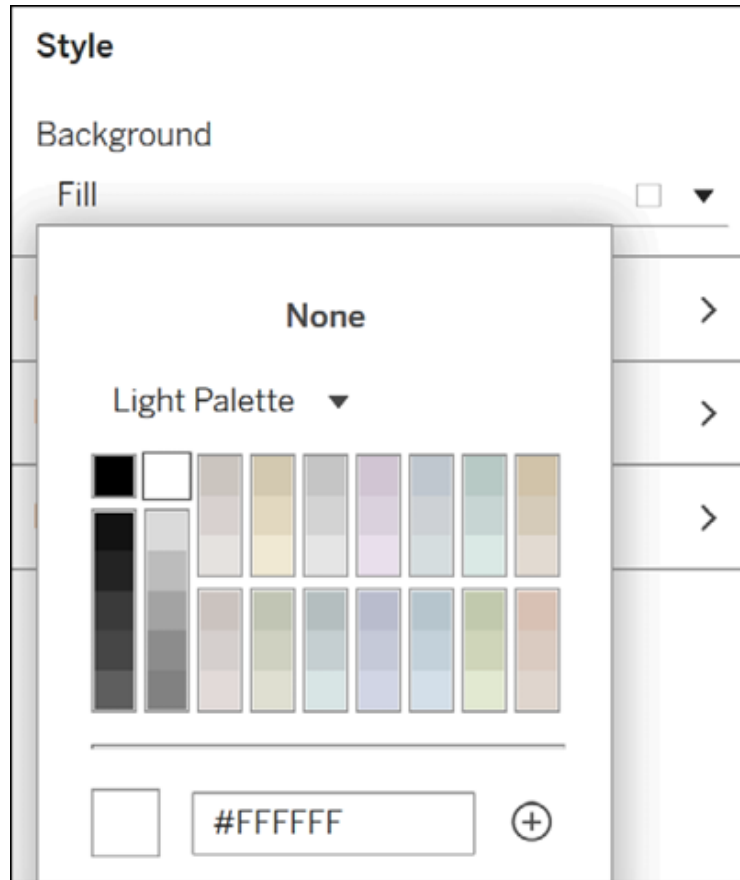


Figure 3.24: Style settings at a worksheet

- **Lines:** In Lines formatting, you can change the settings, including the gridlines and zero lines and the appearance of the view, as shown in the following figure:

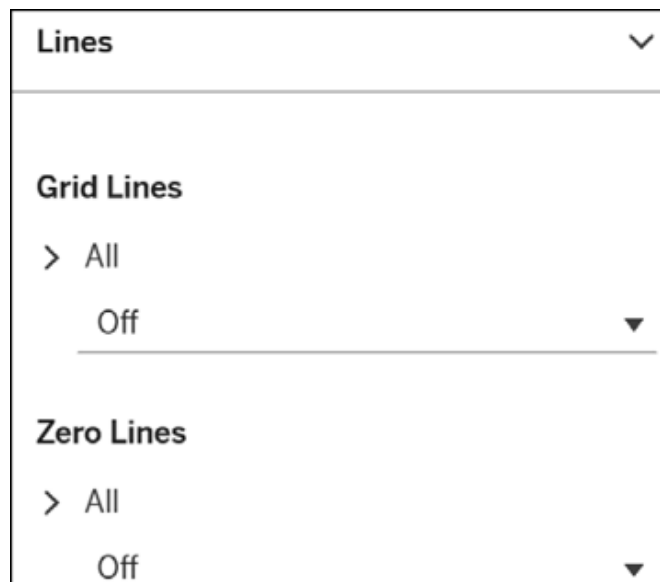


Figure 3.25: Lines Settings at a worksheet

- **Interactive controls:** This section gives control over legends, highlighters, parameters, and filters. Here, you will learn how to control the appearance of interactive controls. i.e., it can change its font size, style, title formats, background color, and color palette, style, borders etc., as seen in [Figure 3.26](#):

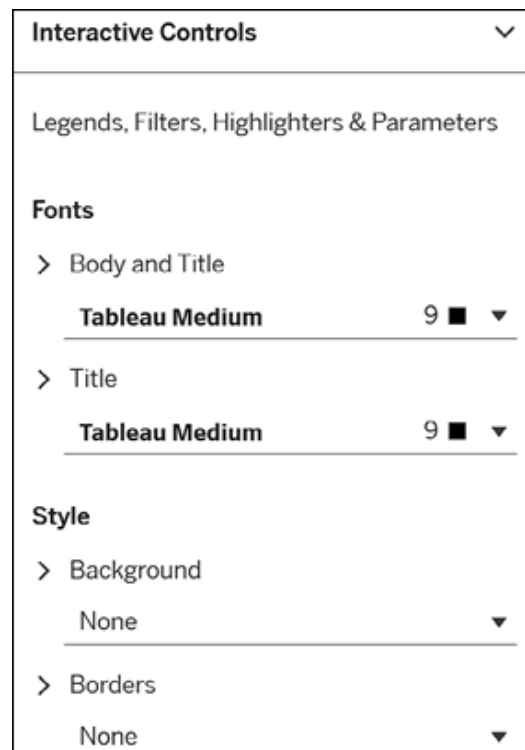


Figure 3.26: Interactive Control settings at a worksheet

- **Borders and dividers:** The lines surrounding the table, cell, pane, and header in the view are called the border. You can control the border/lines - its appearance, color, etc., You can also format the row and column dividers-their appearance, style, color, and level of the borders that divide each row and column via row and column dividers drop-down as shown in the following figure:

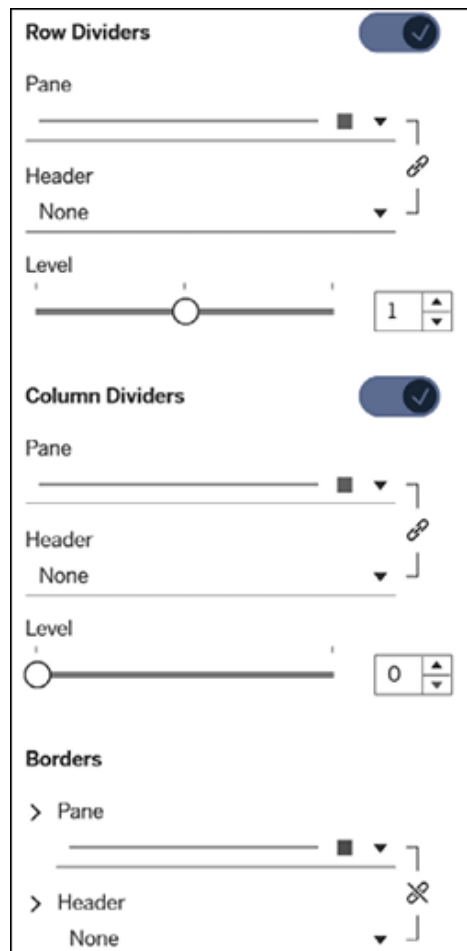


Figure 3.27: Borders and dividers settings at a worksheet

Data can be organized and simplified by using various techniques in Tableau based on different purposes. **Filters** and **Parameters** are an essential part of analyzing data and personalizing the datasheet in view. Let us understand them in detail.

Filters

Filters in Tableau are powerful tools that allow users to control which data is displayed in their visualizations. By applying filters, users can focus on specific subsets of data, enhancing their ability to perform detailed analysis and derive insights. Filters can be applied to dimensions (e.g., categories, regions, dates) and measures (e.g., sales, profit) to include or exclude data based on specified criteria. They can be used at various levels, such as worksheet, dashboard, or data source levels, providing flexibility in how data is refined.

Additionally, Tableau offers various types of filters, including categorical, range, relative date, and context filters, each serving different analytical needs. Interactive filters enable viewers to dynamically adjust the data view, fostering a more

engaging and customized analytical experience. Overall, filters are essential for isolating relevant data, uncovering trends, and facilitating precise data-driven decision-making.

Filters are created and applied on the worksheet level, and they are specific to the data source. Filters can be customized based on the user selection. This will modify the data dynamically in the worksheets and dashboards.

There are various types of filters used based on different purposes in Tableau Online. They are as follows:

- Extract filters
- Data source filters
- Context filters
- Dimension filters
- Measure filters

Let us explore each of them in detail in the upcoming sub-sections.

Extract filters

Extracts in Tableau are snapshots of data taken from a data source that can be stored and accessed locally. They are used to improve performance and enable offline access to data.

The extract filters can extract the data from the data source. However, this filtering works for the extracted data connection and not for the live data connection.

Extract filters are applied when creating a data extract. These filters determine which subset of the data source is included in the extract file. They help in reducing the size of the extract.

First, create a local copy in the Tableau repository by changing the data connection from Live to **Extract**. In other words, click on **Extract** as shown in [Figure 3.28](#):

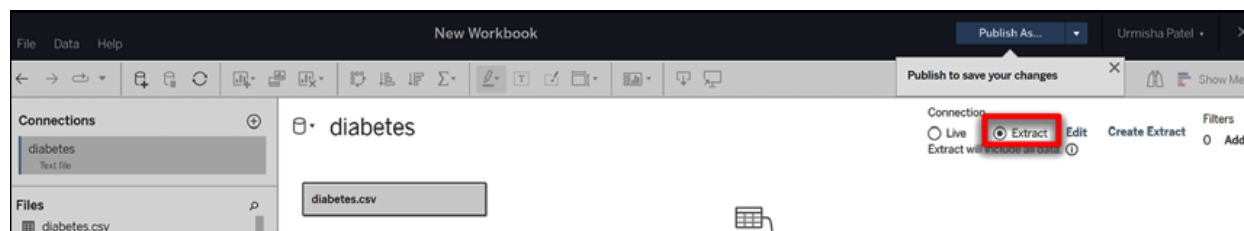


Figure 3.28: Extract data connection Follow these steps to apply **Extract Filter** on **Diabetes** data source:

1. Click on **Edit** next to **Extract**. This will open the **Extract Data** dialog box.
2. Now, click on **Edit** in Filter section, select the field by clicking **Add Filter** and add as extract filter.
3. Select **Outcome** field from the list. Then **OK**:

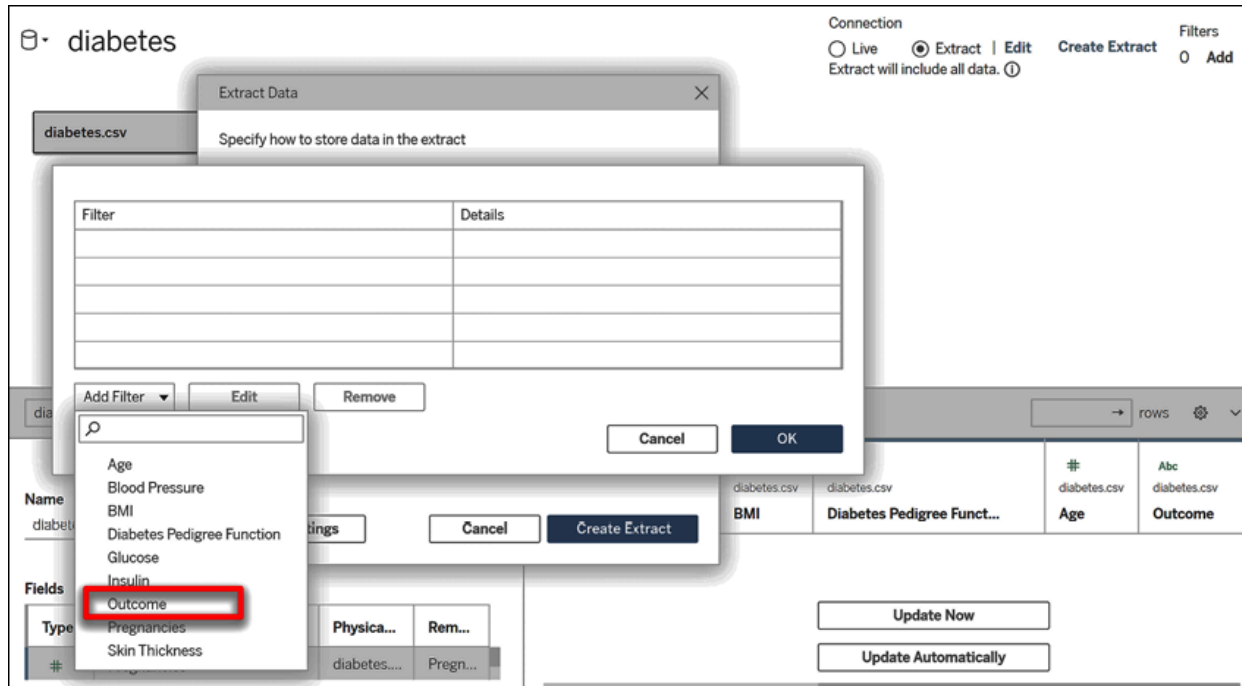


Figure 3.29: Add Filters in Extract Filtering This filter window has multiple options to filter the field Outcome on various use cases as shown in the following figure:

Filter [Outcome]

▼ **General**

List

Select from list ▼

Search

☐ (All)

☐ 0

☐ 1

☐ Exclude selected values

> Wildcard

> Condition

> Top/Bottom

Summary

Selection: Selected 0 of 2 values

Wildcard: All

Condition: None

Limit: None

Reset Cancel OK

Figure 3.30: Extract filter condition *Use case 1: Select from the list*

You can include or exclude the members present in the list. By default, the filter winder opens the **List** option. Follow these steps:

1. Select the member from the list to include, then **OK**.

For Example, select member 0 from the list, as shown in the following figure then ok. This means the data is filtered based on Outcome 0. Therefore, Outcome 1 based records were excluded.

2. Click on **OK**.
3. Click on **Create Extract**.



***Figure 3.31:** Include members from the list Exclude the member from the list by following these steps:*

1. To exclude the member from the list, click on the **Exclude selected values** checkbox, as shown in the following figure.
2. Select the member from the list to exclude, then **OK**.
3. Click on **OK**.
4. Click on **Create Extract**.

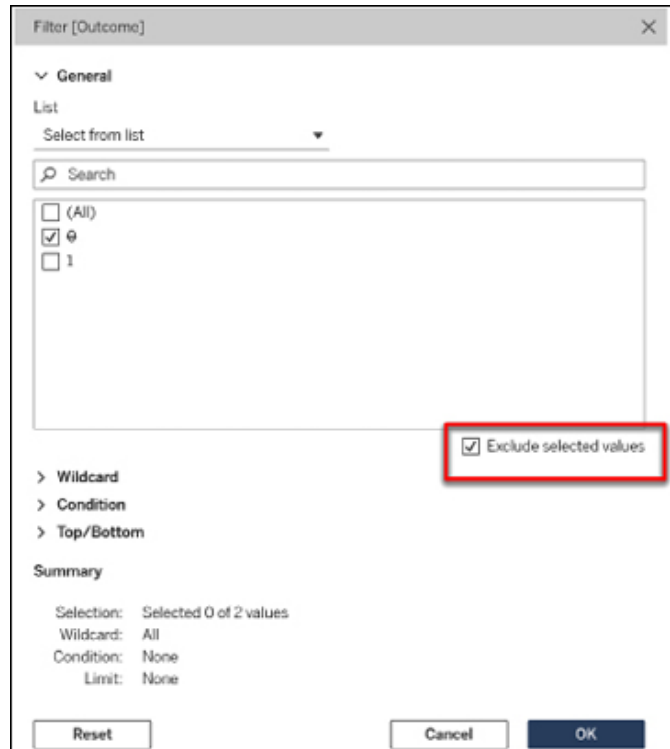


Figure 3.32: Exclude member from the List

For Instance, Check **Exclude Selected Values**. Now, select **‘0’** member from the list to exclude from the dataset then **OK**. This means the data is filtered out the Outcome **‘0’**. Therefore, Outcome **‘1’** based records in the new filtered dataset.

Use case 2: Custom value list

You can also type the member’s name, filter the field accordingly, and create a custom value list.

Follow these steps to create a custom value list:

1. Choose **Custom Value list** from the List drop-down menu.
2. Type the member’s name. Then click on the **Add** button to add the member to the list and filter the field accordingly.
3. Here, you can also add multiple member’s names in the following *step 2*.
4. Then, click on **OK**, as shown in the following figure:

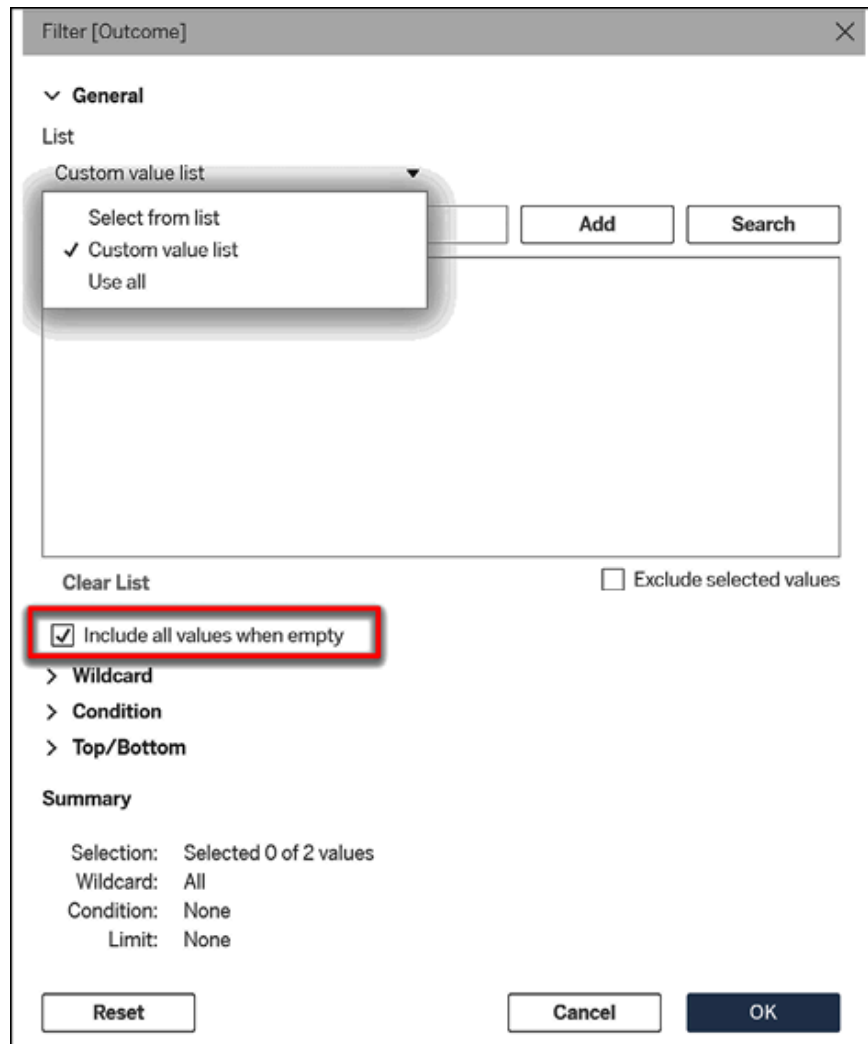


Figure 3.33: Custom value list in extract filter

You can also include all the values when the selected member has no data.

You can clear the custom value list by clicking **Clear List** or select custom values by clicking **X**:

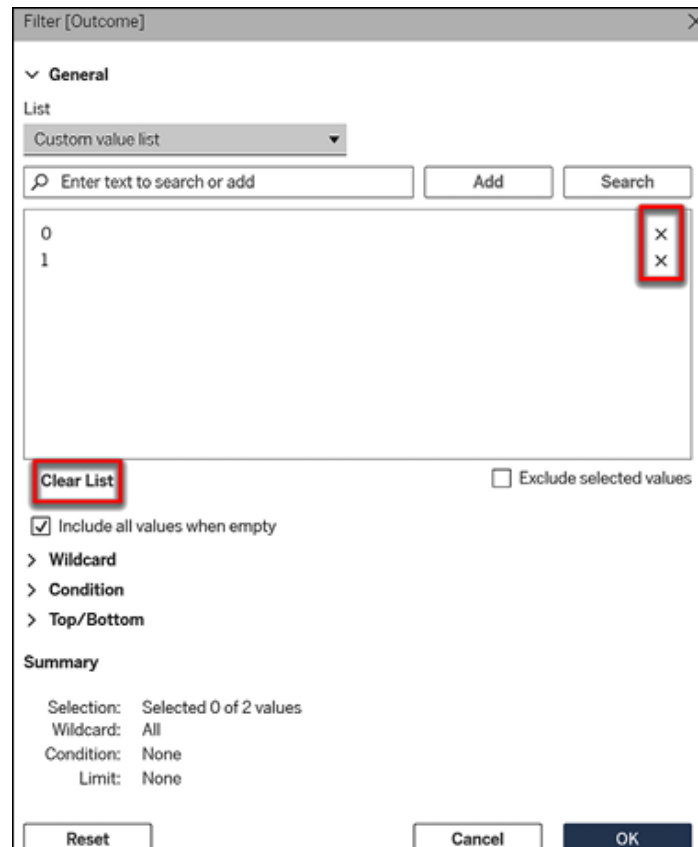


Figure 3.34: Clear custom value list in extract filter *Use case 3: Use all*

This option will select all the members in the field.

1. Choose **Use all** from the List drop-down menu.
2. Then, click on **OK**.

Use case 4: Wildcard

This option will filter the field based on the given wildcard match. Here, the user can type the character and filter the field based on the match value:

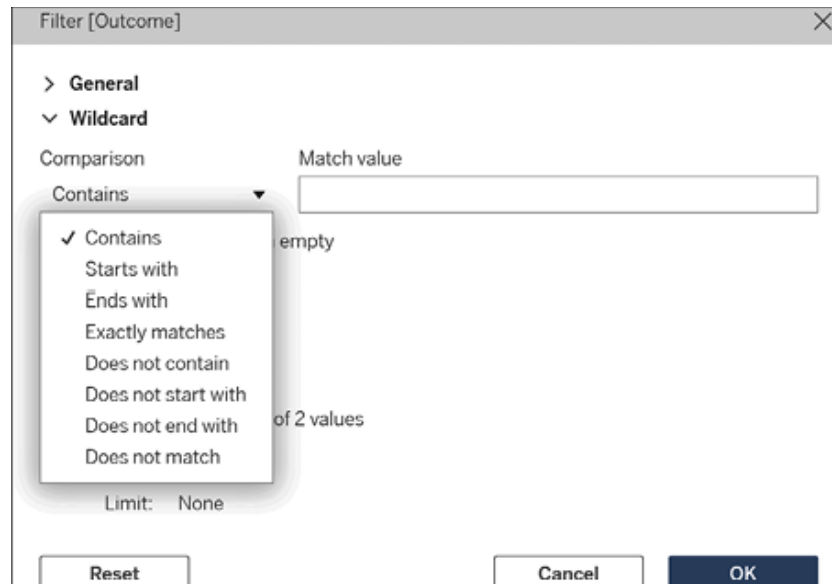


Figure 3.35: Wildcard in extract filter Follow these steps to filter the field based on the given wildcard match:

1. Select the **Wildcard** option.
2. Choose the **Comparison** option from the drop-down menu and type the characters to match in the **Match Value** textbox.
3. Then, click on **OK**.

Use case 5: Filter on condition

This option is used to filter the field based on condition, either by field or by formula.

Let us see each of the following:

By field

1. Select the **Condition** tab in the filter window.
2. Click on the **By Field** radio button to apply a conditional filter based on Field.
3. Choose the field **"BMI"** from the dropdown menu.
4. Now, select the aggregation type **"Median"** from its drop-down list.
5. Choose the **operator ">"** from the list.
6. Enter the **value "23.5"** to evaluate the field.
7. Then, click on **OK**.

Here, the dataset is filtered on the condition where the Median BMI is greater than 23.5 as shown in the following figure:



*Figure 3.36: Condition filter by field **By formula***

1. Select the **Condition** tab in the filter window.
2. Click on the **By Formula** radio button to apply a conditional filter based on Field.
3. Enter the formula in the box as shown in the following figure.
4. Then, click on **OK**:

Filter [Outcome]

> General

> Wildcard

▼ Condition

☐ None

☐ By field

Field: Age Aggregation: Sum

Comparison: = Value: 0

☒ By formula

AVG([BMI]) > 20.5

> Top/Bottom

Summary

Selection: Selected 2 of 2 values

Wildcard: All

Condition: None

Limit: None

Reset Cancel OK

Figure 3.37: Condition filter by formula

Use Case 6: Top or bottom filters

This option is used to filter the Top or Bottom ‘N’ number of records either by field or by formula.

Let us see each of the following:

By field:

1. Select the **TOP** tab in the filter window.
2. Click on the **By Field** radio button to filter the records by field.

3. Select **Top or Bottom**, then enter the number of records.
4. Now select the **Field** and **Aggregation** type.
5. Click on **OK**.

The screenshot shows a dialog box titled "Filter [Outcome]" with a close button (X) in the top right corner. The dialog is divided into two main sections: "Top/Bottom" and "By formula".

Top/Bottom Section:

- ☒ **By field**
- Rank:** A dropdown menu showing "Top".
- Count:** A dropdown menu showing "Enter a Value..." and a text input field containing "10".
- Field:** A dropdown menu showing "BMI".
- Aggregation:** A dropdown menu showing "Average".

By formula Section:

- ☐ **By formula**
- Rank:** A dropdown menu showing "Top".
- Count:** A dropdown menu showing "Enter a Value..." and a text input field containing "10".
- Formula:** A large empty text area for entering a formula.

Summary Section:

- Selection:** Selected 2 of 2 values
- Wildcard:** All
- Condition:** None
- Limit:** None

At the bottom of the dialog are three buttons: "Reset", "Cancel", and "OK".

Figure 3.38: Top filter by field

For instance, this filter restricts the data set to show top ten records based on average BMI.

By formula:

1. Select the **TOP** tab in filter window.
2. Click on **By Formula** radio button to filter the records by field.
3. Select **Top or Bottom**, then enter the **number of records**.
4. Enter the formula as shown in the following figure:
5. Hint: This formula cannot be Boolean formula.
6. Click on **OK**.

Filter [Outcome]

☐ None
 ☐ By field

Rank

Count

Top

Enter a Value...

10

Field

BMI

Aggregation

Average

☒ By formula

Rank

Count

Top

Enter a Value...

10

Formula

MEDIAN(BMI)

Summary

Selection: Selected 2 of 2 values

Wildcard: All

Condition: None

Limit: None

Reset

Cancel

OK

Figure 3.39: Top Filter by Formula For example, this filter restricts the data set to show top ten records based on median BMI.

Data source filters

The data source filter is used to filter the records at the data source level. This can restrict the data/records present in the data set. The data source filter is the same as the extract filter.

However, this filtering can work for both the live and extract connections. Unlike extract filters, which work only for extract connections, Follow these steps to apply

the data source filter:

1. Click on the **Add** of the data source tab.
2. After clicking on the **Add**, you can choose any of the fields from the list.
3. Then, click on **OK**.

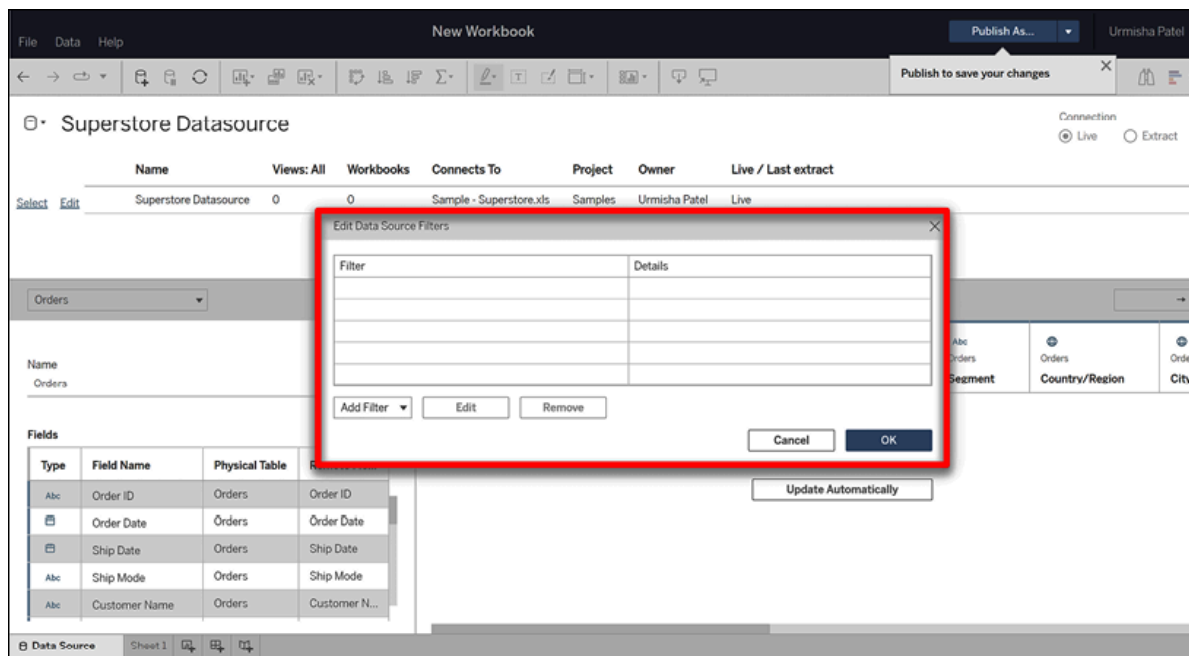


Figure 3.40: Data filters dialog box

4. Now, follow the remaining steps from the topic *Extract filters*.

Context filters

A context filter in Tableau is a special type of filter that sets a context for other filters, which means it creates a subset of data that all other filters will process. When a filter is added to the context, it establishes a data context and makes subsequent filters dependent on it. This can help improve performance and manage complex filtering scenarios more effectively.

A context filter is like a dimension filter. However, the precedence/priority of the context filter is higher than the dimension filter. A context filter will create a separate dataset from the original dataset using one or more categorical filters. All other filters work on the worksheet based on the selected context filter.

Let us understand this filter with an example.

Follow these steps to apply the context filters in the worksheet:

1. Drag the categorical field **Category** to the filter box, as shown in [Figure 3.41](#).
2. Right-click the dimension and select **Add to Context** to apply this field as a context filter.
3. Once the filter is selected as a context filter, the color of the dimension filter changes from blue to grey color. This grey color of dimension defines the context filter.

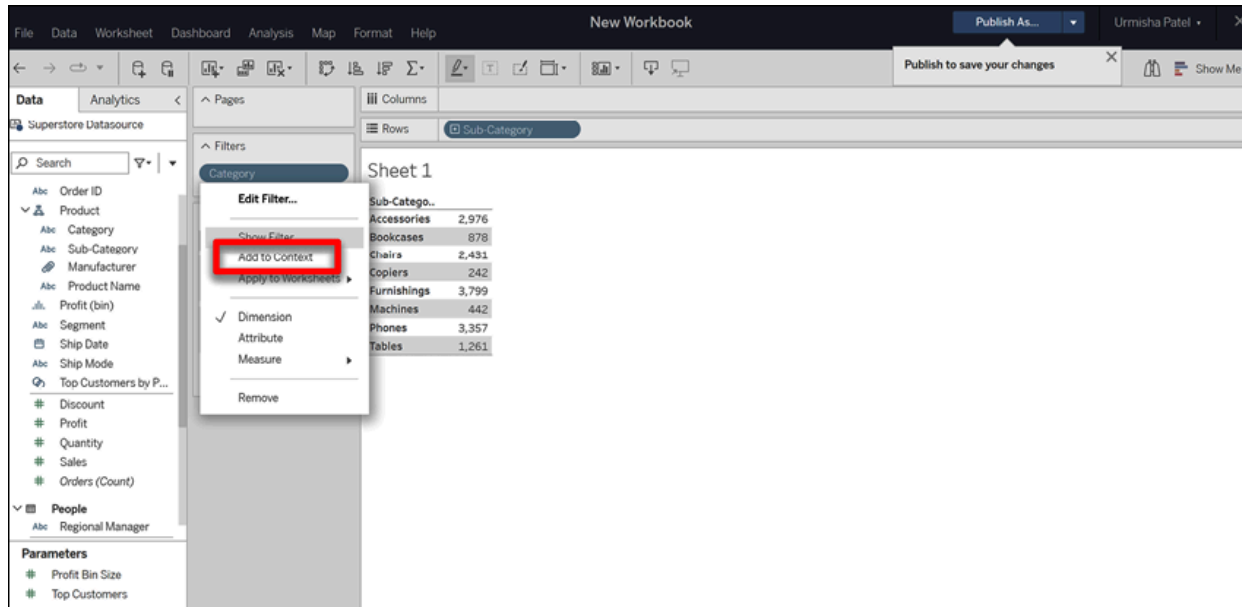


Figure 3.41: Context filtering at a worksheet YYou can also change this context filter to a normal filter by selecting **Remove from Context** by right-clicking **Dimension**.

There are a few advantages of using a context filter:

- **Improves the performance:** Context filter has higher precedence than dimension and measured filters. Hence, selecting the major categorical context filters will improve the performance effectively by creating a temporary dataset from the large dataset.
- **Dependent filter conditions:** When the data source size is larger than the context filter, it can help select the major categories. The other relevant filters can be executed depending on the business requirements.

Dimension filters

In the Filter card, if dimension or categorical/qualitative field is used to restrict the data in the view, then such filtering is defined as a **Dimension Filter**. This can change the filter condition dynamically in the sheet or dashboard.

Follow these steps to apply the dimension filter on the sheet with an example:

1. Connect to the data **Sample-Superstore** data source.
2. Drag the dimension field **Sub-Category** from the **Orders sheet** to column shelves.
3. Drag the measured field **CNT(Orders)** to row Shelves.
4. To apply a filter on the view, drag Category from the dimension section under the data pane to **Filter Card**.
5. It opens the *Filter* window. Select the member from the list. Then click on **Ok** as shown in the following figure:

Filter [Category]

General

List: Select from list Values: All values in hierarchy

Search

- ☒ (All)
- ☒ Furniture
- ☒ Office Supplies
- ☒ Technology

☐ Exclude selected values

> Wildcard
> Condition
> Top/Bottom

Summary

Selection: Selected 0 of 3 values
Wildcard: All
Condition: None
Limit: None

Reset Apply Cancel OK

Figure 3.42: Dimension filter dialog box The dimension filter can be applied through Conditional, Wildcard match, Formulae or Top/Bottom.

To show the filter member list in the view then, right-click the **Dimension** filter on the filter card, and from the drop-down menu, select **Show Filter**, as shown in the following figure:

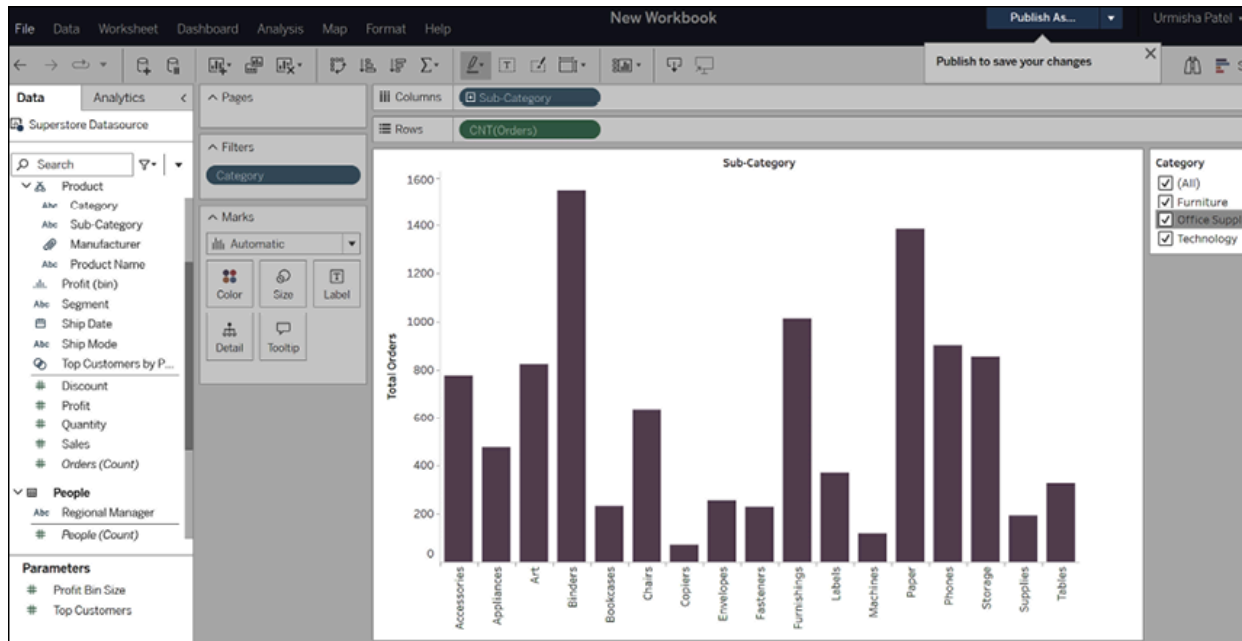


Figure 3.43: Dimension filtering in the sheet You can make the view dynamic by selecting the member from the filter list anytime and see the difference in the view.

Measure filters

The measure filters can filter the data based on the values in the measure fields. The aggregated measured values are used to filter out or modify the data in the view.

Follow these steps to apply measured filters in the worksheet:

1. On the worksheet, select the measured field **Profit** from the measures section under the data pane to the **Filter Card**.
2. This will open the Filter field configuration box. Select the aggregation from the list.
3. Select **Sum** from the aggregation list. Now, click on **Next**.

This will open to a select range of values, as shown in the following figure:

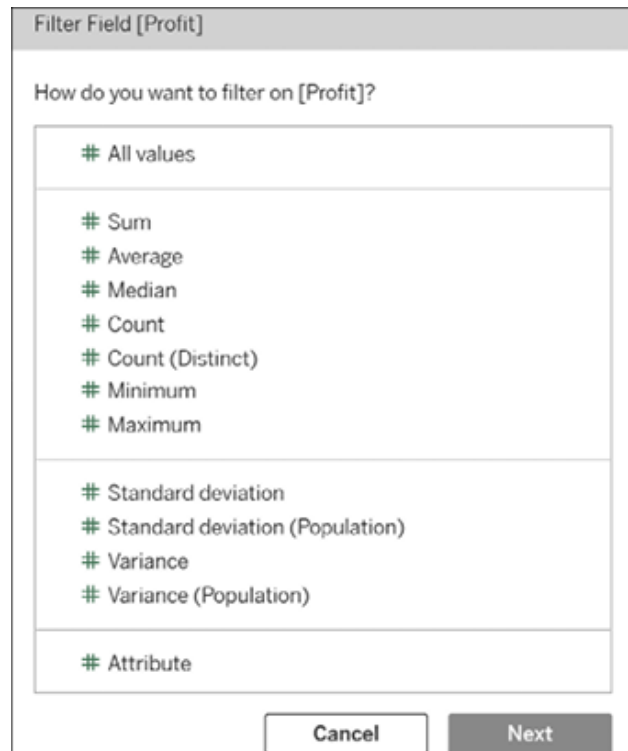


Figure 3.44: Aggregation list on measure filter There are other types of options also present in the window, as shown in [Figure 3.45](#):

- **Range of values:** The measured field minimum and maximum range values is given, and you can modify the upper and lower limit range of values to filter the data.
- **At least:** A minimum value of the measure is given to filter the data that can also modify the range of the values where only the minimum range value can be altered.
- **At most:** The maximum value of the measure is given to filter the data that can also modify the range of values where only the maximum range value can be altered.
- **Special:** The option is to select null or non-null values and filter the data.

Refer to the following figure for a better understanding:

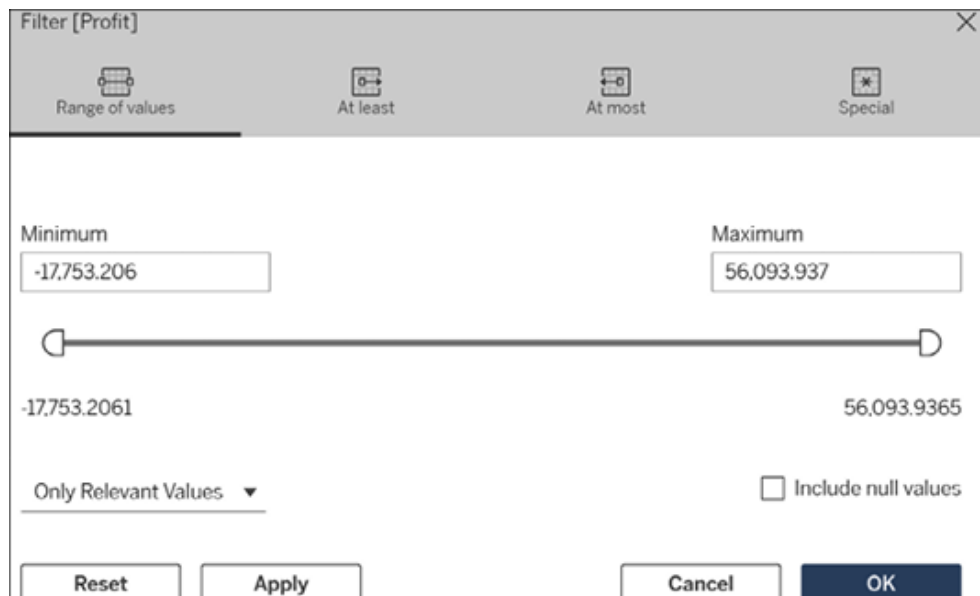


Figure 3.45: Range of values on aggregated values Follow these steps to define the range of values.

1. Select the range of values tab. Then, modify the upper and lower bound for the range of values and click **Ok**.
2. To show a filtered aggregated range of values in the view, right-click the measured filter on the filter card, and from the drop-down menu, select **show filter**.
3. This will show the **Profit range values slider** as shown in the following figure:

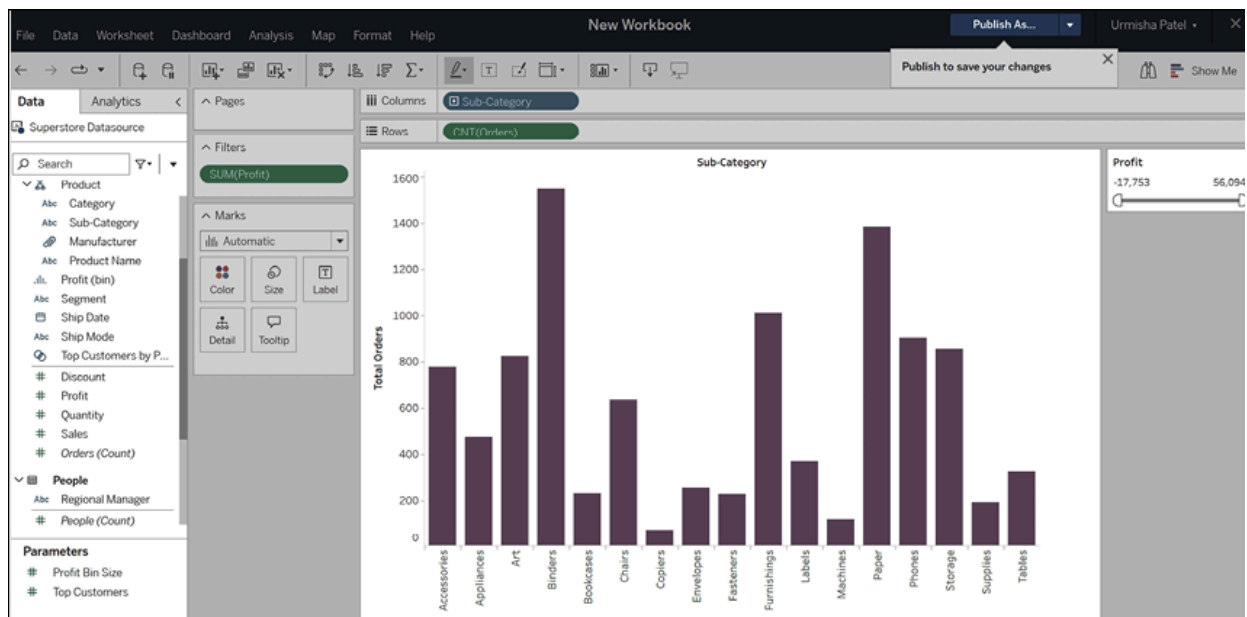


Figure 3.46: Measure filtering in the sheet *Calculated fields*

In the analytical and data-driven business era, you often need to create new data from the existing data. This derived data field is called the **calculated field**.

Calculated fields are used for segmentation, data type conversion, aggregation, filtering, ratios percentage calculations, etc.

Now, follow these steps to create a new field:

1. Connect your data to the **GME Stock** data source.
2. Select **Analysis Create Calculated Field**, in the Tableau worksheet.
3. Enter the formula in the calculation editor for to check whether the stock price is bearish or bullish.
4. Name this field as **Bearish/Bullish**, as shown in [Figure 3.47](#):

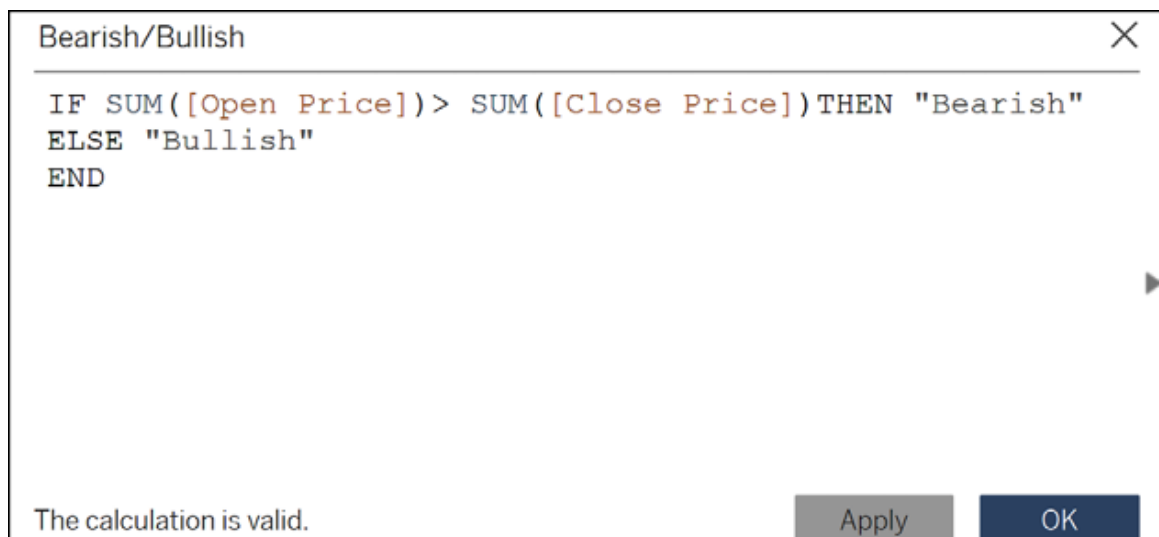


Figure 3.47: Calculated field

5. Hit on **Apply**, then **Ok**.

You will find a new string data type field named **Bearish/Bullish** added under **the Data pane**; it is added to the dimension.

6. Now, you are ready to use this calculated field in the view.

Table calculations

Tableau provides a few built-in table calculations that you can apply quickly to your Viz. called quick table calculations. This reduces your effort in writing the most frequent table calculations.

Note: Quick Table Calculations can be performed on the measure fields only, not on dimensions.

You can use the quick table calculations for the following:

- Running total
- Difference
- Percentage difference
- Percent of total
- Rank
- Percentile
- Moving average
- YTD total
- Compound growth rate
- Year of year growth
- YTD growth

Let us see how to apply quick calculations with an example: **Example:** Suppose you want to know the **ranking of the product category based on the profit** from the **sample-superstore data** using the **Quick Table Calculations**.

Follow these steps:

1. Connect to the data **Sample-Superstore** data source.
2. Drag the dimension field **Category** from the **Orders sheet** to row shelves.
3. Drag the measures field **Quantity, Sales, and Profit** to row shelves next to **Category**.
4. By default, the measured fields are aggregated to SUM.
5. Drag Profit again under the Data Pane to row shelves, Right-Click SUM(Profit), and select **Quick Table Calculation | Rank**. Refer to the following figure:
6. The delta symbol will appear to indicate that Quick Table Calculations are applied to the measured field.

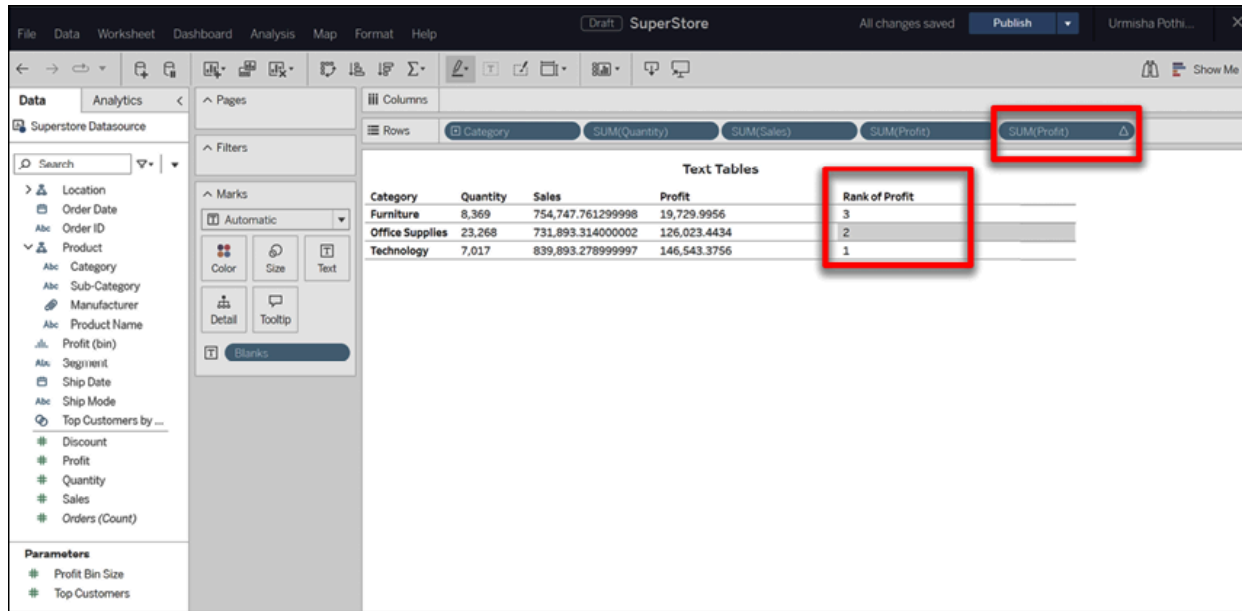


Figure 3.48: Quick table calculations *Editing table calculations*

You can also customize the table calculations. Follow these steps:

1. Right click the measured field, **SUM(Profit)** and select **Edit Table Calculation**.
2. This will open a **table calculation** dialog box where you can configure the following options, as shown in *Figure 3.49*:
 - a. Calculation Type
 - b. Compute using (how to aggregate the values table down, table cross, pane cross, specific dimensions, and at what level).
 - c. Show the table calculations (how to compute the calculations).

Table Calculation

Rank of Profit

×

Calculation Type

Rank

Descending

Competition (1, 2, 2, 4)

Compute Using

Table (down)

Pane (down)

Pane (across then down)

Pane (down then across)

Cell

Specific Dimensions

☐ Blanks

☒ Category

☒ Show calculation assistance

Figure 3.49: Table calculation dialog-box You can also Clear the Quick Table Calculations and remove the calculations applied on the measured field.

Removing table calculations

Follow these steps to remove table calculations:

1. Right click the measured field **SUM(Profit)**
2. Select **Clear Table Calculation**.
3. This will remove the applied table calculation on the measured field and leave it with its original values.

Dashboard

As discussed, at the beginning of this chapter, the dashboard is a collection of worksheets. In other words, it is the collection of views on one page that allows you to compare the views simultaneously. This will display the views at once on a page rather than navigating in separate worksheets.

Creating a new dashboard

Follow these steps to create a new dashboard in a workbook:

1. Select the Dashboard icon in the sheet tab at the bottom of the workbook, as shown in the following figure.
2. This will create a new dashboard named as **Dashboard 1**, as shown, in [Figure 3.50](#):



Figure 3.50: New Dashboard

Note: Similar to Worksheets, you can access the dashboards from the sheet tab.

3. If there are any modifications in a worksheet, then the dashboard will automatically reflect the changes in the view and vice-versa.

Let us explore the **Dashboard** workspace as illustrated in the following figure:

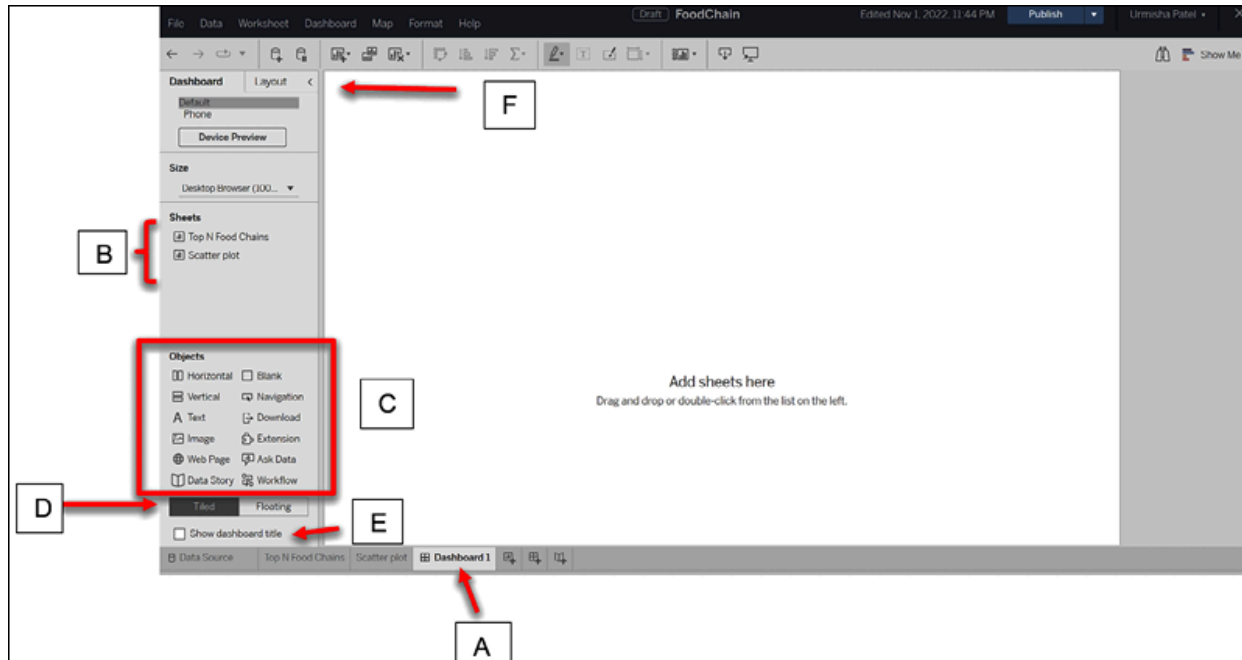


Figure 3.51: Dashboard workspace The preceding figure can be understood as:

- By default, the new dashboard is named **Dashboard N** (where N= 1,2,3... N). For example, the first dashboard in the **Sheet** tab. So, by default, it is named as **Dashboard 1**.
- The Sheet section will display the sheet names available in the sheet tab.
- The highlighted **Object** section is the dashboard objects used to add text areas, insert images and links, create navigation buttons, webpages, etc.
- Tiled and Floating allows you to arrange the worksheets in fixed size or in adjusted size in the display.
- Enable the checkbox to show the dashboard title in the display.
- This section consists of two tabs, **Dashboard** and **Layout**.
- The **Dashboard** tab consists of sheets, objects, titles, devices, display sizes, etc. While in the Layout tab, you can add the border, change the background image or color, adjust the object size, etc.

Adding or replacing the sheet

Follow these steps to add or replace the sheet and make it interactive:

1. Drag the sheets to your dashboard workspace from the **Sheets** section.

For instance, drag Top N Food Chains and Scatter plot to the dashboard on the right.

2. By default, sheets will be arranged in a fixed size, that is, **Tiled**.
3. To swap the sheet, drag the left sheet from the workspace and move it to the right. This will swap the sheet place. If you have noticed, you can also arrange the sheets either in horizontal or vertical halves of the dashboard workspace.
4. To make it more interactive, select the sheet to enable the **Use as Filter** option in the upper right corner of the sheet to use the selected marks in the sheet to filter to the other sheet, as shown in the following figure:

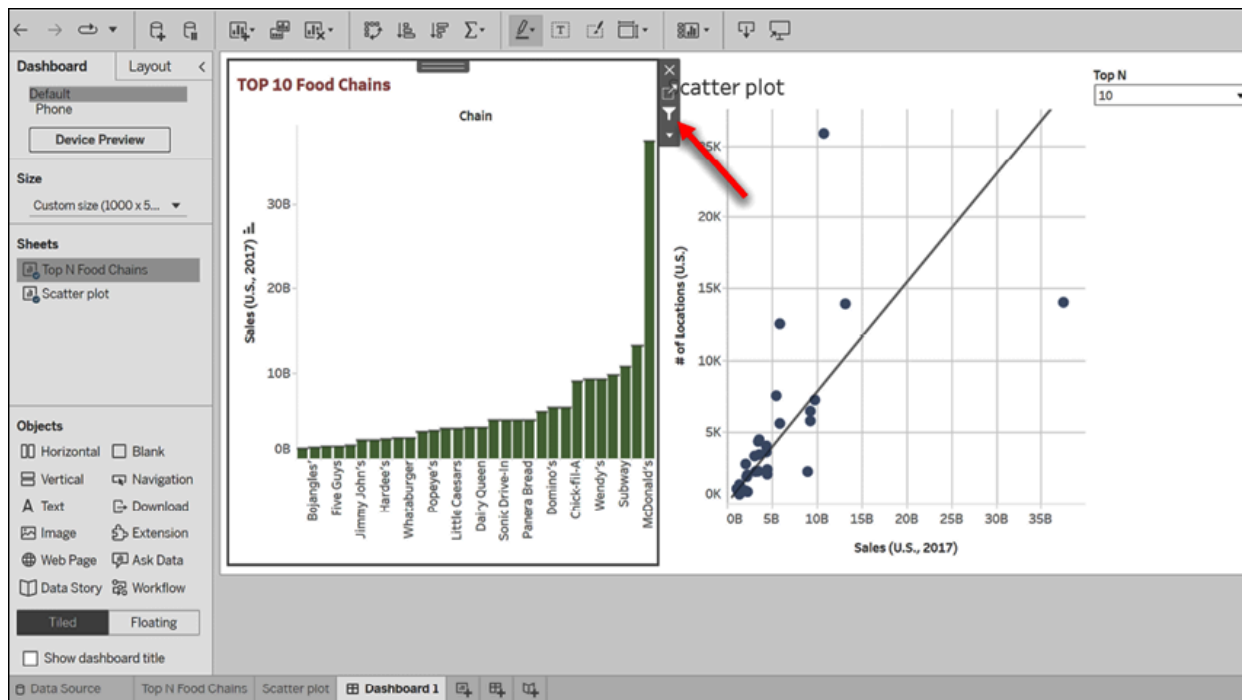


Figure 3.52: Interactive sheets in dashboard workspace *Creating dashboard layouts for different devices*

Dashboards can create layouts for different devices with a wide range of display screen sizes.

Note: You need to create a single dashboard and deliver the URL for different layouts.

As shown in [Figure 3.53](#), the dashboard creates the layout by default. Here, you can change the size of the device, select the device screen sizes available in the drop-down or manually adjust the height and the width of the screen size.

To create any other layouts apart from **Default** or **Phone** then select the **Device** drop-down menu; change its option from **Custom** to any other device as shown in

the following figure:

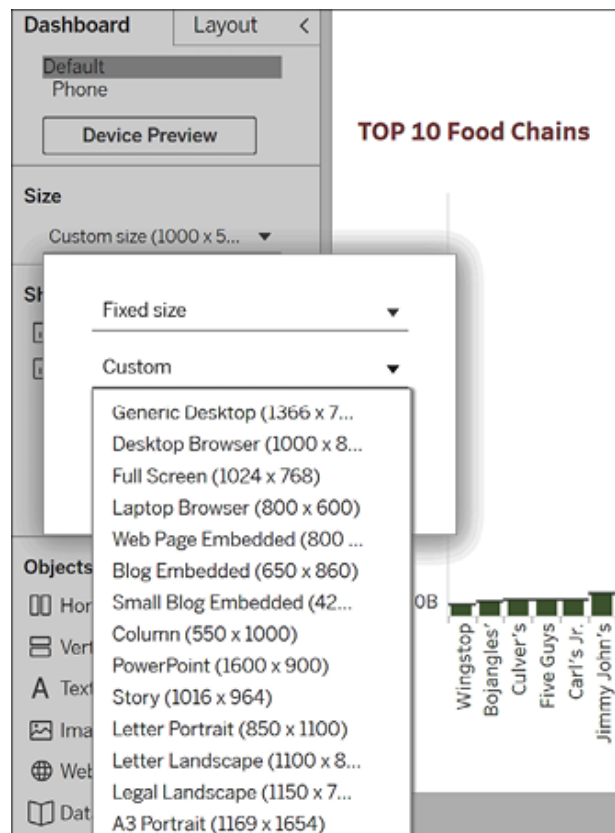


Figure 3.53: Dashboard layouts for different devices

- **Phone layout:** To create a Phone layout, click on the **Phone** option. This will automatically reflect all the changes to the default dashboard.

You can Edit the Phone layout of the dashboard. Hence, right click on the **Phone** option. Now, select the **Edit Layout** option as shown in the following figure:



Figure 3.54: Phone layout of dashboard Now, let us understand the Layout pane to enhance the dashboard visuals.

Dashboard layout

The layout pane provides the options to modify the configuration of the dashboard's selected items, which includes padding, bordering, background colors around the items, and adjusting the size, position, or alignment of the items, or even reordering the hierarchy.

Follow these steps to make configuration changes for the selected dashboard items:

1. Select the dashboard item where you want to make changes and select the Layout tab in the left pane sections for the selected items, which gets enabled under the layout pane.
2. If the selected item is a sheet, you can show the title of the sheet by selecting the checkbox.
3. You can even make this fixed [tiled] item floating and arrange or resize the dashboard item and place it anywhere in the dashboard workspace or even

adjust the X and Y positions very precisely to align your dashboard components.

4. [Optional] Enable the border around the dashboard item and select the bordering style, size, color, etc.
5. [Optional] You can also add the background color or image in the workspace, specifically behind the selected dashboard item.
6. [Optional] To make the visuals clear and cluttered-free, you can also add inner and outer padding to add extra space between visuals.
7. Lastly, you can also change the floating item hierarchy and reorder the arrangement in the dashboard workspace. Refer to the following figure:

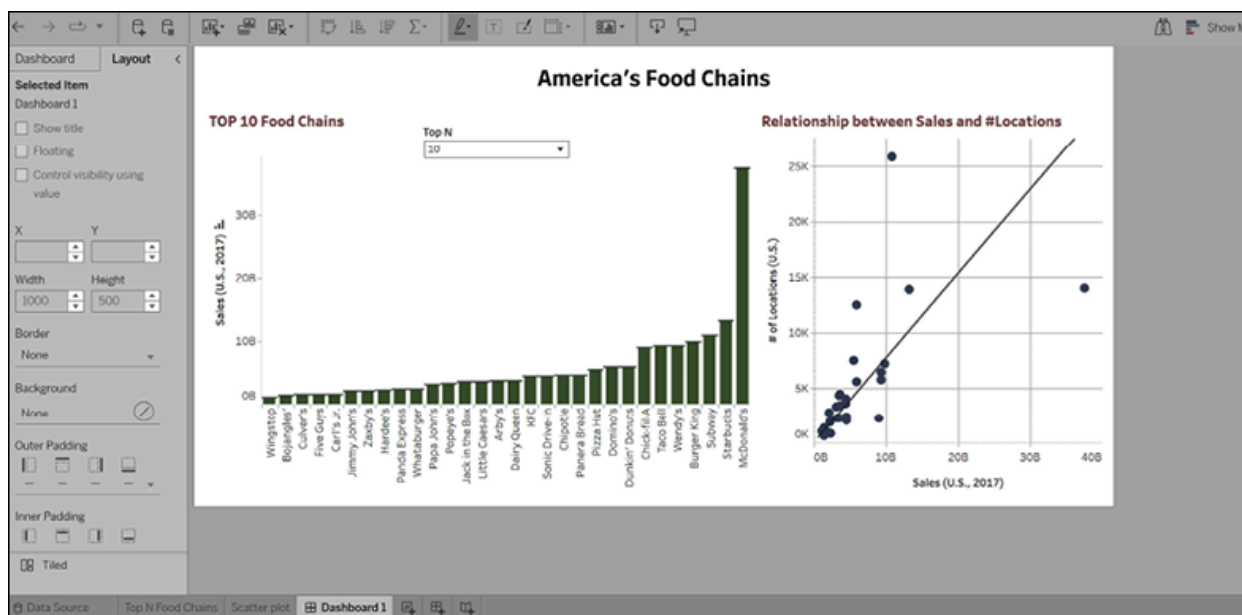


Figure 3.55: Layout pane in dashboard [Story](#)

In Tableau, a **Story** is a sequence of dashboards or worksheets arranged in a specific order to guide your audience through a narrative, effectively conveying information and insights. An individual sheet in the story is called a **Story Point**.

The Story option helps show how the facts are connected and how decisions are related to the outcome.

Creating a new story

Select the Story icon in the sheet tab at the bottom of the workbook, as shown in the following figure. This will create a new story named as **Story 1**:



Figure 3.56: Create a New Story

Note: Similar to Worksheets and Dashboards, you can access the stories from the sheet tab.

If there are any modifications in a worksheet or dashboard, then the respective story points will automatically reflect the changes in the view if it contains and vice-versa.

Let us now explore the story workspace:

Story workspace

Story workspace allows you to edit story points, format the navigation style, or export story points as images using the Story menu. Refer to the following figure for a better understanding:

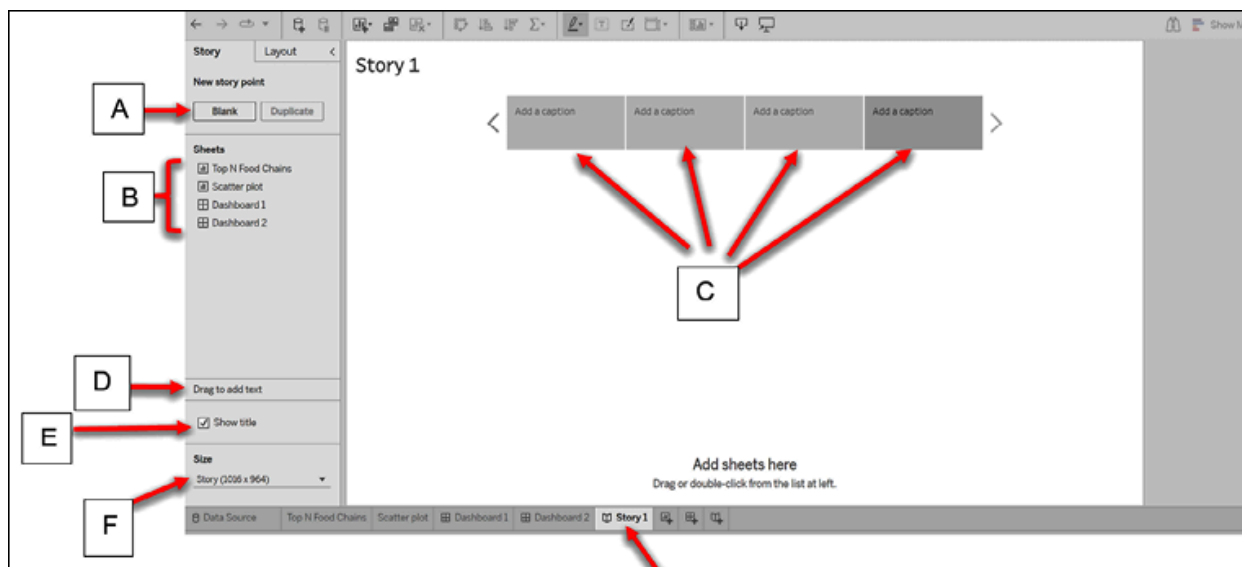


Figure 3.57: Story workspace The **New story point** section allows you to insert a blank new story point in the story or duplicate the already existing story point.

- This Sheets section contains the list of worksheets and dashboards created in the workbook.
- An individual sheet in the story is called story point. For instance, in the preceding figure, there are four story points. You can add the caption or brief description on the story point in the book. Each story point can be edited, updated, duplicated, or deleted. The backward and forward navigation buttons are used in between the story points to navigate.

- Drag to add text, dragging this object to story point will open a text editor.
- If you want to display the story title in the story, then enable the checkbox.
- You can change the size of the display screen of the story.

Layout pane

The layout option refines the look of the story navigation style. By default, the navigation style is Caption boxes, as shown in the following figure:

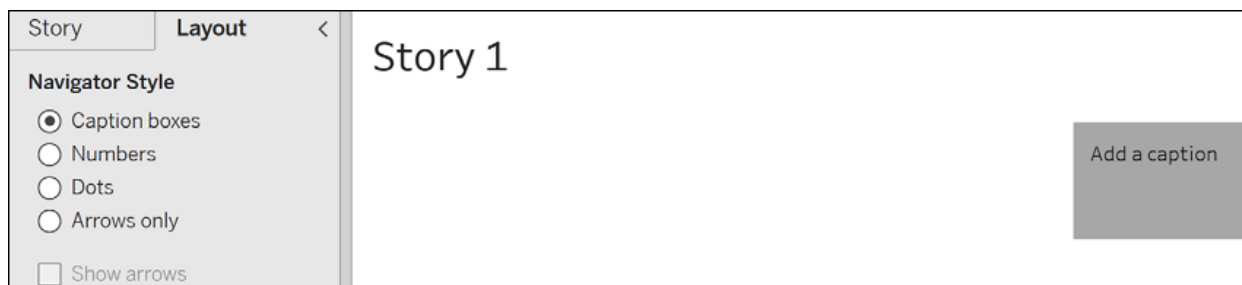


Figure 3.58: Navigation style in story layout pane Follow these steps to create a Story on America's food chain analysis:

1. Click on the **Story** icon on the Sheet tab.
2. A new story named **Story1** will be created.
3. Drag Dashboard 1 from the Sheets section in Story Workspace under Story Point 1.
4. Go to the **Size section and** adjust the display size.
5. Go to the Layout pane next to the Story pane at the top-left corner.
6. Change the Navigation Style to **Numbers**.
7. Again, click on **Story Pane** to add one more story point, then click **Blank**.
8. Drag the **Dashboard 2** from the Sheets section in the Story workspace under Story Point 2.
9. The side arrows give an edge to navigate between the story points.

A sample story on *America's* food chain analysis is created as shown in the following figure:

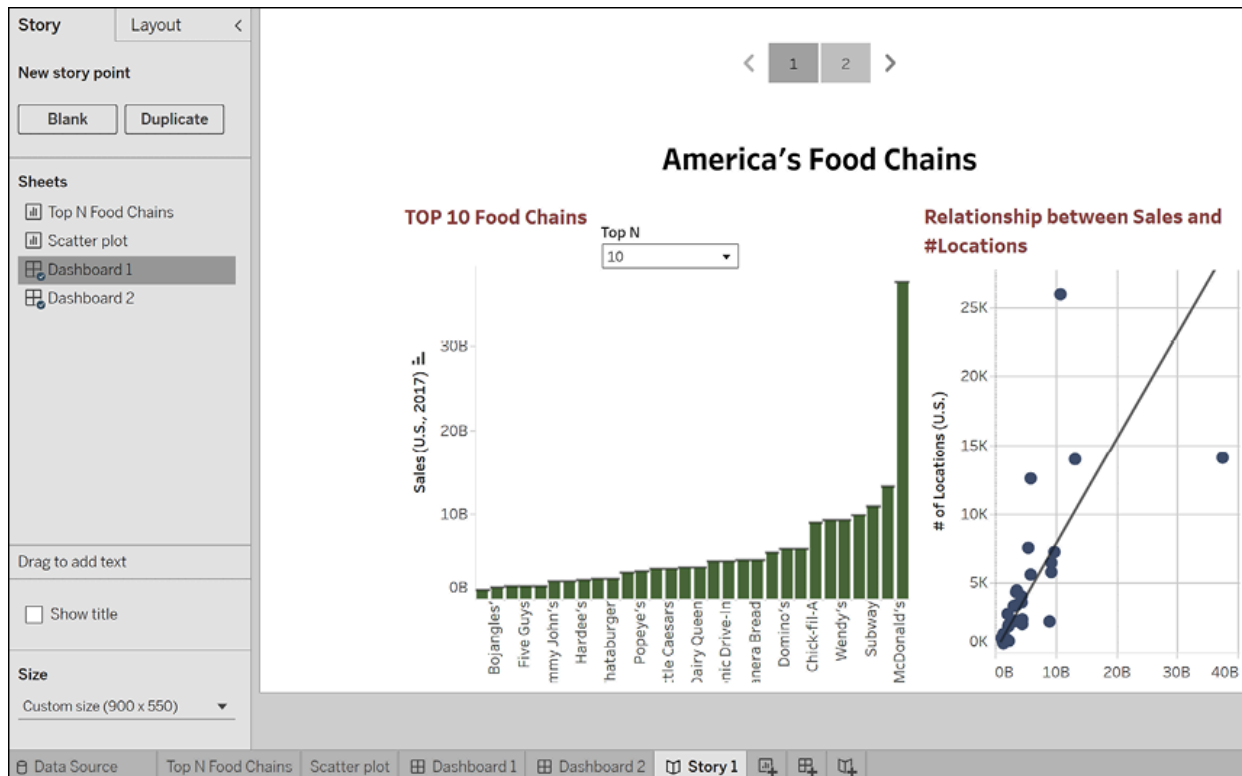


Figure 3.59: A sample story on America's food chain analysis [Conclusion](#)

In this chapter, you learned how to build different graphs and charts from your data and create new fields from the existing fields, apply filters on the data, format the data, create the story, and find out the hidden patterns from the Viz. In the next chapter, you will learn about advanced concepts in building visuals, a few advanced and dynamic charts, level of detail calculations, etc.

Multiple choice questions

1. Which of the following is not a Tableau data field type?

- a. Float
- b. Number
- c. String
- d. Boolean

2. Which type of chart is not available in Tableau?

- a. Pie chart

- b. Speed dial
- c. Bullet chart
- d. Bubble chart

3. What to use for creating a variable size bin?

- a. Sets
- b. Groups
- c. Calculated field
- d. Table calculations

4. Data values available for the visualization?

- a. Table calculations
- b. Basic expressions
- c. LOD expressions
- d. None of the above

5. How do you identify a continuous field in Tableau?

- a. When added to the visualizations, it produces distinct values.
- b. It is preceded by a # symbol in the data window.
- c. It is identified by a green pill in the visualization.
- d. It is identified by a blue pill in the visualization.

6. Which are NOT the Filter types?

- a. Context filter
- b. Dimension filter
- c. Highlight filter

d. Measure filter

7. Which Tableau chart resembles crosstab - pivot tables?

a. Histogram

b. Text Table

c. Line

d. Bar

Questions

1. What is a filter, and what are its types? Explain each of them.
2. How do you build a histogram based on age from a sample-superstore data source?

Answers

1.	a
2.	b
3.	c
4.	a
5.	c
6.	c
7.	b

Points to remember

Here are the key takeaways from this chapter:

- Tableau workbook consists of a sheet-like structure. They are worksheet, dashboard and story.
- Graphs and charts are built on worksheets. The basic charts are:
Bar charts, line charts, pie charts, text table charts, scatter plots, filled maps, and histogram charts.
- User interactions on the visuals can be applied based on filters. There are multiple types of filters and their precedence over each other. They are as follows: Extract Filter > Data Source Filter > Context Filter > Dimension Filter > Measures Filter
- Formatting allows you to make changes in your worksheet, workbook including font size, style, alignment, colors, header, borders, gridlines, graph lines, axis, legends, highlighters, filters, parameters, borders, and dividers, etc.,
- A new calculated field is created from the existing data field.
- Dashboard is a collection of worksheets. In other words, the collection of views on one page lets you compare the views simultaneously.
- A story is a sequence of dashboards or worksheets that work together to convey information, a collective insight from visualization.

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CHAPTER 4

Advanced Data Visualization

The greatest value of a picture is when it forces us to notice, what we never expected to see - John W. Tukey

Introduction

In the previous chapter, we learned the basics of data visualization. It is also important to give the user flexibility to interact with the visuals and analyze the view.

In this chapter, we will learn the advanced concepts in tableau online data visualization. These concepts help create dynamic visuals as well.

Structure

In this chapter, we will cover the following topics:

- Individual, dual, and combined axes
- Advanced charts
- Level of detail
- Parameters
- Groups
- Sets
- Dashboard objects
- Actions
- Tableau accelerators
- Animation in Tableau Online

- Analytics in Tableau Online

Objectives

This chapter illustrates how to create dynamic visuals. By the end of this chapter, you will learn a few advanced graphs that are widely used in the business world. You will also learn how to create animated visuals, apply Tableau's built-in analytics models to your Viz, and learn a few advanced topics such as parameters, groups, sets, and actions. After reading this chapter, you can create dynamic, user-interactive, and performance-optimized visuals.

Individual, dual, and combined axes

Adding axes in the view defines a measuring axis, the blend of two measures, or gives a glimpse of two independent axes in the same pane.

- **Individual axes:** For each continuous measure, adding it to a row or column shelf will add an axis to it. This is called individual axes.

Let us understand individual axes with an example: *Dataset: Sample Superstore.xlsx*

Follow the steps to add an individual axis:

1. Drag a continuous measure field Shipping Cost to row or column shelves in sheet1. This will add the axis to a row or column, respectively.
 - a. Here, add **Shipping Cost** in row shelf as shown in [Figure 4.1](#). The highlighted vertical axis is called the individual axis of Shipping Cost:

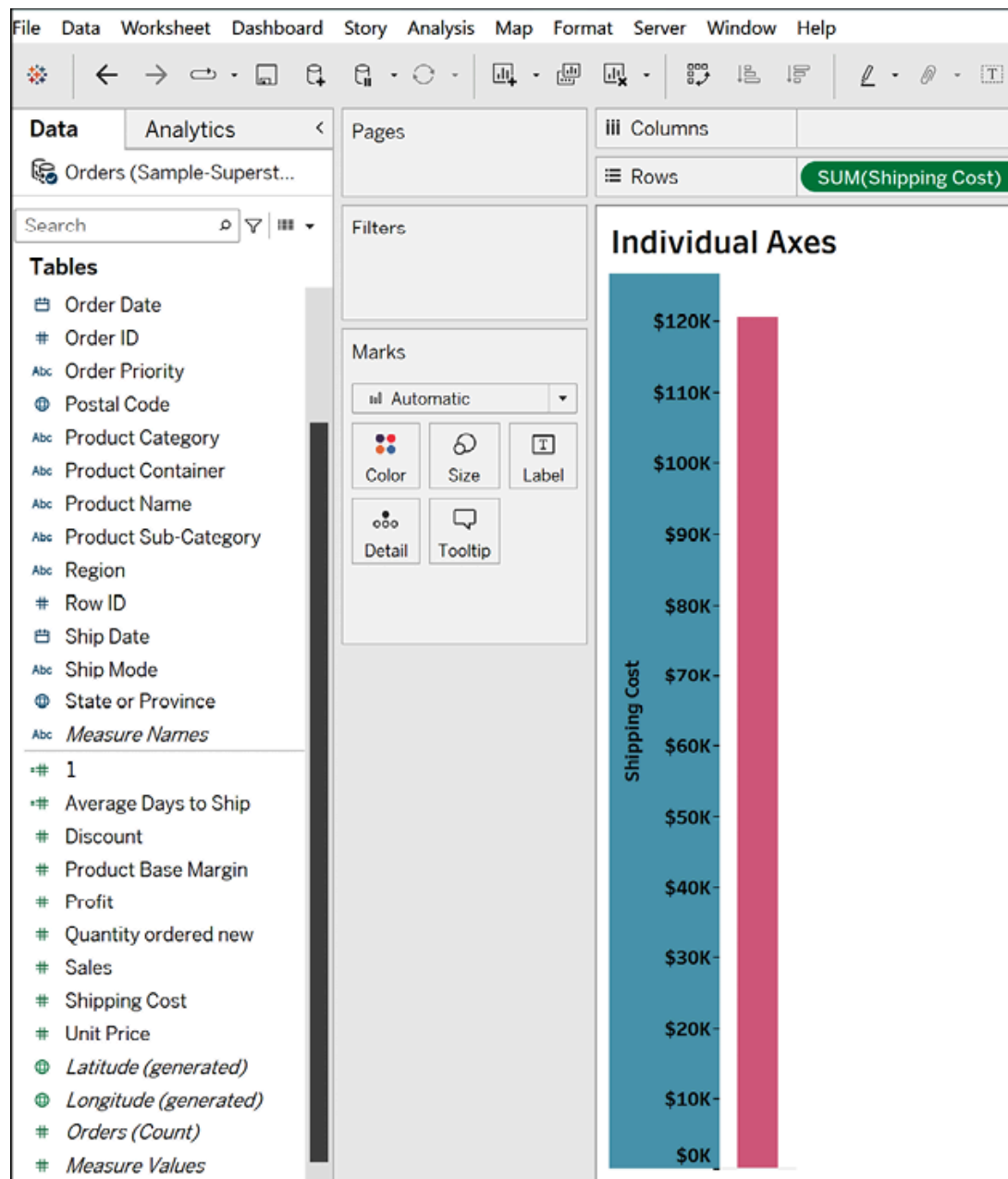


Figure 4.1: Individual axes chart

- **Dual axes:** On comparing two independent measures with different scales in the same pane, then gives a glimpse of two independent axes layered on top of each other. Let us understand dual axis with an example: Dataset: Sample Superstore.xlsx

1. Drag the measured field Sales to the rows shelf and the dimension field Order Date to the **Columns** shelf.

2. Put the measured field Shipping Cost next to Sales on the rows shelf.
3. Now, select the **Bar** chart on the **Marks** card for the **Sales** field and select the **Line** chart on the **Marks** card for the Shipping Cost field.
4. Then, choose the **Dual Axis** option from the drop-down menu of **Shipping Cost**, as shown in the following figure

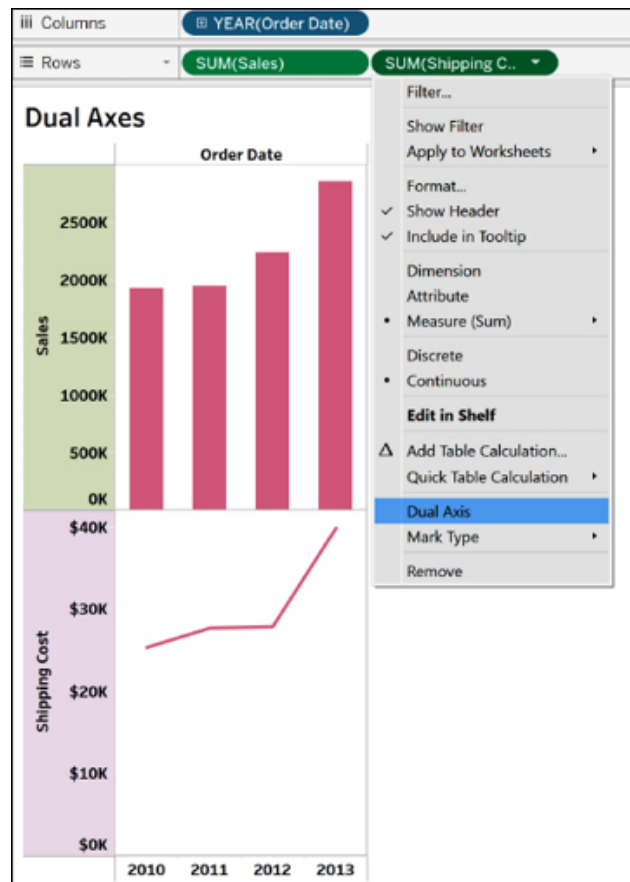


Figure 4.2: Dual axis option The highlighted axis, in green and pink, shows independent measures **Sales** and **Shipping Cost** with different scales on the same pane, as shown in [Figure 4.3](#):

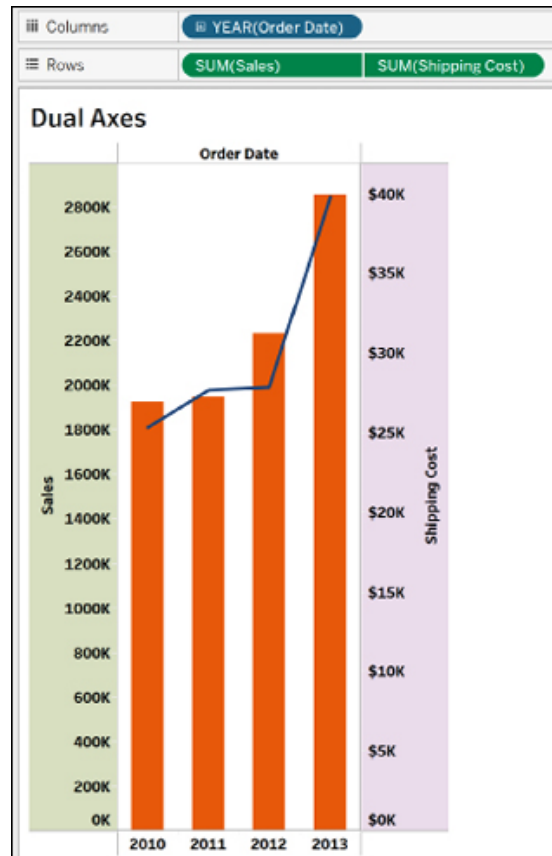


Figure 4.3: Dual axis chart However, these different scales' independent measures can be aligned/synchronized with each other to use the same scale as well.

Follow these steps:

1. Right-click **Shipping Cost** axis.
2. Then, select **Synchronize Axis**, as shown in [Figure 4.4](#):

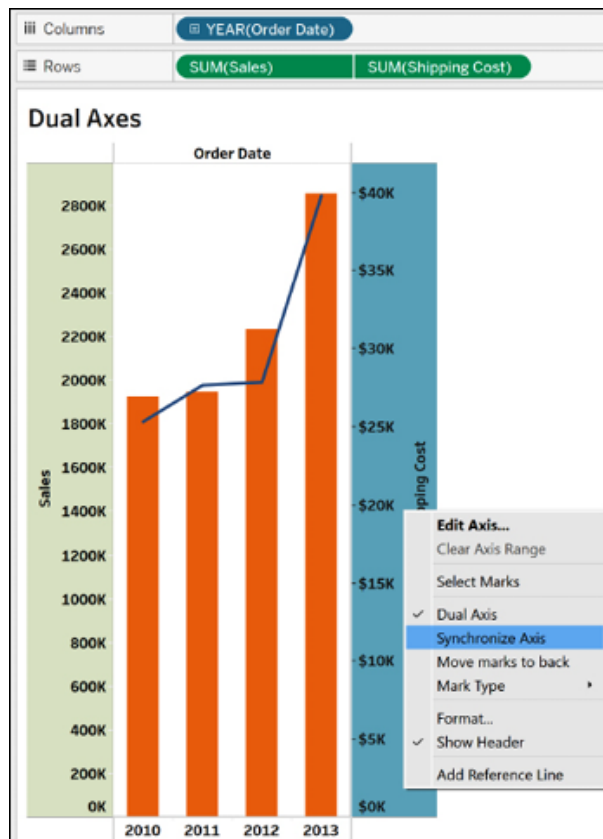


Figure 4.4: Synchronize axis We will see dual-axis applications under the topic of **Advanced charts**.

- **Combined axes:** Combined axes are the blend of two continuous measures in a single axis. Charts based on combined axes are used to compare two measures.

It is important to note that the blending is possible only when both the measure fields share a similar scale and unit used to define the values. Hence, with less distortion, trends between the measures can be considered.

Let us understand with an example. For instance, if you want to compare profit and sales in a combined axes w.r.t. the order date from the Sample-SuperStore dataset, then follow these steps:

1. Drag the dimension field **Order Date** to the **Column** shelf. Now, click the dimension field and choose the **Exact Date** option from the drop-down list.
2. Drag the **Measure Names** to the Filter box and select **Profit** and **Sales** from the listed measured fields.
3. Press *Ctrl* + drag the filtered **Measured Names** to the **Color Marks Card**.
4. You will find that the Profit and Sales fields are now compared with the same

scale and units.

Note: Blend measures used Measure Names and Measure Values, which consist of all the measured fields and values, respectively, in the data source.

Refer to [Figure 4.5](#) for an illustration of the combined axes chart:

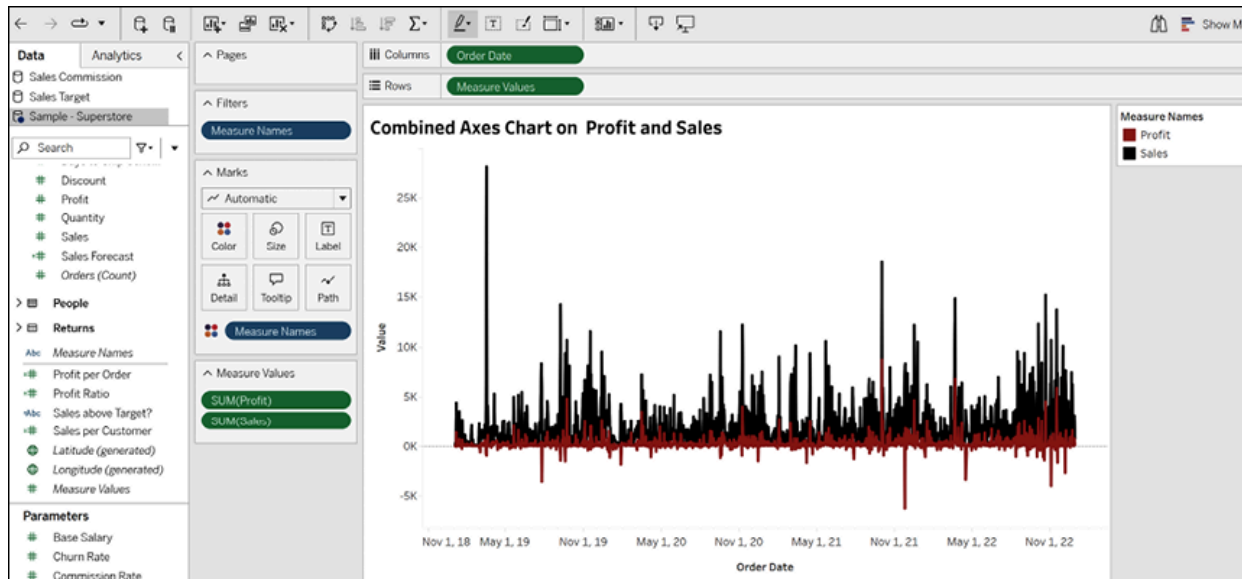


Figure 4.5: Combined Axes Chart [Advanced charts](#)

Many of you must be pondering about advanced charts. Let us look at how they differ from basic charts or the ones we have covered so far. Let us address such questions. Advanced charts can be categorized as extensions of basic charts. There may be the possibility of two or more charts combined or shapes to define the same marks on your visuals. In other words, visualization beyond **Show Me**.

Let us explore a few of them with an illustration.

Lollipop chart

The Lollipop chart is a combination of circles and bars. This chart is a type of dual-axis application chart. It can be used as an alternative to the Gantt Chart. Usually, these charts are used as alternative visuals to side-by-side bars.

Prerequisite: To create a lollipop chart, you need at least one dimension field or more measured fields. For instance, suppose you want to build a lollipop chart on #Locations and highlight the food chains based on total sales. Follow the instructions mentioned:

1. Connect the data Food Chains data source.
2. Drag the #Locations measured field from the data pane to the row shelf and duplicate that same field.

By default, the measured field is aggregated to SUM.

- Now, drag the **Chain** dimension field from the **data pane** to the **column shelf**. These two bar charts for the same field will be created with two separate individual axes.
- Right-click one of the SUM(#Locations) field. Select **Dual-Axis** from the option. This will create the dual axis.
- Select the right axis and then check **Synchronize Axis** to synchronize two axes.
- Since the two y-axes are synchronized, now choose to hide the right axis by unchecking **Show Header**.

Here, Tableau might have automatically converted bars to circles. So, you need to convert the first axis back to bars by opening the first marks card and changing the chart type to **Bars**. Adjust the size of the bar, select **Size marks** card, and reduce the size of the bar size to make the chart look like a lollipop.

- Drag the **Sales** measured field to the **Color marks** card on the second axis bar. i.e., of circle type. This will highlight the Lollipop by colors, and those having the highest sales will be in dark colors.

Color legend will also be seen in the Viz. This will create a basic lollipop chart. Now, make custom format changes to the Viz.

Refer to [Figure 4.6](#) for an illustration of the **Lollipop Chart**:

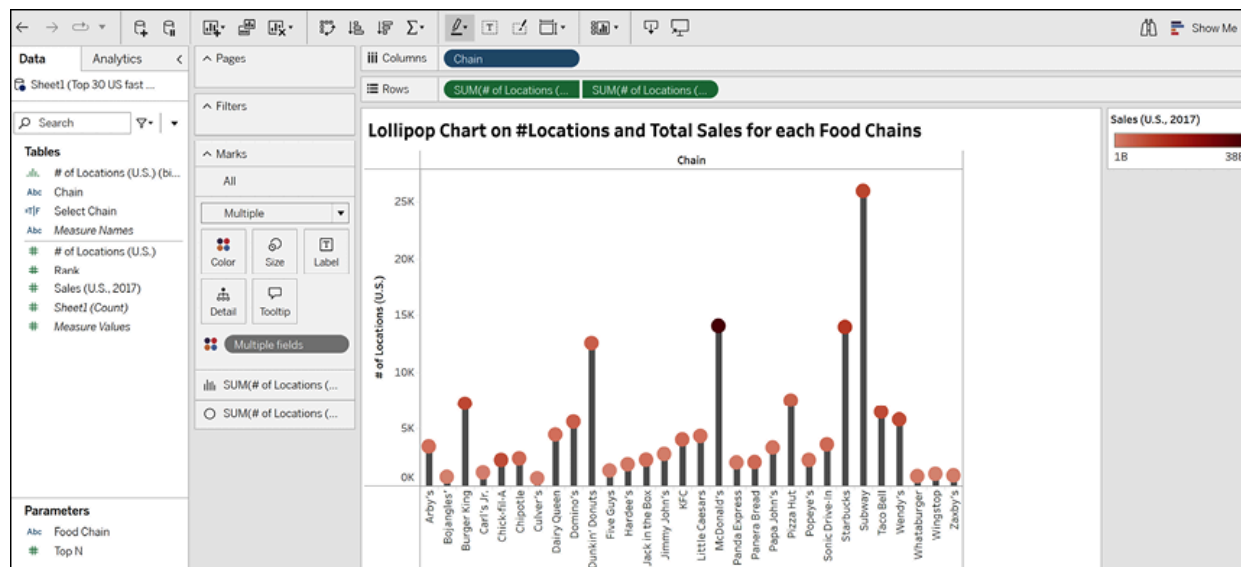


Figure 4.6: Lollipop chart *Donut chart*

Donut charts look like donut-shaped figures. They are modified pie charts that have a hole in the middle. These charts are highly preferable to pie charts because of the data

intensity ratio and space efficiency.

Prerequisite: To create a donut chart, you need at least one dimension field and one or more measured fields.

For instance, you want to create a donut chart on the **Product Category** based on the orders from the **Sample SuperStore** dataset. Follow these instructions.

1. Connect the data **Sample-SuperStore** data source.
2. Set the base:
3. Enter the following in the row shelf **twice**, as shown in [Figure 4.7](#).
MIN(0)
4. This will aggregate automatically.

[Figure 4.7](#) demonstrates what to enter to set the base in a Donut chart:

Columns	
Rows	AGG(min(0)) AGG(min(0))

Figure 4.7: Set the base in donut chart

5. Click the right-entered base field and choose **Dual Axis** from the drop-down menu.
6. Change the **Marks Type** from the Automatic to **Pie**.
7. Here, both the pie charts overlap with each other.
8. Now, click the second **AGG(MIN(0))** under the marks card. Now, change the **Color marks** card | **White**.
9. Now, click the first **AGG(MIN(0))** under the marks card. Drag the dimension field **Category** to **Color marks** card and **Label marks** card.
10. Click the first **AGG(MIN(0))** under the marks card. Drag the measured field **Orders** to the **Angle marks** card and **Label marks** card.
11. If necessary, format the label settings (including color, font style, font size, etc.,) Refer to [Figure 4.8](#) for an illustration of a donut chart:

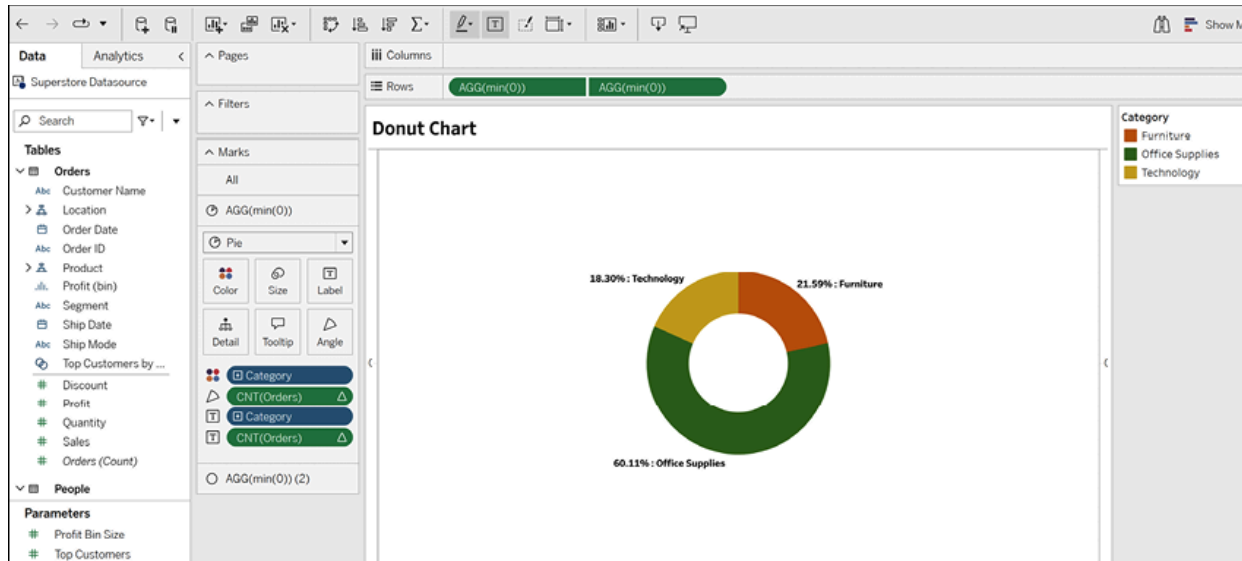


Figure 4.8: Donut chart *Pareto chart*

Pareto charts contain both line and bar charts. It is based on the principle of 80-20 rule to check the efficiency of business plans. This chart is named after *Vilfredo Pareto*. In these charts, the line graph represents the cumulative total, and the bar chart represents the other data values in descending order.

Hint: Application of Dual Axis.

Prerequisite: To create a Pareto chart, you need at least one dimension field and one or more measured fields.

For instance, suppose the manager wants to make a business plan based on sales for the food chain. Then, follow these instructions.

1. Connect the data **Food Chain** data source.
2. Drag the **Sales** measured field from the data pane to the row shelf. Now, duplicate this same field and add to its right.
3. Drag the **Chain** dimension field from the data pane to the column shelf.

By default, Tableau generates two bar charts in the same Viz.

4. Now, click on the **Sort icon** (descending order) from the toolbar to sort the graph in descending order. This will rearrange the food chains based on SUM(Sales).
5. Click the right SUM(Sales), check the **Dual-Axis**.
6. Here, the Tableau might have automatically converted bars to circles. So, you need to convert the first axis back to bars by opening the first marks card and changing the chart type to **Bars**. Change the second axis to **Line** chart type by opening the second marks card.

7. Now, open the drop-down menu of the second Sales row and select **Add Table Calculation**.
8. A dialog box will appear and change the configuration as shown in the following figure:

Figure 4.9: Table Calculation in Pareto chart

9. Change the **Primary Calculation Type** to **Running Total** and click on Add secondary calculation. Now, change the **Secondary Calculation Type** to **Percent of Total**.
10. Finally, change the color of the bars and line by clicking on **Color Marks Card**.

Refer to [Figure 4.10](#) for an illustration of a Pareto chart:

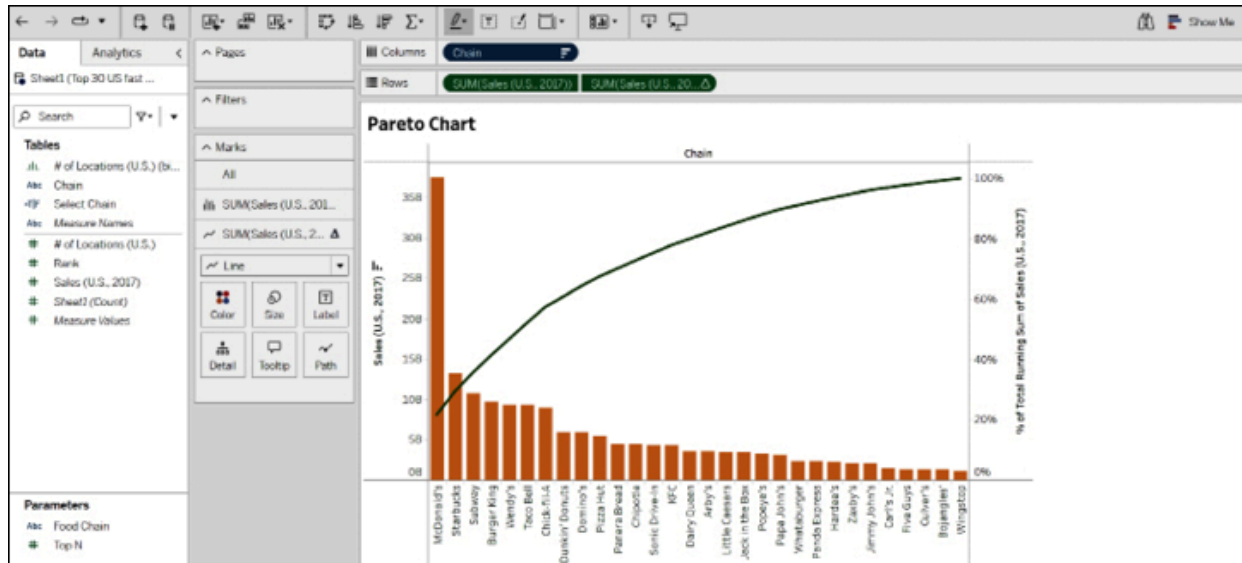


Figure 4.10: Pareto chart Word Cloud

Word Cloud is a collection of words shown in different sizes. The bigger and bolder words are the most frequently or more often used. Word Clouds are highly preferable when working with non-structured datasets, mainly textual.

They are used to make datasets on Twitter tweets, reviews on online portals, etc.

For instance, suppose you want to build a word cloud on a Food Chain based on several locations. Then, follow these instructions.

1. Connect the data **Food Chain** data source.
2. Drag the dimension field **Chain** to the **Text marks card**.
3. Drag the measured field **#Locations** to the **Size marks card**.
4. If necessary, change the mark type from automatic to Text.
5. Now, drag the same dimension field chain to the **Color marks card**.

Refer to [Figure 4.11](#) for an illustration of Word Cloud:

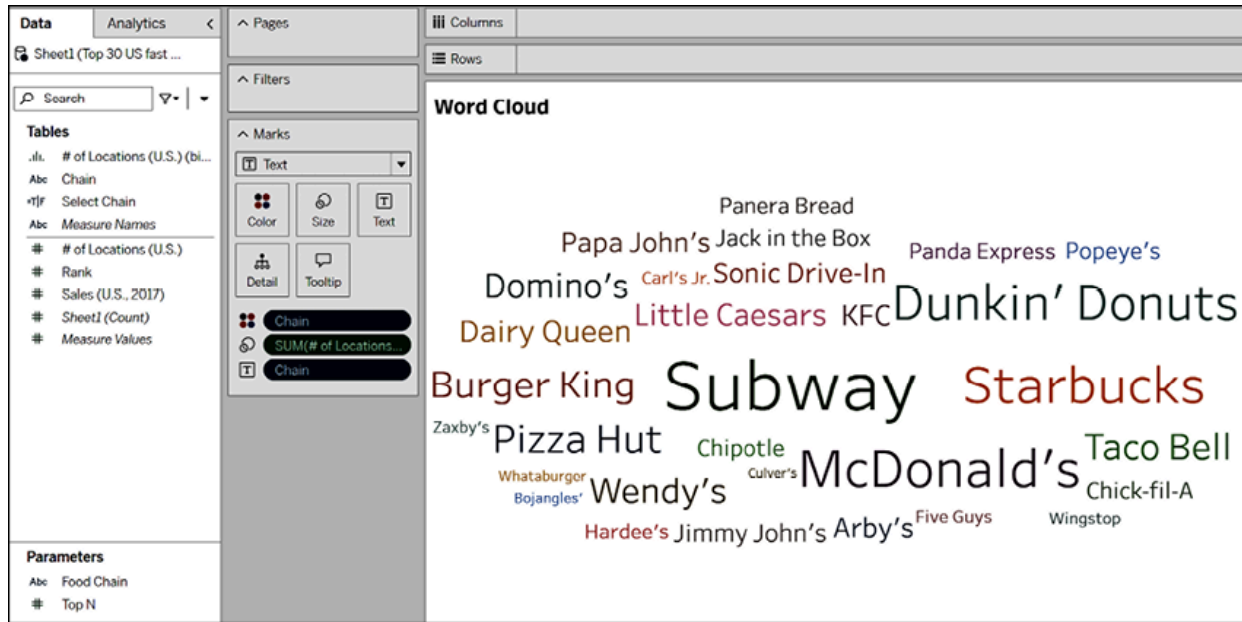


Figure 4.11: Word Cloud *Level of Detail*

Previously, we learned about creating a calculated field where a basic level of computing can be performed. However, to compute the values on the level of granularity, certainly at the data source level and the visualization level, is called the **Level of Detail (LOD expressions)**.

These LOD expressions can be performed at *three* different granular levels:

- **INCLUDE** (More Granular Level)
- **EXCLUDE** (Less Granular Level)
- **FIXED** (Totally Independent Level)

Refer to the following table for a brief idea of the structure and syntax of LOD expressions: **{INCLUDE | EXCLUDE | FIXED <dimension1, dimesnion2, ... n>: <aggregate expression>}**

Element	Description
{ }	The entire LOD expressions is enclosed between { } curly braces.
INCLUDE EXCLUDE FIXED	The LOD Keywords.
<dimension1, dimesnion2, ...n>	Specifies one or more dimensions to aggregate the expression to be joined.
:	This colon will separate the dimension and aggregate expression.

<aggregate expression>	The calculation is to be performed based on the dimension specified.
------------------------	--

Table 4.1 : *LOD expression syntax Now, let us understand how to apply each of these LOD expressions.*

INCLUDE

INCLUDE LOD expressions are more granular level. They are useful when we need to compute the values using the specified dimensions. Fields based on INCLUDE LODs will change the results as you add or remove the dimensions from the view. For instance, creating a visualization on averages of the sum of sales by state across categories from the Sample-Superstore data source.

1. Connect the **Sample-Superstore** data source.
2. Drag **Category** from the Data pane to the Column shelf.
3. Select **Analysis | Create Calculated Field**.
4. Name the Calculation as **Sales per State**.
5. Write the LOD as shown in [Figure 4.12](#):

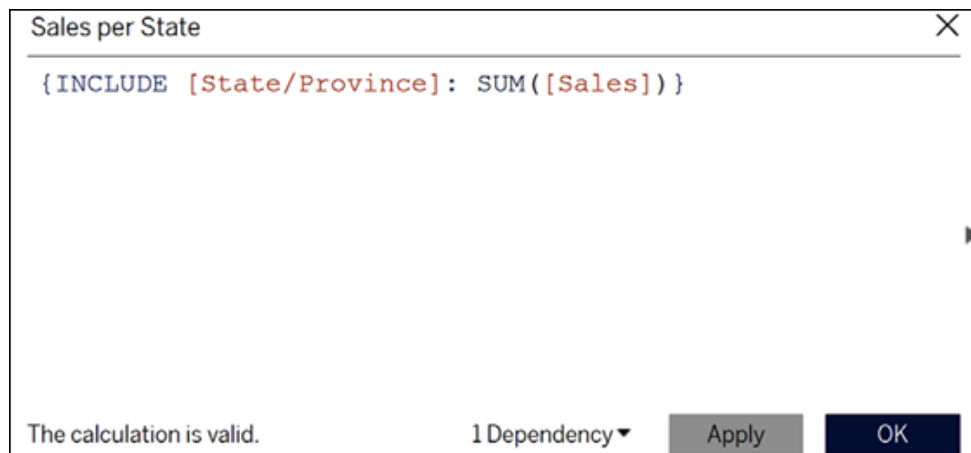


Figure 4.12: *INCLUDE level of detail expressions*

6. Click on **OK**.
7. Now, drag **Sales per State** to the Column shelf from the Data pane, under the Measures, and is aggregated as an average. Refer to the following figure for an explanation of the same:

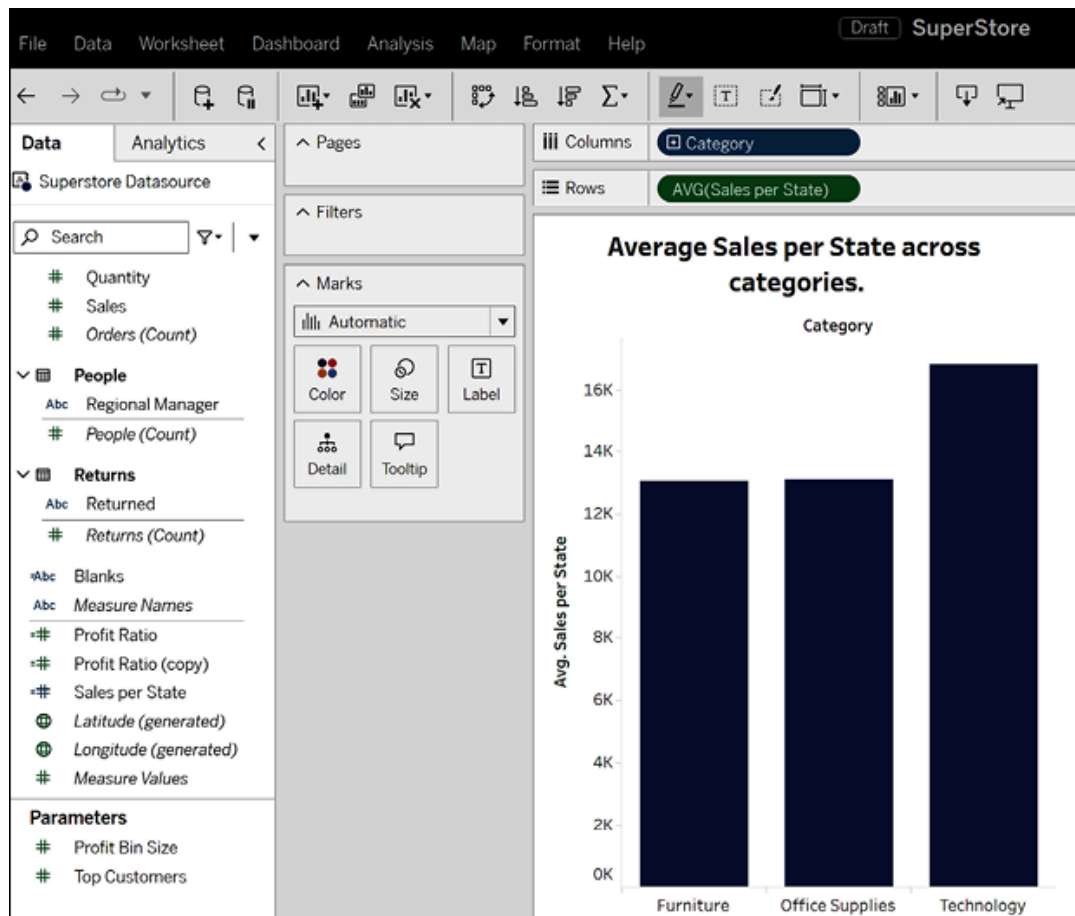


Figure 4.13: Include LODs in view

8. If you add one more dimension to the column shelf, then the LOD expression results will update.
9. Let us get the **Average of Sales per State across Categories and Ship Mode**. The following figure is in reference to the same:

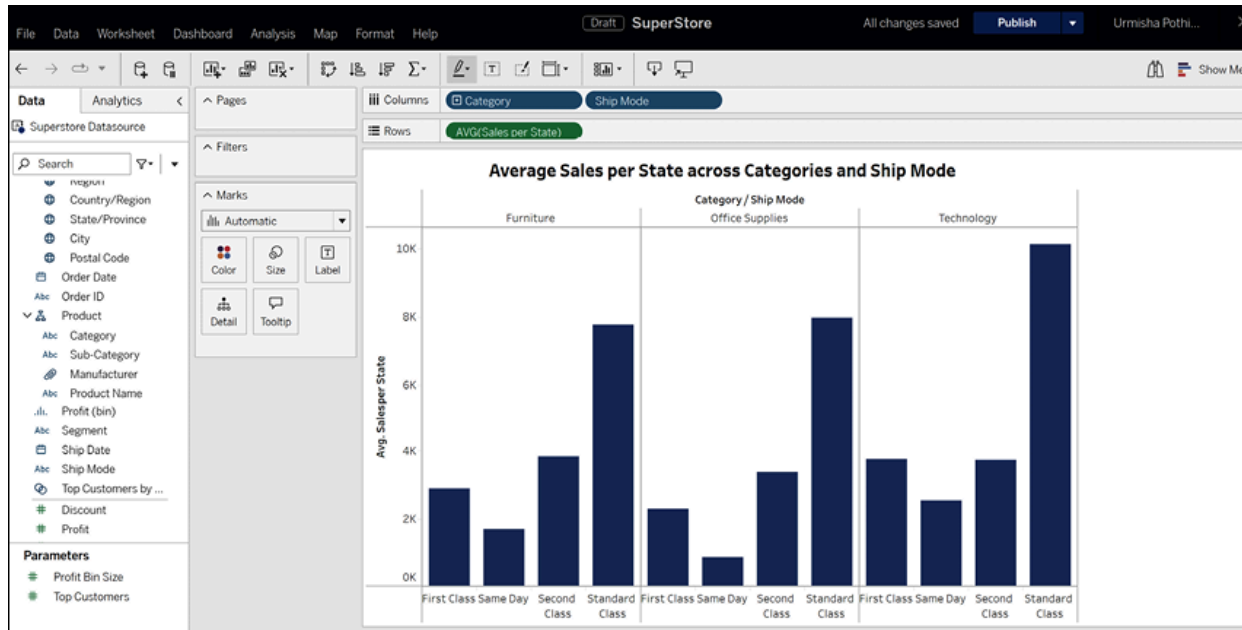


Figure 4.14: Include LODs in view with multiple dimensions.

EXCLUDE

EXCLUDE LOD expressions are less granular level. EXCLUDE Level of Detail expressions are useful when you want to omit the specified dimension from the view of the level of detail or modify the view level calculation. These LODs can be used when there are no dimensions to omit (row-level expressions) from the view.

These expressions can be useful for checking the differences between the overall average scenarios or the percentage of the total.

For example, LOD expression excludes **Region** from the **SUM(Sales)** by **Order Date Month** from the **Sample-Superstore** data source follow these steps:

1. Connect the **Sample-Superstore** data source.
2. Drag **Region** from the **Data** pane to the **Column** shelf.
3. Drag **Sales** from the **Measures** under the **Data** pane to the **Column** shelf and aggregate this automatically to SUM.

SUM[Sales]

4. Now, drag **MONTH [Order Date]** to **Row** shelf from the **Data** pane. Refer to the following figure for an illustration of the steps as mentioned above:

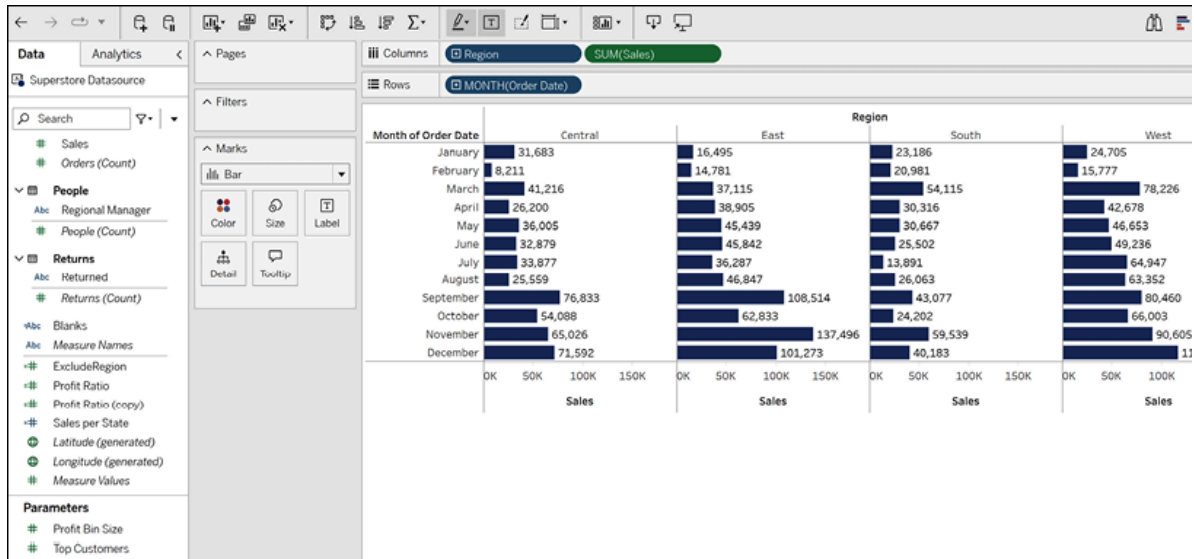


Figure 4.15: Sum of Sales by Order Date Month on a regional basis

5. Select **Analysis | Create Calculated Field**.

6. Name the Calculation as **ExcludeRegion** and write the following LOD, as shown in [Figure 4.16](#):

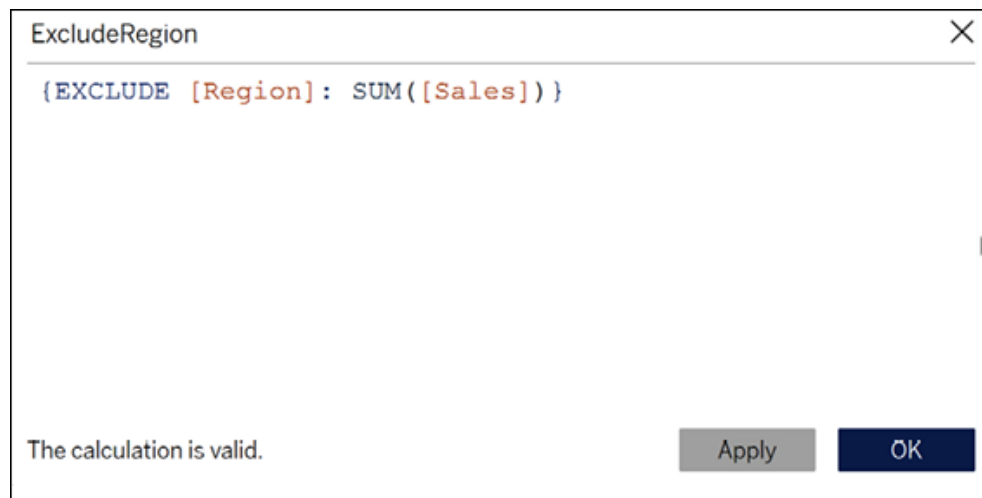


Figure 4.16: EXCLUDE level of detail expressions

7. Click on **OK**.

8. Now, drop this **ExcludeRegion** to **Color Mark Card** and **Labels Mark Card** to the view to show the total sales by month without considering the region and see the difference in the following figure:

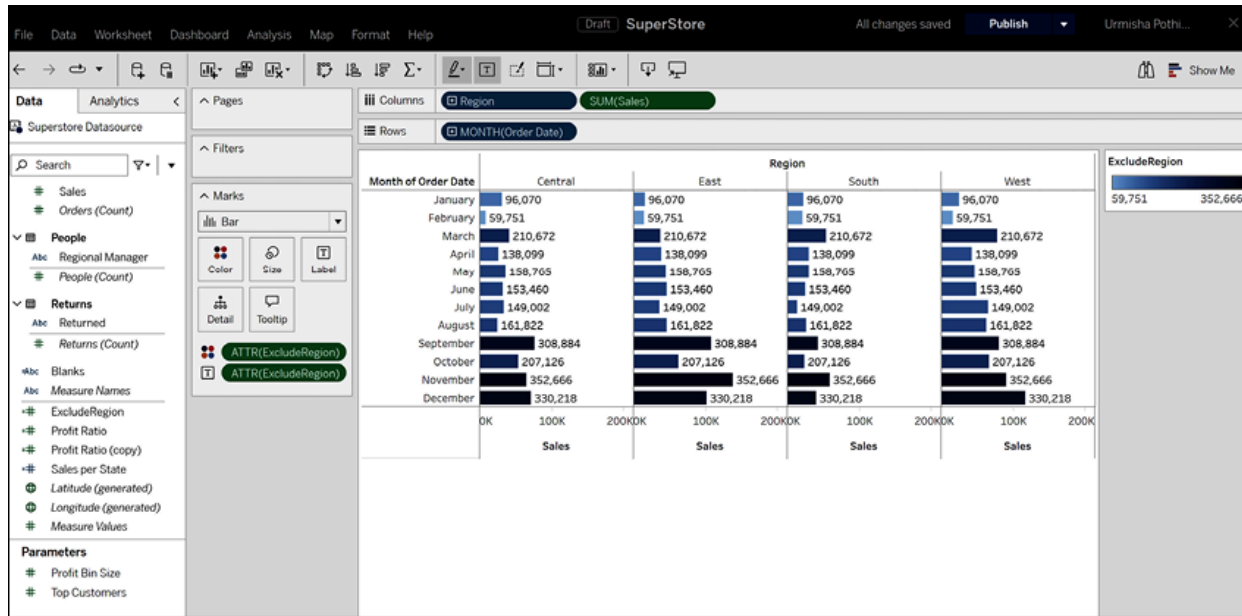


Figure 4.17: EXCLUDE LODs in view *FIXED*

FIXED LOD expressions are totally independent level. This type of LOD can compute the values per the specified dimensions without considering the dimensions on the view. Let us understand this LOD with an example.

Suppose you want to calculate the **SUM(Sales)** by region and State/Province from the **Sample-Superstore** data source, then follow these steps:

1. Connect the **Sample-Superstore** data source.
2. Drag the **Region** from the **Data** pane to the **Row** shelf.
3. Drag **Sales** from the **Measures** under the **Data** pane to the **Column** shelf and aggregate this automatically to SUM.

SUM[Sales]

4. Drag **State/Province** from the **Dimension** under the **Data** pane to the **Row** shelf next to **Region**. Refer to the following figure for a clearer demonstration of the same:

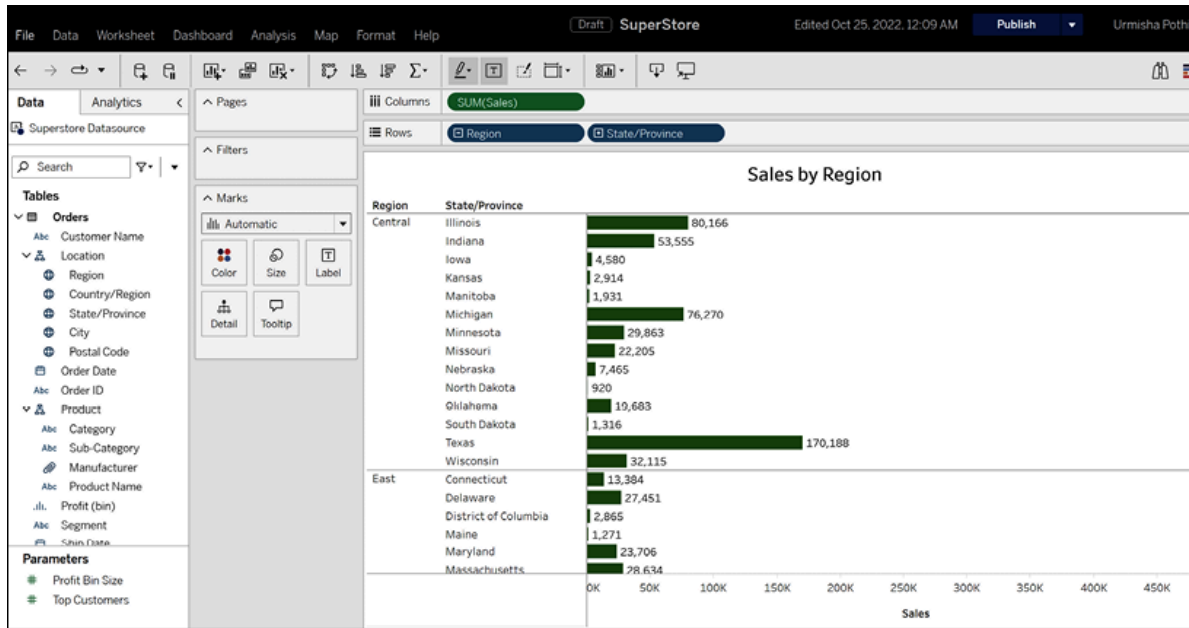


Figure 4.18: Sum of Sales by Region and State

5. Select **Analysis | Create Calculated Field**.

6. Name the **Calculation** as **SalesByRegion** and write the following LOD as shown in Figure 4.19:

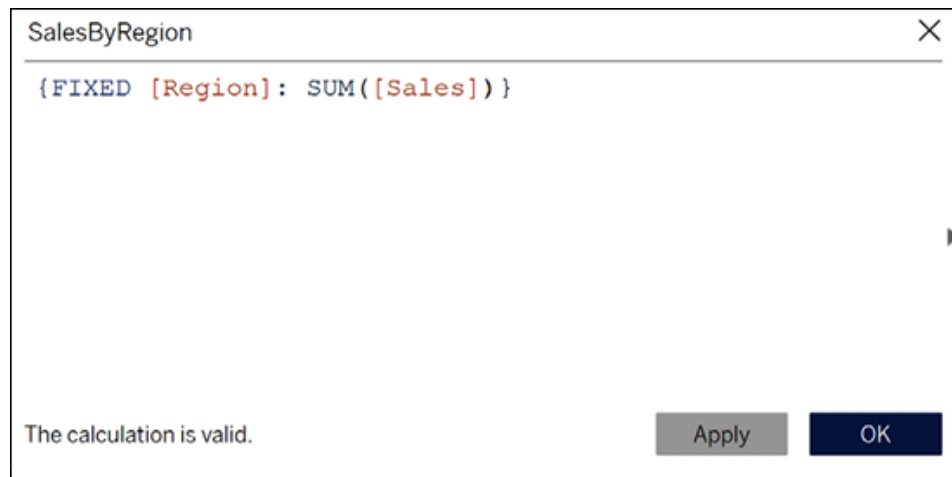


Figure 4.19: FIXED level of detail expressions

7. To see the difference after applying LOD, drop the **SalesByRegion** to **Color Marks Card** and **Labels Marks Card**.

You will see that after applying fixed LODs **SalesByRegion**, the State/Province shows no difference in the sales. Hence, the values for the individual states under each region are identical, as shown in the following figure:

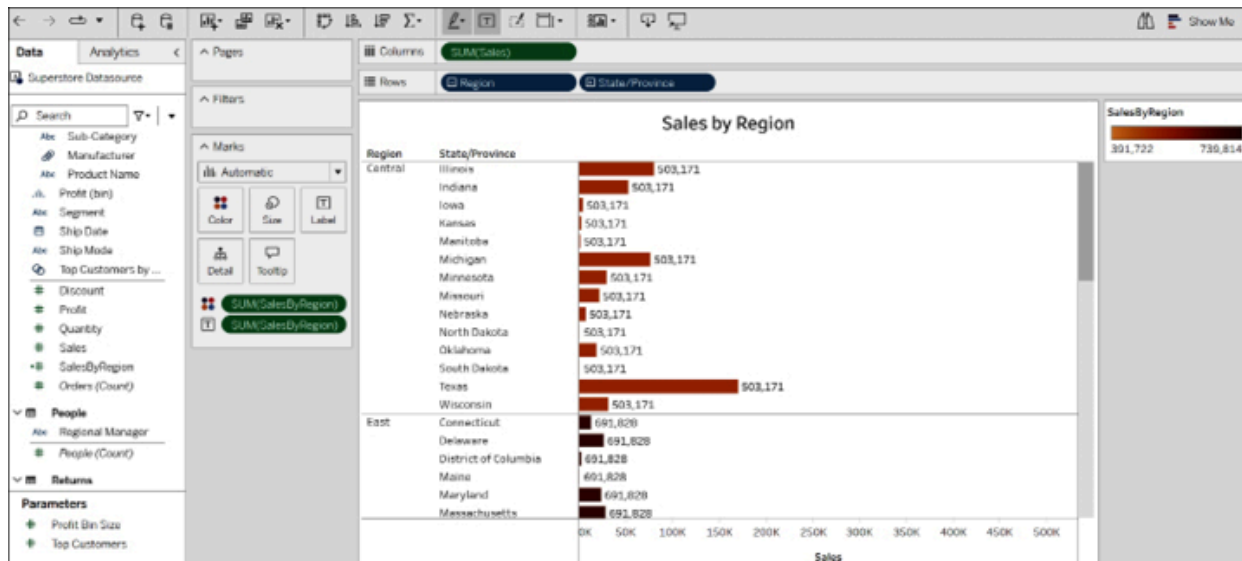


Figure 4.20: FIXED LODs in view *Parameters*

A workbook variable that can replace the filter, calculation, or any reference line is called a parameter. The parameter could be a number, string, date, datetime, or constant number. You can create static as well as dynamic parameters. The dynamic parameters have their advantages, mainly to set the automatic refresh on the current values, list of values, or range of values. This can happen each time the workbook is opened, data refreshes, or Tableau is connected to the data source referenced by the parameter.

We will explore all these use cases in detail. Before that, let us first understand how to create a parameter.

Creating a parameter

Follow these steps to create a new parameter:

1. Right-click the upper right corner drop-down in the data pane and then, click on **Create Parameter**.
2. In the **Create Parameter** dialog box, give the field a **Name**. However, by default, the field name is **Parameter1**, as shown in the following figure:

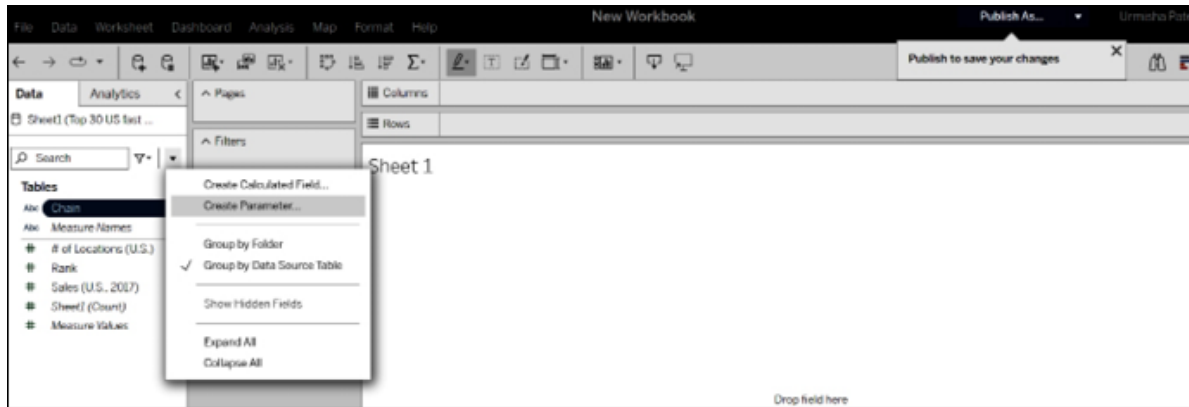


Figure 4.21: Create a new Parameter

- Now, specify the data type of this field for the values you want to create. For example, select the data type as **Date**, as shown in the following figure:

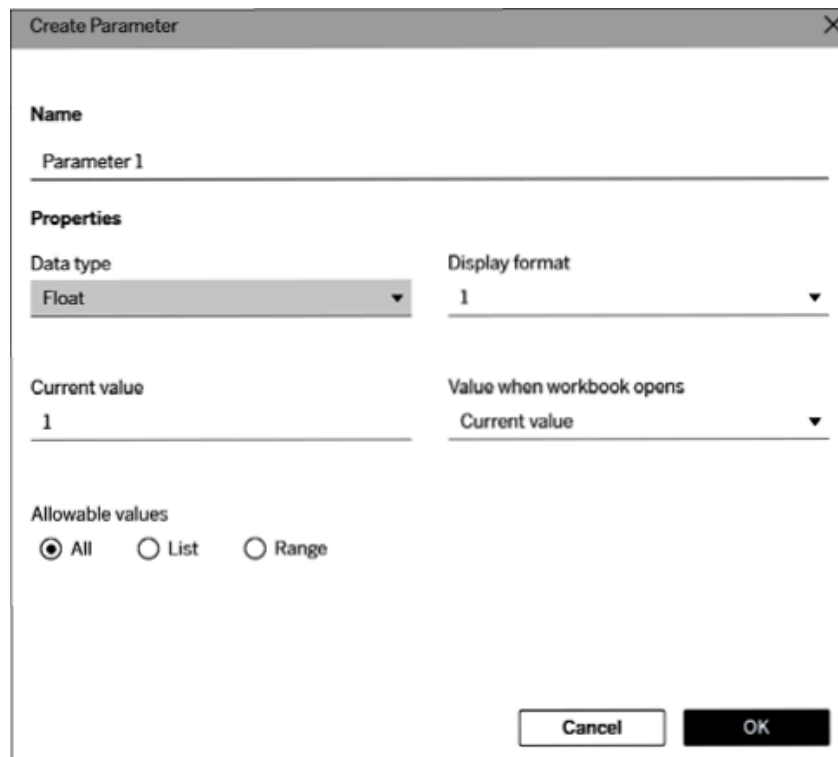


Figure 4.22: Parameter control dialog box Refer to the following figure for further understanding:

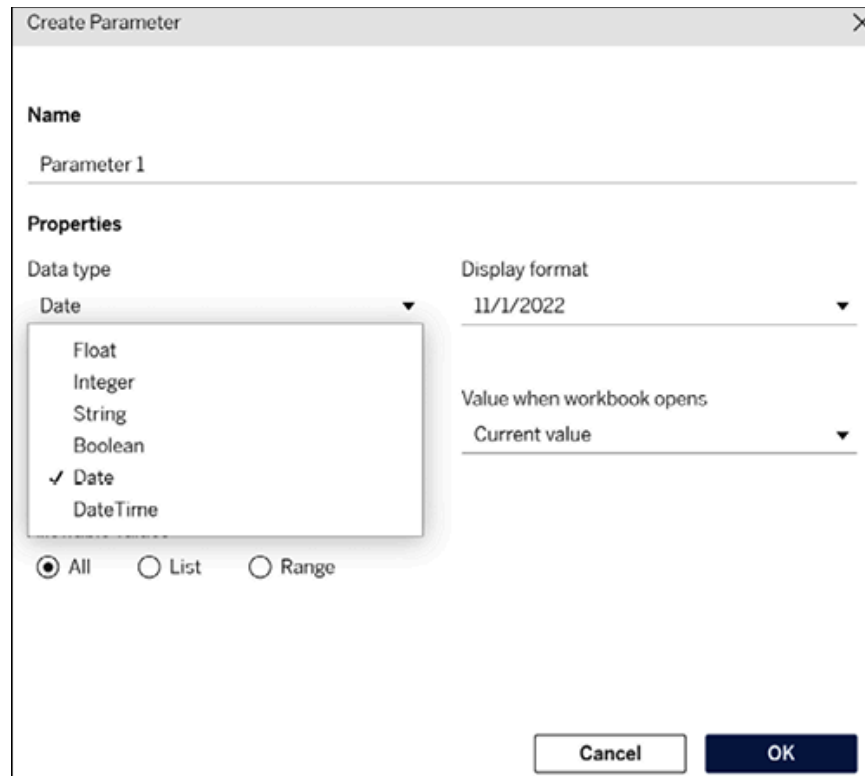


Figure 4.23: Data type control in parameter dialog box

4. Select the display format from the drop-down to view the field values in a particular format. Refer to the following figure for a demonstration of the same:

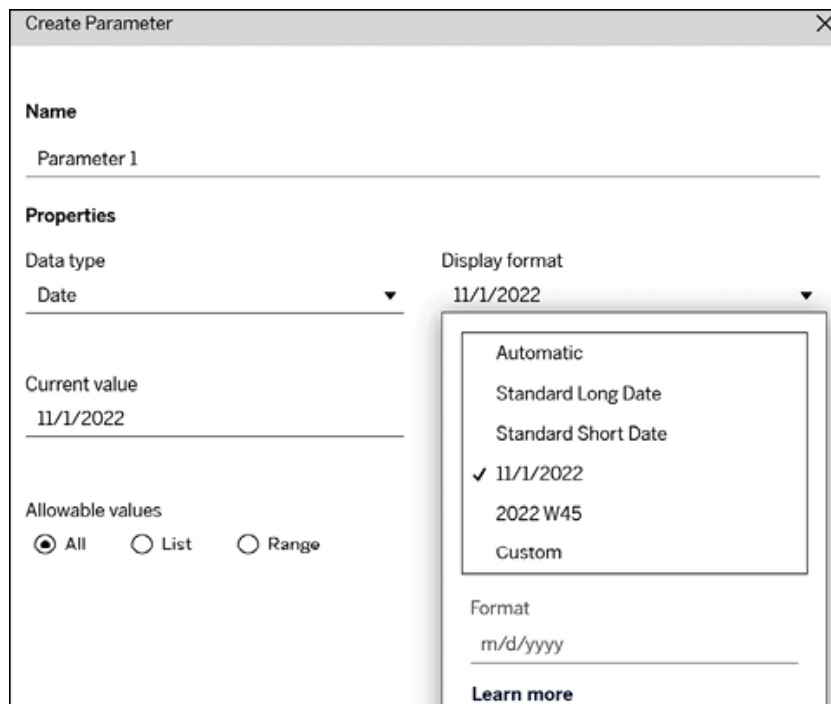


Figure 4.24: Display format control in parameter dialog box

5. Now, you can specify the current value. By default, it will show the most recent value or current value of the date or latest date:

a. You can specify the value when workbook opens on choosing the options from the drop-down menu **Value when workbook opens**.

b. Here, you can create a dynamic parameter. This means if the connected data source is updated and when the workbook opens, the parameter will automatically update the value of the field.

6. Choose the option from the following to accept the values in the parameter:

a. **All**: This parameter control is a simple text field.

b. **List**: This parameter will provide a list of possible values for you to select from. If you select a list, then you need to specify the list of values. You can add the custom values or even from any field by selecting **Add values from** as shown in [Figure 4.25](#):

The screenshot shows a dialog box titled "Allowable values". At the top, there are three radio buttons: "All", "List" (which is selected), and "Range". Below these is a table with two columns: "Value" and "Display As". The first row of the table contains the date "10/28/2022" in both columns. Below the table is a button labeled "Click to add". To the right of the table, there are two radio buttons: "Fixed" (selected) and "When workbook opens". Below these is a button labeled "Add values from" with a downward arrow. At the bottom of the dialog, there are three buttons: "Remove Selected", "Cancel", and "OK".

Value	Display As
10/28/2022	10/28/2022

Figure 4.25: Listing in parameter

c. **Range**: This parameter will provide the values in the specified range that lie within minimum to maximum values, as shown in [Figure 4.26](#).

Note: The Range option will not be supported by the String Parameter. For example, the parameter values lie between 11/2/2022 to 2/10/2023, with the step size set to 1 day to create the parameter control.

7. If you want to refresh the parameter whenever the workbook opens, then select the radio button when the workbook opens; otherwise, the values will remain fixed by default:

Figure 4.26: Range in parameter

8. When finished, click on **OK**.

Editing the parameter

You can edit the parameter from the data pane or the parameter control. Follow these steps to edit the parameter:

1. Right-click on the parameter in the data pane and select **Edit** from the drop-down menu.
2. This will open the **Edit Parameter** dialog box. Make necessary changes as needed.
3. When finished, click on **OK**.

Deleting the parameter

You can delete the parameter from the data pane or the parameter control. Follow these instructions to delete the parameter: Right-click the parameter in the data pane and select **Delete** from the drop-down menu.

Note: If any calculation field uses the deleted parameter, then the calculated field becomes invalid.

Use case 1: Use a parameter in a calculation

The parameter can be used in the calculation to modify the values dynamically rather than manually editing the calculations. To use the parameter in the calculation, then drag the parameter from the data pane and drop it in the calculation formula.

For instance, create a string parameter named **Food Chain** from the **FoodChain** dataset based on the **Chain** field and follow these instructions to use the parameter in a **Calculation**:

1. Right-click the upper right corner drop-down in the data pane and then click on **Create Parameter**.
2. Name the parameter **Food Chain** and select **String** datatype. Now, select **List** radio button and fetch the values from the **Chain** field as shown in the figure.
3. When finished, click **OK**.

Refer to the following figure for a demonstration of the same:

The screenshot shows the 'Create Parameter' dialog box for a String parameter named 'Food Chain'. The 'Name' field is 'Food Chain'. Under 'Properties', the 'Data type' is 'String' and the 'Display format' is 'Arby's'. The 'Current value' is 'Arby's' and the 'Value when workbook opens' is 'Current value'. Under 'Allowable values', the 'List' radio button is selected. A table lists values and their display format, all set to 'Subway'. The 'Fixed' radio button is selected for the allowable values. Buttons for 'Add values from', 'Remove Selected', 'Cancel', and 'OK' are present.

Value	Display As
Subway	Subway
Taco Bell	Taco Bell
Wendy's	Wendy's
Whataburger	Whataburger
Wingstop	Wingstop
Zaxby's	Zaxby's
Click to add	

Figure 4.27: String parameter example

4. Go to **Analysis | Create Calculated Field**.
5. In the **Calculation** editor, write the code as shown in the following figure.
6. When finished, click **OK**:

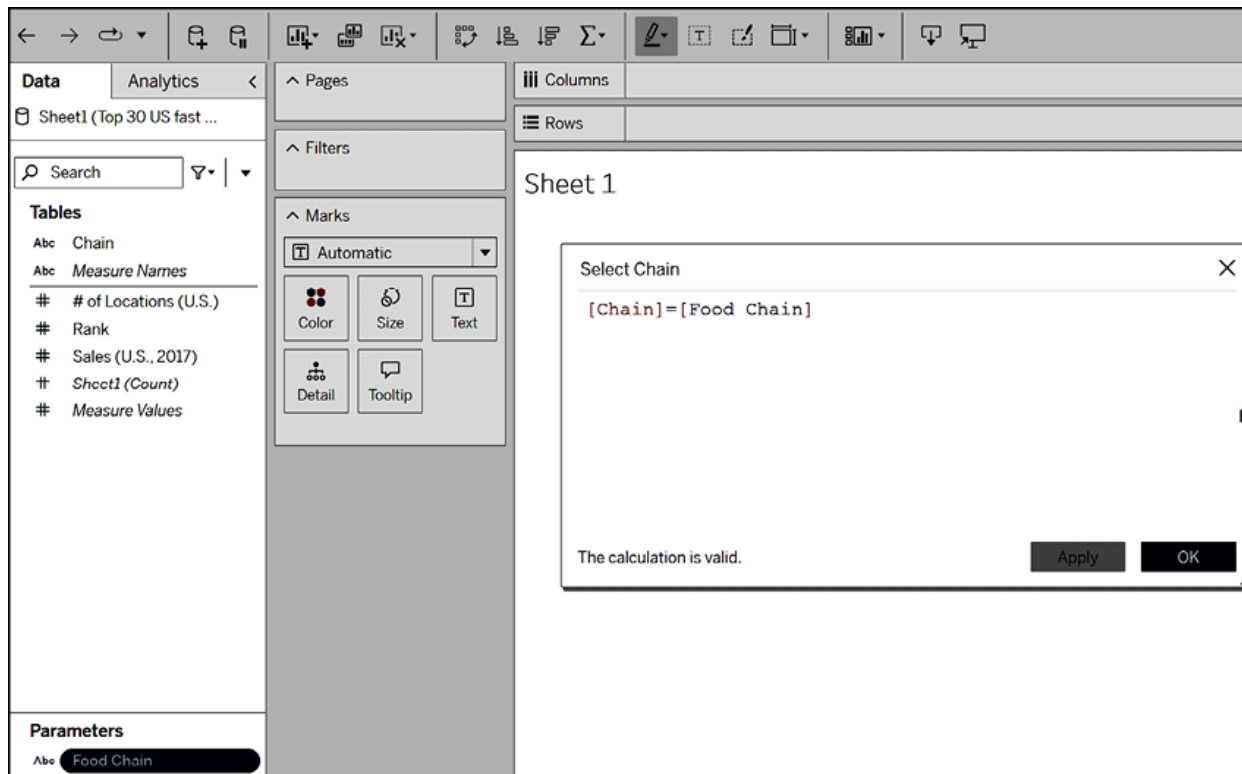


Figure 4.28: Parameter in a calculated field *Use case 2: Use a parameter in a filter*

The parameter can also be used to modify the values dynamically in a Top N filter rather than manually setting the number of values you want to see in the filtered results. Let us understand the usage of the parameter as a filter with an example. Suppose you want to know the Top three, five, or ten food chains by giving users the flexibility to choose the number of Top filtered food chains.

Follow these instructions to create a dynamic filter using a parameter:

1. Drag the dimension field **Chain** to the filter box. Now, select the **Top/Bottom** option to filter the records.
2. Click on the **By Field** radio button to filter the records by field.
3. Select **Top or Bottom**, then instead of entering the number of records. In the **Count drop-down**, choose **Create a New Parameter**, as shown in [Figure 4.29](#).
4. This will open a **New Parameter Dialog Box** and allow the create integer type parameter only, as shown in the following figure:

▼ **Top/Bottom**

☐ None

☒ **By field**

Rank	Count
Top ▼	Enter a Value... ▼ 10

Field	Aggregation
# of Locations (U.S.) ▼	Sum ▼

☐ **By formula**

Rank	Count
Top ▼	Enter a Value... ▼ 10

Formula

Figure 4.29: *Parameter in filter dialog box*

5. Now, create the parameter and named as **Top N**, as shown in the following figure:

Create Parameter

Name

Top N

Properties

Data type

Integer

Display format

3

Current value

3

Value when workbook opens

Current value

Allowable values

☐ All
☒ List
☐ Range

Value	Display As
3	3
5	5
10	10
Click to add	

☒ Fixed
☐ When workbook opens

Add values from

Remove Selected

Cancel

OK

Figure 4.30: Custom values list in parameter

6. Now, click **OK**. You will see **the Top N parameter** in the **Count drop-down list** under the **By Field** Filter editor.
7. Select the **field Sales** and **aggregation type as SUM**.
8. Click on **OK**.

Refer to the following figure for an illustration of the same:

Filter [Chain]

Top/Bottom

None

By field

Rank

Count

Top

Top N

Field

Sales (U.S., 2017)

Aggregation

Sum

By formula

Rank

Count

Top

Enter a Value...

10

Formula

Summary

Selection:

Selected 0 of 30 values

Wildcard:

All

Condition:

None

Limit:

None

Reset

Apply

Cancel

OK

Figure 4.31: Applying parameter in Filter

To show the parameter in the Viz, follow these steps:

1. To show the parameter control in the view, right-click the **Top N parameter** and select **Show Parameter**.
2. Use the parameter control to modify the filter to show the **Top N** products, as shown in the following figure:

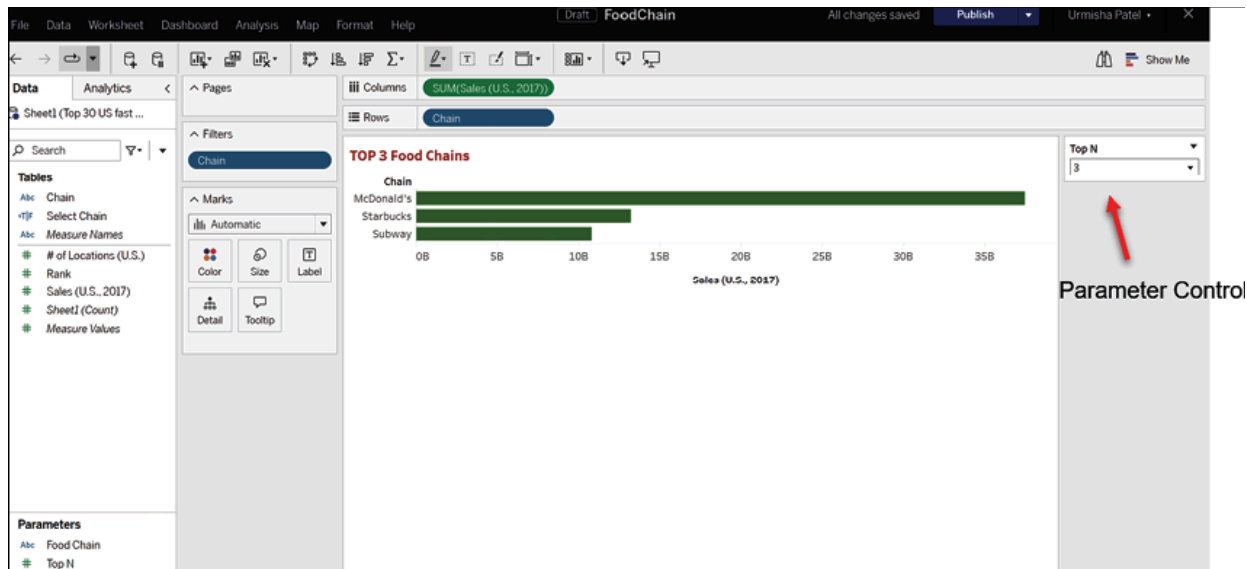


Figure 4.32: Dynamic filter—Parameter Control *Use case 3: Use a parameter in reference lines*

A parameter can be used to dynamically modify the reference lines, band, or box instead of showing the reference lines, band, or box at a fixed location in the axis using a parameter.

Follow the steps to use parameter in reference lines:

1. You will find the list of parameters in the Value drop-down menu in the Add reference dialog box.
2. Select the parameter you want to use to refer to the lines, Bands, or box as shown in the following figure:

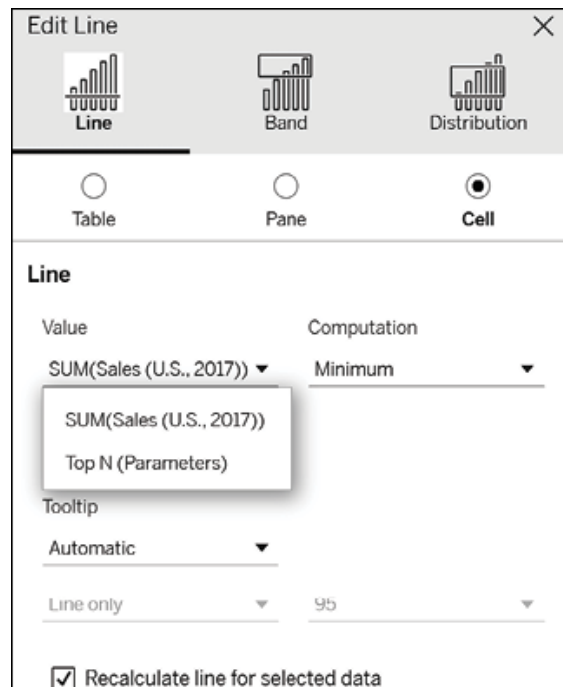


Figure 4.33: Parameter in reference dialog box

Groups

Groups: the word itself indicates similar values in a field hence collectively called a group. In other words, when you try to combine relative members in a field, that creates a group. They are primarily useful for correcting data errors, i.e., inconsistency in the data values in a field, or used for custom grouping the field values, such as for answering what-if type of questions. Here are some aspects related to groups:

- **To create a group:** There are multiple ways to create a group. Follow these instructions steps to do the same:

1. Select one or more data points in the Viz; on the tooltip appears a group icon that looks like a paper clip, .

Or,

Right-click on any field in the data pane and select the **Create | Group** option from the drop-down menu to create a group.

2. Then, from the group dialog box, select the members from that field and click on **Group** button to add those selected members in one single group. The default group name is created using the combined members.

3. [Optional] For the rest of the members, if you want to create a single group, then check on **Include Other** to call this group by default as **Other** as shown in the following figure:

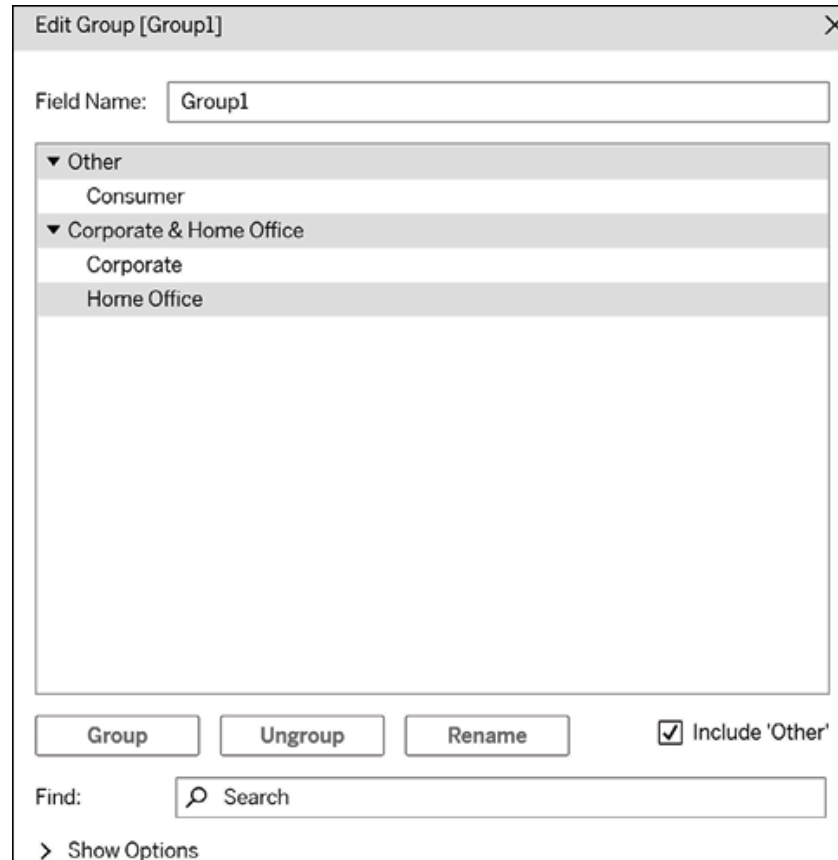


Figure 4.34: Edit Group dialog box

- **To rename the group:** To rename the group, select the group name in the group dialog box and then click on the **Rename** button.
- **To edit the group,** right-click the group field in the data pane and choose **Edit Group** from the drop-down menu. You can now add or remove members from the selected group or change the group field names or default group name.
- **To ungroup:** Right-click the group field in the data pane, click on **Edit Group** from the drop-down menu. Now, select one or more members in the edit group dialog box, select **Ungroup** button to remove the members from the group list.

Sets

Sets are the custom fields used to subset the data based on some conditions. Sets are used to find the top/bottom few records in the measure or selecting the members from

the visualization.

There are two types of sets in Tableau Online:

- Fixed set
- Dynamic set

Note: All Set options are available in the licensed version of Tableau Online. Let us understand each of them in detail.

FIXED Set

When you create a set where one cannot change the members from the set even when the data changes., it is called a fixed set. It can be based on single or multiple dimensions.

Creating a fixed set: Follow these steps to create a fixed set:

1. Select one or more marks from the view.
2. Right-click the marks and select **Create Set**.
3. Assign the name to the set in the create dialog box of set.
4. By default, whatever the members you selected will be included in the set. However, you can check on **Exclude** for the selected members instead.
5. If you want to apply this set automatically to filtering, then check on **Add to Filters Shelf**.
6. When finished, click on **OK**.

A set icon that looks like a Venn diagram will be seen next to the defined set name under the data pane to indicate the field is a set. 

Add or remove the members from a set: You can also add or remove the data points/members from the set. Follow the steps in case to add or remove the members:

1. Select the data points in the Viz you want to add or remove from a set.
2. This will open a tooltip.
3. To add the data point, select Add to [set name]
or
4. To remove the data point, select Remove from [set name].

Show In/Out members in a Set: When a set drags to the Viz, Tableau display the set int two categories as shown in the following figure:

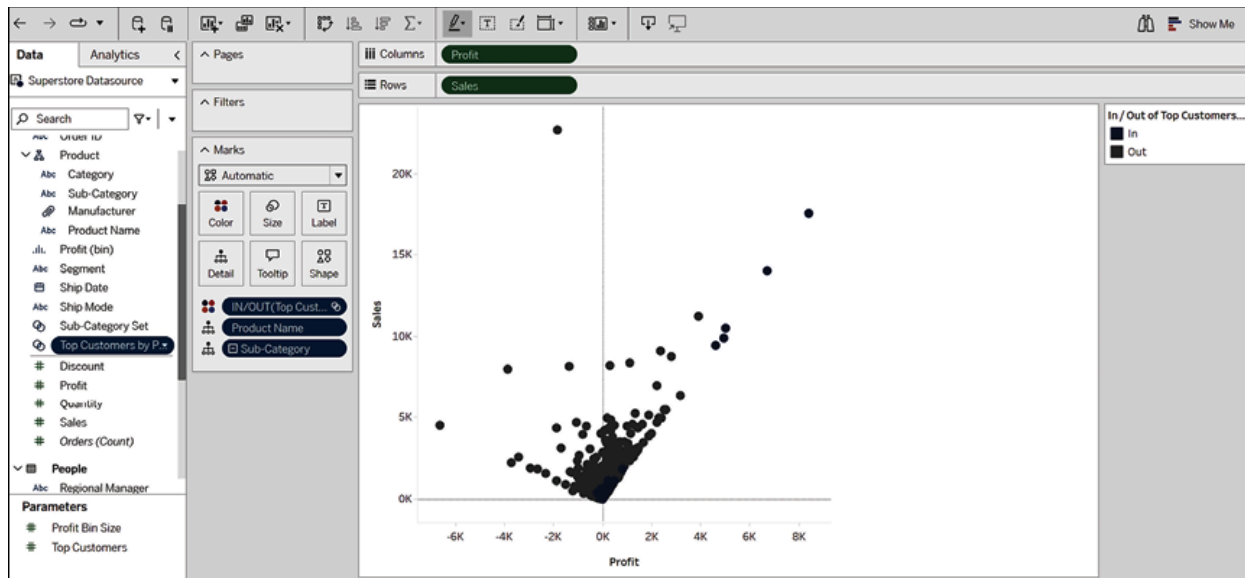


Figure 4.35: Sample set based Viz **In:** The members in the set **Out:** The members excluded from the set.

Here in the given example, based on **Top Customers by Profit Set**, the Viz was highlighted in two categories.

To Show in/out members in a set:

- Click on the **Set** for which you want to see the In/Out members.
- For example, click on Top Customers by Profit Set in the marks card.
- Select **Show Set** option from the drop-down menu.

Now, let users change the set values.

Dynamic set

The Dynamic set can change its members if data changes. These types of sets are based only on a single dimension.

Create a dynamic set

Let us see how to create a dynamic set by following these steps:

1. Right-click the dimension field and select **Create | Set**.
2. By default, in Tableau Cloud; Set will consider all members. A set field will be added in the data pane.

Edit members in the set

To edit the members in the set, follow these steps:

1. Drag the field set to the filter box. Then, a filter dialog box will open based on the set field.
2. Now set values can be adjusted either by **Use all or Conditional** based or using Top to limit the members to include on the set.
3. When finished, click on **OK**.

For example, a **Ship Mode Set** from **Sample-Superstore** data source. Refer to the following [Figure 4.36](#) to edit the members in the set:

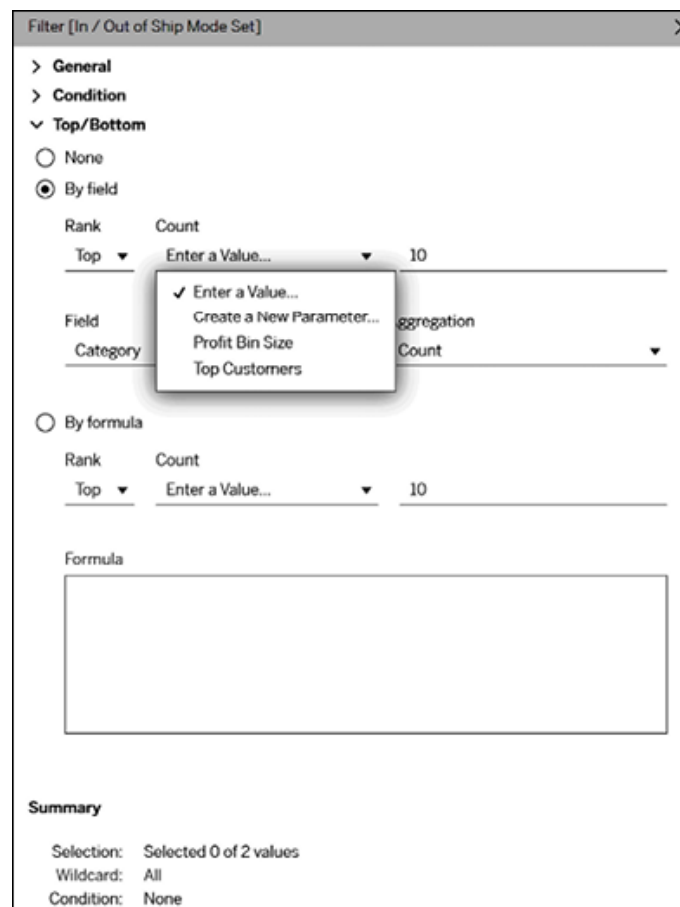


Figure 4.36: Filter dialog box for set field

Note: There is a concept of Combined Set, which combines two sets to compare the members. However, the Tableau Online web environment does not provide this feature.

Dashboard objects

Now, let us understand how to add dashboard objects and enhance the visuals and interactivity: **Horizontal and vertical:** These objects provide the container layouts to arrange the sheets either horizontally or vertically by simply dragging the object to the

dashboard space to fine-tune the workspace size: There are two types of layout container types. They are:

- **Horizontal layout container:** If two or more views arranged in a horizontal layout container, then the width of the views and objects it contains, resizes as shown in the following figure:

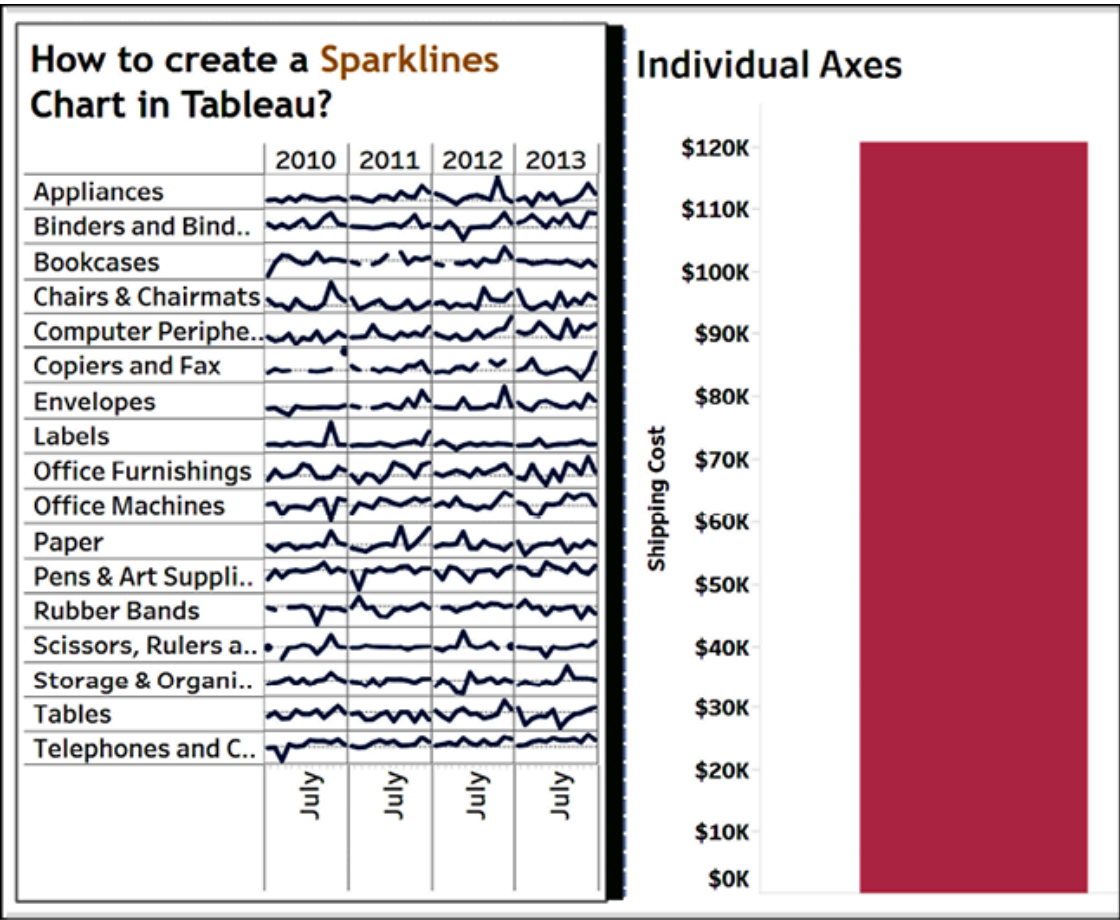


Figure 4.37: Horizontal container layout

- **Vertical layout container:** If two or more views are arranged in vertical layout container then the height of the views and objects it contains, resizes as shown in the following figure:

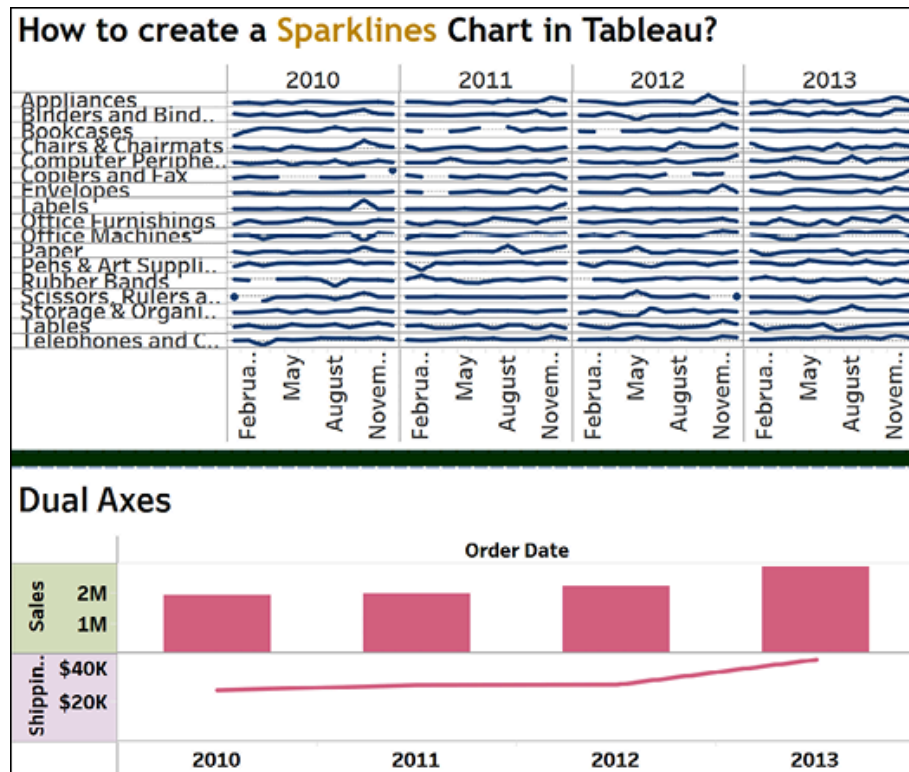


Figure 4.38: Vertical container layout

Add layout containers: Follow these steps to add layout containers and arrange the sheets:

1. Under **Objects** on the dashboard pane. Select **Horizontal** or **Vertical**.
2. Drag the container to the dashboard, as shown in [Figure 4.39](#):

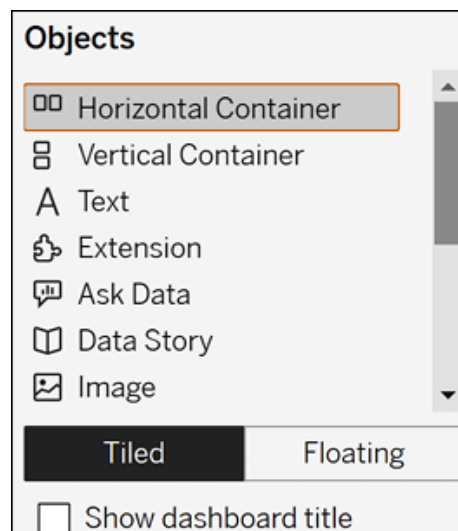


Figure 4.39: Container layout objects

3. Now, add sheets to the dashboard.

- **Evenly distribute items of layout containers:** Follow these steps to evenly distribute items of layout containers:

1. Go to **Layout** pane. Then, under Item hierarchy, select the layout container as shown in Figure 4.40.

2. Now, select **Distribute Contents Evenly**. So, items within that container will be distributed equally:

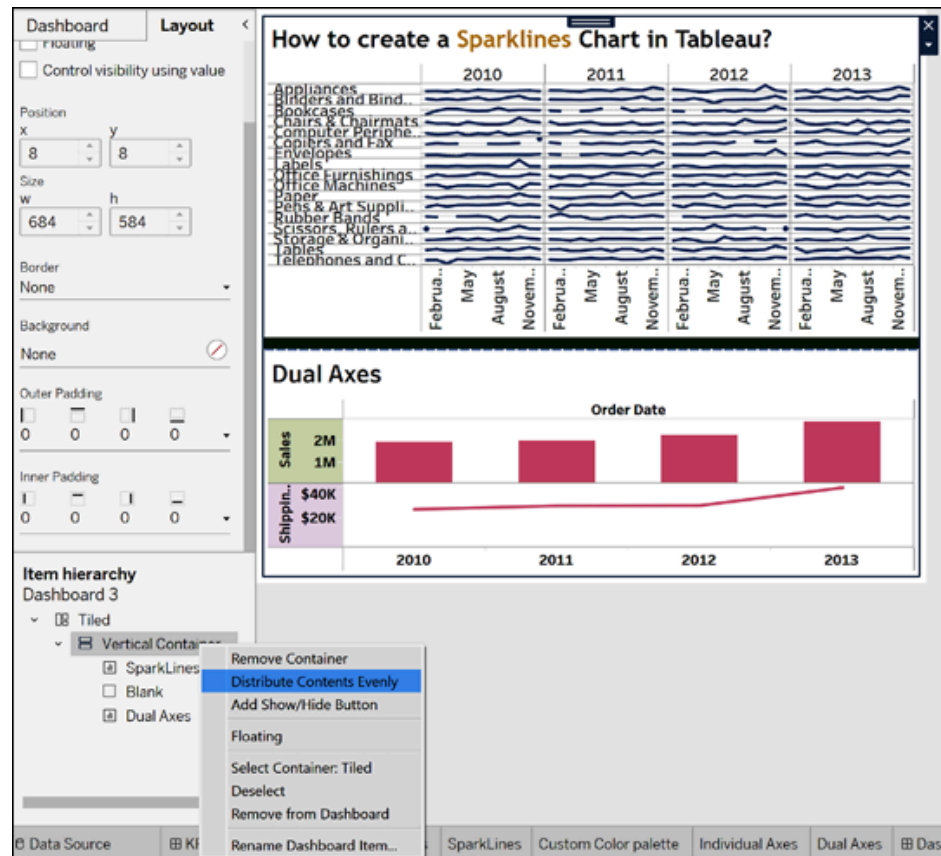


Figure 4.40: Distribute contents evenly

- **Remove the layout container:** Remove the layout container will allow to edit the container items independently. Let us see how to remove the container:

1. Go to **Layout** pane. Then under **Item hierarchy**, select the layout container.

2. Then, select **Remove Container** to delete the container only:

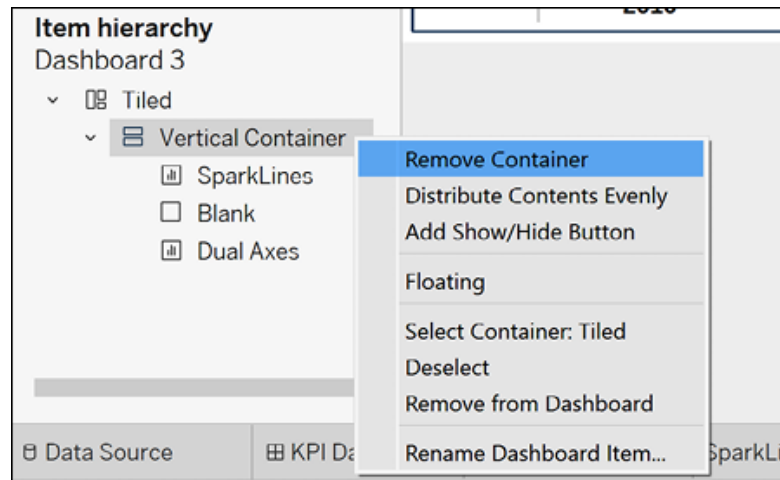


Figure 4.41: Remove Container

- **Tiled vs. Floating Layout:** Each item, layout container, object, or view are placed on the dashboard either tiled (by default) or floating.
 - **Tiled layout:** Items do not overlap each other. The item resizes based on the overall dashboard size:

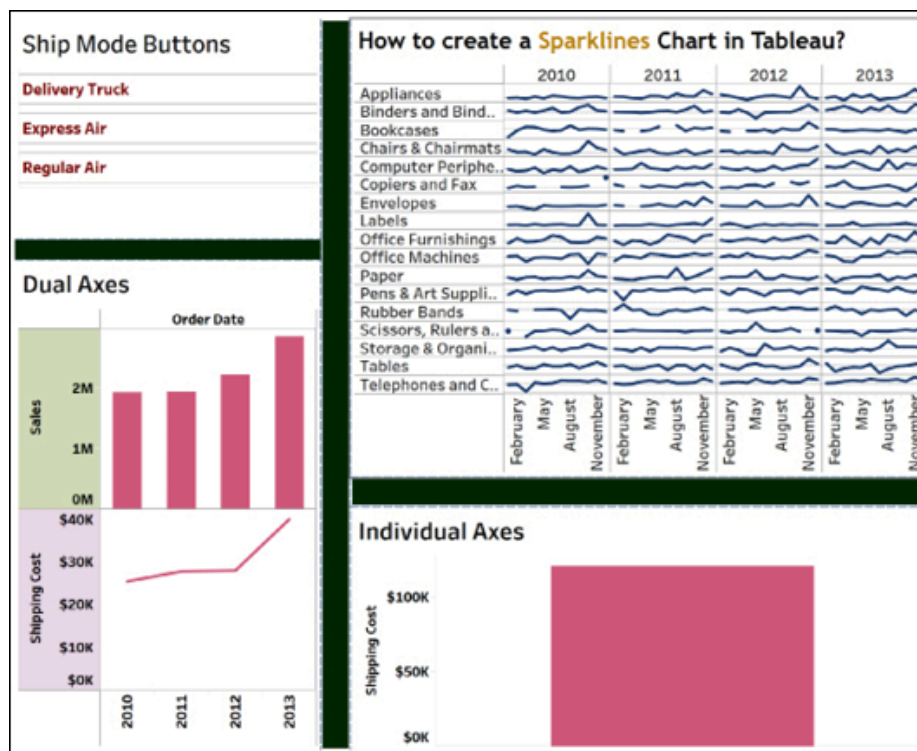


Figure 4.42: Tiled layout container

- **Floating layout:** Items can overlap. The floating view can map over the tiled view, as shown in [Figure 4.43](#).

The floating layout container is highlighted in green background color:

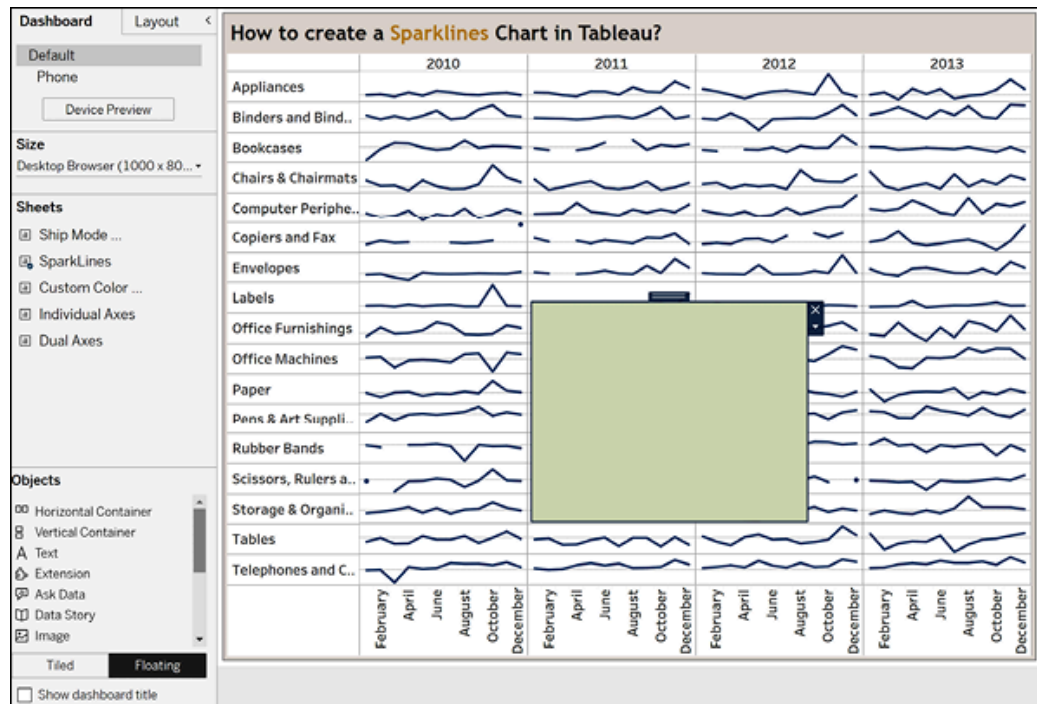


Figure 4.43: Floating layout container maps over tiled

Here, the sparklines sheet is arranged in a Tiled layout container.

The floating layout container maps over that tiled sheet.

Rearrange the floating layout container order: The floating item or container's order can be re-arranged. Follow these steps as shown in [Figure 4.44](#):

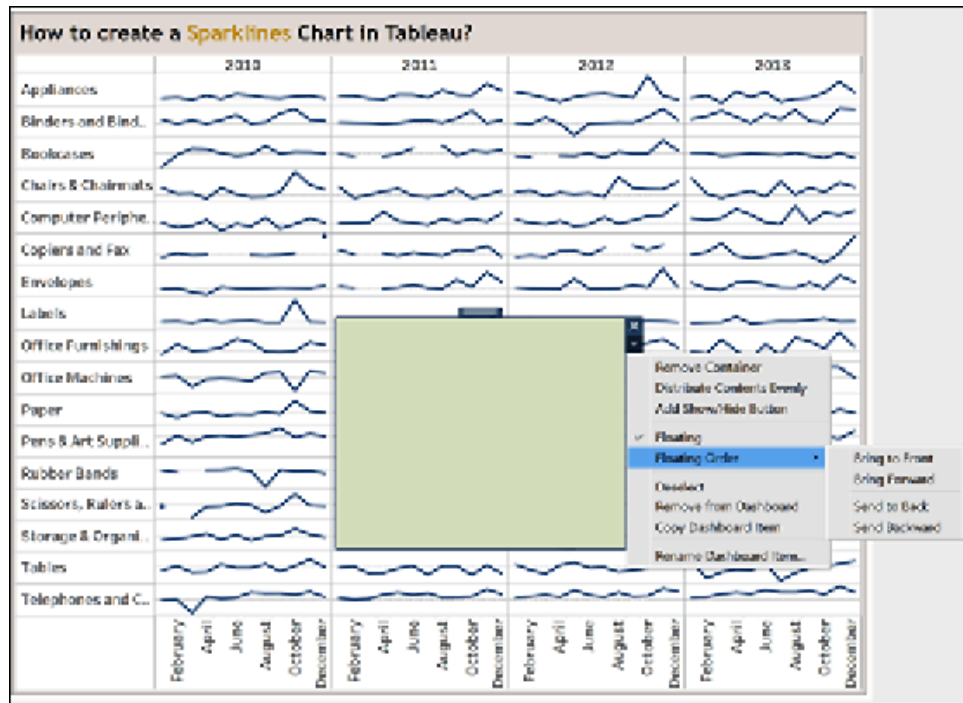


Figure 4.44: Floating Order options

1. Select the floating item or container.
2. Under More options (▼), select **Floating Order**.
3. Then, select the order to re-arrange the item.

Switch the existing item from tiled to float: Follow these steps to switch the existing item from tiled to float:

1. Select the tiled item or container.
2. Under **More** options (▼), select **Floating**.
3. If you want to switch the item or container from float to tiled. Then, select the floating item and then the un-ticked Floating option.

Text

This object provides the text area to insert as a sheet in the dashboard workspace. It also allows to edit the text, font style, size, color, formatting, inserting parameters, etc.

Follow these steps to insert text object:

1. Drag **Text Object** to the dashboard to its right.
2. This will open a **Text Editorial**, as shown in the following figure:

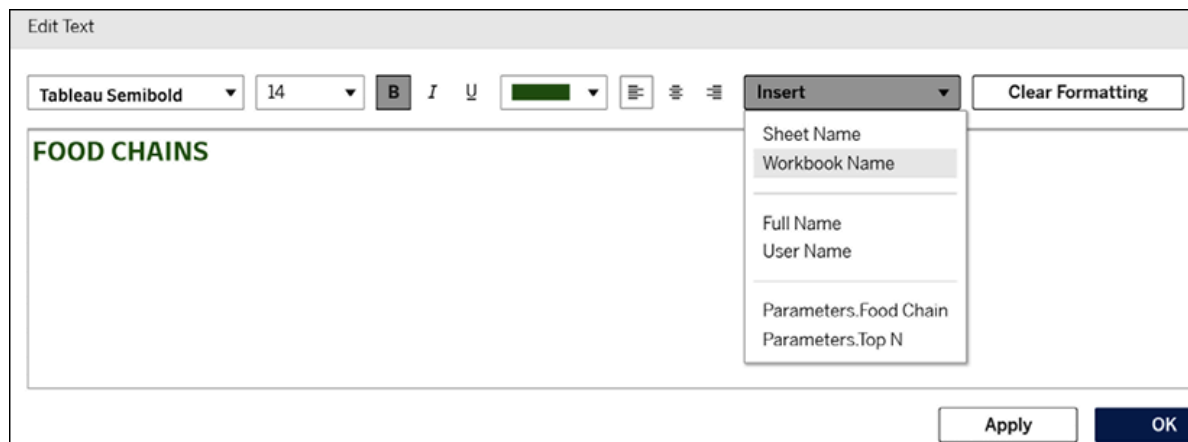


Figure 4.45: Text object editor

3. When finished, click **OK**.

Image

The image object allows you to add more visuals to the dashboard. The images can be imported directly from the device or via specific web URL links. For instance, to insert an image file (includes .jpeg, .jpg, .png, etc.) from the desktop or device, follow these instructions:

1. Drag the **Image** object to the dashboard to the right.
This will open the **Image Object** dialog box, as shown in the [Figure 4.34](#).
2. Click on **Choose** to insert the image file from the desktop.
3. Select **Fit Image** to resize the image within the space or Select **Center Image** to insert the image at the center of the image allotted space.
4. [Optional] Add the URL in value under **URL Opened When Image Is Clicked**. Suppose you want to open a URL when the user clicks on the image.
5. **[Optional] Alt Text:** In case the image does not open, then Alt Text will display. Alt Text is nothing but an alternate text to an image to display.

Note: This Alt Text will display only if the image does not open or display on the screen.

Refer to the following figure for a detailed illustration of the steps above:

Figure 4.46: Image Object editor [Web page](#)

The usage of this object is to embed web page to display the target pages in the dashboard.

For example, Google as a web page.

Note: Site Admins may block the Web Page or Image object from displaying target URLs due to security reasons.

Follow these steps to embed the web page in the dashboard:

1. Drag the web page object to the dashboard workspace.
2. This will open the web page editor dialog box. Enter the target URL, as shown in the following figure:

Figure 4.47: Web page object editor

3. Enter the valid target URL in the text area.

For example, for Wikipedia URL.

4. When finished, click **OK**.

Note: You can also make the URL dynamic by embedding parameter in the URL. This is one more use-case of parameter.

Navigation

The navigation object is a button that can display a text or image to indicate the button's destination to the user. This object is used to navigate between the sheets or, dashboards or stories within a workbook. You can format its style, color, title, border, and background and provide informational tooltips whenever the user hovers over that navigation button.

Follow these steps to add a navigation button in the dashboard:

1. Drag the **Navigation** object to the dashboard work area.
2. Right-click the object and select the **Edit** button from the drop-down menu.

This will open the Navigation button editor dialog box, as shown in the following figure:

3. Perform the necessary changes in the navigation button look and feel. When finished, click **OK**:

Edit Button

Navigate to

None

Button Style

Text Button

Title

Enter optional title

0/80 characters

Font

Tableau Bold

Format

Border

None

Background

100%

Tooltip

Enter optional tooltip text

0/80 characters

Apply OK

Figure 4.48: Navigation Button object editor

Blank

This object is mainly used to adjust the spacing in-between the sheets or any dashboard items to make dashboard cluttered-free. Here are some options:

- **Add a Blank Object:**

1. Drag Layout container (Horizontal or Vertical).
2. Then, drag the **Blank object** to the dashboard:

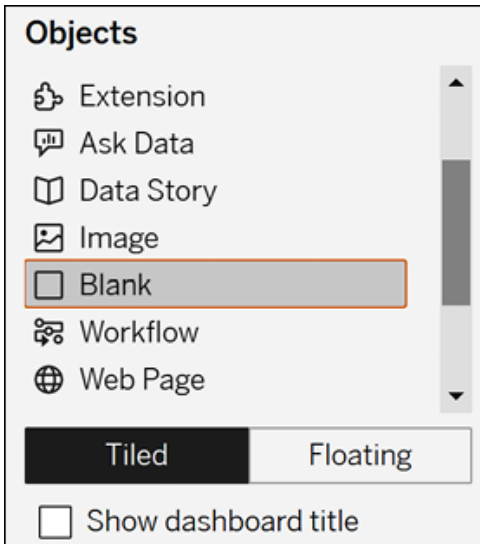


Figure 4.49: Blank object

- **Resize Blank Object:**

1. Select the **Blank Object**.
2. Under **More** options (▼), select **Edit Height**.

Here, the blank object is placed on a vertical layout container. Hence, it is showing **Edit Height** in option. Otherwise, a horizontal layout container will show the **Edit width** option. Refer to the following figure:

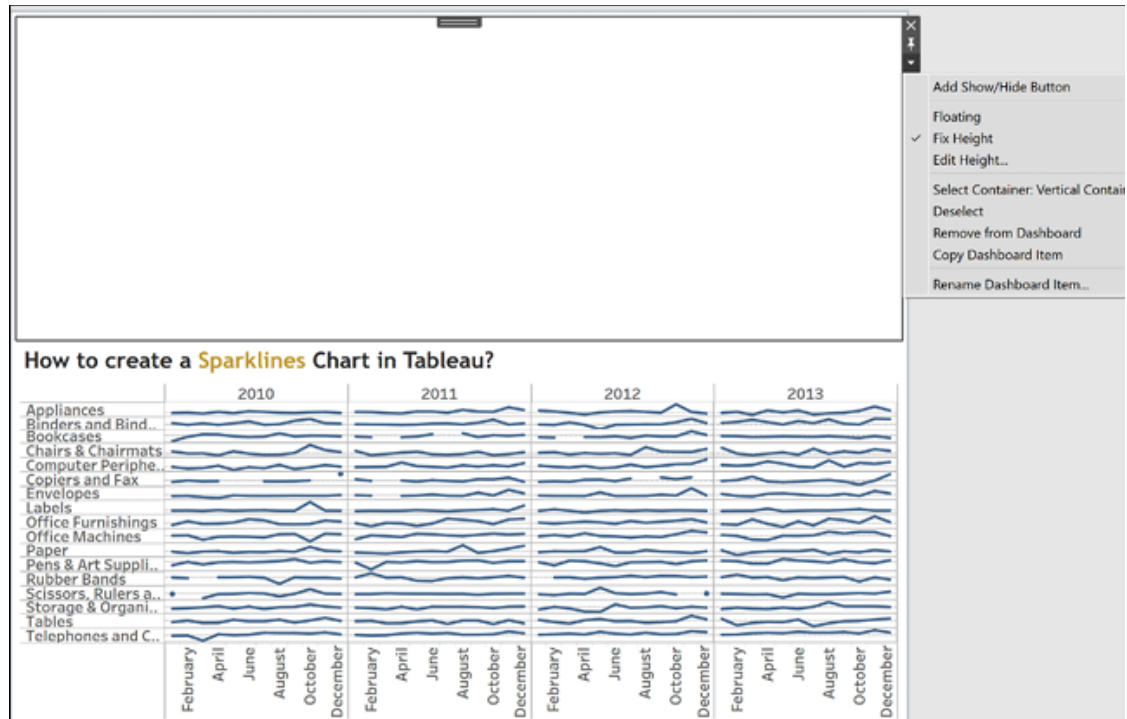


Figure 4.50: Resize blank object

- **Format Blank Object:**

1. Select the **Blank Object**.
2. Now, go to **Layout** pane.
3. Select the format options as shown in [Figure 4.51](#):
 - **Background:** To assign color to the object.
 - **Border:** To assign style or color of the line to the blank object.
 - **Outer Padding:** To add outer padding to the edges to shorten the line.
 - **Inner Padding:** To add space inside of the worksheet border.

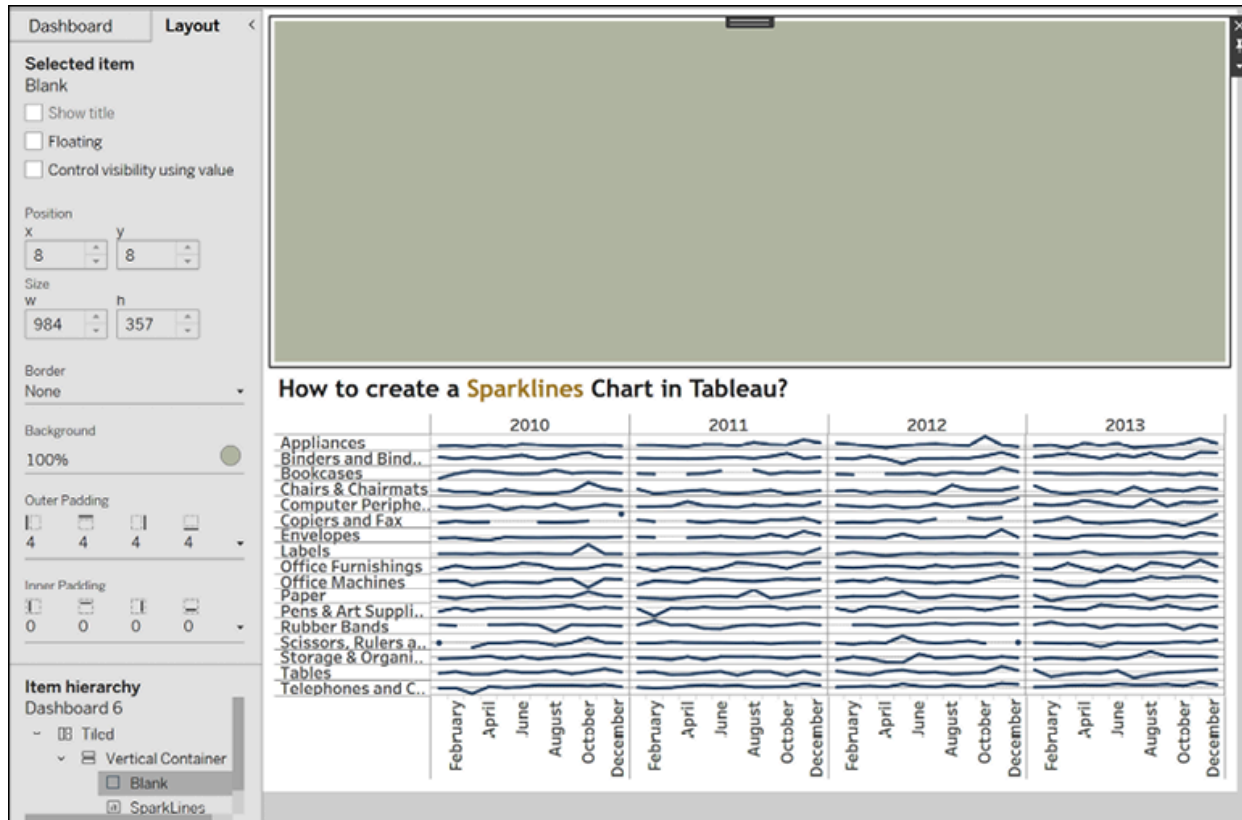


Figure 4.51: Format blank object [Download](#)

This object is also a button used to quickly download or create a **PDF** file, crosstab of the selected sheets or PowerPoint slide, or an image of the entire dashboard. The rest of the formatting options are like the Navigation Object.

For more details, [Figure 4.52](#) displays the download button object editor dialog box:

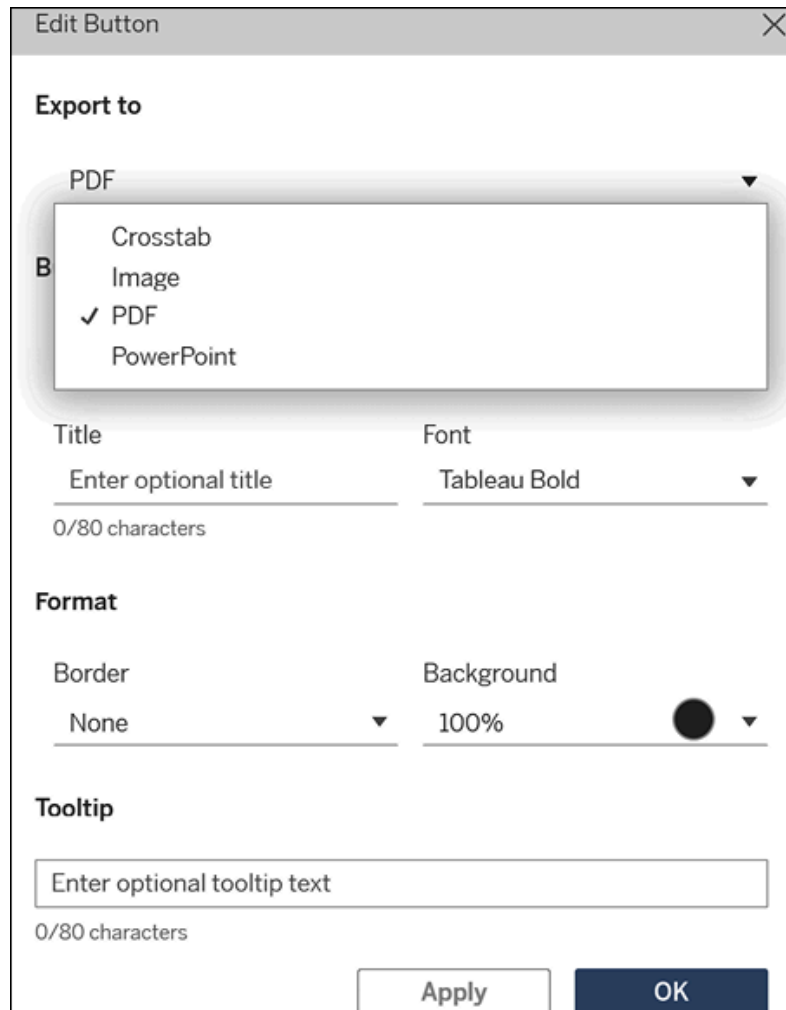


Figure 4.52: Download object editor [Extension](#)

Extension object provides the expanded dashboard functionalities and features created by third-party developers with the help of web applications. This might be disabled by the Site Administrators. Extension object allows you to add the unique features in the dashboard, which directly integrate them with the outside third-party applications.

Follow these steps to integrate third-party applications via Extension Object:

1. Drag the **Extension** object to the dashboard from the **Object**. This will open the **Add an Extension** editor dialog box to integrate third-party applications, as shown in [Figure 4.53](#).
2. To add the extension, select and search for the extension.
3. To insert the downloaded .trex file from the device (laptop), click on **Access Local Extensions**.
4. To add the **Sandboxed extensions** to run the protected web environment without any access to resources or services on the web.

The following figure illustrates the details mentioned above:

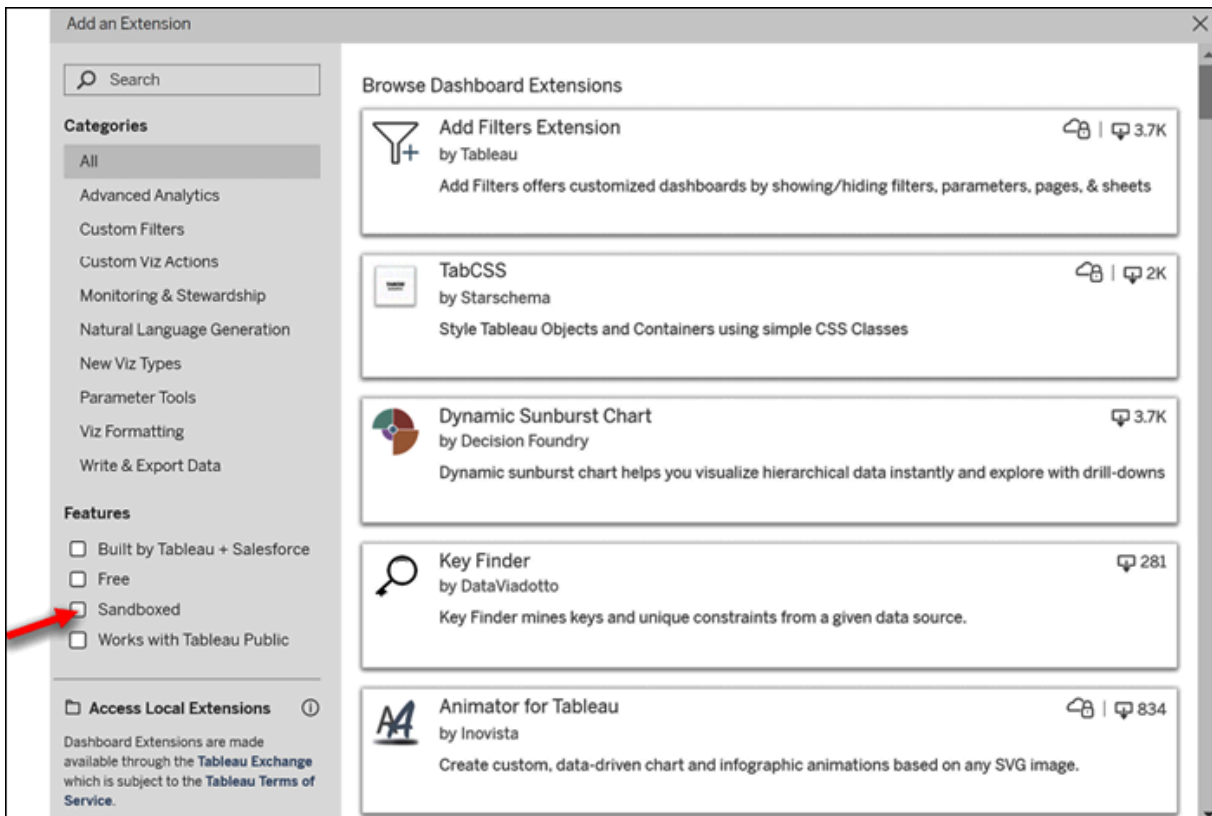


Figure 4.53: Dashboard Extension object [Ask data](#)

The Ask Data object creates a lens page on the Tableau site for specific audiences like marketing, sales, or support to create queries on the selected fields that also recommend visualizations.

Follow these steps to add the Ask Data object to the dashboard:

1. Drag the **Ask Data** object to the dashboard canvas.
2. Then, go to **Server** | **Sign in** and connect to the server.
3. Now, select the Ask object, and from **More options**, choose **Reload**. By default, the Ask Data object is enabled for the site, but site administrators can disable it to preserve the information.

Enable Ask Data object on the site:

1. Go to **General** site settings.
2. In the **Availability of Ask Data** section, choose **Enabled** to enable creation for ask data lenses for all published data sources.

Workflow

This object is the same as the **Extension Object**. Workflow object allows users to create, edit, and run flows to prepare and clean data. This object helps users to answer the questions related with data, sharing insights, and ensuring the data accuracy. However, the object requires authorization to run the extension on the web page.

Follow these steps to add a Workflow object to the dashboard:

1. Drag the **Workflow** object to the dashboard canvas.
2. Then, log in to Salesforce account to authorize the workflow.
3. To re-authorize, choose **Reload** from **More** options.

Data story

It is tedious and time-consuming when to write any executive summary of the Tableau dashboard. Tableau Data Stories will automatically generate the summary or narrate the insights and save you time. As you make any changes in visualization, the data story will automatically adjust its summary, and this helps in getting the key insights faster.

Follow these steps to add the data story to the dashboard:

1. Drag the data story object from the object section to the dashboard.

For example, you have two sheets: Scatter plot and Top N Food Chains. Then, the data story is created on an individual sheet. You can also customize the fields on which summary you want to generate, as shown in the following figure:

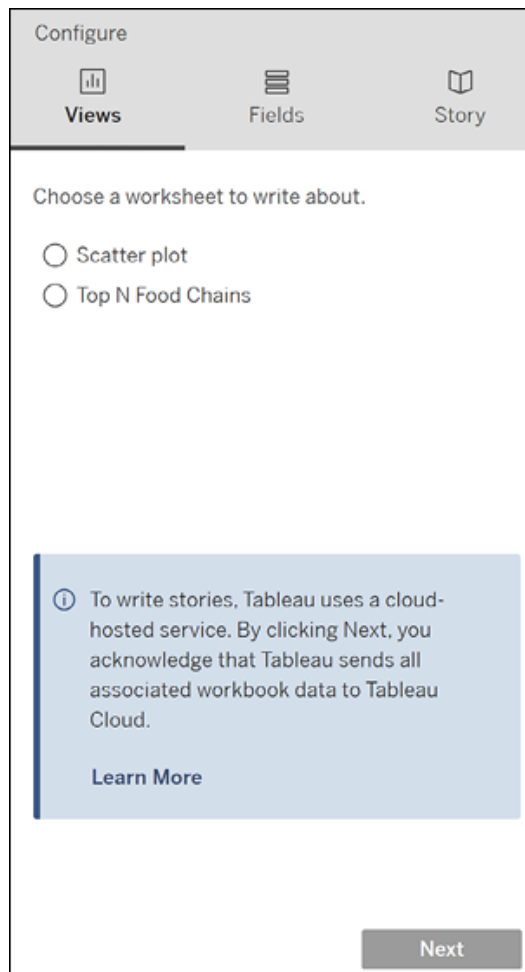


Figure 4.54: Data Story object

Actions

Actions let you create interactive relationships between the dashboard objects, data, other worksheets, and the web. Actions are set up with some response, navigation, or change in view whenever users interact with the visualizations by clicking on the menu, hovering over the visuals, or selecting the marks.

There are various types of actions in Tableau:

- Go to URL
- Filter
- Highlight
- Go to Sheet
- Change Parameter
- Change Set Values

Now, we will see how to add actions and create more interactive worksheets and dashboards.

Inserting an action in Tableau

Rule: The process of inserting an action is the same, but the usage of various action types is different.

Follow these steps to add or edit any action type: To edit an action, follow these steps:

1. Go to **Worksheet | Actions**.

Or,

Go to **Dashboard | Actions**.

This will open **Action Dialog** box, as shown in the following figure:

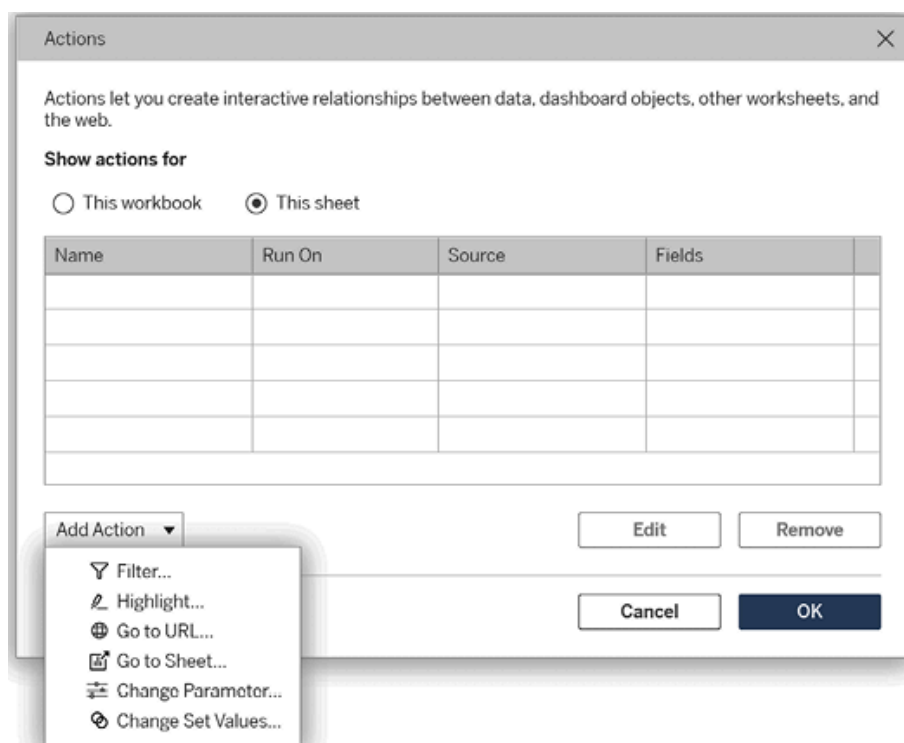


Figure 4.55: Action dialog box

2. To fetch the list of **Actions** in this workbook, choose the **This Workbook** radio button or select the **This Sheet** radio button to view the list of Actions applied on the active sheet.

Add action

To Add action, follow these steps:

1. Select the type of action to apply on the sheet from the **Add Action** drop-down list, as shown in the following figure.
2. When you add an action, you can specify the name for the action under the **Name** textbox.
3. To Edit the existing Action from the list, select the **Edit** button available on the right side of the dialog box.
4. You can also remove the action from the list by selecting the list and then clicking on the **Remove** button.
5. When finished, click on **OK**.
6. Again, click on **OK**.

Now, we will explore each of the actions with an example.

Filter

Filter actions work if the user selects a mark/hovers over Viz or clicks on the menu in the source sheet. The information will be sent to the target sheet, and the target sheet will quickly change and start showing selected marks-related information. For instance, it shows the relationships between #Locations and Sales in various food chains. Let us apply a filter action on the dashboard; follow these steps:

1. Go to **Dashboard |Actions**.
2. Select the **This Sheet** radio button to add the actions on the active dashboard.
3. Choose **Filter** from the **Add Action** drop-down menu.
4. Assign the name to the action filter as **Filter1**.
5. Now, select the source sheet from the active dashboard.

For example, the **Top N Food Chains** worksheet is selected as the source sheet, as shown in the following figure:

Source Sheets

▼

☐ Scatter plot

☒ Top N Food Chains

Run action on

☐ Hover

☒ Select

☐ Menu

☐ Single-select only

Figure 4.56: Source Sheets in filter action Choose how to run the action:

- **Hover:** Runs whenever user hover over the marks in the view of the source sheet.
- **Select:** Runs when user clicks on the marks in the view of the source sheet.
- **Menu:** Runs when user clicks the marks in the view of the source sheet, then in the tooltip, click on the action filter to see the response in the target sheet.
- Mostly preferred is select **Run Action**.
- **Single-select only:** Enabling this checkbox will restrict the user to select only *one* mark from the view. By default, multiple marks can be selected from the view.

Now, select the target sheet from the active dashboard.

For example, **Scatter plot** worksheet is selected as the target sheet as shown in the following figure:

Figure 4.57: Target sheets in filter action Choose the option; now let us understand what happens when the selected view gets cleared to the target sheet:

- **Keep filtered values:** On choosing this option, it will continue to show the filtered results in the target view.
- **Show all values:** This will bring back the target view to normal. Show all the values without any filter. Mostly preferred in real-world scenarios.
- **Exclude all values:** Changes in the filter will exclude all the values. This option is used as an exception if the dashboard shows those sheets that do not include those values.

Now, specify the target sheet data you want to show:

1. Choose either **All Fields** or **Selected Field**, as shown in the following figure:

Filter

☒ All fields ☐ Selected fields

☐	Source Field	Target Data Source	Target Field
☐	Click to add ▼		

Remove

Cancel OK

Figure 4.58: Filter option in Filter action

2. If you choose the **Selected Field** radio button, then specify or choose the fields from the source field drop-down and then select the target data source and field as well.
3. When finished, click **OK**.

Go to URL

Go to URL action is used to create a hyperlink that points to a webpage or any external webpage tableau resource. For instance, to create a dynamic URL based on Chain Field, follow these steps:

1. Go to **Dashboard |Actions**
2. Select **This Sheet** radio button to add the actions on the active dashboard.
3. Choose **Go to URL** from the Add Action drop-down menu.
4. Assign the name to the action filter as **HyperLink1**.
5. Now, select the source sheet from the active dashboard.

For example, the **Top N Food Chains** worksheet is selected as a source sheet.

6. Choose how to run the action: **Hover**, **Select**, or **Menu**, as shown in the following figure:



Source Sheets

Dashboard 1 ▼

☐ Scatter plot

☒ Top N Food Chains

Run action on

☐ Hover

☒ Select

☐ Menu

Figure 4.59: Source Sheet in go to URL action

7. Now, specify where the link to open for URL target
8. **New Tab if No Web Page Object Exists:** This states that the webpage will open in the browser on the sheet if it lacks a web page object.
9. **New Browser Tab:** In all scenarios, the URL will open the webpage in the web browser. Refer to the following figure:



URL Target

☒ New Tab if No Web Page Object Exists

☐ New Browser Tab

Figure 4.60: URL Target Go to URL Action

10. Enter the URL. To create a dynamic URL then, click on Insert and select the field or parameter on which a dynamic URL can be created.
11. For example, static URL: www.wikipedia.com. However, to generate dynamic URLs, click on Insert, then select **Chain**. So, the URL will automatically change to **www.wikipedia.com/<Chain>** <Chain> signifies the field is inserted in the URL. Hence, when the user selects any Chain from the source field, the respective Wikipedia URL gets opened for the same food chain.



URL

www.wikipedia.com/<Chain> Insert ▼

www.wikipedia.com/Arby's

Figure 4.61: URL in go to URL action

12. [Optional] Select any one option:
 - a. **Encode data values that URLs do not support:** If the data values consist of special characters, then enabling this checkbox will translate those special

characters so that the browser can understand and fulfill your request.

- b. **Allow multiple values via URL parameters:** If there are multiple values, then link the web page of multiple values via URL parameters, define the delimiter value, and escape the delimiter character to decode the values.

Refer to the following figure for an illustration of the above:



Figure 4.62: Data Values in go to URL action

13. When finished, Click on **OK**.

Highlight

Highlight actions are used to call the attention of the users by coloring the selected specific marks or dimming all other marks. For example, suppose you want to highlight the records in the target sheet related to selected food chains from the source table. The specific marks get highlighted by colors, shape or size of those marks and dimming all the other marks.

Follow these steps to apply highlight actions:

1. Go to **Dashboard | Actions**.
2. Select **This Sheet** radio button to add the actions on the active dashboard.
3. Choose **Highlight** from **Add Action** drop-down menu.
4. Assign name to the action filter as **Hightlight1**.
5. Now, select the source sheet from the active dashboard.

For example, the Top N Food Chains worksheet is selected as a source sheet.

6. Choose how to run the action: **Hover**, **Select**, or **Menu**.
7. Select the **target sheet** from the active dashboard. For example, the **Scatter** plot worksheet is selected as the target sheet.
8. Now, select the target highlighting fields as shown in the figure.

9. **All Fields:** Selecting this radio button, all the target sheet fields will be highlighted.
10. **Dates and Times:** The target sheet field marks will be highlighted if the date and time match those of the selected marks in the source field.
11. **Selected Fields:** Only selected target sheet fields will be highlighted.
12. When finished, click on **OK** twice:

Name
Highlight1 Insert ▼

Source Sheets
Dashboard 1 ▼
☐ Scatter plot
☒ Top N Food Chains

Run action on
☐ Hover
☐ Select
☒ Menu

Target Sheets
Dashboard 1 ▼
☒ Scatter plot
☐ Top N Food Chains

Target Highlighting
☒ All Fields
☐ Dates and Times
☐ Selected Fields
☐ Chain

Cancel OK

Figure 4.63: Highlight action editor dialog box

Note: The Highlight Actions options are available in the licensed version of Tableau Online.

Go to Sheet

Go to Sheet actions are used to navigate between the worksheets, dashboards, or stories. Let us understand how to apply this with an example. Suppose you want to create a navigation between two sheets from Sample Superstore data source, i.e., category-wise

sales bar chart and category-wise order sales line chart. So, let us apply the **Go to Sheet** action to achieve the same.

Follow these steps:

1. First, create two worksheets:
2. Create a **Category Sales Bar Chart** as shown in the following figure:

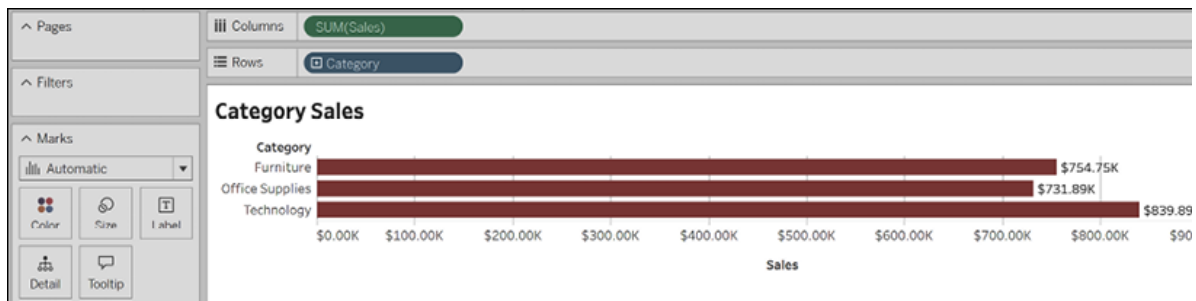


Figure 4.64: Sample Sheet1 for go to sheet action

3. Now, create another **Category Order Sales Chart** as shown in the following figure:

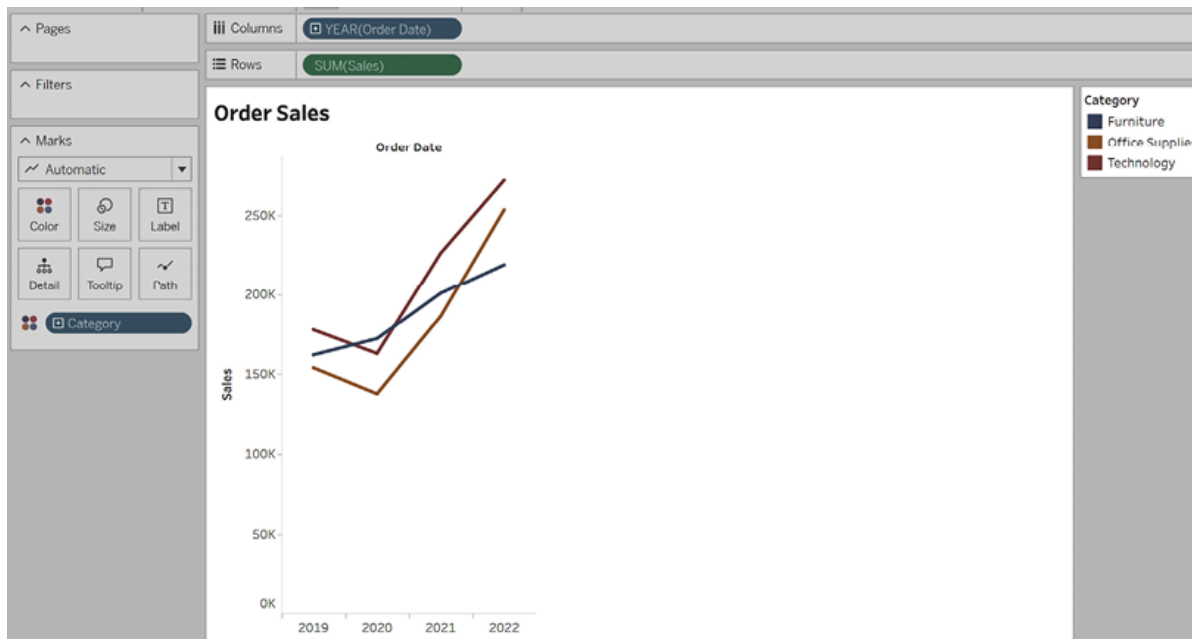


Figure 4.65: Sample Sheet2 for go to sheet action

4. To add Go to Sheet action in Worksheet, Go to **Worksheet | Action Or**,
To add Go to Sheet action in Dashboard, Go to **Dashboard | Action**
5. Select **Go to Sheet** action from the **Add Action** drop-down menu from the dialog box.

6. Now, assign the name to **Action as GoToSheet1**, then select the source sheet from where the user can navigate to another sheet from the drop-down menu.
7. Here, selects Sheet 1 as category-wise **Sales Bar Chart**.
8. Now, specify how to run the action: **Hover**, **Select**, or **Menu**. This works like other actions, as discussed earlier.
9. Next, select the target sheet from the drop-down menu, as shown in the following figure:

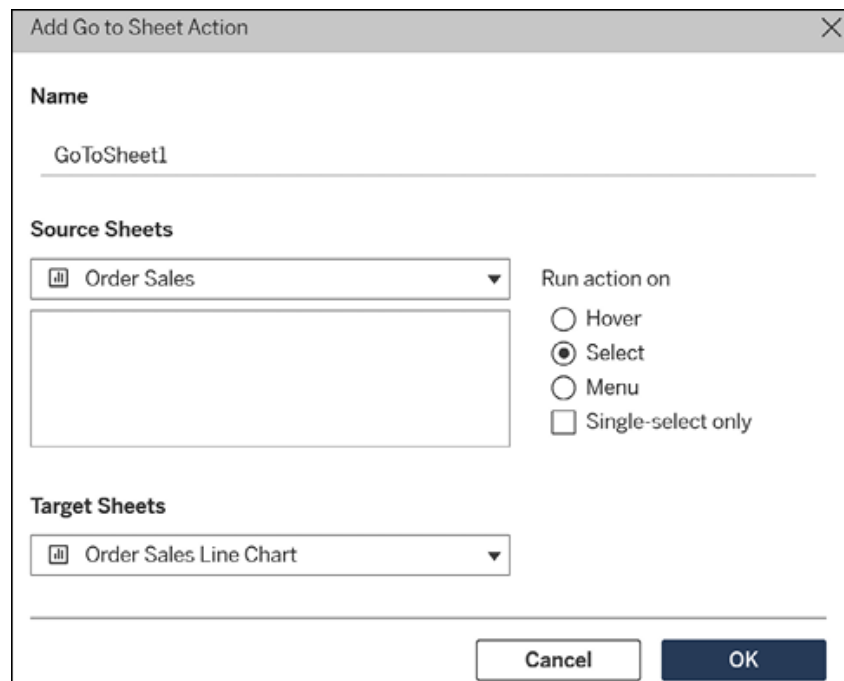


Figure 4.66: Go to Sheet Action dialog box

10. Here, selects Sheet 2 as the category order sales line chart. When finished, click **Ok** twice.

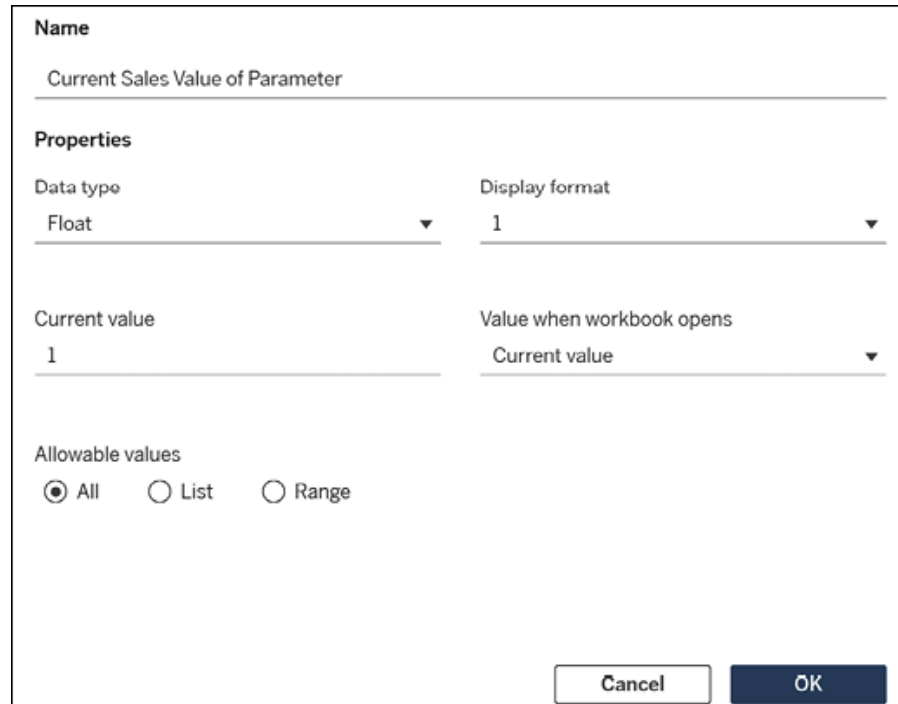
On click any marks in the source sheet, i.e., Sheet1 Viz, then this active sheet (source sheet) will change to the target sheet, i.e., Sheet2 Viz.

Change parameter

Parameter actions is an alternative way for users to directly interact with the marks on a Viz or dashboard to control the aspects of their analysis. This can be achieved on parameters. Let us understand the change parameter action with an example. Suppose you want to change the value of the parameter showing sales for the specific order date year from a line chart on ship mode from the Sample Superstore data source.

Follow these instructions to build such a dashboard from scratch:

1. To create a parameter, click on the right-most drop-down arrow next to the search option under the data pane and select **Create Parameter**.
2. Now, assign the name to the parameter as **Current Sales Value of Parameter** in the dialog box, select data type as Float, set value as one, and select **All** radio button for allowable values.
3. When finished, click **OK**, as shown in the following figure:



The image shows a 'Parameter creation dialog box' with the following fields and options:

- Name:** A text field containing 'Current Sales Value of Parameter'.
- Properties:**
 - Data type:** A dropdown menu set to 'Float'.
 - Display format:** A dropdown menu set to '1'.
 - Current value:** A text field containing '1'.
 - Value when workbook opens:** A dropdown menu set to 'Current value'.
- Allowable values:** Three radio buttons: 'All' (selected), 'List', and 'Range'.
- Buttons:** 'Cancel' and 'OK' buttons at the bottom right.

Figure 4.67: Parameter creation dialog box

4. Drag this newly created parameter to **Label Marks Card** in a sheet. Now, select a parameter and choose the **Format Number** option from the drop-down menu. Change the format to currency in dollars (\$) with two decimal places. Refer to the following figure:

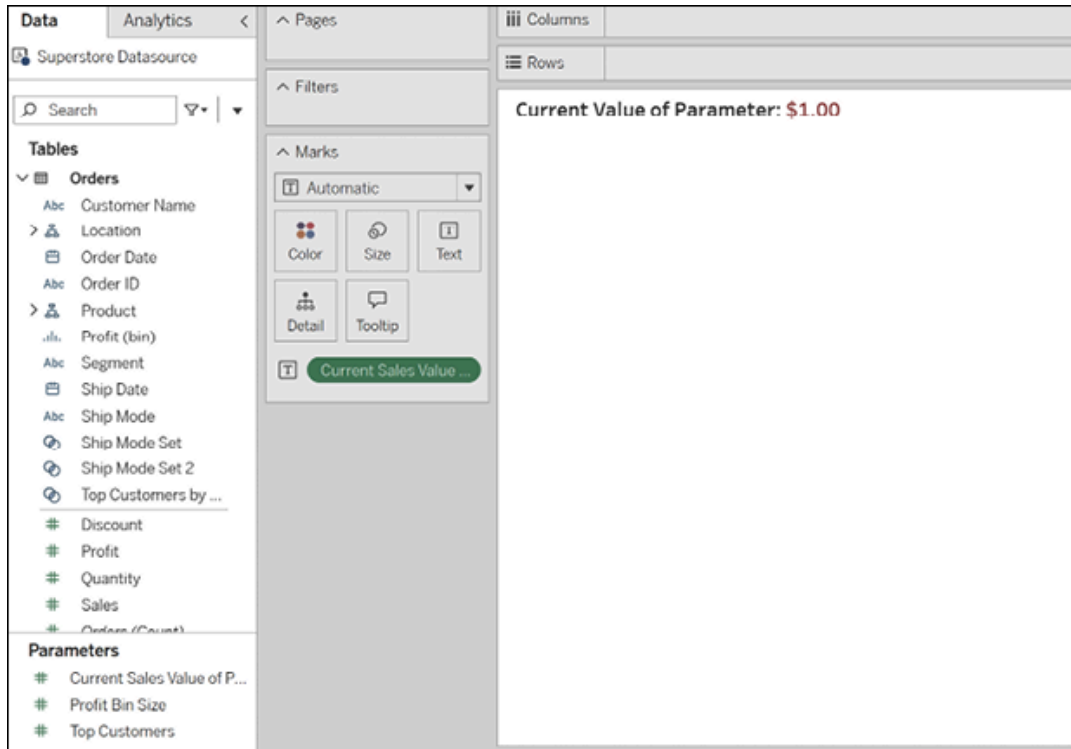


Figure 4.68: Sheet1 in change parameter action example

5. Insert a new sheet and now create a line chart of **Ship Mode** based on **YEAR (Order Date)** and Sales (By default, aggregated to SUM) as shown in the following figure:

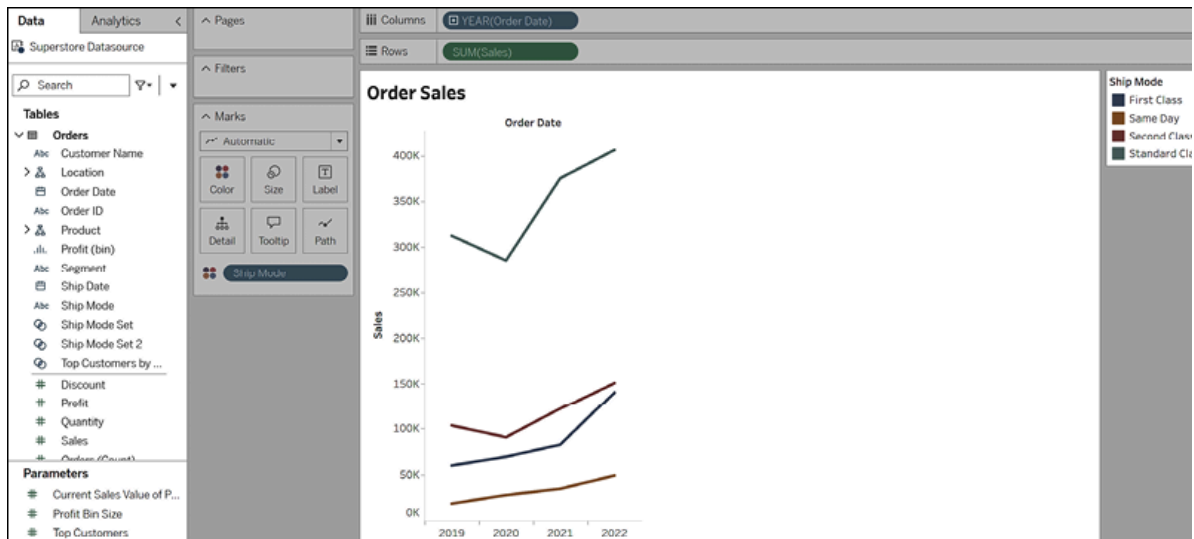


Figure 4.69: Sheet2 in change Parameter action example

6. Now, create a new dashboard, drag both the sheets to the workspace, and insert Change parameter actions to the dashboard.

7. To insert **Change Parameter Action** to **Dashboard**, then Go to **Dashboard Actions**. And select the **Change Parameter** option from the **Add Actions** drop-down menu in the **Action Creation** dialog box. Now, follow the steps as shown in the following figure:

The dialog box is titled "Change Parameter". It contains the following fields and options:

- Name:** A text field with "Parameter1" and an "Insert" button.
- Source Sheets:** A dropdown menu showing "Dashboard 1". Below it, a list of checkboxes: "Current Value of Parameter" (unchecked) and "Order Sales Line Chart" (checked).
- Run action on:** Radio buttons for "Hover", "Select" (selected), and "Menu".
- Target Parameter:** A dropdown menu showing "# Current Sales Value of Parameter".
- Clearing the selection will:** Radio buttons for "Keep current value" and "Set value to" (selected).
- Source Field:** A dropdown menu showing "# SUM(Sales...".
- Aggregation:** A dropdown menu showing "Sum".
- Value:** A numeric input field with the value "1".
- Buttons:** "Cancel" and "OK" at the bottom right.

Figure 4.70: Change Parameter action dialog box

8. Assign a name to the parameter action as **Parameter1**. Select the source sheet from where the user is going to interact on the Viz.
- For example, select the sheet **line chart** based on **Year Order Date** and **Product Sales**.
9. Now, specify how to run the action, such as hover, select, or menu.
10. Now, select the target parameter from the workbook on which the change in parameter values can be applied on.
11. Here, select the **Current Sales Values of the Parameter** from the target parameter list.
12. Select the value to display in the actions from the source field Viz interaction.
13. Here, selects **SUM(Sales)** from the source field drop-down and aggregation to SUM.

14. Now, specify whether you want to retain the selected change parameter value or set any value on selection clearing.
15. Here, select set the value to back to \$1 again when the selection gets cleared.
16. When finished, click **OK**.
17. Now, see the difference when you select any value in the marks in the source field, the change in the parameter value. Refer to the following figure for an illustration:

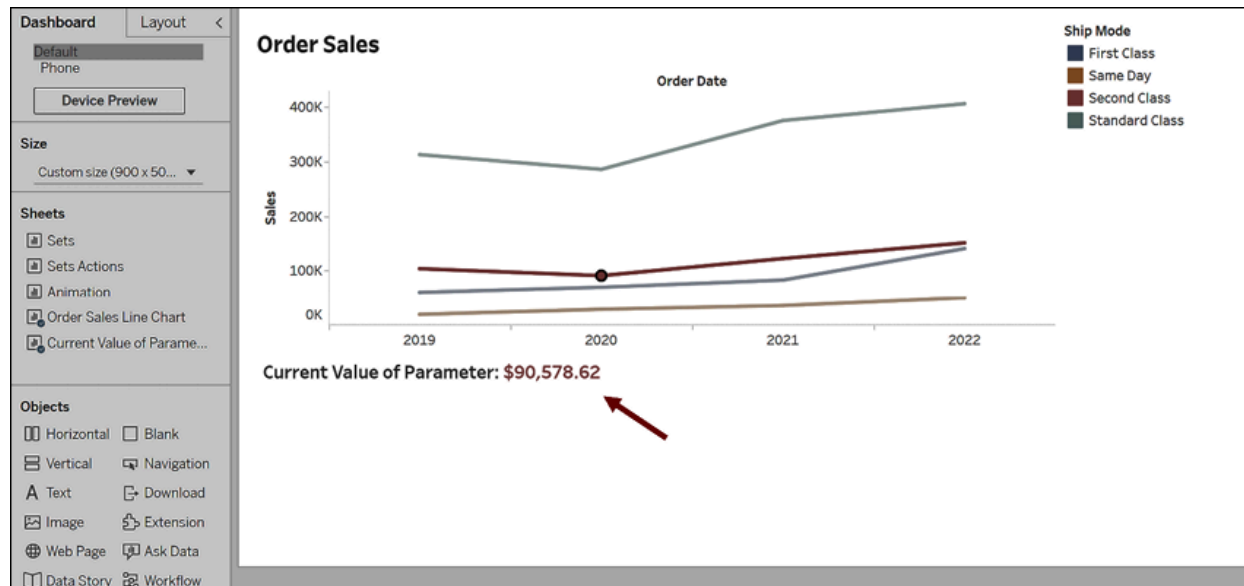


Figure 4.71: Change Parameter action example [Change set values](#)

Set actions gives flexibility to the users to directly interact with the Viz or dashboard to control the aspects of their analysis. In other words, if the user selects any mark point in the Viz, the set actions will change the values in the set. Let us illustrate the set actions with an example.

Pre-requisite: The set actions applied on the existing set in the workbook and the update the values based on the set in the Viz. For instance, suppose you want to change the values in the **Ship Mode** field using **Ship Mode Set** from the Sample Superstore data source and if user selects mark points in the Viz of scatterplot of Sales and Profit.

Follow these instructions to apply set actions:

1. Create a **Ship Mode set** including all members of **Ship Mode** field.
2. Now, create a scatterplot on Profit vs. Sales as shown in [Figure 4.59](#).
3. Drag **Ship Mode set** in the **Color Marks** Card and **Ship Mode** field to the **Label Marks** Card.
4. Now, drag a calculated field **Profit Ratio** in the label **Marks Card**.

5. And click **Ship Mode Set** in the **Marks Card**, choose **Show Set** option to add **Set members**.
6. To add Set Actions to the viz, follow these steps.
7. Create a set action.
8. Go to **Worksheet Actions**, in the worksheet Or,
Go to **Dashboard | Actions**, in the dashboard
9. Click on **Add Actions**, then select **Change Set Values** in the Actions dialog box, as shown in the following figure:

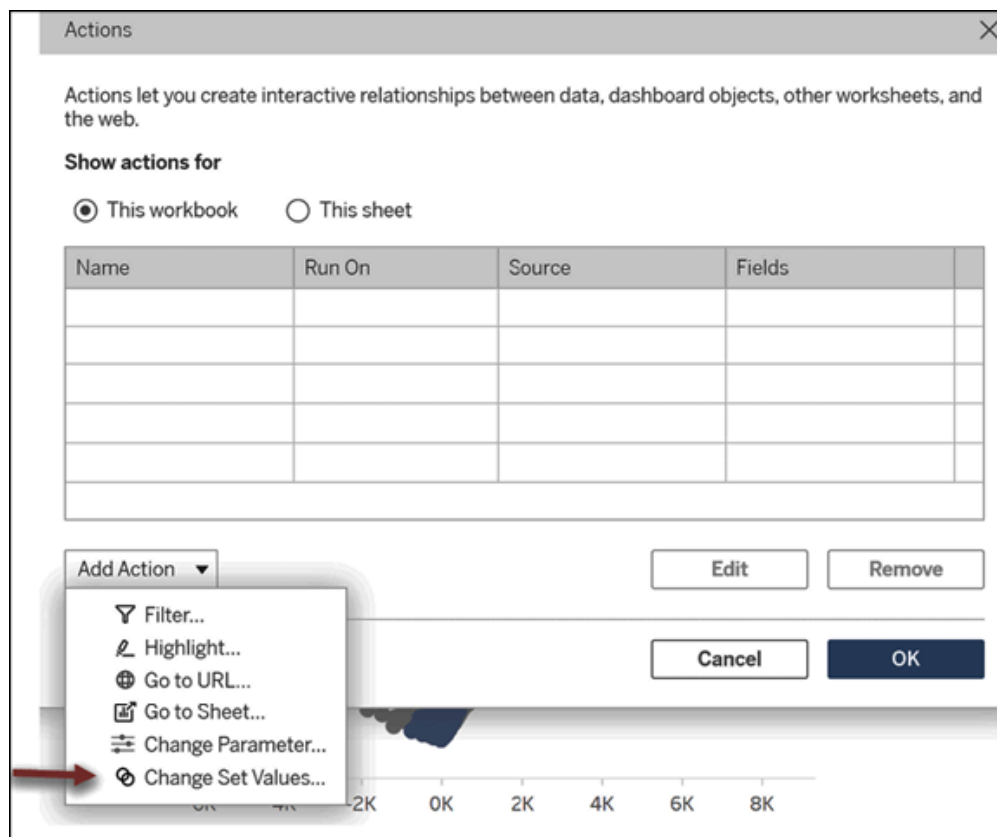


Figure 4.72: Create set action

10. Assign name to **Set Action** and select the source sheet. The current sheet is selected by default.
11. Choose how to run the action: Hover, Select or Menu.
12. Select the target set from the active dashboard or worksheet.
13. Now, when action is applied then what happens to the target values:
 - **Assign values to set:** Only selected values in the set.

- **Add values to set:** Include the selected values in the set.
- **Remove values to set:** Exclude the selected values from the set.

14. Next, specify when the selection gets cleared what can happen to the target values in the view.

- **Keep set values:** The selected value will remain in the set.
- **Add all values to set:** Add all the values to the set.
- **Remove all values to set:** Remove all the selected values from the set.

Refer to the following figure to add set action:

Figure 4.73: Set Action dialog box

15. When finished, click **OK** twice. Refer to the following figure for a better understanding:

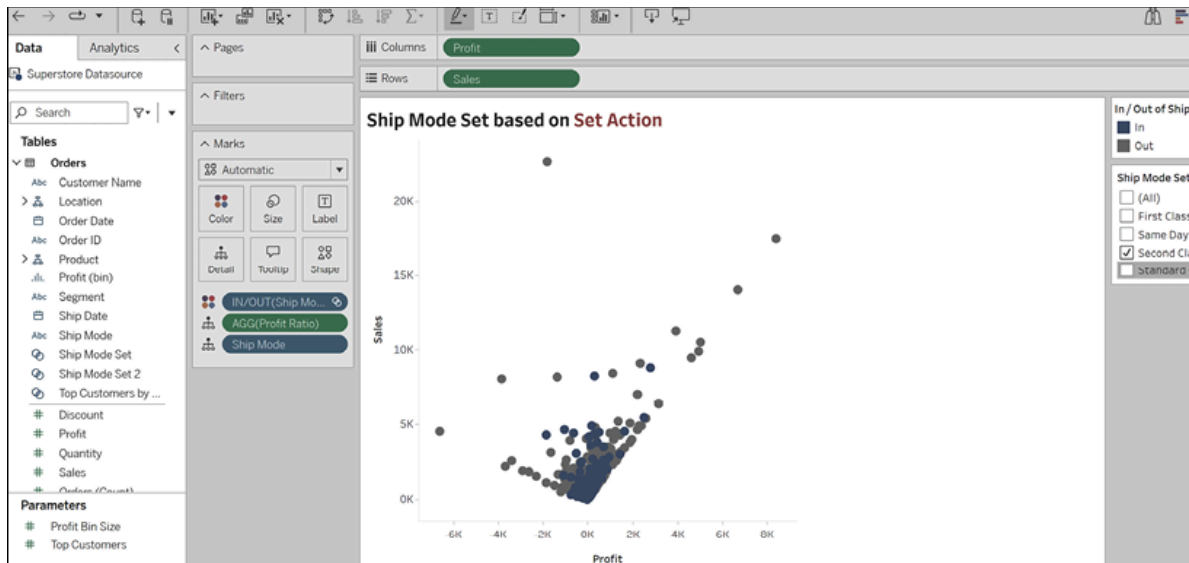


Figure 4.74: Set Actions

Tableau accelerators

In [Chapter 2, Connect to Data to the Web](#), you have already learned about Tableau Accelerators, about how to use pre-built-in dashboards designed to quickly jumpstart the data analysis in minimal steps once you connect your data.

Note: There are only limited number of accelerators available to use when you create the workbook in Tableau Online.

Animation in Tableau Online

Animation is a way to improve communication and drive change. Tableau viz with animation helps to explore, understand, show changes, and present the data, as well as help to make better informed business decisions. The animations in Tableau will highlight the changing patterns in the data, or outliers or spikes, etc., or even help to see the clusters or set members.

Note: Tableau Online supports only few features in animation than Tableau Desktop.

Let us create an animated Viz. For instance, you want to create an animated Viz based on the year of order date to see the variation in profit and sales for the top customers by profit set of members and rest. Then, follow these steps:

1. Select the **Orders** sheet from **Sample SuperStore** data source.
2. Now, create a scatterplot between **Profit** and **Sales**.
3. i.e., Drag the measured fields **Profit** to **Columns** and **Sales** to **Row** shelf. Go to Analysis and uncheck the **Aggregated Measures**.

4. Drag **Top Customers by Profit** set to **Color Marks** card. Make relevant changes in color difference for the set's in/out measures.
5. Now, drag **Order Date** to **Pages** Box. Select the **YEAR** of Order Date.

Refer to the following figure for an explanation of the above:

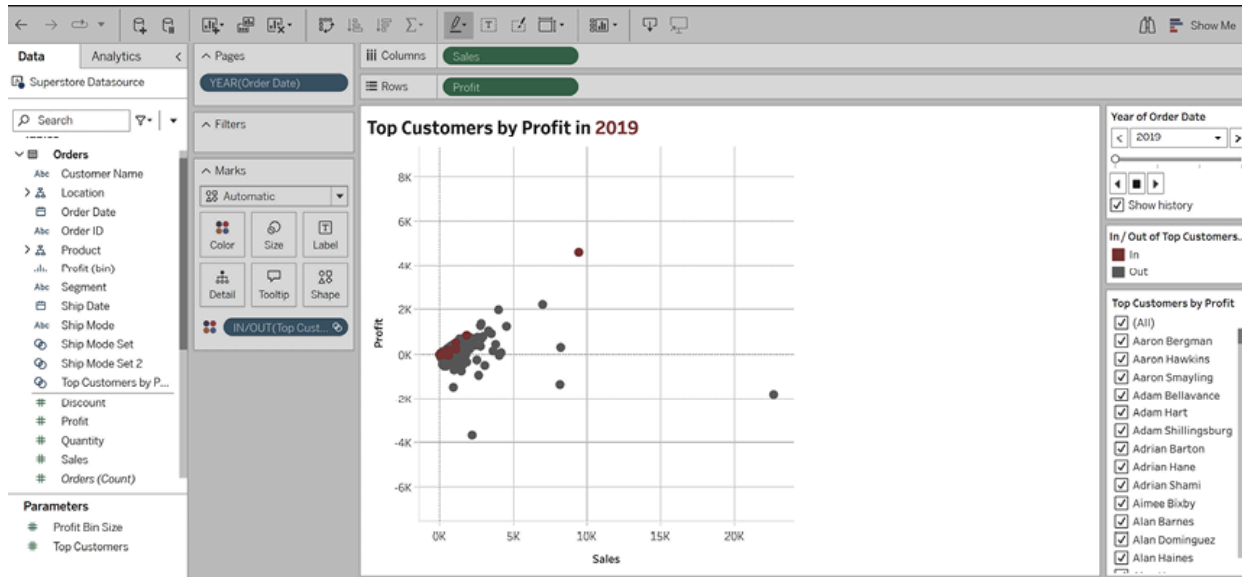


Figure 4.75: Animation viz in Tableau Online Now, click on the run icon to see the animated Viz, as shown in the following figure.

Or make necessary changes to animation settings.

Note: Tableau Online provides the basic animation settings (includes play reverse, stop, play forward, and show history).

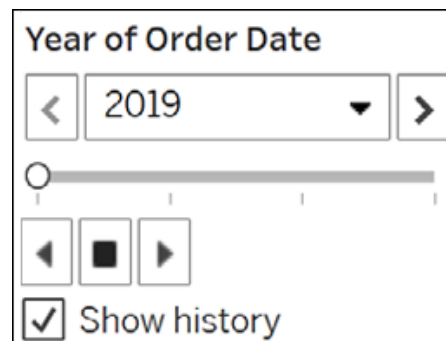


Figure 4.76: Animation Viz in Tableau Online [Analytics in Tableau Online](#)

The analytics pane in the Tableau Online worksheet workspace allows you to apply the advanced analysis to a view. In comparison to Tableau Desktop, Tableau Online provides limited analytics objects to apply to a view, as shown in [Figure 4.77](#):

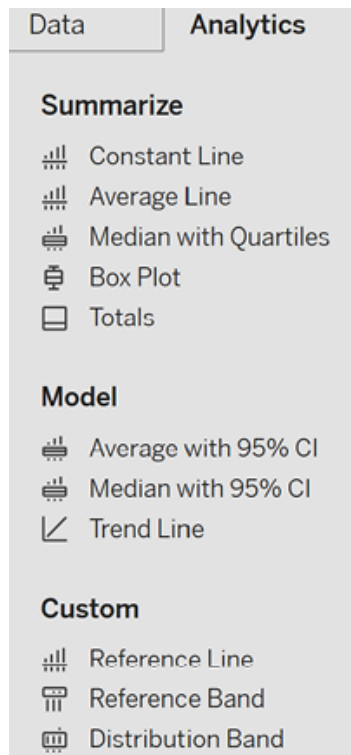


Figure 4.77: Analytics pane in Tableau Online To add an analytics object to a view from an analytics pane, simply drag it to the view.

Note: If analytics objects cannot be applied to the fields in the view, then it will not be available.

To delete the analytics object from the view, follow these steps:

1. Click on the undo icon.

Or,

2. Drag the object from the view.

Or,

3. Select the object in the view, then click on **Remove** from the tooltip.

As you can see, the analytics objects are split into three different parts:

- **Summarize:** These objects give a glimpse of descriptive statistics.
- **Model:** These objects are used to know the trend and pattern of the data.
- **Custom:** You are allowed to create custom analytics on your data where dynamic and user-interactive analysis on data can be performed.

Let us understand each of them in detail.

Summarize

The Summarize section consists of five different objects:

- **Constant Line:** Tableau displays a constant value when a constant line object drags it to the view. This will add a reference line for the measured field to the table. The constant value can be set when you specify the value of the constant. For example, when you drag the object to the order category bar chart. Then, view the display like the following figure:

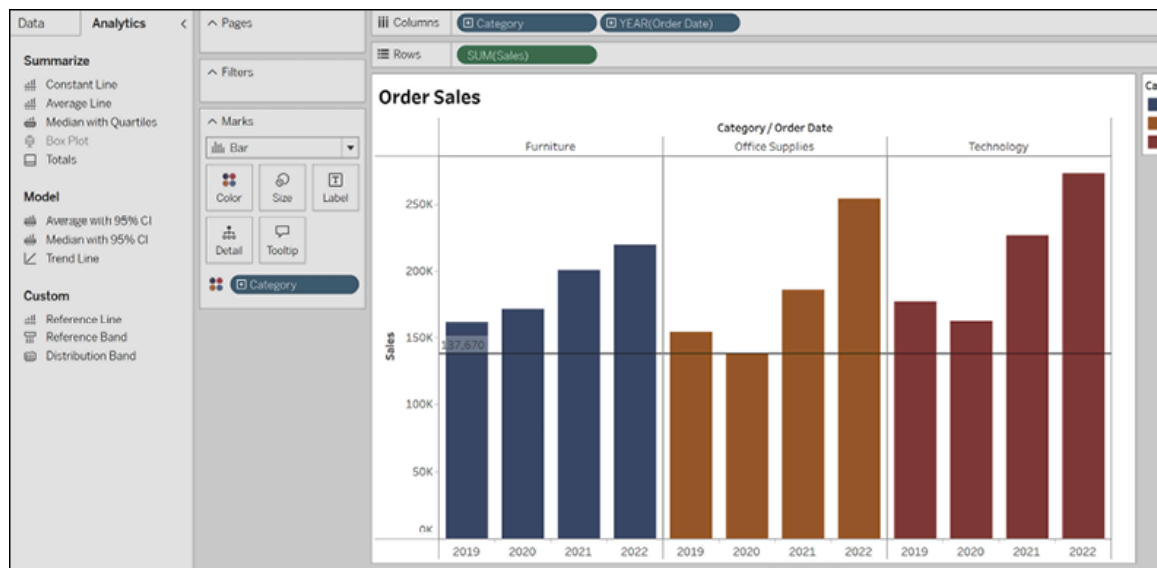


Figure 4.78: Constant Line in Analytics pane

- **Average Line:** The **Average Line** object adds an average reference line at the table level, pane level, or cell level, which defines the scope of the item. These reference lines can be added on the measured field. For example, when you drag the object to the order category bar chart on selecting to add the average **Reference Line** at pane level. Then, view the display like the following figure:

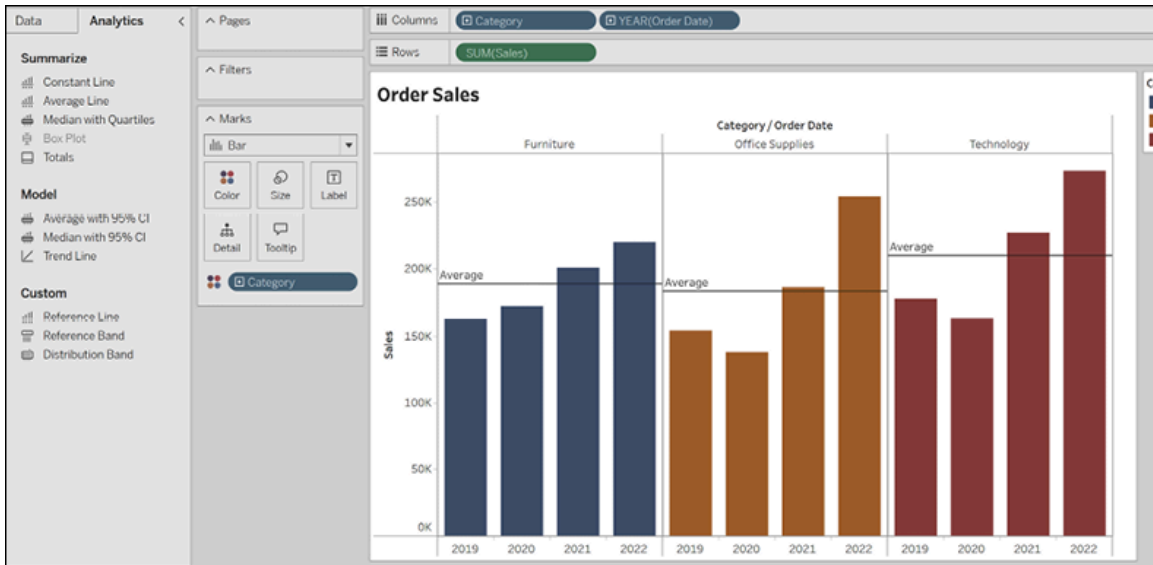


Figure 4.79: Average Line in Analytics pane

- Median with Quartiles:** This object adds median lines and distribution bands to the view for the specific measures. In the case of an average line object, you can add average lines to the view, which was influenced based on outliers in the data. However, the median with quartiles provides the summary of the data and distribution of the data. The distribution bands are computed on quartiles where the middle two quartiles are shaded, as shown in the following figure:

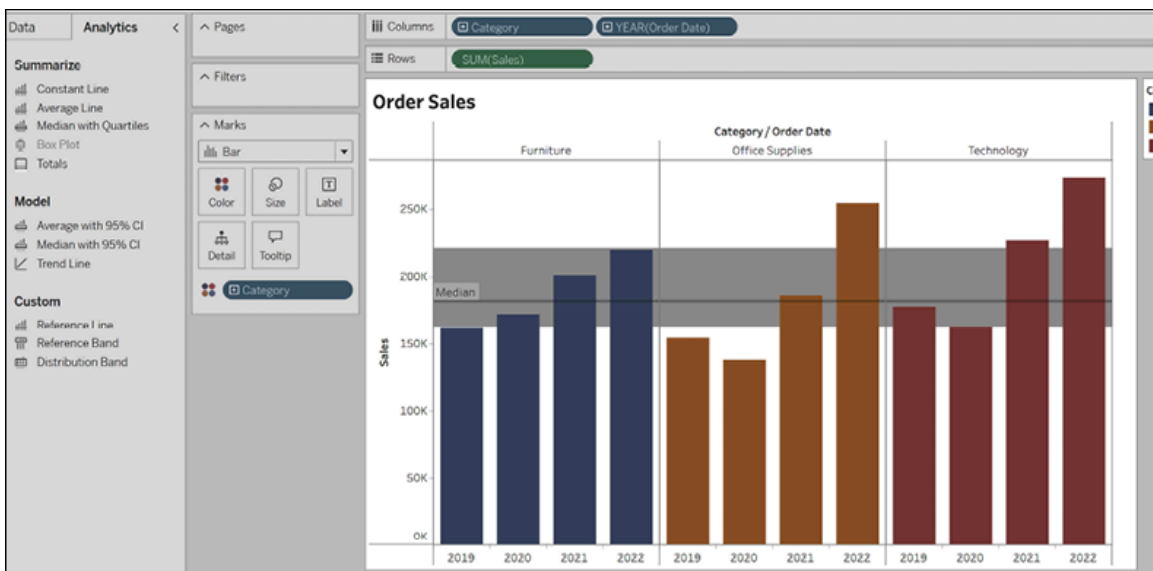


Figure 4.80: Median with quartiles in Analytics pane

- Box Plot:** Box Plot is widely used to provide statistical information for a specific measure about the median, quartiles, and whiskers. This also shows the distribution of data along an axis. Similar to the above-learned summarizing

objects, the scope of the box plot is at cell, pane, and table levels. For instance, in a line chart based on category-wise yearly order sales, you will find the following figure by dragging the box plot object to the view. Hence, the boxplot for each category presents five sample statistics, the minimum, the lower quartile, the median, the upper quartile, and the maximum in the view.

Refer to the following figure:

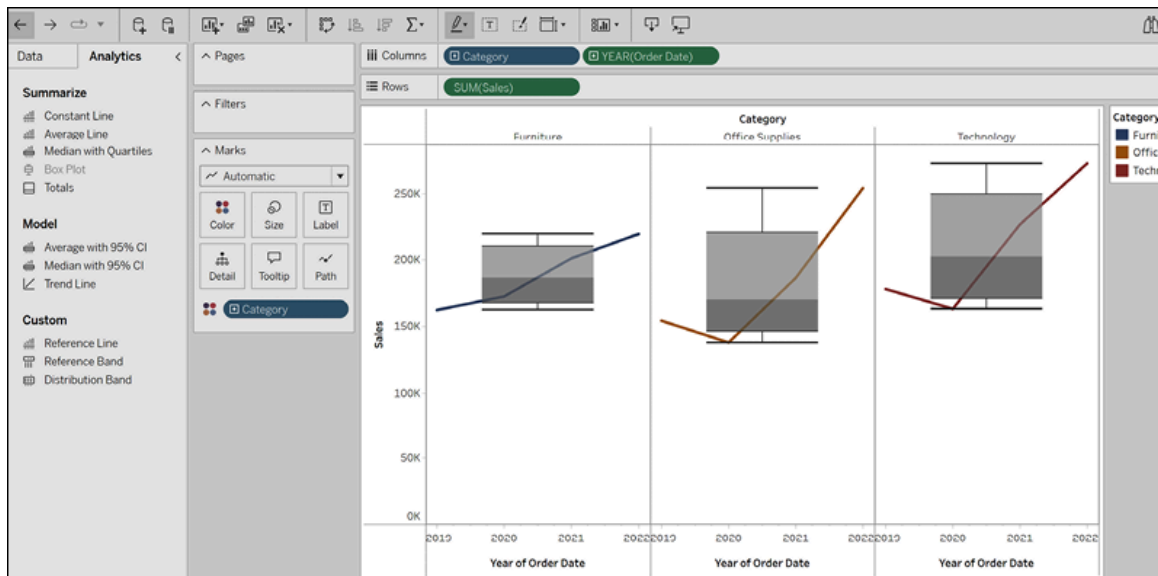


Figure 4.81: Box plot in Analytics pane

- **Totals:** Totals object is used to add the grand subtotal or total of measured columns or rows. To add totals to the view, on dragging the object, drop options are subtotals, column grand totals, and row grand totals.

Model

The model section consists of three different objects:

- **Average with 95% CI:** Add Average with 95% CI to the view; this will add an average line with the distribution band, which is configured as 95% confidence interval. The scope of this object is at the cell, pane, and table levels. Hint: The average line might be influenced by the outliers in the data.
- **Median with 95% CI:** Instead of the average line with distribution bands in the view, the Median Line will show along with the 95% confidence interval distribution band. The scope of this object is at the cell, pane, and table levels.
- **Trend Line:** A Trend Line is used to estimate or predict the trend of the variable to understand the pattern. There are multiple options in trend line model types: linear, exponential, power, logarithmic, and polynomial. To add the trend line to

the view, drag the object and select the dropping option from the available model types in the trend line. Suppose on the scatter plot of profit vs. quantity on Sub-Category, adding the trend line to the view as shown in [Figure 4.82](#). You can select the best-fit trendline based on the highest.

R-Square value from the available options as shown in the following figure:

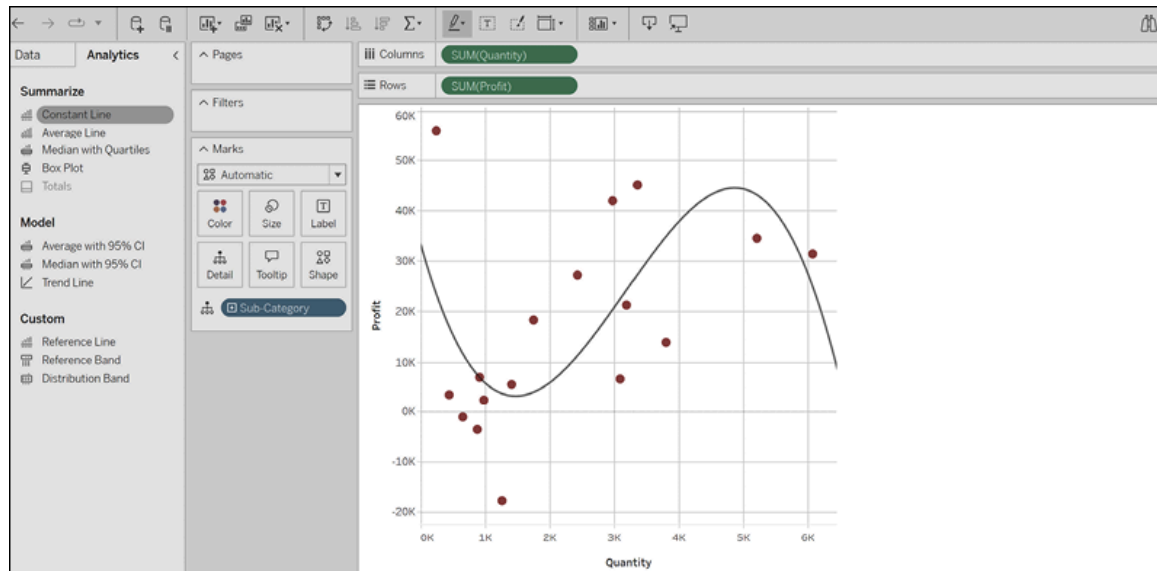


Figure 4.82: Trend Line in Analytics pane

Custom

The Custom section comprises of three different objects:

- **Reference line:**

In the summarize section, you learned to add constant average lines and median lines to the view. Here, you can add reference lines on count, average, median, minimum, maximum, summation etc., on the measured fields. So, the custom section gives more flexibility to add the reference lines at various computation levels. You can also add labels, lines, and even tooltips. You can add a constant line or line with confidence intervals or only confidence intervals. The scope of levels are cell, pane, and table.

- **Reference band:**

Similarly, you can add custom reference bands where bands are computed at various levels: total, average, minimum, maximum, constant, sum, and median. Values can be considered direct measure field values or numeric parameters. You can also add tooltips and labels here.

- **Distribution band:**

The distribution bands are the confidence interval bands that shade the region, which theoretically sampling distribution is based on the given value with the standard error.

The value can be computed at percentage level, percentiles, quartiles, or standard deviation. The scope of the value can be applied at cell, pane, or table level. You can also add tooltips and labels here.

Conclusion

In this chapter, you have learned how to create user-interactive and dynamic visuals, build advanced charts, create parameters, and add actions to the dashboard. Additionally, we have explored the dashboard objects and calculated fields with LODs, animations, and analytics.

In the next chapter, you will learn about content management interaction tools on Tableau Online, including alters, notifications, subscriptions, etc.

Multiple choice questions

1. What are the components of a dashboard?

- a. Image
- b. Horizontal
- c. Data story
- d. All the above

2. An icon associated with the field that has been grouped.

- a. A set
- b. A group
- c. A Paper clip
- d. Hash

3. Which of the following is NOT a trend line model?

- a. Linear trend line

- b. Binomial trend line
- c. Exponential trend line
- d. Exponential trend line

4. A Tableau chart which displays measures over time is called?

- a. Bar
- b. Line
- c. Scatterplot
- d. Histogram

5. A visual transition between filter, zoom settings, different pages, sort, and changes to filter, parameters, and set actions is called?

- a. Filter
- b. Pages
- c. Parameters
- d. Animations

6. What allows you to compare measures that have different scales?

- a. Individual axes
- b. Combined axes
- c. Composite axes
- d. Dual axes

7. What is used to run complex queries involving many dimensions at the source level instead of bringing all the data to the Tableau interface?

- a. Calculated field
- b. Level of detail
- c. Parameters

d. All the above

8. Mention the interactive elements to add a dashboard for users.

a. URL actions

b. Edit tooltip options

c. Filter actions

d. All the above

9. Which chart types always include bars sorted in descending order?

a. Line chart

b. Bar chart

c. Pareto chart

d. Gantt chart

10. What should be used to create a calculated field that returns data independent of the data granularity in a view?

a. A table calculation

b. INCLUDE LOD expression

c. FIXED LOD expression

d. EXCLUDE LOD expression

Questions

1. What is action and its types?
2. How do you change the values of parameters in a view?
3. How do you add animation to the view with an example?
4. What is a dual axis? Create a Pareto chart for your sample data.

Answers

1.	d
2.	c
3.	b
4.	b
5.	d
6.	d
7.	b
8.	a,c
9.	c
10.	c

Points to remember

Here are some key takeaways from this chapter:

- Adding axes in the view defines a measure axis or the blend of two measures and gives a glimpse of two independent axes in the same pane. There are three types of axes:
 - Individual axes
 - Combined axes
 - Dual axis
- There are a few advanced charts which are called the extension of basic charts. These charts apply the dual axis. They are:
 - Pareto chart
 - Lollipop chart
 - Donut chart

- There is a Word Cloud chart built on the frequency of words in the text data with few mouse clicks.
- A calculated field created to compute the values on the level of granularity, certainly at the data source level and the visualization level, is called the Level of detail expressions. They are:
 - INCLUDE
 - FIXED
 - EXCLUDE
- A workbook variable that can replace the filter, calculation, or any reference line is called a parameter. The parameter could be a number, string, date or datetime or a constant number. A parameter can be static or dynamic.
- When you try to combine relative members in a field, then that creates a group.
- Sets are the custom fields used to subset the data based on some conditions. There are two types of sets:
 - Fixed set
 - Dynamic set
- Actions allow you to create interactive relationships between the dashboard objects, data, other worksheets, and the web. There are six types of actions:
 - Go to URL
 - Filter
 - Highlight
 - Go to Sheet
 - Change Parameter
 - Change Set Values
- Animation is a way to improve communication and drive change. Tableau Viz with animation helps to explore, understand, show changes, present the data, and make better-informed business decisions.
- In comparison to Tableau Desktop, Tableau Online provides limited analytics objects to apply to a view. However, the Tableau Online analytics pane consists of summarize, model, and custom sections to add reference lines, reference bands, reference distribution, box plots, trend lines, etc., to add advanced analysis to the view.

- Dashboard lets you add several views, including web page images, and you can even compare the views.

CHAPTER 5

Content Management and Interaction Tools

A good tool improves the way you work - Jeff Duntemann [Introduction](#)

In the previous chapter, you learned how to create dynamic, user-interactive, and performance-optimized visuals. In this chapter, will explore some amazing features and tools of this interactive Tableau web platform to make managing the content on the web easy for users.

Structure

In this chapter, we will cover the following topics:

- Tableau content hierarchy
- Tableau file types and folders
- Content page type
- Basic interactions
- Custom views
- Enable alert
- Subscription in Cloud
- Metrics

- Exporting content
- Comment and share
- Enabling notifications if extracts fail
- Personal space

Objectives

By the end of this chapter, readers will learn how to manage the content (including views, workbooks, data sources, etc.) using the basic and advanced interactive tools available on Tableau Cloud. Here, they will learn how to get notified if the schedule extracts fail, download the content, add alerts, comments, metrics, etc.

After reading this chapter, the reader, as a site admin, can give the users an edge depending on the user's permission to manage, get alert, subscribe, or notify in the exception cases.

Tableau content hierarchy

Tableau Cloud uses hierarchies to organize the content, its access, and permissions on-site through projects, permissions, or content movement.

Refer to the following list to understand Tableau content hierarchy:

- **Projects:** The site administrators can create a top-level project where project leaders, owners, or content administrators can have full-fledged access to the project's contents and any nested projects.
- **Permissions:** The content administrator or owner of the project/workbook can assign permissions to individual users or groups. Users can be provided with fewer capabilities if assigned to multiple groups.

Note: Groups are lower in the hierarchy than individual user permissions.

- **Content movement:** The content owner or administrator can move contents, including a workbook, views, data sources, etc., between

projects or folders. If moving a project, then contents within it will also move along with child projects.

Tableau file types and folders

In Tableau Cloud, your work can be saved using different Tableau file types: workbooks, data connections, data sources, package data files, bookmarks, etc. The following list explains the types of Tableau files and folders:

- **Workbooks (.twb):** The Tableau workbook files saved in **.twb** file extension. A workbook comprises of one or more worksheets, dashboards or stories without supporting local files and background images. Also, it does not contain the actual data.
- **Packaged workbooks (.twbx):** The Tableau packaged workbook files are saved in the **.twbx** file extension. It is a zipped file that contains worksheets, dashboards, and stories, as well as supporting local files and background images and any supporting local data files, such as extracts, text files, Excel files, etc. It is useful for sharing workbooks with others who do not have access to the original data.
- **Data sources (.tds):** The tableau data sources are saved in the **.tds** file extension. This file contains the information necessary to connect to the actual data, along with any modifications made, including information about fields, aggregations, and calculated fields. It does not contain the actual data.
- **Packaged data sources (.tdsx):** A zipped file that contains the data source file (**.tds**) as well as local data files such as text files, Excel files, access files, hyper extract files, or local cube files. This Tableau zipped packaged data source file is saved in **.tdsx** file extension.
- **Extract (.hyper):** To work anytime offline or to improve the performance of Tableau, **.hyper** is an extract file that contains the local copy of a subset or entire data set. The hyperdata files can easily be shared with others.
- **Bookmarks (.tbn):** The Tableau bookmarks files are saved in **tbn** extension. Basically, bookmarks contain a single worksheet that can

be quickly shared with others.

When you install Tableau Desktop, these files can be stored automatically as a local copy in the My Documents | My Tableau Repository directory. You can also save your work files in the network directory depending on content type; various Tableau file types are stored in Workbooks, Data Sources, Projects, etc., respectively.

Content page type

Tableau Cloud enables organizations to manage content through sites, projects, and content such as workbooks, flows, metrics. Refer to the following for an explanation of the same:

- **Sites:** Sites are independent content collections contained with the same server. In short, within your Tableau site.
- **Projects:** A content folder containing other projects, workbooks, flows, and data sources.
- **Content:** Workbooks, flows, metrics, collections, views, and data sources.

As shown in the following figure, the content can be categorized into multiple types:

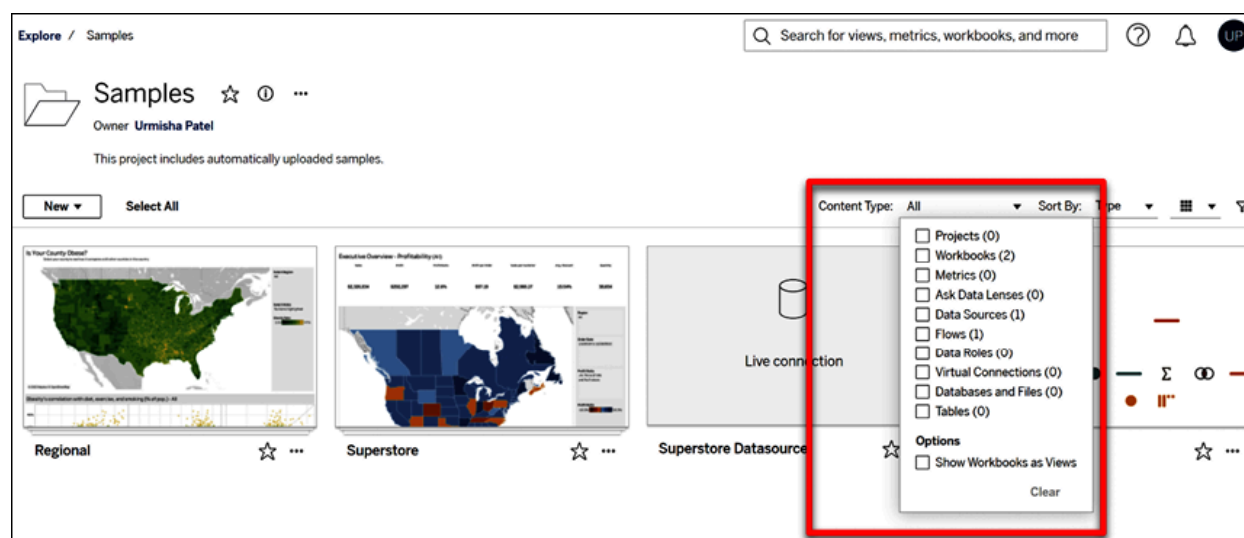


Figure 5.1: Content type The most frequent content type user uses to categorize the Tableau contents can be accessed by simply clicking on the search bar as shown in the following figure:

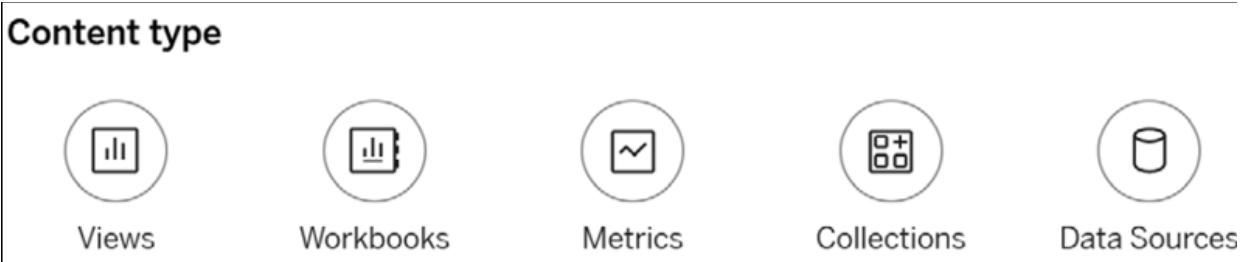


Figure 5.2: Content type on search bar You can access the contents and interact based on the user permission level. Now, let us understand the permission concepts briefly: [Understanding permission rules](#)

Tableau Cloud permission rules determine capabilities based on the user roles and can be applied at various levels (projects, workbooks, data sources, etc.). Let us look at them briefly:

- **Group users:** Capabilities assigned to individuals or groups.
- **Effective permissions:** Final permissions based on site and group/user capability settings. Refer to the following figure:

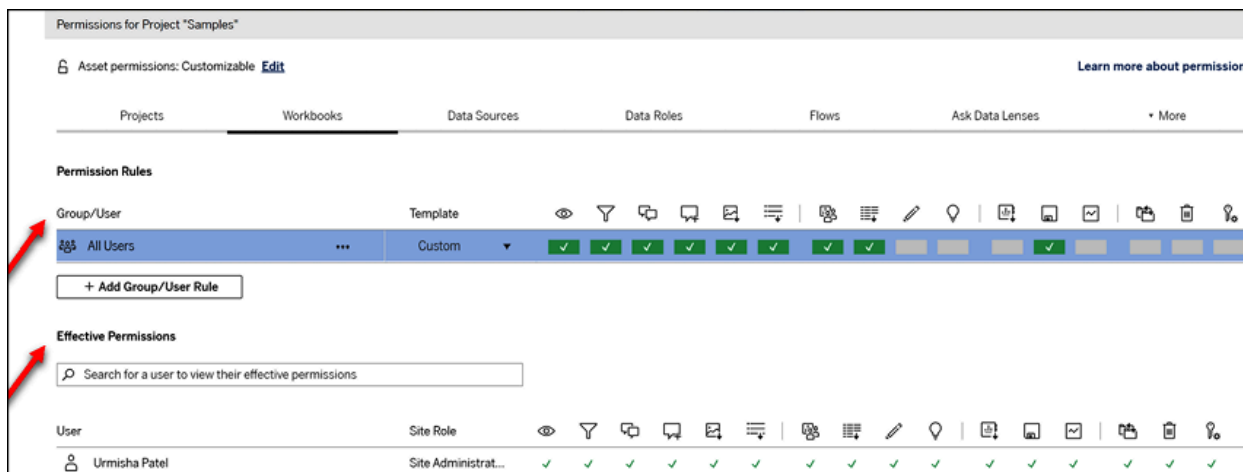


Figure 5.3: User Permissions We will cover this topic in detail in [Chapter 6](#), Site Management.

Basic interactions

Tableau Cloud users interact with the content by filtering, highlighting, and using toolbar actions. This includes Undo, Redo, Replay, Revert, Refresh, Pause, Data Details, Full Screen, Filtering, and many more actions, as shown in the sample dashboard in [Figure 5.4](#):

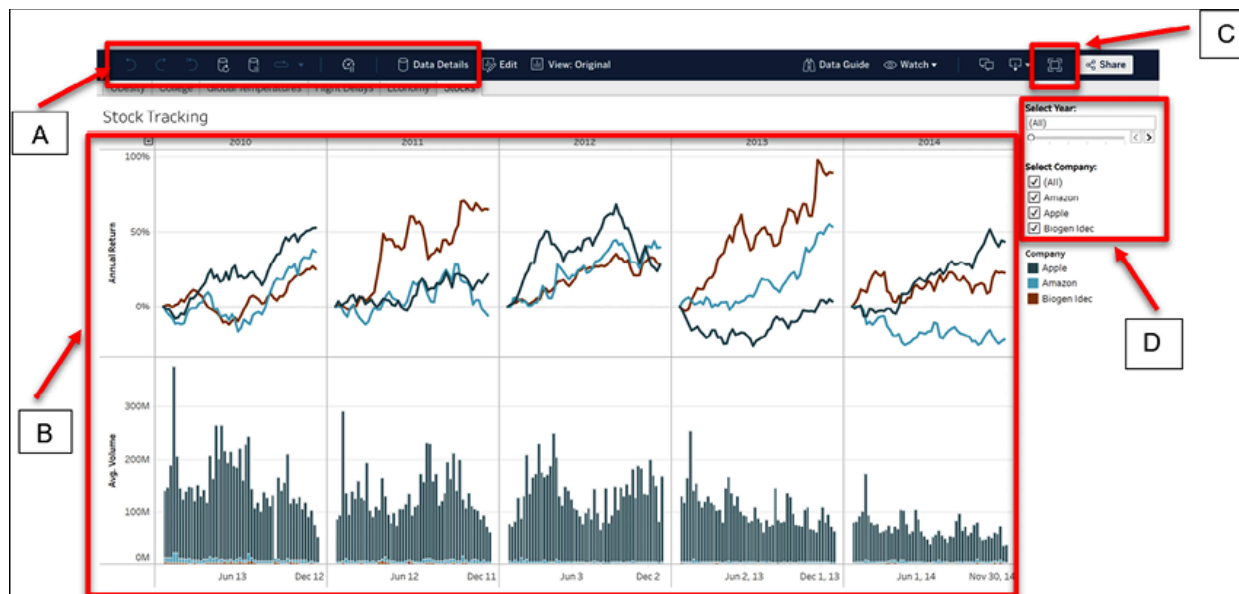


Figure 5.4: Basic interactions on sample dashboard Let us understand the basic interactions: **A:** The list of toolbar actions used to interact with the content are as follows:

- **Undo:** Reverse the action on the dashboard.
- **Redo:** Take back the reversed action on the dashboard.
- **Revert:** This will bring the dashboard back to the default view.
- **Refresh:** To clear the cached data/view.
- **Pause:** This will pause any data updates temporarily.
- **Replay:** Replay an action on the dashboard.
- **Acceleration Pause:** This will pause the acceleration on the dashboard.
- **Data Details:** To view the information on data usage, owner, data, etc.

B: Actions and Highlighting: Canvas area where you can interact with visuals in sheets/dashboards, to perform actions and highlight visuals.

C: Full Screen: To view the dashboard in full-screen mode.

D: Filtering: To filter dashboard data points.

Custom views

Users can create custom views on the cloud from the **View: Original** menu to be shared or used privately, as shown in the following figure:

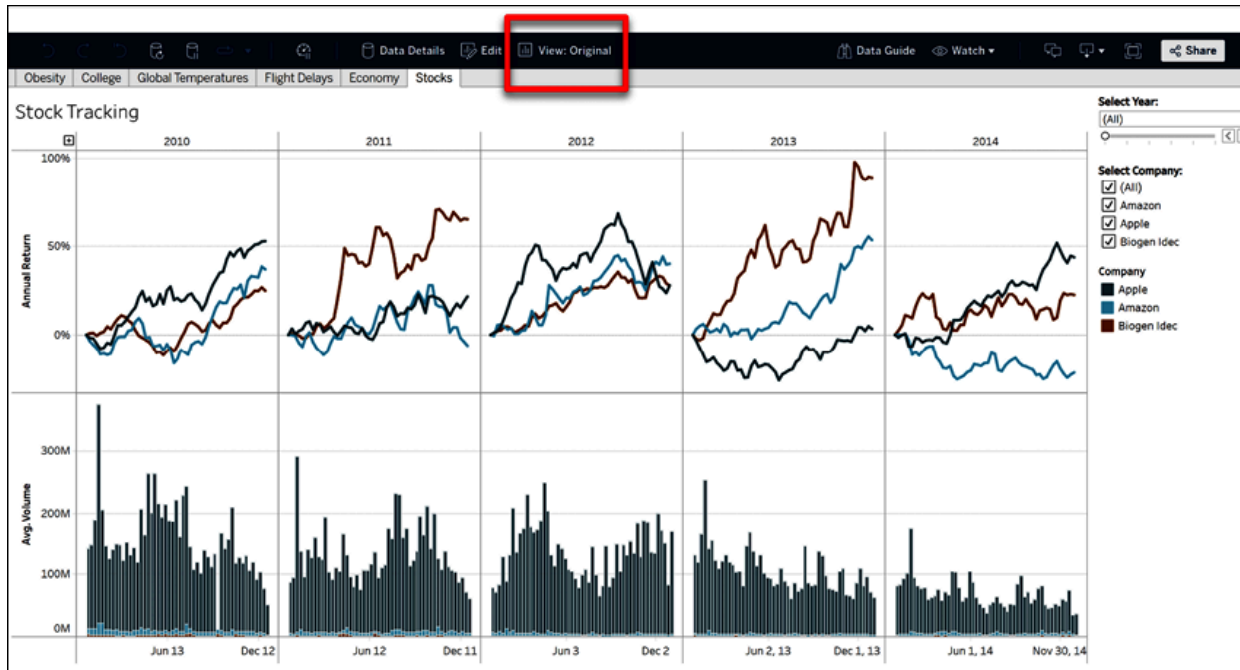


Figure 5.5: Custom views on the sample dashboard Follow these steps to create a custom view:

1. Click on the **View Original** menu on Tableau Online. This will open a **Custom View** dialog box, as shown in [Figure 5.6](#).
2. Choose a name for your custom view and write under the textbox **Name this view**.
3. Check the box **Make it my default** if you want to make this custom view as default, making this the custom view that will always open first when accessed.
4. Check the box **Make visible to others** if you want to make custom views listed and accessible to others.
5. Click on **Save** to commit your changes to be saved as a custom view.
6. Under **My Views**, you will find the list of saved custom views of this dashboard.
7. Under **Other Views**, you will find the list of other public custom views on this dashboard:

Custom Views

Save Custom View

Name this view

☐ Make it my default ☐ Make visible to others

Save

My Views

Nothing saved yet

Other Views

✓ Original (default)

Urmisha Patel

Figure 5.6: Custom views dialog box Now, let us help you understand how to add an alert to your dashboard.

Enable alert

Users can create data-driven alerts on Tableau Online from the **Watch** drop-down menu once a visualization with an axis has been selected, as shown in [Figure 5.7](#). Alerts help you to notify when data reaches the set threshold value.

Note: Alert can be enabled if either the X or Y axes have a numeric value. i.e., alert works on measured fields.

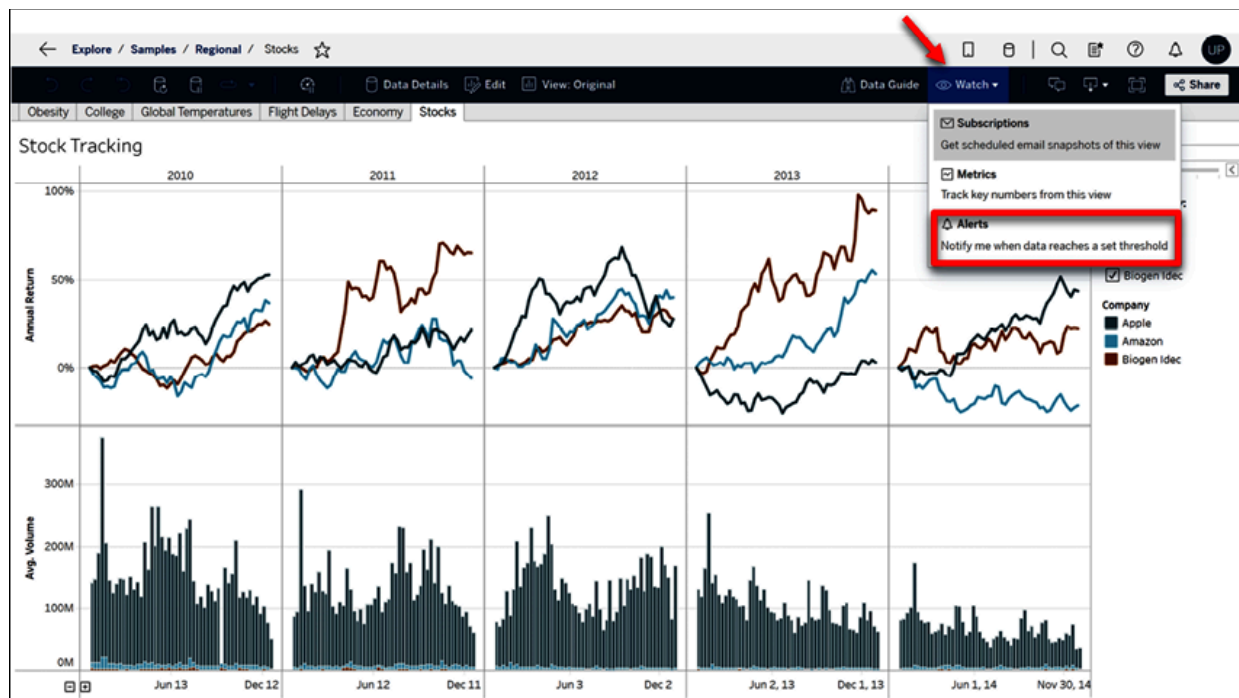


Figure 5.7: Alert in sample dashboard Follow these steps to create Alerts:

1. Choose the **Alerts** option from the **Watch** drop-down menu on your dashboard.
2. Click on the **Create** button after selecting a numeric axis of a chart.
3. For example, select the **Annual Return** axis of the **Stock Trading Chart** to create an alert on the numeric field and then click on the **Create** button.
4. This will open a **Create Alert** dialog box.
5. Choose the logical operator to configure the alert from the **Condition** drop-down menu.
6. Under **Threshold**, assign the value determining when an alert is triggered.
7. Assign a name to this alert under **Subject**.
8. Now, choose how frequently you want to alert delivery from **When the condition is True, Send Alert** drop-down menu.
9. Now, add the users who will receive the alerts in **Recipients**.

10. You can also click on the **Make visible to others** box if you want anyone with access to view or add themselves to this alert.
11. When finished, click on **Create Alert**, as shown in the following figure:

Create Alert

Send alert if 'Annual Return ' is:

Condition	Threshold
Above or equal to ▼	0.52

Condition currently true

Subject

Data alert - Stocks

When the condition is true, send alert:

Daily at most ▼

Recipients

Urmisha Patel ×

☐ Make visible to others ⓘ

Create Alert

Figure 5.8: Create an alert dialog box [Subscription in Cloud](#)

Users can create a subscription on Tableau Online from the **Watch** drop-down menu, delivered via email, as shown in [Figure 5.9](#).

On enabling the subscription, you will receive either PDF snapshots, images of the view or workbooks at regular intervals of time to your specified email address. Once you enable a subscription, you are not required to sign in to Tableau Online.

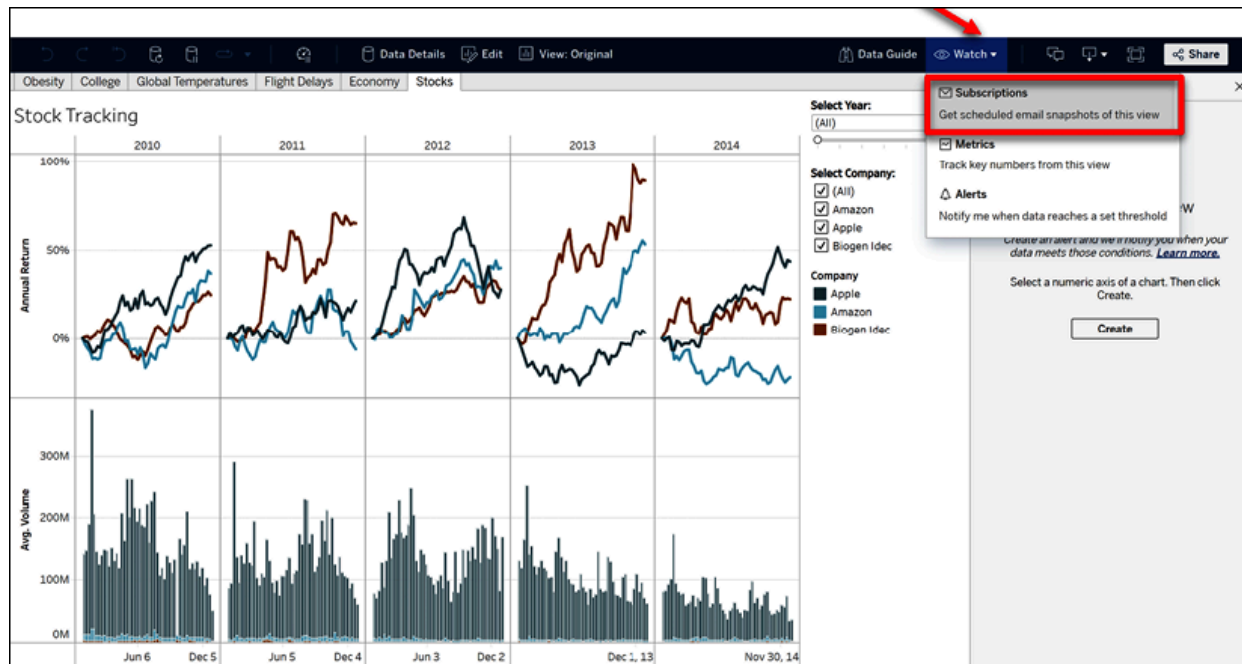


Figure 5.9: Subscription in sample dashboard Now, let us see how to add a subscription in the cloud for yourself and others. Follow these instructions to enable a subscription:

1. Choose the **Subscriptions** option from the **Watch** drop-down menu on your dashboard.
2. List down all the users who will receive the subscriptions under **Subscribe Users**.
3. Check the box **Subscribe Me** if you want to add a subscription for yourself.
4. Choose the option to include components (this view, entire workbook) from the **Include** drop-down menu.
5. You can tick the checkbox **Don't send if the view is empty** in case the view has no records to display. Refer to [Figure 5.10](#).
6. Now, choose the image, PDF, or image + PDF from the **Format** drop-down menu.
7. Enter the subject line of the subscription email under the **Subject** textbox.
 - a. [Optional] Add the message to the text added to the body of the email under **Message**.

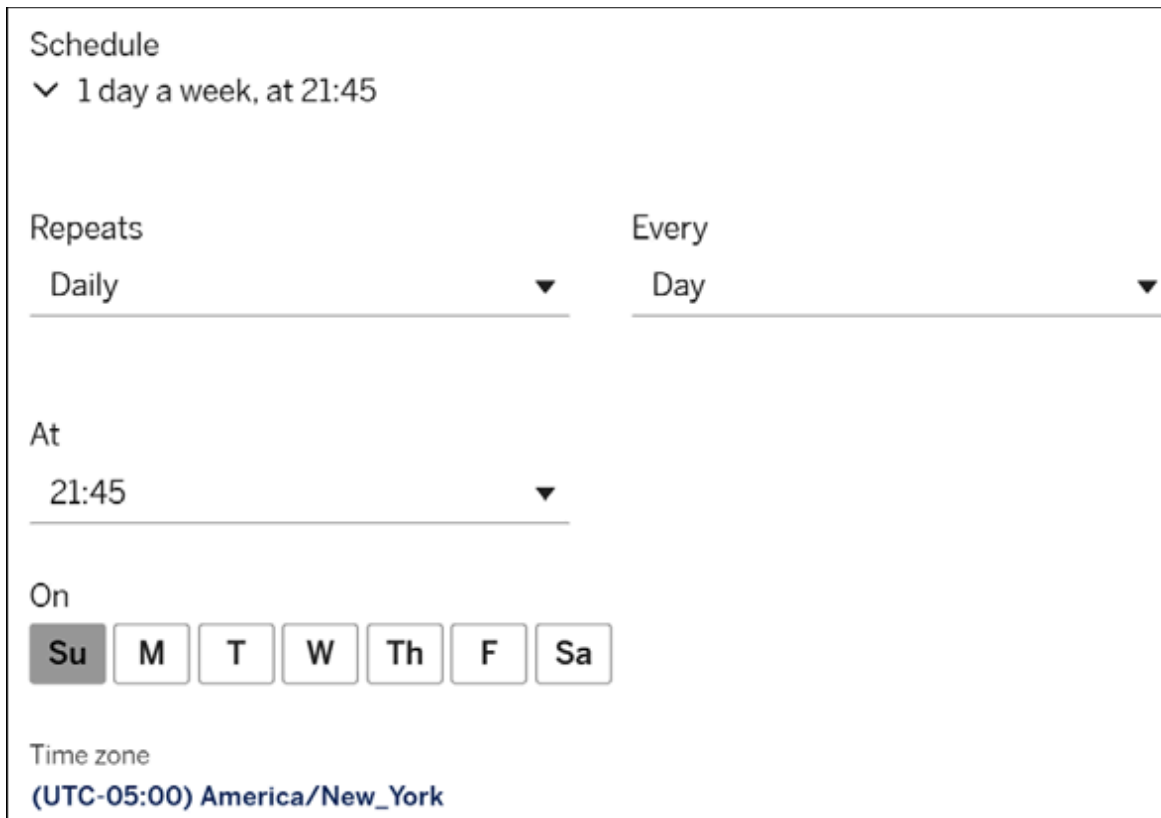
8. Choose the subscription run type (schedule or upon refreshed) from the **frequency** menu which is available only if the data is connected to any data source server.



The image shows a 'Subscribe' dialog box with a close button (X) in the top right corner. The dialog contains several sections: 'Subscribe Users' with a text input field labeled 'Type to add users or groups'; a checkbox labeled 'Subscribe me'; an 'Include' section with a dropdown menu currently showing 'This View' and a downward arrow; a checkbox labeled 'Don't send if view is empty'; a 'Format' section with a dropdown menu currently showing 'Image' and a downward arrow; a 'Subject' section with a text input field containing 'Stocks'; and a 'Message (Optional)' section with a text input field labeled 'Add a custom message'. At the bottom, there is a 'Manage Subscriptions' link, a 'Cancel' button, and a 'Subscribe' button.

Figure 5.10: Subscribe dialog box

9. You can specify the schedule for the subscription to run under **Schedule** as shown in the following figure:



Schedule

▼ 1 day a week, at 21:45

Repeats

Daily ▼

Every

Day ▼

At

21:45 ▼

On

Su M T W Th F Sa

Time zone

(UTC-05:00) America/New_York

Figure 5.11: Schedule in Subscription dialog box [Metrics](#)

Users can create metrics on Tableau Cloud from the **Watch** drop-down menu and share it with other users. **Metrics** help track the key members from the current view, as shown in [Figure 5.12](#):

Note: Metrics cannot be applied to the chart including the date parts and date values.

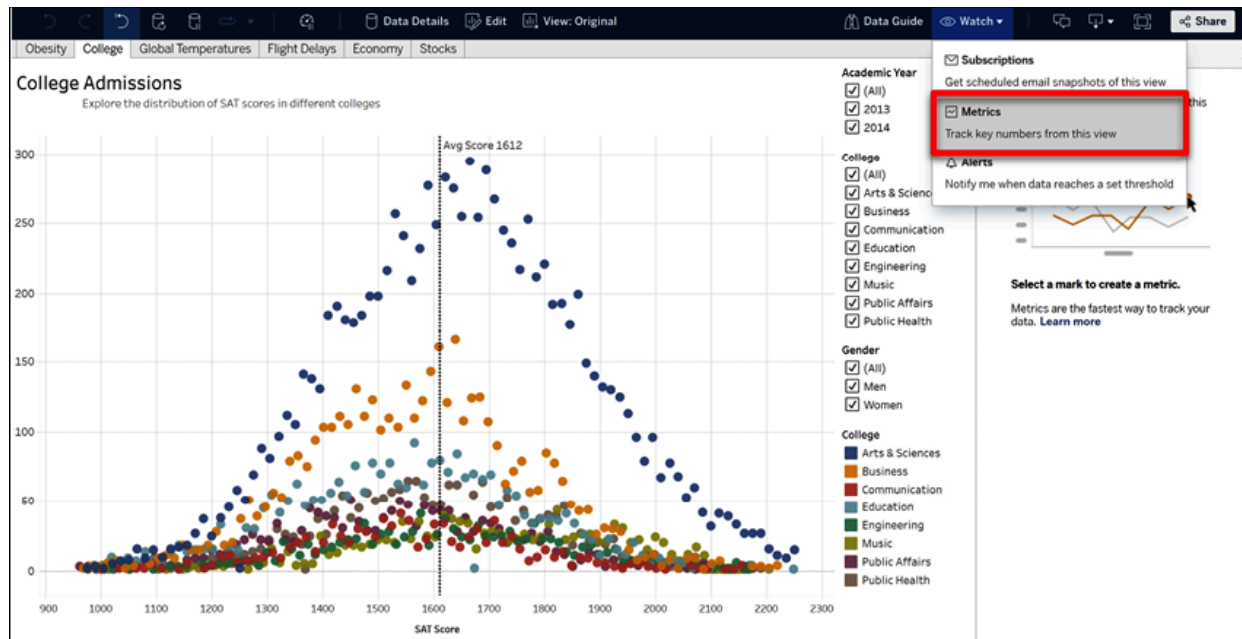


Figure 5.12: Metrics in sample dashboard Follow these instructions to add metrics in the current view:

1. Choose the **Metrics** option from the watch drop-down menu on your dashboard.
2. Select any one-mark point from the dashboard. The metrics editor will display automatically to the right side of the dashboard.
3. In the **Metrics** dialog-box as shown in [Figure 5.13](#), review will show the sample view of what your metric will look like/what mark you selected.
4. You can assign a name to your custom metric under **Name**.
5. Now, in the **Description** box, you can add the description of what your metric represents.
6. Choose the **Comparison Type** to compare against the historical period or constant value.
7. Now, select the measure from the measure drop-down menu to use from the selected mark/sheet.
8. Select the **Date dimension** (for views with dates only) from the **Dimension** drop-down menu.
9. The **Project** will display the destination project name for the metric.

Metrics

Preview

254

Refreshed now

Name

1 (Arts & Sciences, 1680)

Description (optional)

Comparison type

No comparison

Definition

Measure

SUM(1)

Date dimension

None

Project

Samples

Change location

Metrics display data from the owner's perspective. Users with permission to view this metric will see the data as it appears to you.

Learn more

Create

Figure 5.13: Metrics dialog box [Exporting content](#)

Content can be exported from the dashboard on Tableau Cloud using several methods, which can be controlled from the **Download** menu in the toolbar.

There are multiple file formats to download. Users can choose the option from the **Download** menu as follows:

- **Image:** This option will download the current view in **.png** format.
- **Data:** This option opens the new browser window, which contains the measured and categorical fields used in the current view.

- **Crosstab:** This option will download the full data in Excel or **.csv** file format.
- **PDF:** This option will download the dashboard, story, view, or specific sheet in PDF format, either in portrait or landscape orientation, page size, and scaling.
- **PowerPoint:** This option allows you to download the view or specific sheet in a powerpoint file.
- **Tableau Workbook:** This option allows you to download a **.twbx** file, including the data in a selected version. Refer to the following for an illustration of the same:

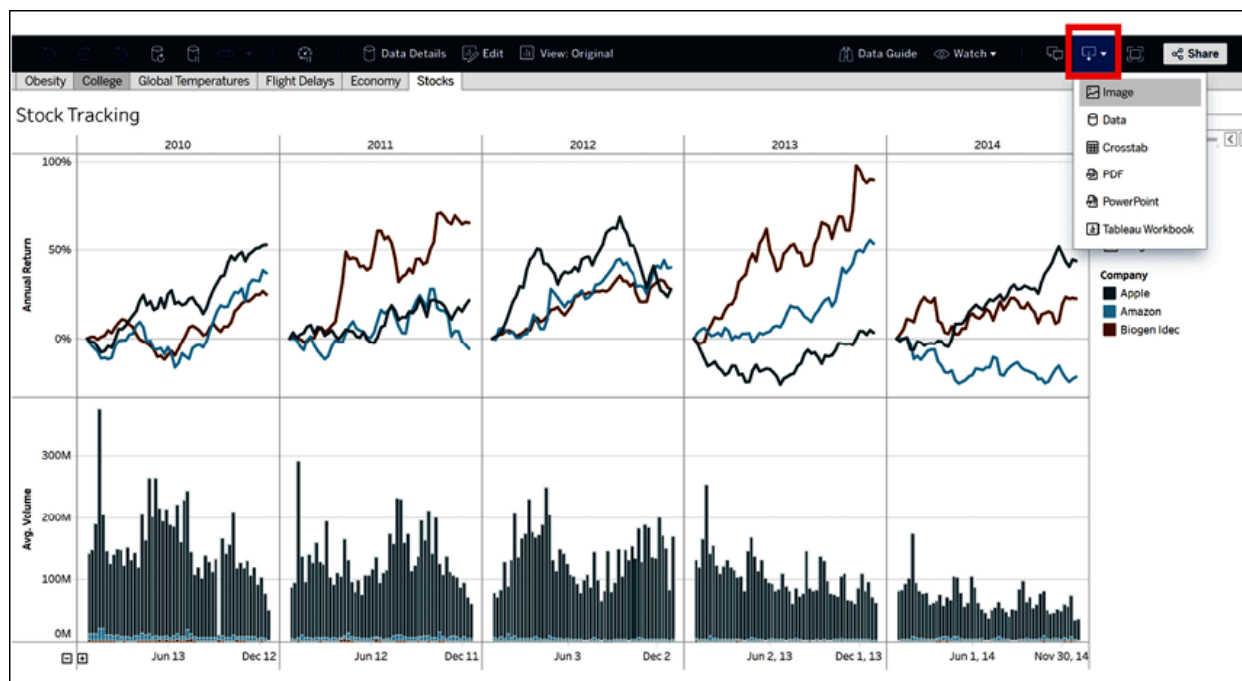


Figure 5.14: Export in sample dashboard *Comments and share*

Users can share and comment on the dashboard content with Tableau Online. This is one of the ways to review the report and share the reviews on the comment section to other users. Refer to *Figure 5.15*.

Now, let us explore each of these interaction tools in detail.

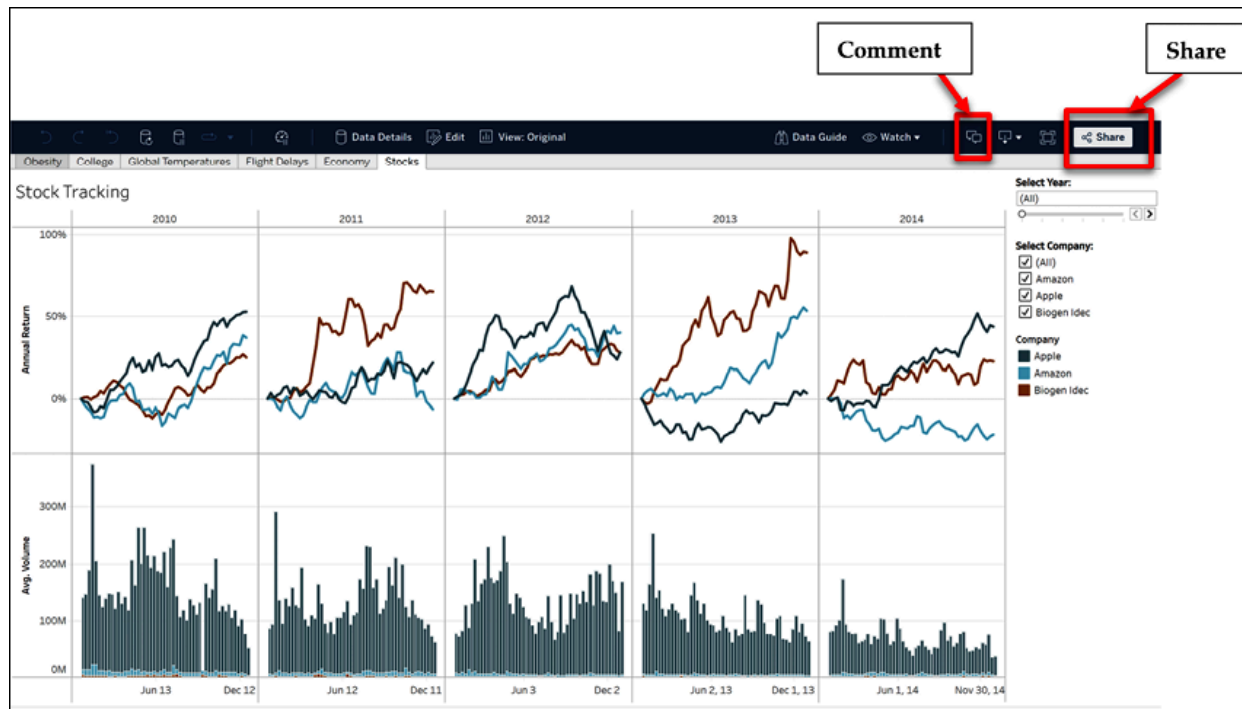


Figure 5.15: Share and Comment in sample dashboard [Comments](#)

Follow these steps to add comments to the dashboard:

1. Click on the **Comment** menu, as shown in [Figure 5.15](#), to add or view the comments.
2. Select Comments and type the comment message to compose under **Add a Comment**.
3. You can insert the image of the current view on selecting the image icon to embed the image in the comments, as shown in [Figure 5.16](#).
4. Insert the user with the @ symbol in your text to notify them under the **Notify Someone**.
5. When finished, click on **Post** button.



Figure 5.16: Comment dialog box [Share](#)

Follow these steps to share the content with the users:

1. Click on **Share** menu, as shown in *Figure 5.15*, to share the view with the users or embed it to the webpage.
2. Select **Share** and enter the username to share views through Tableau Online. Only the specified users are allowed to view it.
3. User can share or embed the link provided or copy the embed code as shown in the following figure:

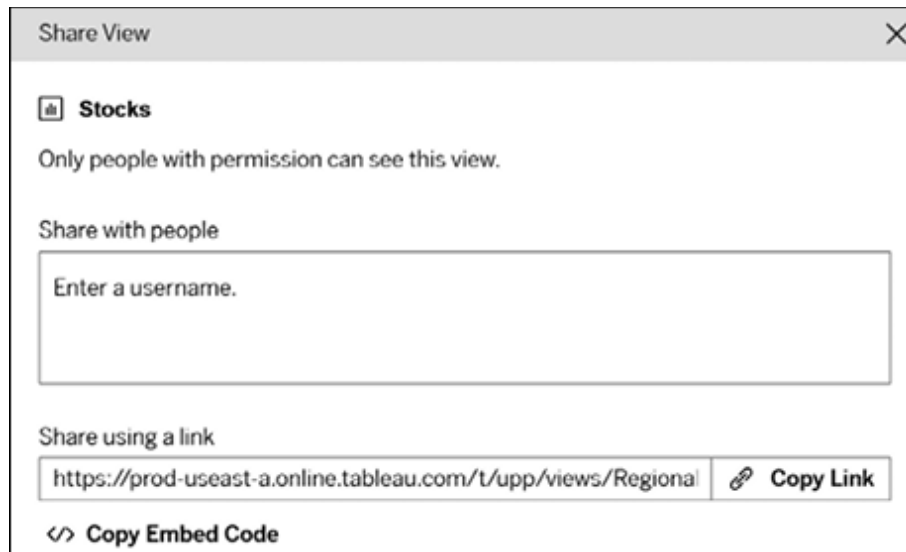


Figure 5.17: Share dialog box *Enabling notification if extracts fail*

It is important to notify the owner of the data sources or workbooks that the data extract refresh is unsuccessful. Tableau Cloud allows the owner to receive an email notification.

Follow these instructions to enable notification if extracts fail:

1. Sign in to Tableau Cloud as a site-administrator.
2. Go to the **Settings** page of your site, then click on the **General** tab.
3. Scroll to **Manage Notifications** and check the **Extract jobs**.

The data source owner would receive an email if the extract failed, and not the owners of the workbook that connects to that data source. By default, **Extract jobs** notifications are enabled. You can manage other notifications related to system status or collaboration to receive the notification or enable-disable the notification as shown in the following figure:

Manage Notifications

Allow or disable notifications for all of your site users

	On Tableau	Email
Collaboration		
Comment mentions	☒	☒
Share	☒	☒
Data alerts		☒
Systems Status		
Flow runs	☒	☒
Extract jobs	☒	☒
Webhooks		☒
View Acceleration	☒	

If grayed out, the notification option is disabled for use.

Figure 5.18: Enable Notifications *Personal space*

Personal space provides a secure and private location to publish content on Tableau Cloud. The personal space is a secured location for the contents before it is ready to share with others. Personal space will only support the workbooks. There are a few limitations to keeping the environment lightweight:

- You cannot create a new extract (you can use the existing extracts) from within the personal space.
- You can connect to published or embedded data sources.
- Alerts and notifications will be temporarily disabled so that others can set up the content (workbooks, data sources, projects, etc.).

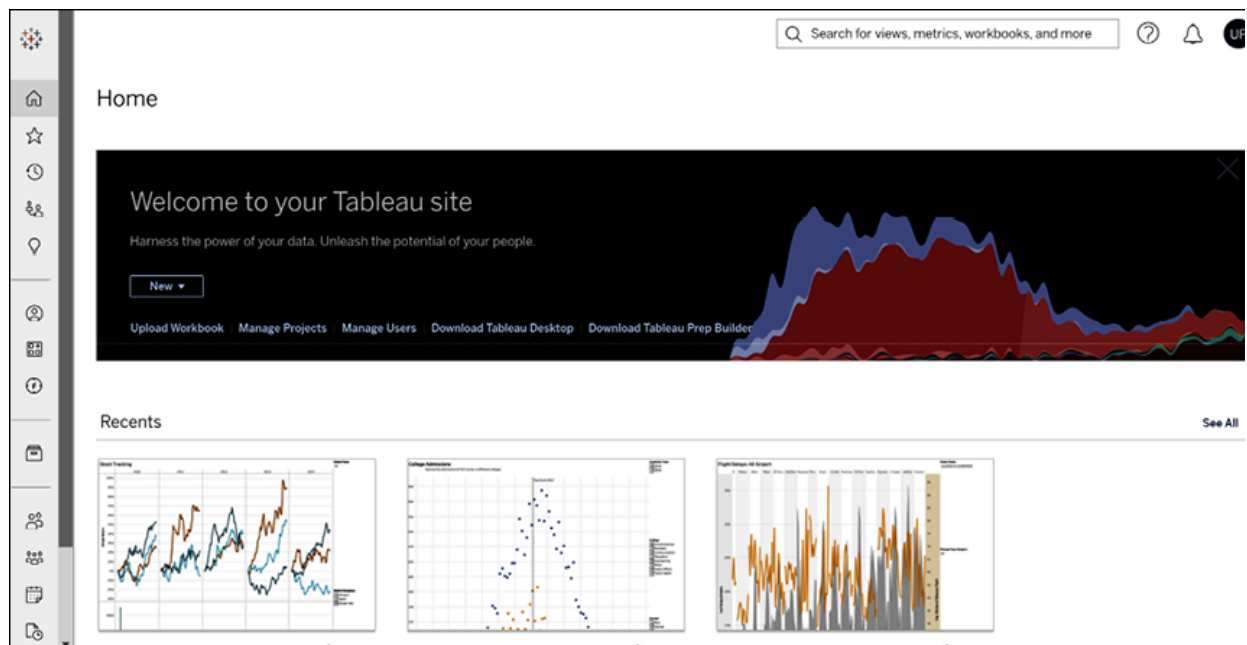


Figure 5.19: Personal space in homepage Now, let us explore the personal space page:

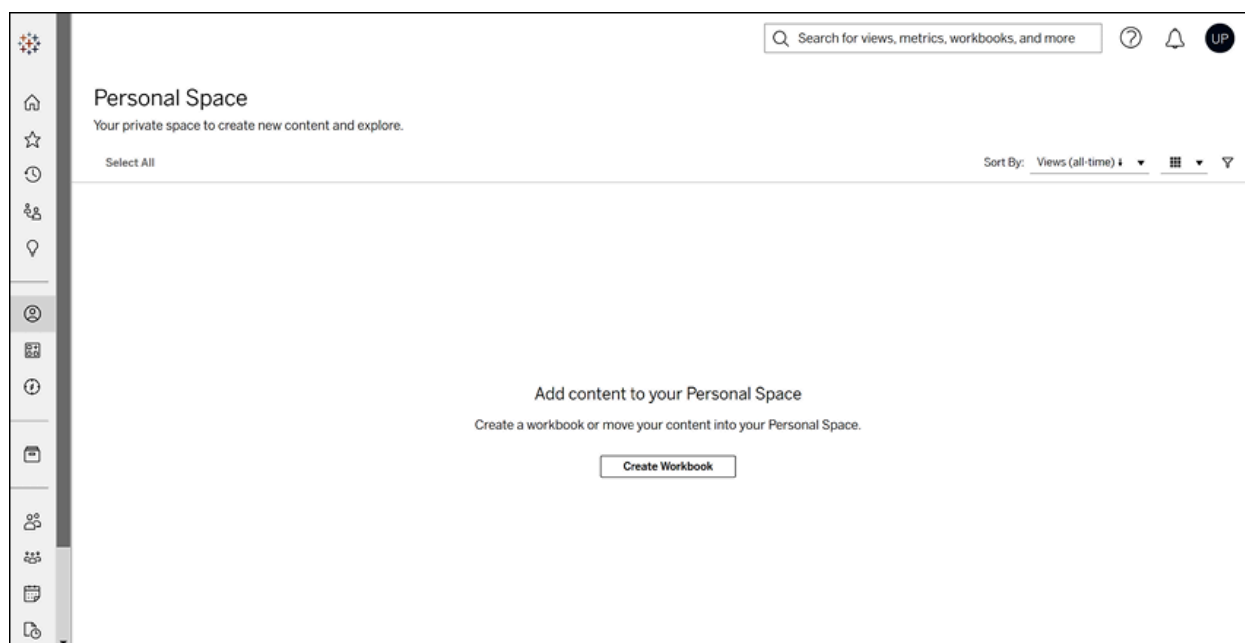


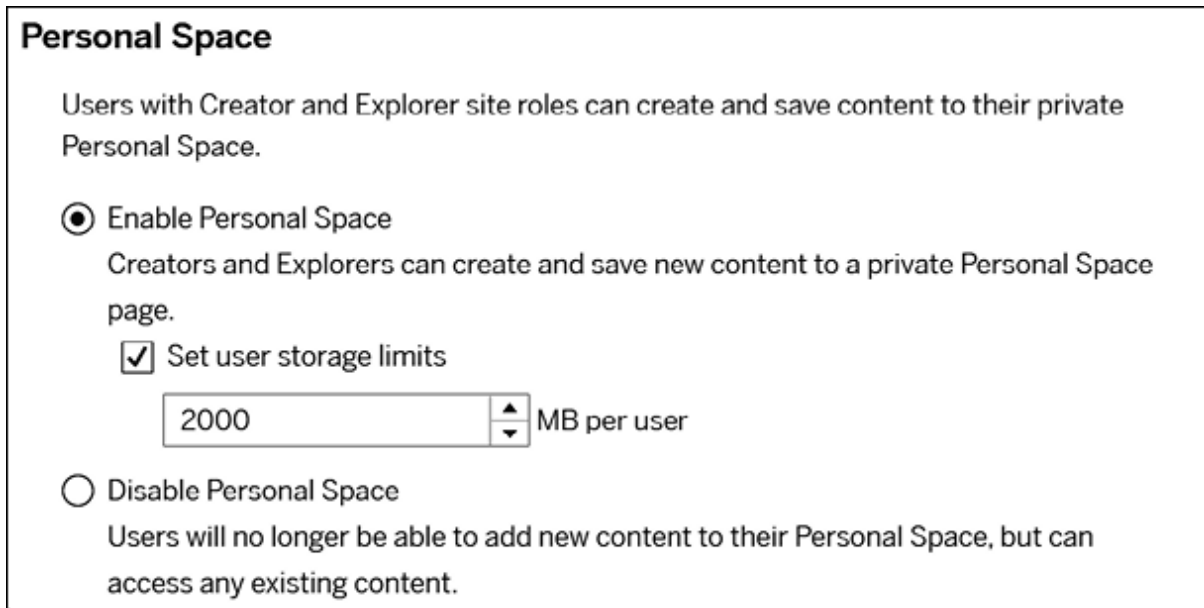
Figure 5.20: Personal space page

1. Click on **Create Workbook** to add content to your personal space, connect your data source, and start building the visuals.
2. This workbook can be saved in personal space and moved to other projects or locations. You can change the settings of Personal Space storage limits and enable or disable the personal space.

Follow these steps to set up personal space storage:

1. Click on the settings icon on the leftmost menu of the home page.
2. Scroll down the page to **Personal Space**, as shown in the [Figure 5.21](#).

Make the necessary changes as per requirements.



Personal Space

Users with Creator and Explorer site roles can create and save content to their private Personal Space.

☒ **Enable Personal Space**
Creators and Explorers can create and save new content to a private Personal Space page.

☒ Set user storage limits

2000 MB per user

☐ **Disable Personal Space**
Users will no longer be able to add new content to their Personal Space, but can access any existing content.

Figure 5.21: Personal Space settings [Conclusion](#)

In this chapter, readers have learned how to notify users via email or message if extraction failed in any exception and how to add an alert if system usage is beyond its limit defined.

We have also learned how to subscribe to the users or groups. We also discussed how users can create their own custom views, add comments, or even download the contents in .pdf, .jpeg, or .ppt, etc. formats.

In the next chapter, readers will learn how to control the users' ability to view and access the workbooks or data, how to secure the data, create user groups, etc.

Multiple choice questions

1. Which is the secured and private location to publish workbooks, and views in Tableau Cloud?

- a. Home
- b. Personal Space
- c. Explore
- d. All of the above

2. Select the most appropriate option: Which file formats can be downloaded from the Download menu?

- a. Image
- b. PowerPoint
- c. Crosstab
- d. All of the above

3. Which interaction tool helps to track the key members from the current view?

- a. Alert
- b. Notifications
- c. Metrics
- d. Custom views

4. Metrics cannot be applied to the chart includes what type of data?

- a. Numeric
- b. Geographical
- c. Date
- d. Categorical

5. Which interaction tool helps you to notify when data reaches the set threshold value?

- a. Custom views
- b. Subscription
- c. Metrics
- d. Alert

Answers

1.	d
2.	c
3.	c
4.	d

Questions

1. How to enable email notification if extract fails?
2. How to download the current view in .pdf format?

Points to remember

Here are the key takeaways from this chapter:

1. Tableau Cloud users interact with the content with basic interaction, including filtering, highlighting, and using toolbar actions.
2. Users can create custom views on the cloud to be shared or used privately.

3. Users can create data-driven alerts on Tableau Online once a visualization with an axis.
4. Users can create a subscription on Tableau Online delivered via email.
5. Users can create metrics on Tableau Cloud and share them with other users.
6. Contents can be exported from the dashboard on Tableau Cloud using several methods, which can be controlled in the toolbar.
7. Users can share and comment on the dashboard contents with Tableau Online.
8. It is possible to notify the owner of the data sources or workbooks that the data extract refresh is not successful. Tableau Cloud allows the owner to receive an email notification.
9. Personal Space provides a secure and private location to publish content on Tableau Cloud.

**Join our book's Discord space Join the book's Discord
Workspace for Latest updates, Offers, Tech happenings around
the world, New Release and Sessions with the Authors:**

<https://discord.bpbonline.com>



CHAPTER 6

Site Management

Your old site is the best prototype of your new site - Hoa Loranger

Introduction

In the previous chapter, we learned how to create dynamic, user-interactive, and performance-optimized visuals. In this chapter, readers will explore, as a site administrator, how to manage the sites, contents, monitor activity, permissions, and users.

Structure

In this chapter, we will cover the following topics:

- Site and site planning
- Row-level security
- Permissions
- Manage content ownership
- Upload a custom logo for your site
- Admin insights or site status

Objectives

By the end of this chapter, readers will learn about key concepts for effective site management. After reading this chapter, you, as a site admin, will understand how to manage users or groups of users, how permissions work at different levels in Tableau Cloud, and many more.

Site and site planning

Tableau Cloud enables organizations to manage content through sites, projects, and contents. Now, we will try to understand the definition of site and what counts as site administrative tasks:

- **Site:** A site is an abbreviation of website, which means a collection of contents including projects, workbooks, and data sources, users, or groups of users. Each site has its own URL and its own group of users and content. All contents can be accessed, managed, and published as per the permissions on the site defined by the site administrator.
- **Site administrative tasks:** A site administrator provides unrestricted access to the Tableau Cloud site. The site administrator gives access to the users with unrestricted or limited restricted permission to access at the site level.

A site administrator creates and maintains the framework that allows Tableau Desktop users to share, publish, manage, and connect to data sources and workbooks.

The site planning starts with the job responsibilities of site administrators. So, let us find out the site administrator's tasks.

The following list details a site administrator's task in setting up a site.

- **Uploading the logo:** Branding the organization logo on the site and configuring the site settings.
- **Configuring site access:** Setting up user authentication and sign-in and checking whether to allow access on Tableau Cloud site.
- **Creating projects:** Managing the Tableau Cloud site contents (data sources and workbooks), apart from default projects, creating more projects on the site as per organizations' requirements.
- **Set up the permissions structure:** Managing users' or groups' site roles and capabilities.
- **Add users:** Adding or importing multiple users on the site based on their authentication type and license roles.
- **Get your data to Tableau Cloud:** Tableau desktop users will publish the data sources. So, as a site administrator, managing the data sources refreshes, access permissions and maintaining the central connection information.
- **Analyse site usage and performance:** Monitoring and tracking the user activity, extract refresh tasks, monitoring the usage of publish data sources and

workbooks.

Row-level security

When you publish a workbook on Tableau Cloud, by default, all users are allowed to access the workbook and can see all your data in the view. You can limit the access by applying a filter at the data row level, which allows only the given person signed in the Tableau Cloud to see in the view. This approach for securing data at the row level applies to the live or extracted data sources.

Before we begin with **row-level security (RLS)** concepts, let us understand how user-based filtering works.

Suppose your manager asked you to create a quarterly sales report for certain category products in different regions, as shown in the following figure:

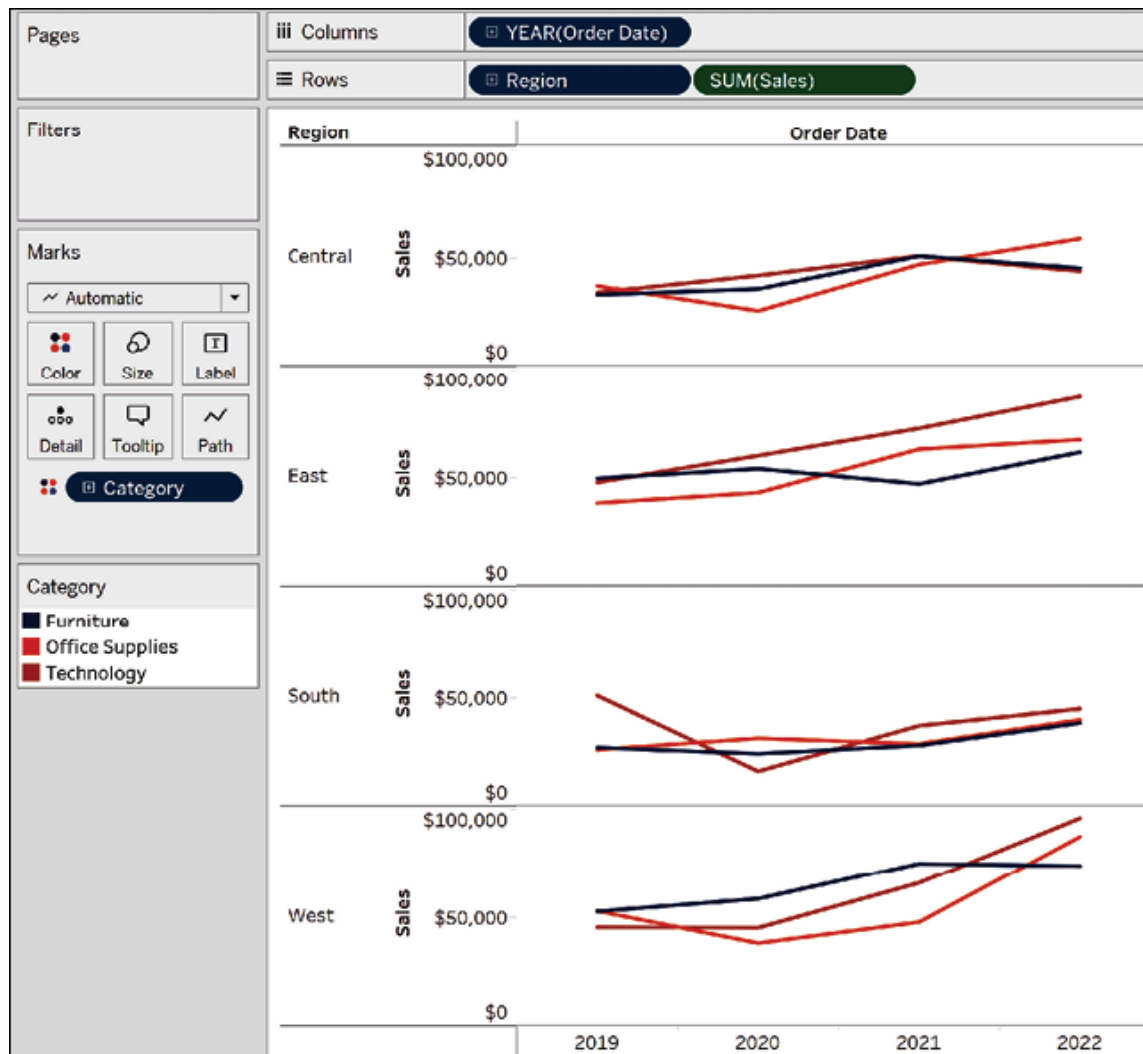


Figure 6.1: User-based sales report When you publish this report, each regional manager can view this same report. However, you must only allow the relevant data to be shared with each regional manager in the region. Here, you can restrict access to the data and view the w.r.t region and manager by applying a user filter instead of creating a separate view for each manager.

Restricting the data at the row level is referred to as **row-level security (RLS)**.

Tableau offers two methods for RLS. Let us look at them in detail: [Method 1: Create a user filter and map users to value manually](#)

This user filter based RLS approach helps to secure your data sources and workbooks. This approach works best for a static set of users or groups.

Follow these steps to apply user filter-based RLS method: In the example discussed earlier (under method 1), we wanted to filter the view based on region and depending on the user filter. i.e., regional manager.

1. Open the workbook on the Tableau Desktop and connect the Sample SuperStore data on which you want to apply the filter.
2. Now, go to the worksheet/view, and you want to apply the filter.
3. Click on **Server | Create User Filter | Region** to filter the view based on region, as shown in the following figure: Hint: Select the field on which you want to filter the view.

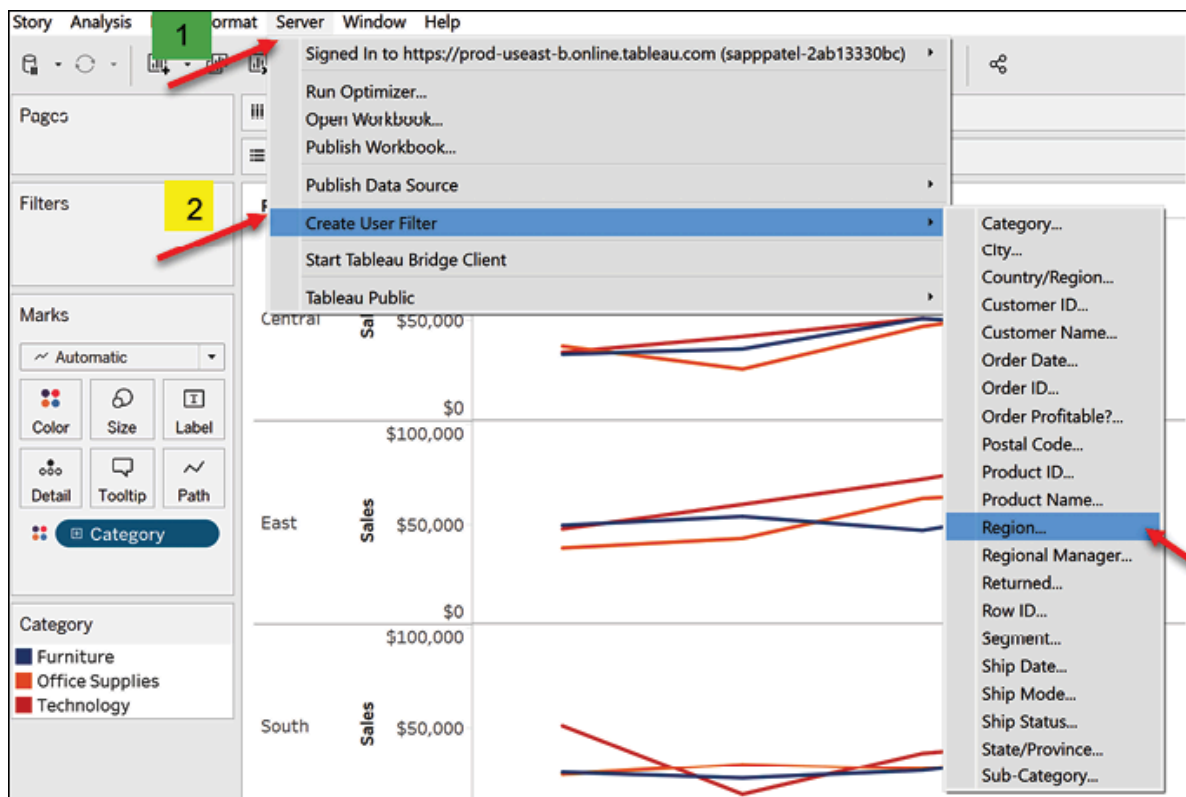
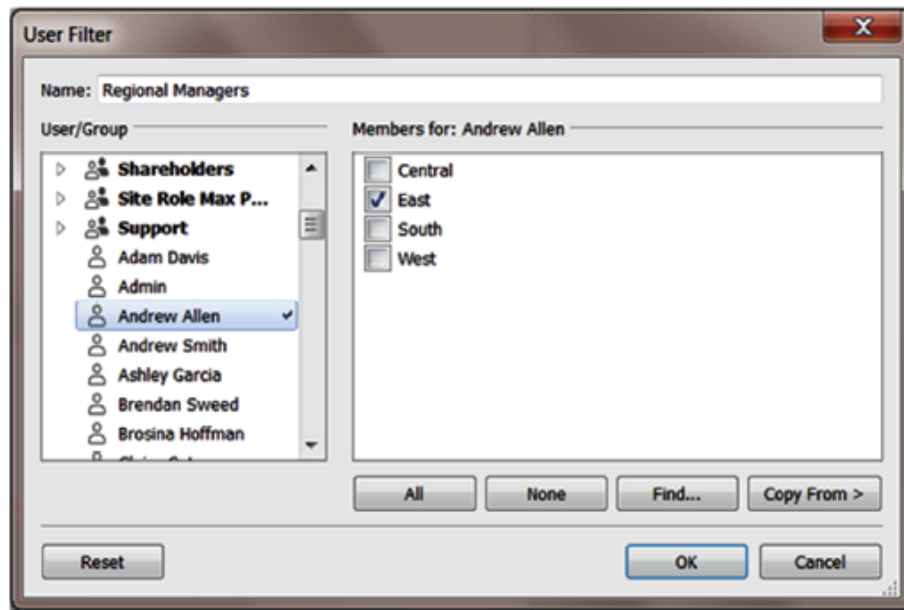


Figure 6.2: Create user filter based on region

4. In this **User Filter** dialog-box, select users or groups. Here, select the **Regional Manager** from the username list on the left and select the Region on the right.
5. Repeat this process for each user or group. Once the mapping is done, click on **OK**. This will create a user filter set. In [Figure 6.3](#), *Andrew Allen* is allowed to see only the records of the eastern region:



*Figure 6.3: User filter dialog-box Here, Andrew Allen is allowed to see only **East Region Records**.*

6. Drag this **User Filter Set** to filter shelf. The filter can be applied as a **Context Filter**, as shown in the following figure:

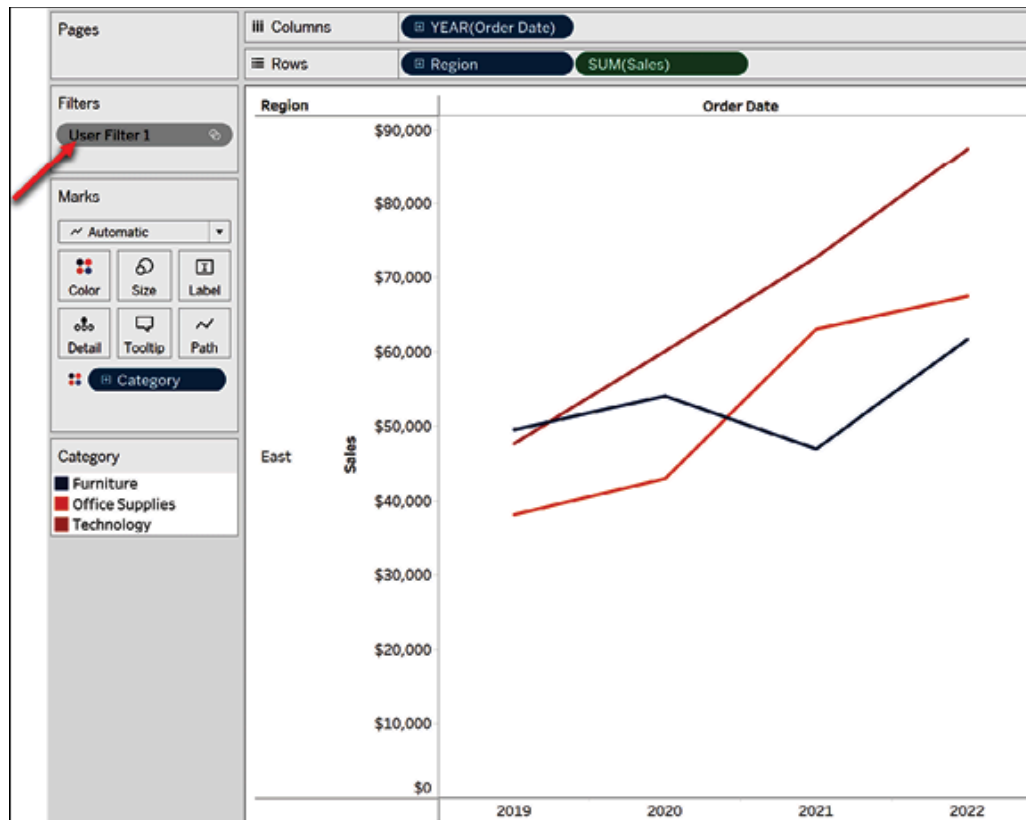


Figure 6.4: User filter method (RLS -1)

This user filtering must be applied to each workbook, and if any changes occur in the user base, then it needs to update the filter and re-publish the data source. Though it is convenient, it requires high maintenance and security.

Method 2: Create a dynamic filter using a security field in the data

Now, let us understand another RLS method. To restrict access at the data row level, this approach requires a database to use a security field for filtering. This automates the process of mapping users with the data values. Let us apply this approach to the **Sample Superstore** data source by following these steps:

1. Open the workbook in Tableau Desktop and connect the data on which you want to apply a filter.
2. On the worksheet, select **Analysis | Create calculated field**, following field named as **User Filter** as shown in the following figure:

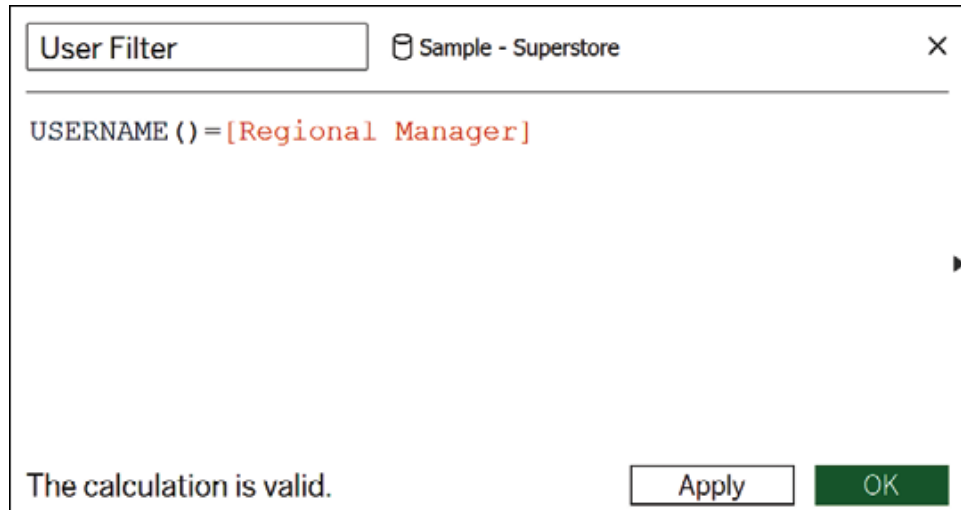


Figure 6.5: User filter - calculated field This calculation states those users who are managers can see the data in the view.

3. Click **OK**.
4. Now, add the **User Filter** field to the **Filter** shelf.
5. In the **Filter** dialog box, select **True** and click **OK**, as shown in the following figure:

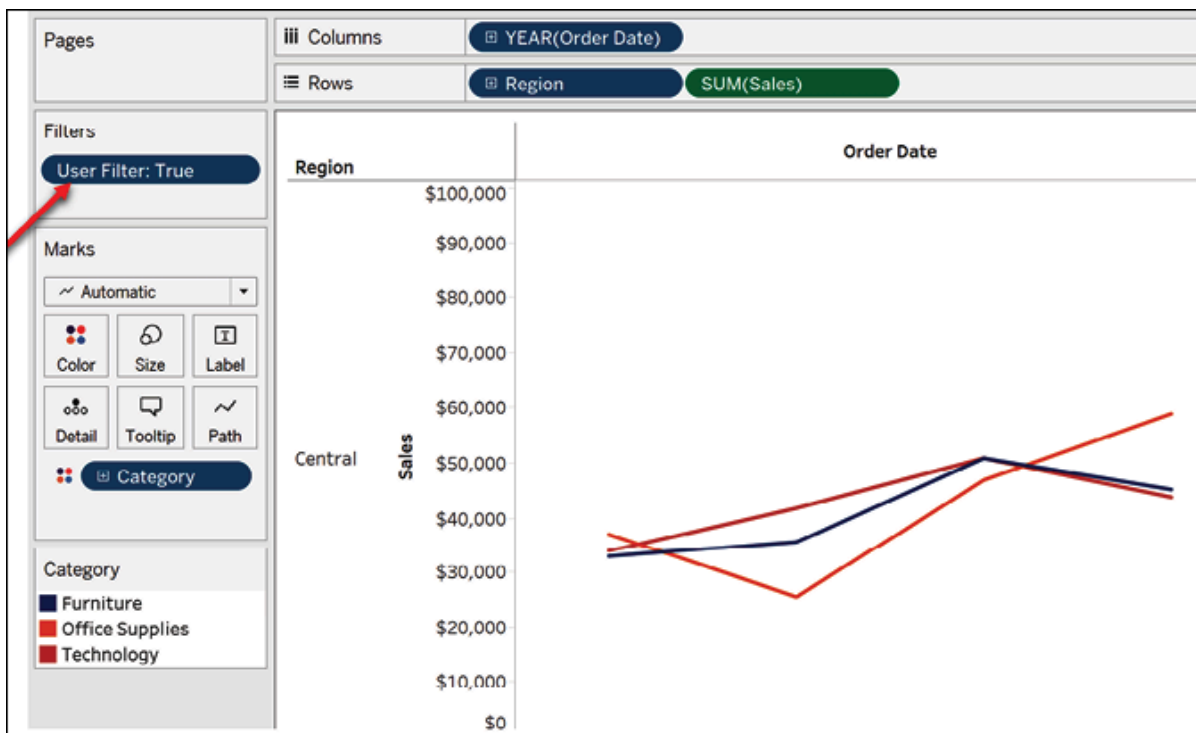


Figure 6.6: Dynamic User Filter method (RLS -2) *Create a dynamic filter directly at data source level* Instead of maintaining and applying a user filter at each worksheet level, it is better to apply the user filter at the data source level. Follow these steps:

1. Follow the steps in Create Dynamic User Filter.
2. Now, select the **Data Source** tab.
3. In the **Data Source** page, Under Filter, click on **Add**, as shown in the following figure:

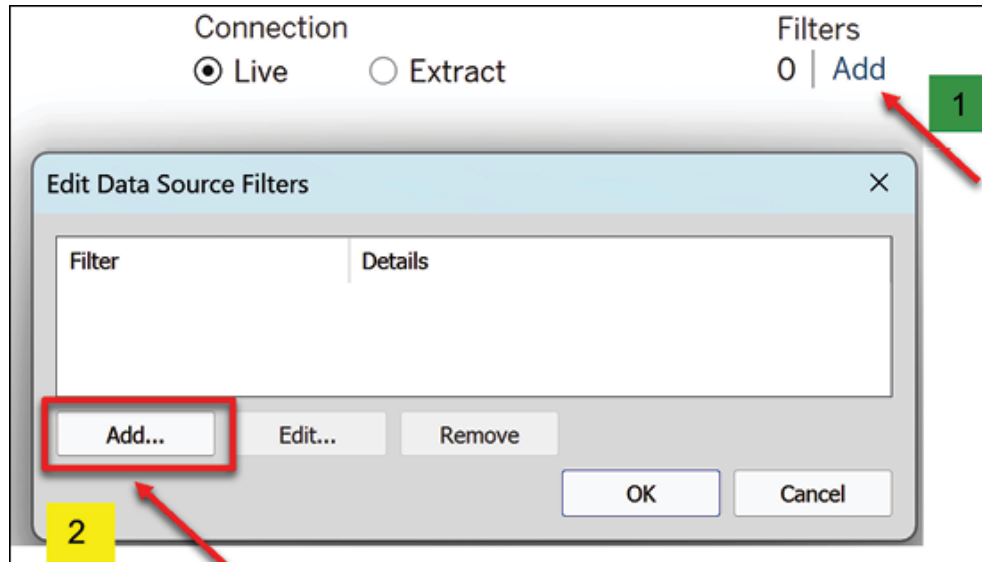


Figure 6.7: Filter data source

4. On the **Edit Data Source Filters** dialog box, click on **Add** button.
5. Next, select that calculated field: **User Filter**. Then, click **Ok**.
6. Select **True**. Then **OK**.
7. Again, **OK**.

Permissions

Permissions can be established for projects, users, groups, or assets which are already existing. These permissions are made up of the capabilities that allow users or groups, whether they can access or deny an asset, to view, publish, explore, edit, download data sources, or delete contents. The next topics will help manage permissions on the Tableau Cloud. Let us learn them first.

Users and groups

Tableau Cloud allows you to add more users or groups of users to the Tableau site and set their site roles, which define each user's or group's access levels. It is advisable to organize multiple users by creating groups to manage them more easily.

The process of giving user permission begins with adding the user to the Tableau Cloud site if it does not exist. Let us learn how to add users to the site.

Add users to the site

Anyone who wants to publish, edit content, view, explore, or even administer the site must be added. The administrators can use the following two options to add users to the site: **Option 1:** Add users to the site by user's email

1. Sign-in to the Tableau Cloud site, then click on **Manage Users** on Home page.

Or,

You can also navigate to the Users Page when you sign in to the Cloud site.

2. Now, click on **Add Users | Add Users by Email** on the **Users** page, as shown in the following figure:

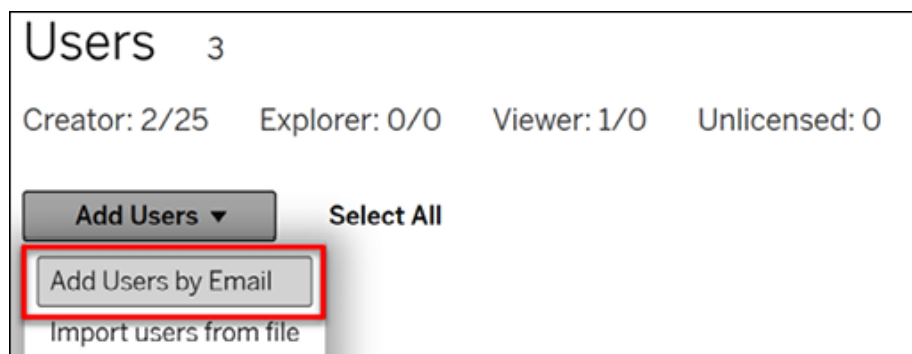


Figure 6.8: Add users by email on user's page

3. This will open the add users dialog box, as shown in [Figure 6.9](#):

Add Users

☐ Tableau with MFA
Users will receive an invite email that contains a link to the site and instructions to set up their Tableau ID.

☒ Tableau
Users will receive an invite email that contains a link to the site and instructions to set up their Tableau ID.

Enter email addresses

Click here and type email addresses separated by semicolons

Site role: Explorer (can publish)

Cancel Add Users

Figure 6.9: Add users by email dialog-box

4. Select the authentication method through which you want to add users to the Tableau Cloud site.
5. Tableau with multi-factor authentication (through an external identity provider).
6. Tableau with default user email address and password authentication.
7. Enter the user's email address in the textbox, and if you want to add more users, then separate each user's email address with the semicolon. For example:
urmishaeduwor1d@gmail.com; abc@example.com; def@yahoo.co.
8. Now, select and assign the **site role** to all users from the drop-down list.
9. Finally, click on **Add Users**.

Option 2: Add users to the site by importing users via CSV file Automating the process of adding users to the site by importing users through a CSV file that contains the user's information.

Follow these steps to import users:

1. Sign in to the Tableau Cloud Site, then click on **Manage Users** on the **Home** page.
Or,
Navigate to the **Users** page when you sign in on Cloud site.
2. Now, click on **Add Users | Import users from file** on the **Users** page, as shown in the following figure:

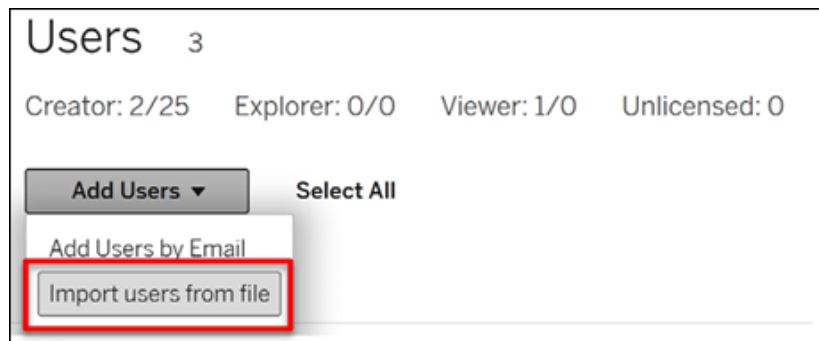


Figure 6.10: Import users from CSV file on users' page This will open add **Users** dialog box, as shown in Figure 6.11:

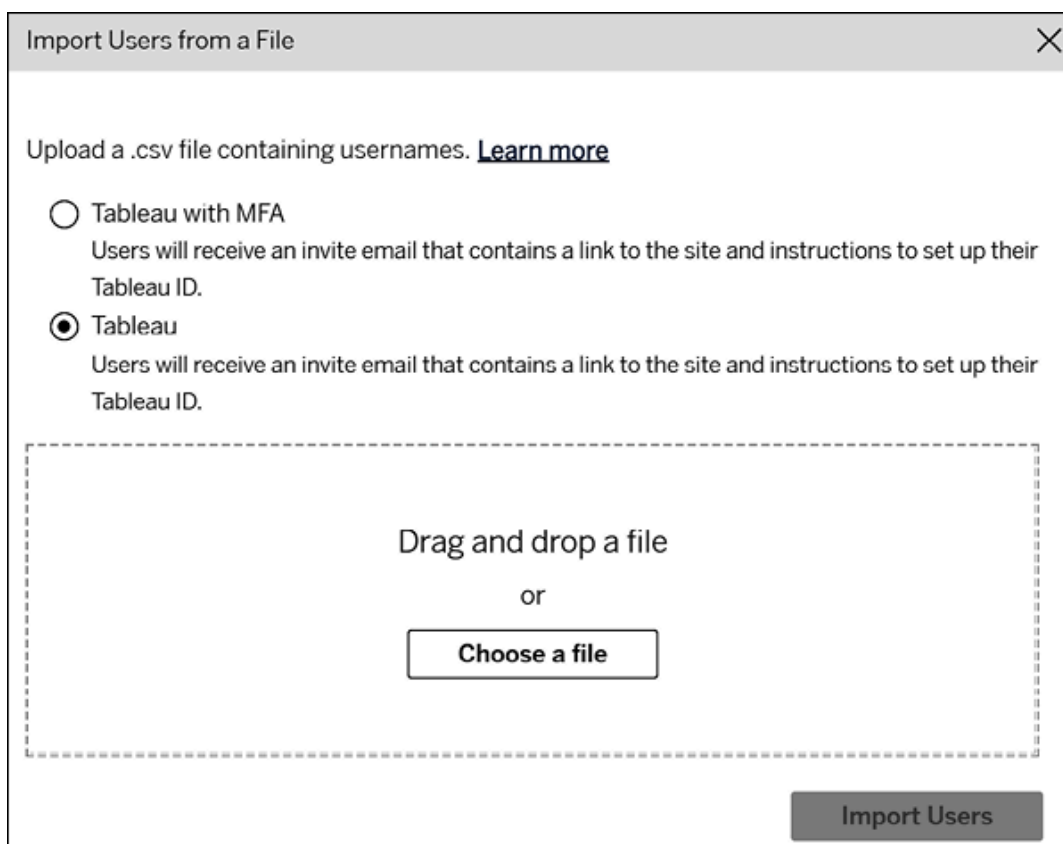


Figure 6.11: Import Users from a File dialog-box

3. Now, select the authentication method by which users can be verified on the Tableau site. You can choose either option:

- a. Tableau with multi-factor authentication (through an external identity provider).
- b. Tableau with default user email address and password authentication.

Note: Authentication Type can be change later. Follow the path: Go to Users Page. Now, Select the User then click on **Actions (...)** | **Authentication**. Finally, choose the authentication type and click on Update.

4. Now, click on **Choose a File** to navigate the CSV file, then click **Open**.

5. Further, click on import users.

The following are important CSV import file guidelines:

- The CSV file does not include any column header.
- Every row/line represents a user.
- The required user information for each user:
 - Username = user's email address.
 - The file is in UTF-8 format.
 - If user's email address contains '@' character other than in domain separator. Then '@' symbol must refer with hexadecimal format: \0x40.
 - For example: **abc@def@gmail.com** should be abc\0x40def@gmail.com

By default, the user's site role would be considered as 'Unlicensed' if not defined in the file.

The CSV file may contain the following information for each user w.r.t positions:

- **Username:** The user's email address. This is the only required information.
- **Password:** Tableau Cloud does not use this information. You can delimit the position for each user to define correctly. However, even if you specify values, Tableau Cloud will ignore them.
- **Display Name:** Tableau Cloud does not use this information. You can delimit the position for each user to define correctly. However, even if you specify values, Tableau Cloud will ignore them.

- **License Level:** Define the user's site role: **Creator, Explorer, Viewer, or Unlicensed.**
- **Administrator Level:** Define the user, whether **Site** or **None.**
- **Publishing capability:** Define whether the user is allowed to publish content on the site. Here, the allowed values are Yes/True/1 or No/False/0.

Let us create a dummy file **users1.csv** based on the CSV import file guidance.

Sample file: **users1.csv**

1. **abc@gmail.com,, ,Creator,Site,True**
2. **def@gmail.com,*****,Def Xyz,Viewer,None,0**
3. **xyz@biz.com,, ,Explorer,None,1**
4. **pqr@gmail.com,, ,Explorer,None,0**

Set user site roles

In the previous topic, we understood how to add users to the Tableau Cloud site independent of license type. Now, let us see how to apply the user's site roles.

The site role will determine the maximum capability of users on the site, whether the user can publish the contents, explore, view, administrate, or manage the contents (projects, data sources, workbooks) on the site.

Let us look at the user site roles and license types of capabilities.

The license type is associated with the user, and the site role is what you assign to the user, which determines the license type they require to attain the maximum capabilities on the site.

Let us explore the association between various license types, user site roles, and their capabilities on the site, as shown in the following table:

License Type	Site Role	Tableau Clients Access	Capabilities in Tableau Cloud
Creator	Site Administrator	Tableau Prep, Desktop, Bridge, Mobile	The highest level of access on Tableau Cloud. The unrestricted user access at the site level.
	Creator	Tableau Prep, Desktop, Bridge, Mobile	The maximum level of content access at the non-administrator level.

Explorer	Site Administrator Explorer	Tableau Mobile or Cloud from browser	Same access as the site administrator. However, it cannot connect to external data sources or virtual connections from Tableau Cloud (web version).
	Explorer (can publish)	Tableau Mobile or Cloud from browser	Can own or publish the contents from tableau Desktop. However, cannot save or edit the workbook from the standalone data sources with embedded connections.
	Explorer	Tableau Mobile or Cloud from browser	Can view or browse the content. But it cannot save work and cannot connect with virtual connections.
	Viewer	Tableau Mobile or Cloud from browser	Such users have read-only access and can subscribe to publish views. But it cannot save custom views or interact.
Viewer	Viewer		Can view the publish contents, subscribe the publish views, download images or summary data. But cannot set alert, create, edit, publish, or save content.
Unlicensed	Unlicensed		Cannot sign in the Tableau Cloud.

Table 6.1 : User site roles and license types in Tableau Cloud

Note: For multiple Tableau Cloud sites, users must have a license type for each site they belong to.

To change a user's site role, follow these steps:

1. Navigate to **Users** page on Tableau Site.
2. Select the user and click on **Actions (...)** | **Site Role**, as shown in the following figure:

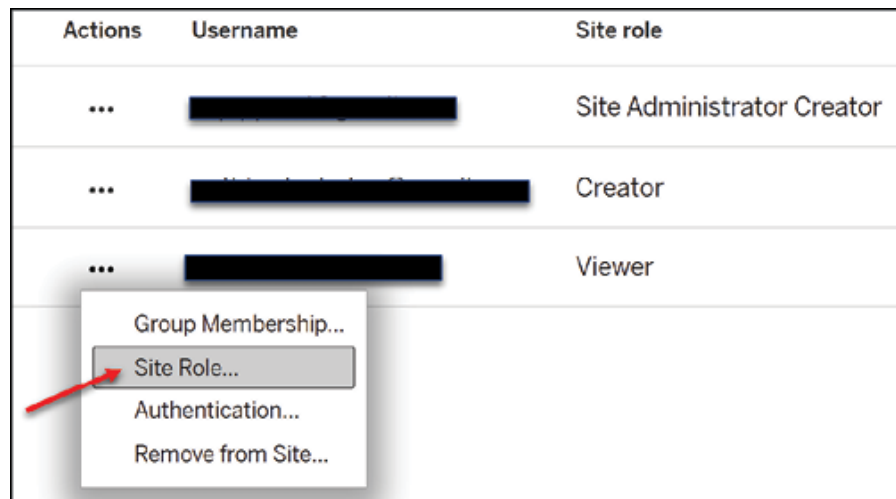


Figure 6.12: Change Site Role

3. Assign the new role to the user from the drop-down list w.r.t license type the user hold.
4. Click on **Change Site Role**:

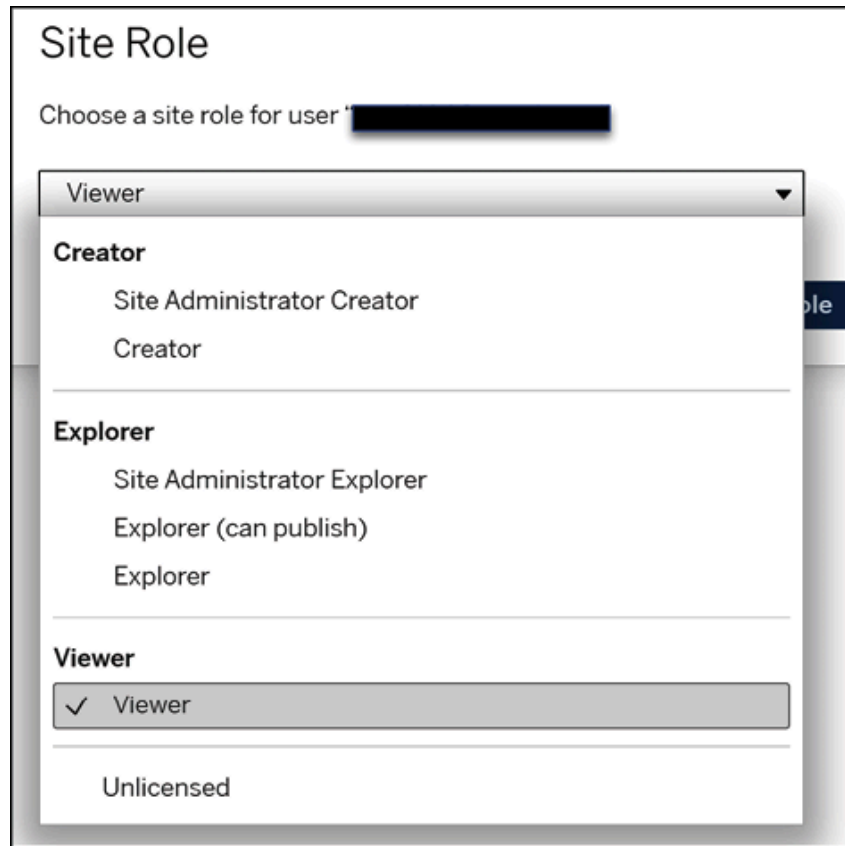


Figure 6.13: Site Role dialog-box

Removing users from the site

You can remove a user from the site, but this is possible only if that user does not own any content on the site. If the user owns the content, then set the user's site role to unlicensed.

Follow these steps to remove users from the site:

- Navigate to the **Users** page on the Tableau Cloud site.
- Select the user and then, click on **Actions (...)** | **Remove** from site as shown in the following figure.
- Click on **Remove User**.

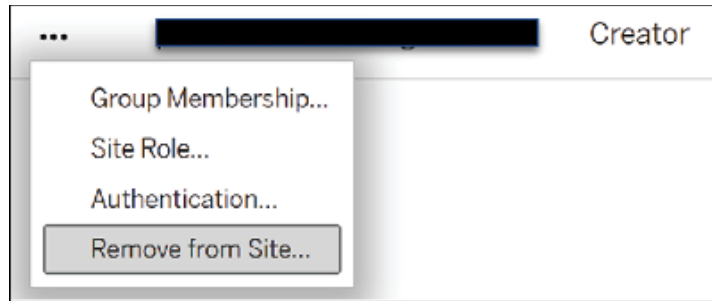


Figure 6.14: Remove user from site

Creating and adding groups

To manage multiple Tableau Cloud users in an easier way than as a site admin, it is advisable to create groups on the site.

Note: Users can belong to multiple groups.

Follow these steps to create a group:

1. Navigate to **Groups** page on Tableau site. Then, click on **Add Group**.
2. Assign the name for the Group.
3. Select **Grant role on sign-in** and then assign **minimum site role** for the group.
4. Click on **Create**, as shown in [Figure 6.15](#):

*Figure 6.15: Create Group dialog-box **Add users to groups**: Follow these steps to add users or groups:*

1. Go to the **Users** page. Then, select the users you want to add in the group.
2. Select **Actions | Group Membership...** This will open the Group Membership dialog box, as shown in the following figure:

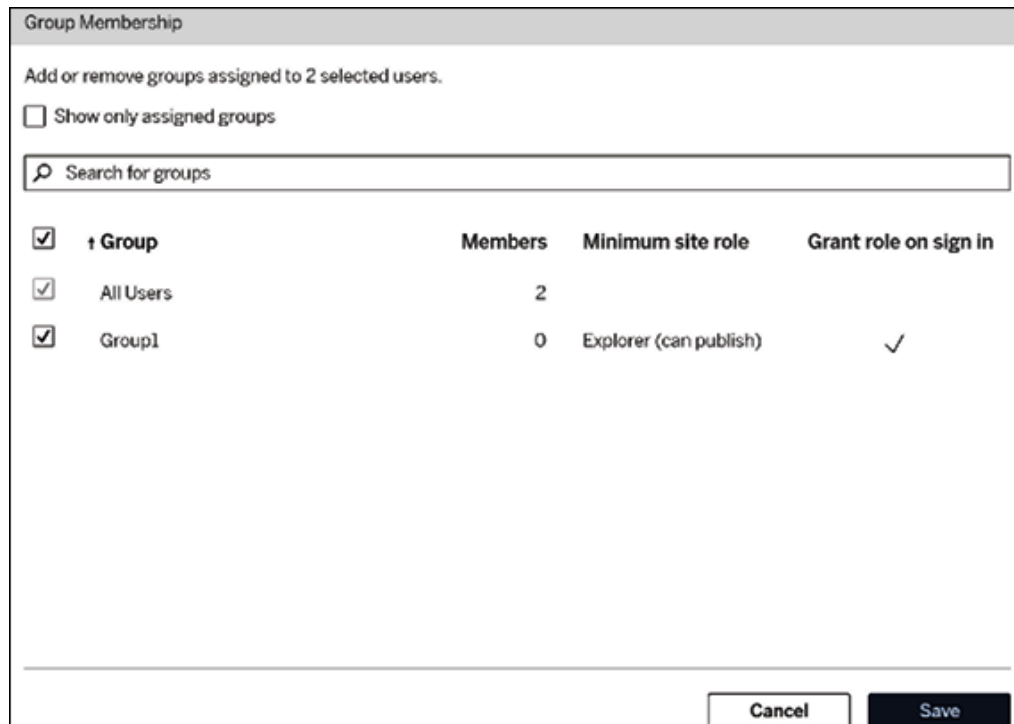


Figure 6.16: Group membership dialog-box

3. Now, select the **Group** name you want to add to those selected users.
4. Click on **Save**.

Deleting group

You can delete the custom groups anytime where users can be removed from the group, but users are not deleted from the site. However, you cannot delete All Users Groups.

1. Go to the **Groups** page. Now, select the group you want to delete.
2. Select **Actions | Delete Group...** This will prompt the dialog box to delete the selected group, as shown in the following figure.
3. Click on **Delete**.

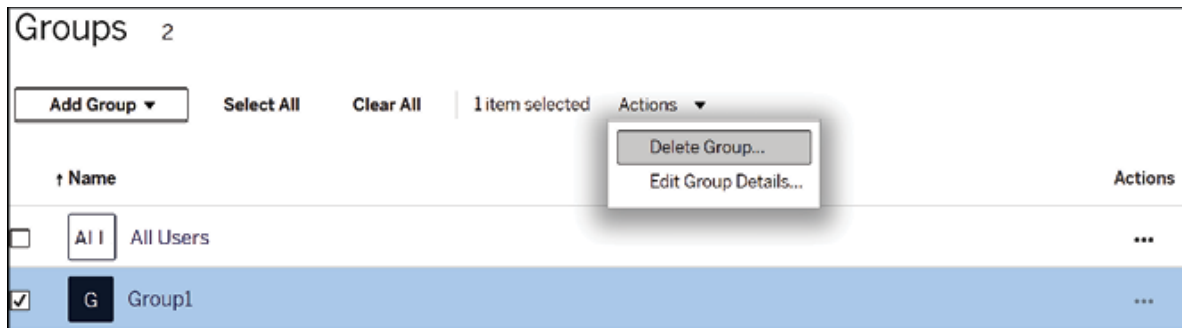


Figure 6.17: Delete group [Set permissions](#)

It is crucial to understand the permission rules to set at project level, at content level, or set when content publishing from Tableau Desktop. Let us explore each level.

- **Project level permissions:** Follow these steps to set permission at the project level:

1. Open your Cloud site and navigate to the **Project**.
2. Click on **More Actions (...)** and select **Permissions**, as shown in [Figure 6.18](#):

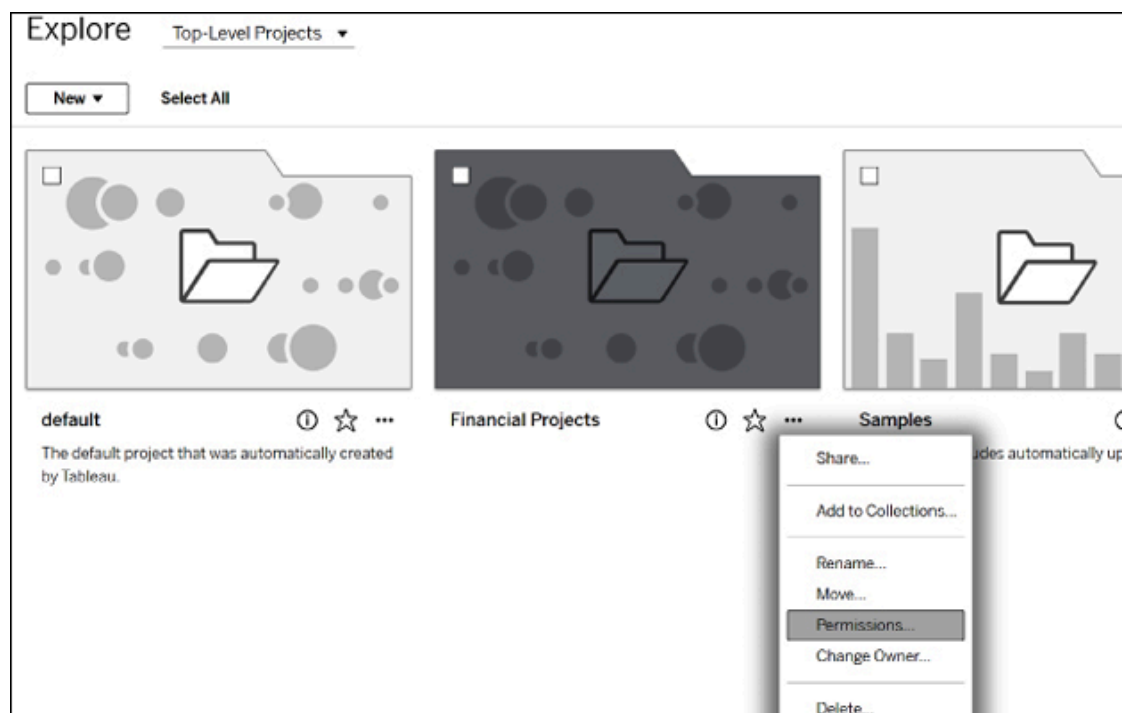


Figure 6.18: Project level

3. This opens the **Permissions** dialog box. This dialog box has two main areas: Permission Rules and Effective Permissions, as shown in the

following figure:

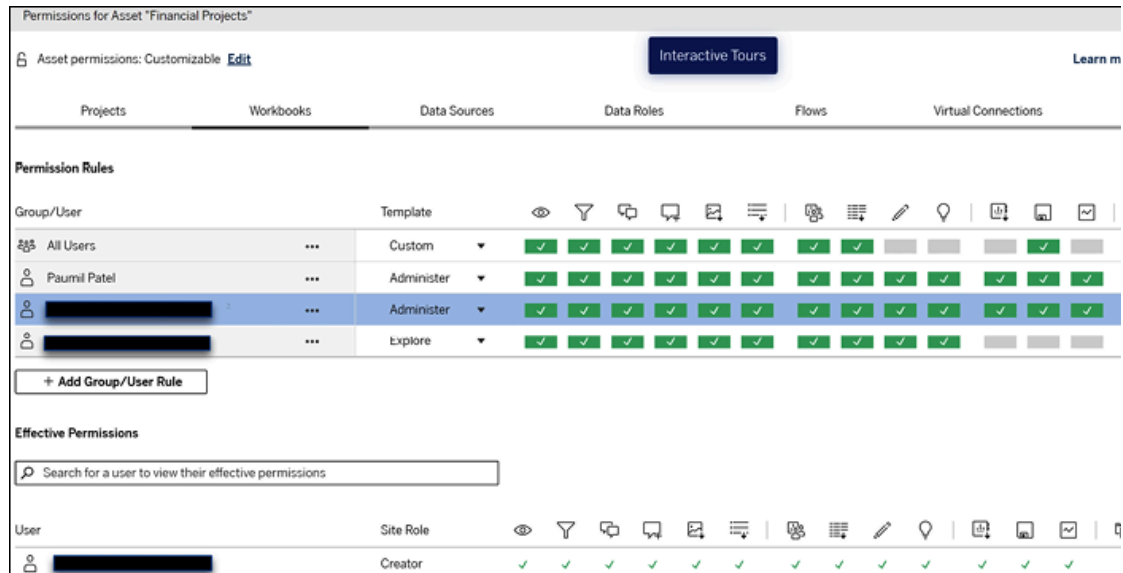


Figure 6.19: Permission rules at project level

4. Here, you can select the content type at which you want to set the permission rules. Each content type has a separate tab such as workbooks, data sources, projects, etc.
5. To modify the existing permissions rules, select the appropriate tab at what level you want to set the permissions for the users or group of users.
6. To create a new rule, click on **+Add Group/User Rules** and select or search the group or user and then choose the template from the drop-down list or create your own custom rules by clicking the capabilities.
7. When finish making updates, click on **Save**.
8. If **None** is selected in the template, then it will delete that rule.

Note: Only the site administrators, project owners, and project leaders are allowed to set the permission rules at the project level.

- **Content level permissions:** The Permissions dialog box at the Project level contains tabs for each content type. While creating a permission rule at the project level, you make sure to set the permissions at each content type. On leaving user permissions or a set of capabilities unspecified, then capabilities will be considered as denied access.

Note: The permission rules are set as a default at the project level for the contents saved under that project and any nested projects.

- **Publishing level permissions:** There are some specific scenarios under which the action needs to be taken in the combination of permission capabilities and site roles. Except explorer and viewer site roles are allowed to publish, save a copy (download), or overwrite (edit) the content on site.

Manage content ownership

When you publish any content such as workbooks or data sources on the Tableau Cloud site or even when you create a project on the Tableau Cloud site, then you are called an owner of those contents and projects.

Note: Only the administrator, the content owner, or project leader with the appropriate site role can change the ownership of the content.

Here are some considerations for changing content ownership:

- If you want to remove any user from the Tableau Cloud. Make sure you re-assign any content that the user holds to someone else. Otherwise, that user's site role will be set to unlicensed, and then the administrator needs to intervene and take certain actions on that content.
- It is recommended not to assign ownership to viewers or explorer site roles. Otherwise, the metrics schedule will be suspended. Assign the ownership to creators or explorer(publish) site roles that are required to overwrite, refresh, or delete metrics.
- When you change the ownership of the content, such as workbooks or data sources, the embedded credentials will be deleted. Hence, for flows, embedded credentials need to be preserved when changing the ownership. You must update the embedded credential by editing the connection information on Tableau Cloud. Alternatively, the new owner must download the content data sources or workbooks and manually update the embedded credentials by re-publishing the content.

Now let us look at how we can change the owner of the content resource. Steps to change the owner of the content:

1. Navigate to the **Explore** page on your Tableau Cloud site.
2. Now, navigate and select the content you wish to assign to someone else.

3. If you want to re-assign all the contents within the same project, then navigate to the **Project** page and select that project.
4. If you want to re-assign those contents with the same owner, then go to the **Users Page** and select the owner.
5. Select the contents you want to re-assign and then select **Actions (...) | Change Owner**. This will open the **Change Owner** dialog box, as shown in the following figure:

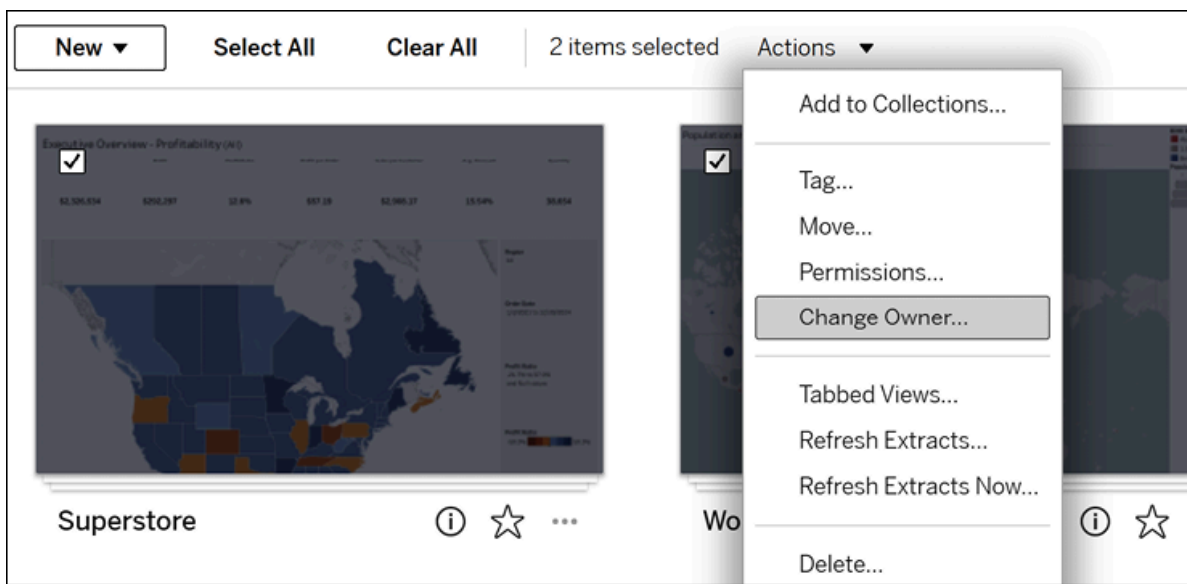


Figure 6.20: Change content ownership

6. Now, select the owner to whom you want to assign the ownership of the contents.
7. Click on **Change Owner**.

Note: After the ownership is re-assigned, the original owner no longer has ownership of the contents, and project permission and site role determine the access to those contents on the Tableau Cloud site.

Upload a custom logo for your site

To embrace your site header with brand identity, you can change the Tableau Cloud Site Logo to your custom logo.

Follow these steps to upload a custom logo on your site:

1. Sign in to the Tableau Cloud Site as a site administrator.
2. Navigate to the **Settings** page.

3. Select the **General** tab. Then, scroll down to the **Site Logo** section.
4. Click on **Select File**. This will open the dialog box. Now, browse your logo from the device. Then, click **OK**.
5. (Optional) If you want to revert the settings. Then click the **Revert** button to go back to the last saved settings.

Or,

Click **Reset to Default** to show the original logo.

Finally, click on **Save**, as shown in *Figure 6.21*:

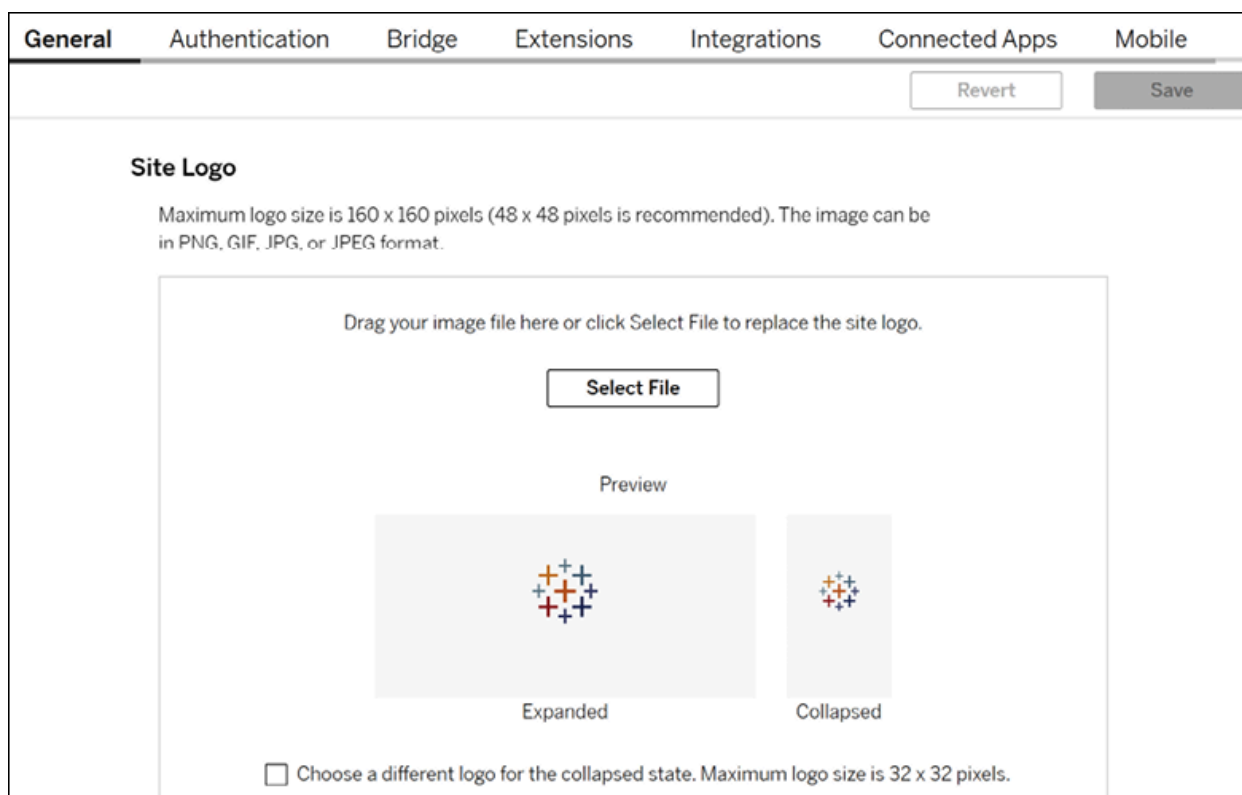
The screenshot shows the 'General' tab of the Tableau Cloud settings. The 'Site Logo' section is active, displaying instructions on logo size and format. A 'Select File' button is present. Below it, a 'Preview' section shows two versions of the logo: 'Expanded' and 'Collapsed'. At the bottom, there is a checkbox to 'Choose a different logo for the collapsed state'.

Figure 6.21: Upload site logo [Admin insights or site status](#)

Admin insights/site status are pre-built admin views in your Tableau Cloud site workbook. They help you monitor your site's health and activity and answer a range of questions.

Let us explore admin insights in detail. Admin insights will help in providing the following:

- Site traffic, including site activity and site performance views, disk space usage, extract performance, and so on.

- User roles and sign-in activity
- Publishing related activities

There is a pre-built dashboard that helps in monitoring site activity as follows:

- **Traffic to Bridge connected to data sources:** Analyzing the data sources via Bridge connected and its usages.
- **Background tasks for extracts:** Analyzing pending or completed extracts and schedule extract refreshes.
- **Bridge extracts:** Analyzing pending or completed extracts and schedule extract refreshes by Bridge.
- **Background tasks for non-extracts:** Analyzing the non-extracts pending background tasks.
- **Flow performance history:** Analyzing the flow performance.
- **Stats for space usage:** Space utilization by publishing contents—extract or live connections.
- **Stale content:** Analyzing content that has not been used or reviewed for a long time.
- **Data quality warning history:** Reviewing the data quality warnings in use, edited or removed.

You, as a site admin, can create your own admin insights data views by following these steps: To connect to admin insights data, follow these steps:

1. Sign in to Tableau Cloud site and navigate to **Admin Insights**.

Or,

Navigate to **Site Status**. Select **Create | Workbook** and then select the **Data Source** to get started with.

2. Now, you can start creating your custom views. For example:

- Who are the active users? How many are active users?
- Who has not signed in to the site for more than 90 days?

Conclusion

In this chapter, we have learned about site administrators and tasks, how to manage users, groups, site roles, permissions, etc. In the next chapter, we will explore various

methods of publishing content and refreshing extracts on Tableau Cloud.

Exercise

1. Do explorers have permission to publish the contents on Tableau Cloud? True or false
2. Can project leaders add users to the site? True or false
3. Does the dynamic user filter method help to automate the column-level security? True or false
4. Can viewers allowed to save the Tableau Cloud report? True or false
5. Can the user authentication type not be changed? True or false
6. Does changing content ownership revoke the user's access permission to that content? True or false
7. Can the same users belong to two or more groups at the same time? True or false
8. Can a custom template be created on Tableau Cloud to provide users with content permissions? True or false

Questions

1. How to create groups in Tableau Cloud?
2. How do you create a dynamic user filter in row-level security?
3. How to manage the content ownership on Tableau Cloud?

Answers

1. True
2. True
3. False
4. False
5. False
6. True
7. True

8. True

Points to remember

Here are some key takeaways from this chapter:

- Tableau Cloud enables organizations to manage content through sites, projects, and content.
- A site is a collection of contents, including projects, workbooks, data sources, users, or groups of users.
- A site administrator is a person who creates and maintains the framework that allows Tableau Desktop users to share, publish, manage, and connect to data sources and workbooks. A site administrator manages a lot of responsibilities on sites.
- Restricting the data at the row level is referred to as RLS. Tableau offers two methods for RLS:
 - User-based filtering manually mapping values
 - Dynamic user filtering using a security field
- Permissions can be established for projects, users, groups, or assets which are already existing.
- The process of giving user permission begins with adding the user to the Tableau Cloud site if it does not exist.
- The license type is associated with the user. The site role you assign to the user determines the license type they require to attain the maximum capabilities on the site.
- You can add, update, or remove the users or groups from the sites.
- The permission rules are set at the project level, at the content level, or set content publishing from Tableau Desktop.
- Only the administrator, the content owner, or the project leader with the appropriate site role can change the ownership of the content. After the ownership is re-assigned, the original owner no longer has the ownership of that content, and project permission and site role determine the access to those contents on the Tableau Cloud site.
- You can enhance your custom Tableau site logo by following a few simple steps on the Tableau Cloud site.

- Admin insights/site status is a pre-built admin views workbook for your Tableau Cloud site. It is helpful for monitoring its health and activity.

CHAPTER 7

Publishing and Scheduling

The future of publishing is about having connections to readers and the knowledge of what those readers want - Seth Godin [Introduction](#)

In the previous chapter, readers learned how to manage users and groups, manage the site data, and control content access on the web. In this chapter, they will explore various methods to publish their workbooks and data sources with other users and setup extract refreshes.

Structure

In this chapter, we will cover the following topics:

- Tableau Bridge
- Publish a workbook
- Publish a data source
- Data refreshes and scheduling data extract refreshes
- Autosave workbook

Objectives

By the end of this chapter, you will learn how to publish content (including workbooks and data sources) on your Tableau Cloud site, schedule the data extract refreshes and autosave a workbook. After reading this chapter, you will

gain substantial knowledge on how to share and keep your content fresh on Tableau Cloud.

Before we look at publishing and scheduling extracts concepts, it is crucial to understand Tableau Bridge, a software that helps to keep your data fresh. Let us cover this topic in detail.

Tableau Bridge

Tableau Bridge runs on the local computer/machine in your network. Hence, it facilitates secure communication for private and virtual data sources to connect with Tableau Cloud. Tableau Bridge is a great option to keep your data fresh. It also comes in handy in case you are facing difficulty connecting directly to Tableau Cloud with your data sources. [Figure 7.1](#) defines how the Tableau Bridge in a secure way communicates with Tableau Cloud:

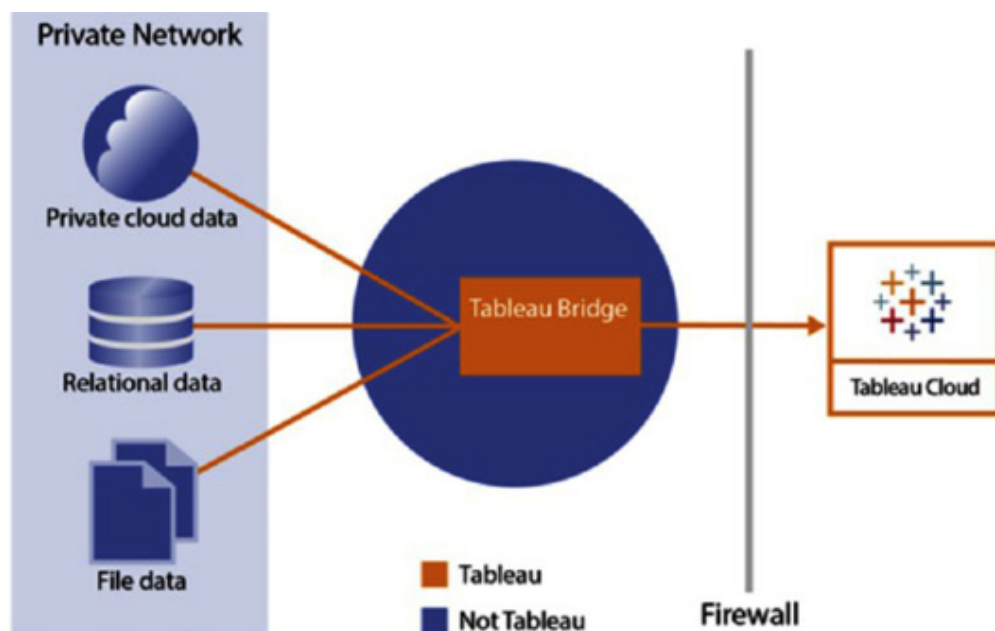


Figure 7.1: Tableau Bridge security from [Tableau.com](https://tableau.com)

The Tableau Bridge communicates with Tableau Cloud and creates an encrypted connection to enable connectivity between private or virtual data, behind a firewall and on your Tableau site.

Note: Only the site administrators, creators, and explorers (can publish roles) have the authority to use Tableau Bridge and client software to share and manage their own content on Tableau Cloud site.

Follow these steps to install Tableau Bridge on the machine:

1. Download the latest version of the installer from the download page on the Tableau website.

Or,

In Tableau Desktop, go to **Server | Install Tableau Bridge Client**. This will help you navigate to the download page on the Tableau website.

2. Run the Windows installer on your system.
3. Accept the checkbox for license agreement, then click on **Install**.
4. Click on **Yes**.
5. Allow your computer to **Restart to complete the installation** process.
6. Now, open Tableau Bridge software on your machine.
7. Log in with your Tableau Cloud credentials. Refer to the following figure:



Figure 7.2: Tableau Bridge Sign-In Next, we will explore how to use Bridge to keep the data fresh in the data refreshes and scheduling data extracts refreshes, section.

Publish a workbook

Tableau Cloud enables organizations to share and publish their contents. In other words, a creator or site administrator can publish their workbooks with a few simple clicks, and others can view, edit, or interact with them if the users have cloud permissions.

There are two ways to publish a workbook. Let us explore them individually. Here, we will publish a workbook titled *Mental Disorder by Sex* from Tableau Desktop to Tableau Cloud.

Publish the workbook through Tableau Desktop

Follow these steps to publish the workbook through Tableau Desktop version:

1. Go to **File** in the toolbar and select **Share**.

Or,

Click the **Server** in the toolbar and select **Publish Workbook**.

Or,

Click the **Share**  button in the toolbar.

2. Click on **Quick Connect** Tableau Online. Refer to the following figure:

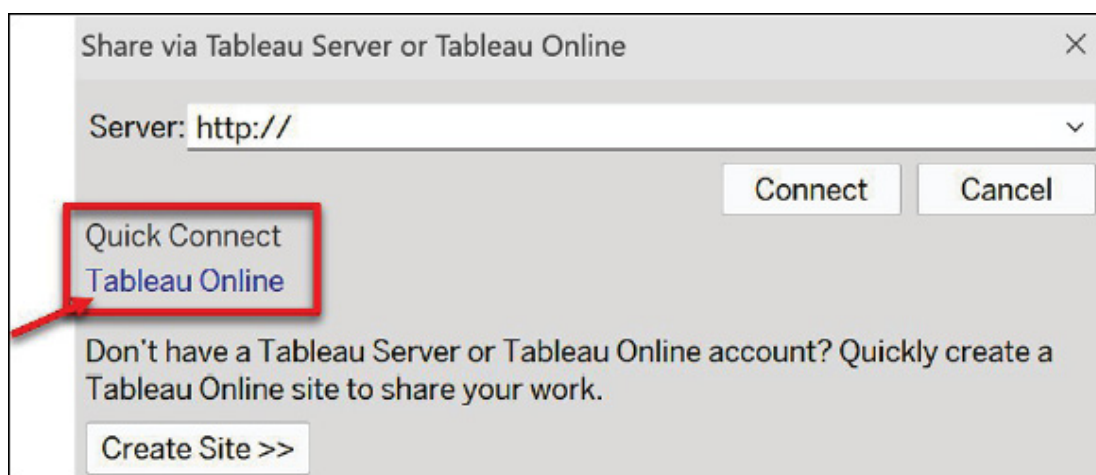


Figure 7.3: Share dialog box

3. If you do not have a site yet, create one on Tableau Cloud. Otherwise, sign-in to the site on Tableau Cloud using your site credentials.

4. Now, select the **Project** from the drop-down option where to publish in the **Publish Workbook** dialog box, as shown in [Figure 7.4](#):

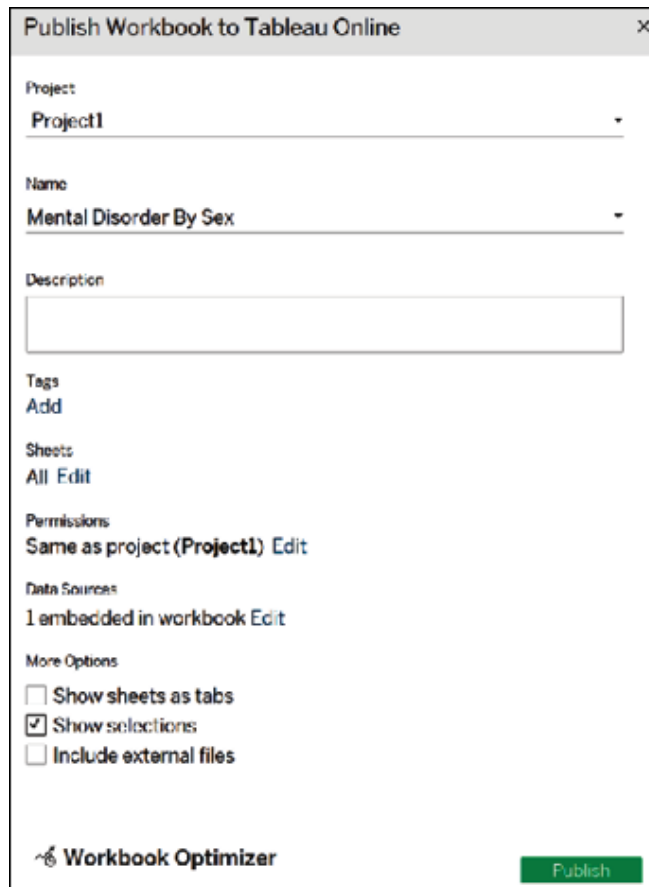


Figure 7.4: Publish Workbook dialog box

Follow these steps to publish workbook via Tableau Desktop:

1. Assign a name to your workbook or publish it in the existing one.
2. Provide a description of the workbook. (optional)
3. In the next step, (optional) search Tags.

Add tags that contains space, comma, or put tags in the quotation marks (**"Mental Disorder"**). Hence, users can easily relate to the workbook when they browse the site.

4. Click **Edit** on **Sheets** to select the sheets or dashboards from the list to publish. By default, all the sheets and dashboards will be published.

5. Now, for **Permissions**, click on **Edit** to give access to the users or keep as the default project settings.
6. Generally, a site administrator manages the permissions on the cloud.
7. For data sources, click on **Edit** if you want to change the data source authentication and publish type.
8. By default, publish type will be **Embedded in workbook** where data is embedded in the workbook along with the password. Refer to the following figure:

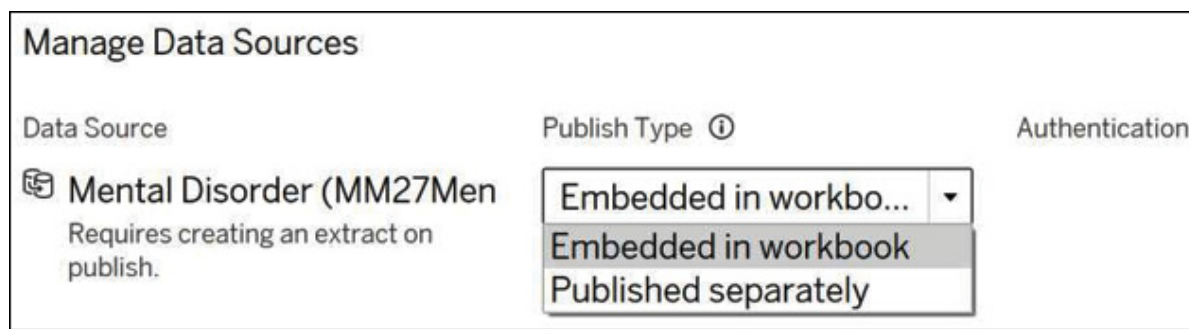


Figure 7.5: Manage Data Sources in publish workbook

Follow these steps to schedule the extract refresh:

1. For more options, configure the publishing options.
2. Select **Show Sheets** as tab, if you want user to navigate tab based.
3. By default, **Show Selections** are selected. This option will highlight the view when user opens the workbook.
4. Select **Include external files**, if your views contain information that is not available to cloud or users then use this option to publish.
5. Here, you can include Excel, CSV or other data source files local to your computer.
6. Click on **Publish**.
7. Set up the extract refresh schedule to the extract to published. (Optional, this will keep refreshing the extract on selected interval, without any manual interventions.)

Publish the workbook through Tableau Cloud

Now, let us explore another method of publishing the same workbook directly through the Tableau Cloud site (web version). Follow these steps to publish the workbook through Tableau Cloud (web version):

1. Select **New** from your site's homepage.
2. Now, choose **Upload Workbook** from the menu as shown in the following figure:

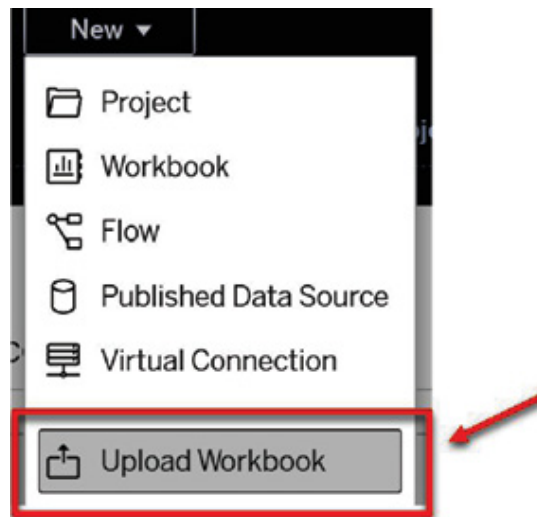


Figure 7.6: Upload Workbook directly on site Or,

Click on **Upload Workbook** button on the site homepage, as shown in [Figure 7.7](#):

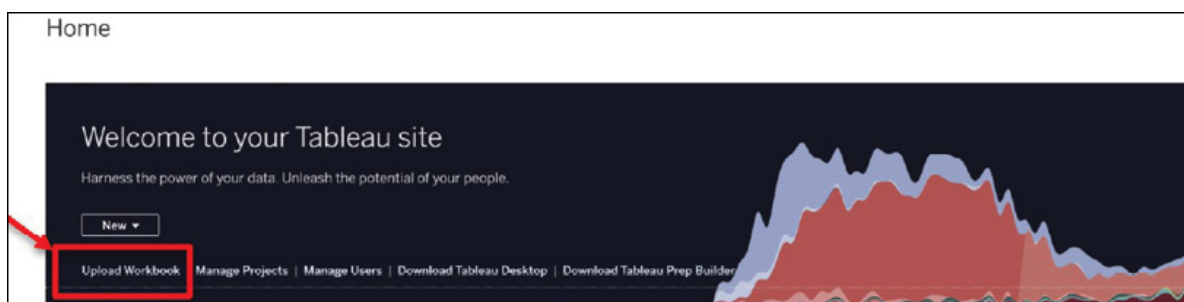


Figure 7.7: Upload Workbook directly from Tableau Cloud site

3. This will open the **Upload Workbook** dialog box, as shown in [Figure 7.8](#).
4. Choose a **File**, browse your workbook from the desktop or drag or drop a file.
5. Now, assign a name to the workbook under **Name** textbox.

6. For **Project**, select the project to publish.
7. For **Description**—provide the information to the workbook (Optional).
8. By default, All Sheets are selected. Now, configure the publishing option to navigate each sheet via selecting on tabs if you want to give user the flexibility to navigate each sheet via selecting on tabs.
9. Now, click on **Upload**. Refer to the following figure:

Upload Workbook

Drag and drop a file
or
Choose a file

Upload a workbook from a TWB or TWBX file. Maximum file size: 50 MB.

Name

Project
default

Description (Optional)

0 / 4,000

Sheets
All

☐ Show sheets as tabs

Cancel Upload

Figure 7.8: Upload Workbook dialog box [Publish a data source](#)

When you want to make a data source available to other users, you can publish it via Tableau Cloud or Tableau Desktop. If the workbook is embedded with the data source, then it can be saved. However, if the data source needs to be published separately then follow these steps. There are *two* ways to publish the data source like workbook publishing: [Publish data source through Tableau Desktop](#)

Follow these steps to publish the data source through Tableau Desktop:

1. Go to **Server | Publish Data Source** on the toolbar and select the data file as shown in [Figure 7.9](#):

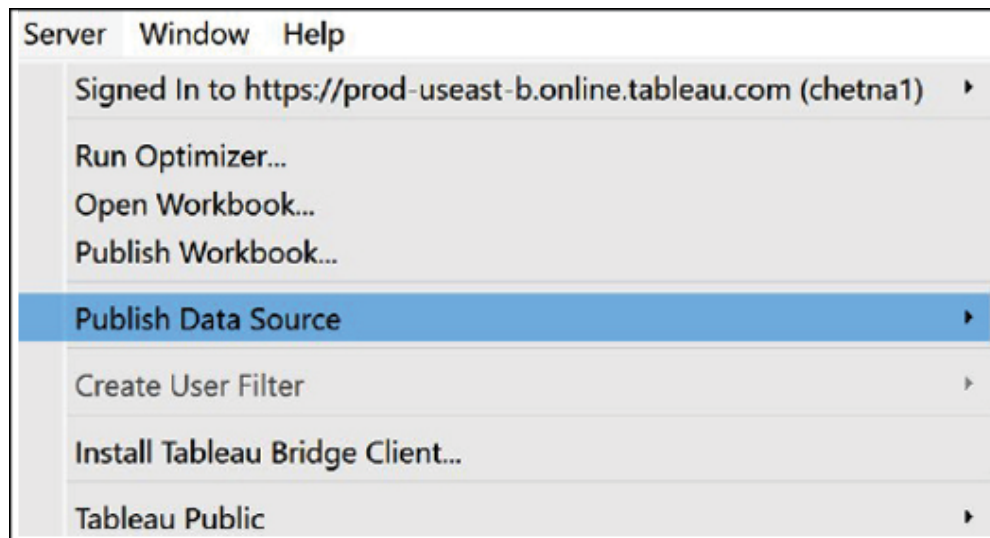


Figure 7.9: Publish Data Source from Tableau Desktop This will open **Publish Data Source** to **Tableau Online** dialog box, as shown in [Figure 7.10](#):

Publish Data Source to Tableau Online ×

Project
Default ▾

Name
Mental Disorder (MM27MentalHealth) ▾

Description

Tags
[Add](#)

Permissions
Same as project (**Default**) [Edit](#)

Authentication
Refresh not enabled [Edit](#)

Tableau Bridge required for on-premises data
If Tableau Online can't connect directly to this data source, it will use a Tableau Bridge client to keep this data fresh.

More Options
☐ Update workbook to use the published data source

ⓘ Tableau Cloud will temporarily access the credentials provided for [] to confirm it can maintain a live data connection.

[Publish](#)

Figure 7.10: Publish Data Source dialog box

2. For **Project**, select the project you want to publish.
3. For **Name**, assign a name to the data source.
4. For **Description**, provide the information and description of the data source.
5. For **Tags**, assign appropriate tags to the data source that will help others find the data source.
 - a. Separate Tags either by using a comma, space, or putting tags in quotation marks. (Mental Disorder Data)

6. For **Permissions**, either accept the default settings or provide access permissions to users.

Note: A site administration manages the user permissions on the cloud.

7. For **Authentication**, provide credentials to access the data and select the type to authenticate the data source. Here, you can specify when the data is published and how the authentication should be handled.
 - a. If **Allow refresh access** is selected, then the data source will be refreshed after publishing.
 - b. If **Refresh not enabled** is selected, then the data source refresh will be disabled after publishing. Refer to the following figure:

Data Source	Connection	Authentication	Extract	Username
Mental Disorder (MM27MentalHealth)	MM27MentalHealth	Allow refresh access Refresh not enabled Allow refresh access	Yes	

Figure 7.11: Authentication options

8. Under more **Options**, you can select whether to continue using the local data source or update the workbook connection with the new published data source connection, to use instead of local data source.
9. Click on **Publish**.

This method of publishing a data source works best. However, you still have one more option to publish on-premises data (Tableau Cloud Only).

Publish data source directly from Tableau Cloud

Follow these steps to publish the data sources directly from Tableau Cloud (web-version):

1. Go to Site Home Page. Then click **New Publish Data Source** as shown in *Figure 7.12*:

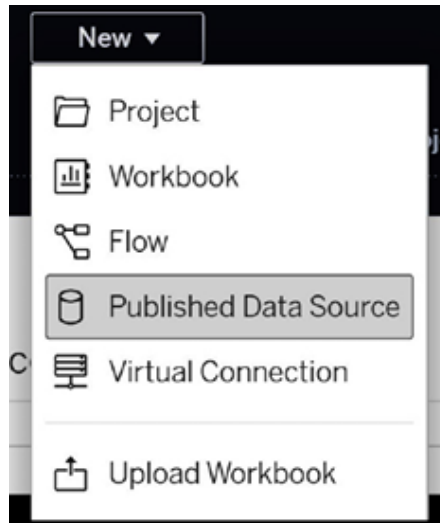


Figure 7.12: Publish On-premises data (Tableau Cloud Only) Or,

Go to **Explore**. And then click **New | Published Data Source**.

Or,

Go to **Explore**, select **Project** folder. Then, click **New | Published Data Source**.

This will prompt Connect to Data dialog box as shown in [Figure 7.13](#):

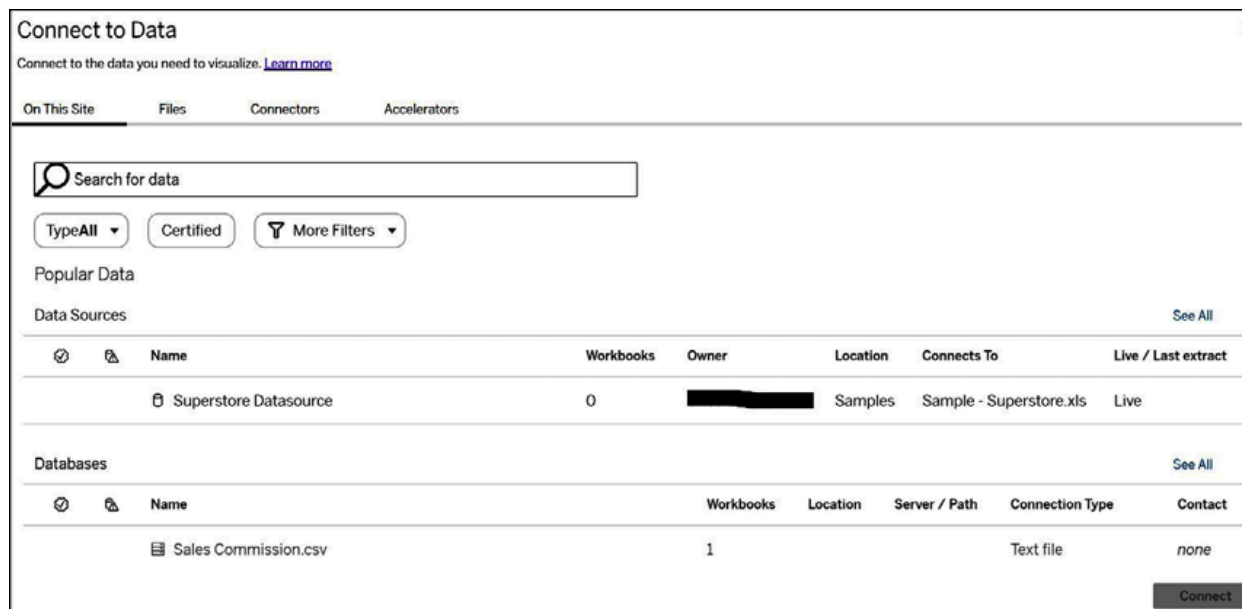


Figure 7.13: Connect to Data dialog box Follow the steps to connect data to the web as discussed in [Chapter 2](#) under the topic: Connect data to the web.

1. Once you connect your data file, click on **Publish As...**

2. By default, the data source will be published as **Live Connection**. However, to publish the data extract, click on **Data Source | Select Extract**, as shown in [Figure 7.14](#):
3. Then, choose **Create Extract**.
4. Select the sheet tab. Now, click **Publish As**.

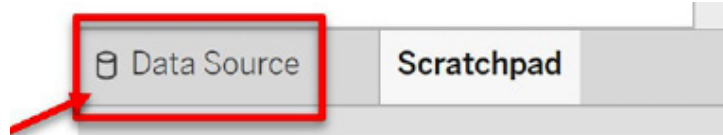


Figure 7.14: Data Source

5. Now, assign a name to the data source and choose the project to save it. Then, publish the data file as shown in [Figure 7.15](#):
6. Click on **Publish**.

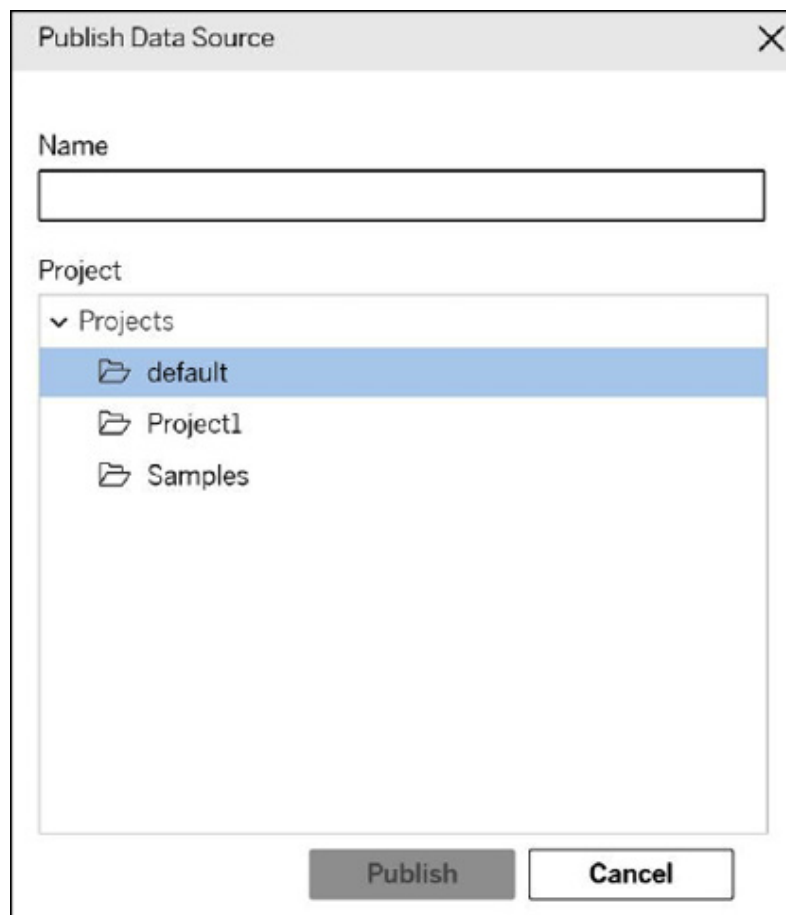


Figure 7.15: Publish Data Source [Publish the data sources from Tableau Cloud using Tableau Bridge](#)

Follow these steps to publish the data sources from Tableau Desktop using Tableau Bridge:

1. Go to **Server | Start Tableau Bridge Client** on Toolbar.
2. Once your machine is connected to Tableau Bridge, select **Server Publish Data Source** to begin the publishing process. If you have not signed in to your tableau cloud site, then, it will be prompted.
3. Now, follow the *steps 2-9* as discussed in the topic: Publish the data source through Tableau Desktop.

Note*: After publishing the data source using bridge on site, Schedule Extract Refresh dialog box will prompt.

Edit the data source

The option **Edit Data Source** can be used to create a new data source or edit the existing published data source. Here, you can make necessary changes in the schema, or create a join from the data source, then follow the steps mentioned to edit and test the data source change:

1. Navigate to the **Data Source Page**. Then, select the data source you want to make changes.
2. Click on **Edit Data Source**.
3. Then, click **Data Source** to make necessary changes or edit the schemas or joins.
4. Now, click on the **Scratchpad** and save the published data source.
5. Click on **Publish**.

Roll back changes

If you want to revert to the last published data source version follow these steps:

1. Select the data source you want to revert to.
2. Then, click on **File**.
3. Now, choose **Revert to Published**. This will revert to that data source to its latest published version.

Data refreshes and scheduling data extracts refreshes

In Tableau Cloud, you can manually refresh the data extracts as well as schedule data refreshes. Let us explore them individually in the upcoming sub-sections.

Manual data refresh

We can start refreshing the extract data directly on Tableau Cloud (web-version). Follow these steps to manually run the data refresh:

1. Go to the **Data Source Page** on Tableau Cloud site where the data source is published.
2. Select the data source and then click on **More Options (...)** next to it.
3. Now, select **Refresh Extracts Now** from the menu.
4. Now, select **Full Refresh** as shown in [Figure 7.16](#):

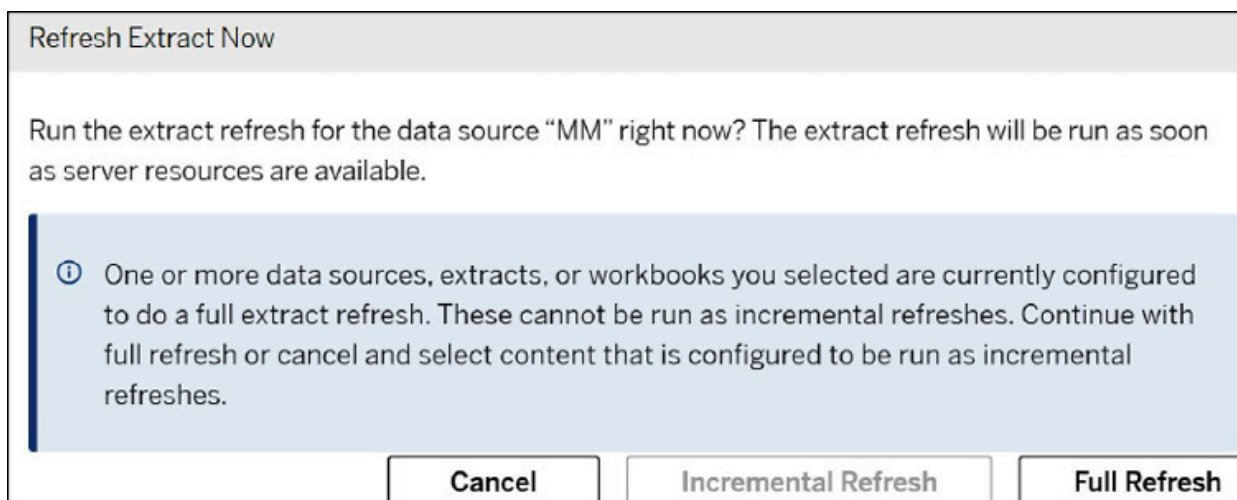


Figure 7.16: Refresh Extract Now [Schedule data refresh on Tableau Cloud](#)

In Tableau Cloud, you can schedule a refresh for the cloud-hosted data extracts such as, SQL based cloud platforms, [Salesforce.com](https://www.salesforce.com), Google Analytics or other data hosted on cloud platforms like Amazon RDS, Microsoft Azure, or Google Cloud Platform.

Note:** In case you do not want to use live connection to Microsoft Excel, SQL based cloud platforms, or other cloud platforms then you must use Tableau Bridge.

Let us learn more about creating a refresh schedule:

1. Go to the **Data Source** page on Tableau Cloud site where the data source is published.
2. Select the data source and then, click on **More Options (...)** next to it. Now, select **Refresh Extracts...** from the menu.
3. For **Refresh Type**, select the type of refresh you want to perform as shown in Figure 7.17.
4. A full refresh is selected by default. The incremental refresh is only enabled when configured in Tableau Desktop before publishing the extract.
5. Full refresh will reload all the data in the dataset/data source. However, Incremental refresh updates the new or recently changed data since the last refresh.
6. Refresh **Frequency**, set up the refresh frequency, to run/refresh extract at desired intervals.
7. You need to select whether the refresh is hourly, daily, weekly, or monthly. Then depending on the repeat frequency, select the time, day, and date intervals.
8. Now, click on **Create**:

Create Extract Refresh

Refresh Type

☒ Full Refresh
☐ Incremental Refresh

One or more data sources, extracts, or workbooks you selected are currently configured to do a full extract refresh. These cannot be run as incremental refreshes. Continue with full refresh or cancel and select content that is configured to be run as incremental refreshes.

Refresh Frequency

Every Fri, at 10:05 PM

Repeats: Every:

At:

On:

Time zone: **(UTC-05:00) America/New_York**

Cancel **Create**

Figure 7.17: Create Extract Refresh dialog box [Schedule data refresh using Tableau Bridge on Tableau Cloud](#)

At the beginning of this chapter, we covered the concept of Tableau Bridge. Here, we will see how to use bridge to keep our data refreshed.

Follow these steps to refresh data on Tableau Cloud via Tableau Bridge:

1. Go to **Data Source** page on Tableau Cloud site where the data source is published.
2. Select the data source and then, click on **More Options (...)** next to it.
3. Now, select **Refresh Extracts** from the menu.
4. Follow *steps 3-5* as discussed in topic: **Schedule Data Refresh** on Tableau Cloud to successfully setup the data extract refresh.

Or,

Go to **Project Page | Select Data Source File | Extract Refreshes** tab.

5. Then click on the **New Extract Refreshes** button. Now, follow *steps 3-5* as discussed in **Schedule Data Refresh** on Tableau Cloud to schedule the refreshes.

Or,

From **More Options** on **Data Source** page, select **Schedule with Bridge (Legacy)** from the menu and follow the next steps to configure a schedule for refresh.

6. Now, select the computer through which you want to connect as shown in [Figure 7.18](#):
7. Next, set up the refresh frequency. Select the date, time, day intervals and refresh frequency: hourly, weekly, daily, or monthly.
8. Click on **Create**. This will create a secured connection between your machine and tableau cloud site for the data source. Refer to the following figure:

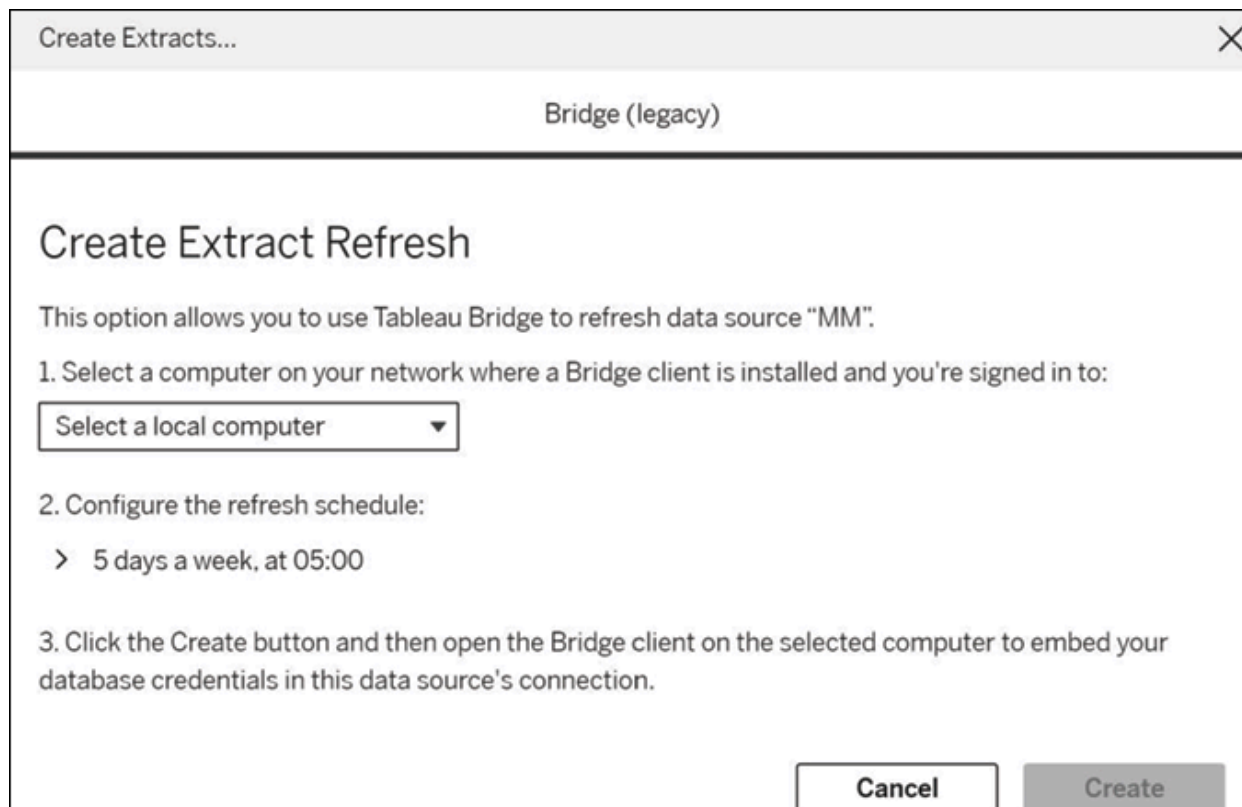


Figure 7.18: Create Extract Refresh Dialog Box using Tableau Bridge

Note: For a successful data refresh, keep your Tableau Desktop connected with Tableau Bridge. For this, select Server | Start Tableau Bridge Client. Once your machine connected with Bridge, you can proceed for data refresh.

Add or update an existing schedule

For instance, you want to add another refresh schedule on an existing one or wish to update the scheduled refresh for whatever reason. Then, follow these steps:

1. Sign in to your Tableau Cloud site and then navigate to **Data Source** page.
2. Now, select the data source and then click on **Extract Refresh** tab.
3. To **Add a new schedule**, click on **New Extract Refresh** button and configure the schedule for the refresh.
4. To Update the existing schedule, select the checkbox next to the schedule you want to update, then click on the **Actions** menu, select the **Change Frequency**, and reconfigure the schedule for the refresh.
5. Important points:

- a. Tableau Bridge supports data sources embedded in workbooks except live connections and extract refreshes for Excel, Text, Statistical (sas7), and CSV data files.
- b. Tableau Bridge can be used to refresh the extracts if extracts for the data sources and virtual connections relate to the **Private Network Type** instead of the Tableau Cloud network type.
- c. To successfully schedule the refresh, it is better to embed the database credentials in the connection through bridge client.
- d. Support for Schedule with Bridge (Legacy) will be decommissioned in future release.
- e. Hence, it is highly recommended to use **Online Refresh** schedule.

Autosave workbook

In Tableau Cloud, while you work on a new or existing workbook, **Autosave** functionality saves the changes you make to it. In other words, as soon as you start working on the workbook, autosave saves the changes to a private draft version of your workbook. Once you are ready to share the updated version with others, click on the **Publish** button. It will replace the last published version of the workbook with the revised draft.

To enable Autosave, you need specific permissions. Users with web-editing capabilities at the workbook level, such as Creators and Explorers (who can publish), can turn on Auto-save. However, users with view-level access cannot enable this feature.

Note: Autosave is available for workbooks which are smaller than 5MB in file size. For larger workbook files, you need to manually save the workbook changes and publish them.

In case there are multiple users editing the same workbook (more than one user works on the same workbook simultaneously). In such situations, each user can start editing the workbook from the last published version and save their own version in private draft. If another user publishes their draft before you publish your own version, your changes will overwrite theirs. However, you will receive a notification when you begin editing the workbook version from another user's

published version. Here, you will be offered a choice either to continue working on your private draft or discard the draft or edit the current revised version.

Let us say you want to check when the workbook was last published. Then, follow these steps to check the workbook **Revision History**:

1. Go to **Workbook**. From the actions menu.
2. (...). Click **Revision History** as shown in *Figure 7.19*:

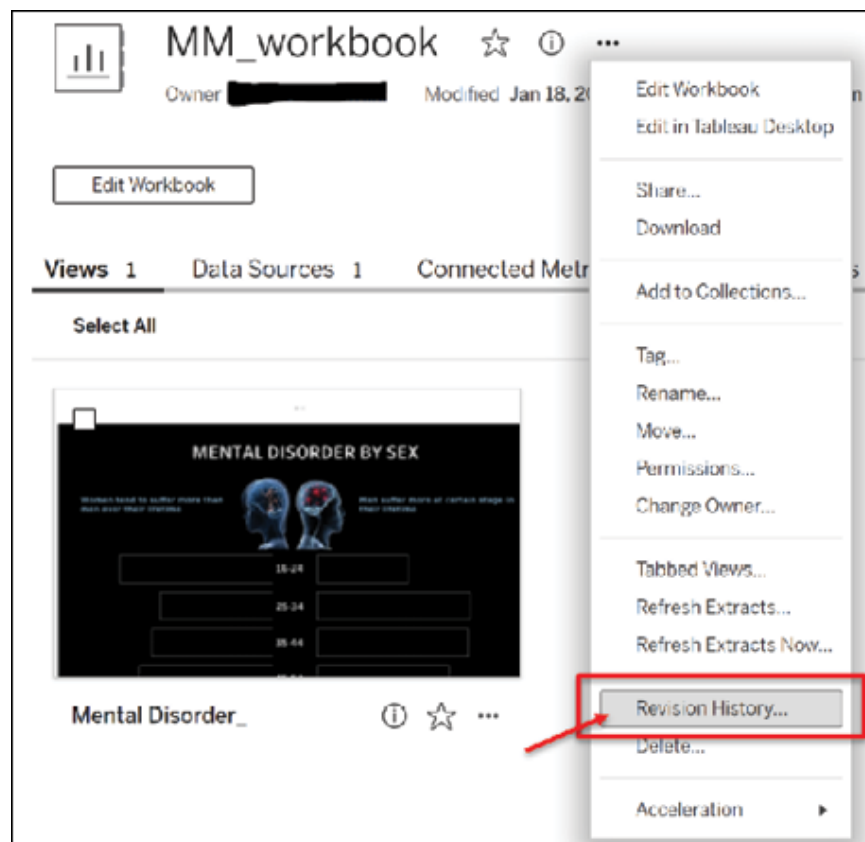


Figure 7.19: Revision History

3. Now, click the actions menu on revision history, as shown in *Figure 7.20*.
4. To preview, click on **Preview** the previously published version.
5. To download, click on **Download** the published version.
6. To delete, click on **Delete** the previous versions.
7. To restore the specific version, click on the **Restore** button to restore the specific published version.

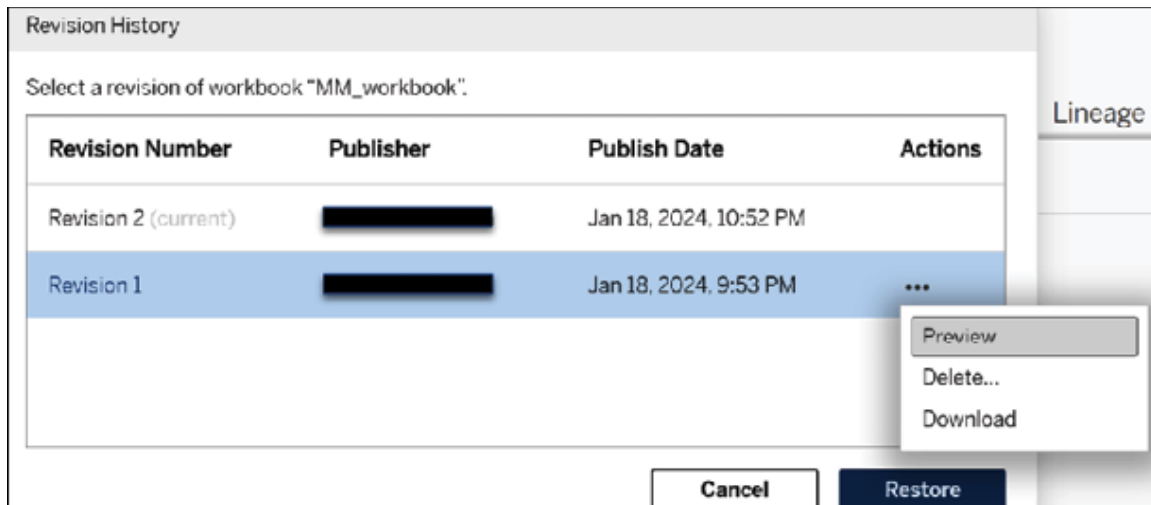


Figure 7.20: Manage revisions Conclusion

In this chapter, we learned how to keep your data (private or virtual data sources) fresh in different ways and share your content (including workbooks, data sources) in a secure way.

In the next chapter, readers will get the hands-on experience on Tableau Cloud from scratch i.e., starting from building views to sharing and scheduling refresh on Tableau Cloud.

Multiple choice questions

- 1. Which software facilitates the secured communication between Tableau Cloud and Tableau Desktop?**
 - a. Tableau Reader
 - b. Tableau Explorer
 - c. Tableau Bridge Client
 - d. None
- 2. Select the appropriate option: Which options are available to set up the extract refresh on Tableau Cloud?**
 - a. Refresh extracts
 - b. Refresh extracts now

- c. Schedule with Bridge (legacy)
 - d. All of the above
- 3. While running Tableau Bridge, which network type is used to extract refreshes?**
- a. Tableau Cloud
 - b. Private Network
 - c. None
- 4. Select the appropriate option: Which options are available to publish the data source on Tableau Cloud?**
- a. Tableau Desktop
 - b. Tableau Bridge
 - c. Tableau Cloud
 - d. All of the above
- 5. Are auto-save workbooks allowed on Tableau Cloud?**
- a. True
 - b. False
 - c. None
- 6. Auto-save workbooks are available up-to what file size on Tableau Cloud?**
- a. 5MB
 - b. 2MB
 - c. 1GB

d. 1TB

Answers

1.	c
2.	d
3.	b
4.	d
5.	a
6.	a

Points to remember

Here are some key takeaways from this chapter:

- Tableau Bridge is a software that helps keep your data fresh. It facilitates secured communications for private and virtual data sources to connect with Tableau Cloud.
- Site administrators, creators, and explorers (can publish is a separate role defined in Tableau Online) have the authority to use Tableau Bridge.
- Tableau Cloud enables organizations to share and publish their workbooks and data sources.
- There are various ways to publish the data sources and workbooks.
- Publish the workbooks and data sources from Tableau Desktop.
- Publish the workbooks and data sources from Tableau Cloud directly.
- Publish the workbooks and data sources from Tableau Desktop using Tableau Bridge.
- You can edit, roll back, or add a new workbook or data source on the Tableau Cloud through a few simple steps.

- Autosave is available for workbooks which are smaller than 5MB in file size.
- **Scheduling:** There are various ways to keep your data fresh on Tableau Cloud:
 - Schedule data refresh using Tableau Bridge on Tableau Cloud directly.
 - Manual data refresh.
 - Schedule data refresh directly on Tableau Cloud.

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CHAPTER 8

Troubleshooting Tableau Challenges

A problem well stated is a problem half solved.

- John Dewey

In the previous chapter, readers learned how to publish the reports, schedule, and extract refresh. This chapter will cover those topics to resolve the common issues while working with Tableau Cloud.

Structure

In this chapter, we will cover the following topics:

- Refreshes data connectivity
- Refreshes user credentials issues
- Extract refresh failures
- Issues in subscription
- Live query issues
- Other common issues

Objectives

This chapter will illustrate how to resolve the most frequent and common issues while refreshing the extracts, connecting the data sources, working on live queries, issues in subscriptions, and other common issues with Tableau Cloud.

After reading this chapter, the user will know what steps need to be taken to resolve the issue with Tableau Cloud when performing certain tasks.

Refreshes data connectivity

To resolve the refresh issue related to data connectivity, there is the possibility of following errors such as.

- Error related to user credentials.
- Database unreachable.

Follow these steps to resolve the issue:

1. Re-check the user credentials, reset the password, and then test the Tableau connection.
2. Now, sign in/log in to the database and reset the database connection with Tableau Cloud.

Refreshes user credentials issues

When you recently changed your password or user credentials, sometimes Tableau Cloud displays abnormal errors, such as ***Your account or password is incorrect. Please try again or Click Forgot Password.***

This type of error occurs when sometimes your web browser cache's/cookies automatically enter the old password during login.

To resolve this issue, follow these steps:

1. First, you must clear your web browser history, cache, and cookies.
2. Login to the website: www.online.tableau.com with your correct credentials. Then Sign In.

Extract refresh failure

The most common issue any user/developer/site administrator will face is the issue of extract refresh failure. This happens because five times scheduled extracts failed. Hence, Tableau Cloud suspends the refreshes. Tableau Cloud will not try to run the scheduled extract until any action is taken to resolve the issue of failure.

Let us see a few errors and how to resolve them:

- **Error that indicates the database was unreachable.**

Usually, such an error triggers if the database is online. You need to log in/sign in the data again to access its data.

- **Error related to a data connection or access token validation.**

This type of error can be resolved only if you update the data connection information.

Follow these steps to resolve such error:

1. Select the **data source** whose scheduled extract failed.
2. Go to the **Connection** tab or **Select Edit Connection** from the **Actions** drop-down menu.
3. Update the data connection details i.e. Server, Port, Password, and so on.
4. Click **Ok**.

Tableau Cloud will restart the extract refresh. However, if data connection credentials are embedded when you publish the workbook or data source from the Tableau Desktop, then in such case, you need to republish the workbook or data source after updating the data connection information. Here, you can also set a new refresh extract. Otherwise, Tableau Cloud will restart as per the existing schedule.

Issues in subscription

The user might receive a subscription with the error message in their email, such as: ***The View Snapshot in this email could not be properly rendered.***

Multiple reasons trigger such a message. They are as follows:

1. Database temporarily down

If Tableau Viz has a live database connection. Then, sometimes, a subscription does not generate, and the user might receive such an error.

2. Missing credentials

In case if your Viz is published with embedded credentials and the embedded credentials are outdated, or if that same Viz is republished without the embedded credentials, then the user will receive the above error.

3. Cannot see the images in email.

If the user does not have permission to download images, then even if the user has subscription permission, still, images cannot be viewed in the email.

To resolve such error, follow these instructions by the site administrator:

- Go to Workbook/View, click on **More actions(...)**
- Select **Permissions**.
- Now, change the permission rule as shown in the figure under **Image Download Icon** from disable (x) to enable (✓):



Figure 8.1: View Permissions

- Then, **Save** the changes.

4. Missing data quality warnings

Users will receive the error if data quality warnings are in view or the workbook is not included in the subscription email. Hence, to resolve this issue, follow these steps:

- Go to **Site Settings** and select the **General** tab.
- Now, select the checkbox under the Data Quality Warnings in Subscriptions as shown in the following figure.

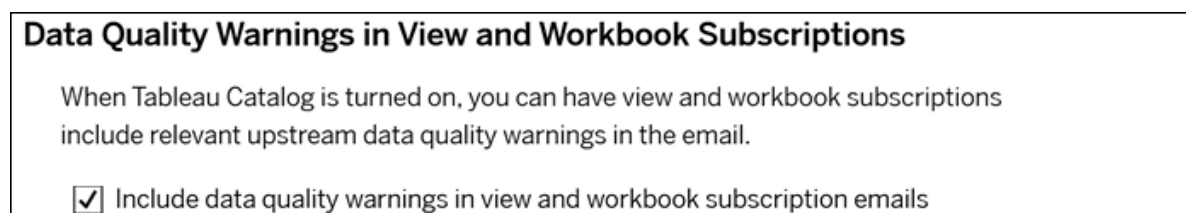


Figure 8.2: Data Quality Warnings

5. Suspended subscriptions

If subscriptions fail for five consecutive times then by default, subscription get suspended and generate the defined error.

6. Resume suspended subscriptions

Users/Owners can resume their suspended subscriptions by following these instructions:

- a. Go to the Account Menu, select My Content
- b. Select the Subscription tab and resume the subscription from the list.

7. Missing PDF attachment

Like Images failing to display in the email, PDF downloading/attaching in the email can be disabled by the site administrator based on permission rules set and sometimes if the size of the PDF exceeds the email size limit.

Note: In Tableau Cloud, the maximum email size limit is 2MB per site.

To resolve the issue, follow the same steps as enabling **Downloading Image/PDF** from the permission sets, as shown in the figure.

8. Cannot subscribe or No Subscription icon.

If Viz uses a live data connection, then it is most important to embed credentials. The Tableau Cloud either requires embedded database credentials or data that does not require credentials. If any of these conditions are not satisfied, then an error will display.

You will not see a subscription icon or cannot subscribe.

Live query issues

Tableau Cloud will prompt live query issues due to the options not being available.

Let us identify the possible causes for live query issues:

1. You are prompted for database credentials.
2. No *Live* data sources are listed in the client.
3. Tableau Bridge does not support the data source type

Other common issues

Tableau Online sometimes has issues related to data refresh, and the refreshed data does not match the actual data source. Hence, it is highly recommended that you do a manual data refresh from the Tableau desktop instead of relying on an online data refresh. Such issues create a bad impression on the business user's trust and confidence in using Tableau reports. However, you can avoid such a situation if you avoid scheduling the extract refresh whenever any daemon processes or production/maintenance tasks are running simultaneously.

Conclusion

In this chapter, we learned how to troubleshoot the most frequent issues while working with Tableau Cloud in a few simple steps.

In the next chapter, readers will get a glimpse of real scenarios and the hands-on exercise on building charts and dashboards and then publish and extract refreshes.

Multiple choice questions

1. What is the maximum email size limit per site in Tableau Cloud?

- a. 1MB
- b. 5MB
- c. 1GB
- d. 2MB

2. Which permission does user required to download an image?

- a. View
- b. Image download
- c. Filter
- d. None

Answers

1.	d
2.	b

Points to remember

Here are a few key takeaways from the chapter:

- The user may encounter a few common issues while working with Tableau Cloud, such as refresh data connectivity, user credentials, extract refresh failure, issues in subscriptions, permissions, live query issues, and other common issues.
- **Refreshes data connectivity:** An issue related to the data connectivity with Tableau Cloud. Sometimes, tableau loses connection with the data source, and vice-versa.
- **Refreshes user credentials issues:** An abnormal error message starts prompting if the user has recently changed their credentials.
- **Extract refresh failures:** One of the most common issues users face whenever consecutive five times extracts fail while trying to refresh the extracts online.
- **Issues in subscription:** The user might face issues in the subscription related to data quality warnings, suspended subscriptions, missing credentials, inability to see the email, etc.
- **Live query issues:** Such issues erupt if the user is void of options available or access permissions.
- **Other common issues:** If tableau reports do not reflect the same records as the data source. In such a situation, it is better to perform a manual refresh from the desktop than rely on online extract refreshes.

CHAPTER 9

Miscellaneous Projects

A good tool improves the way you work - Jeff Duntemann [Introduction](#)

In the previous chapters, readers learned how to create views, manage sites and contents, keep data fresh, and data management b. In this chapter, they will apply those learnings to sample data and explore Tableau Cloud with hands-on experience. These exercises give a glimpse into the real world of Tableau Cloud and Tableau Desktop usage.

Structure

In this chapter, we will cover the following two case studies:

- Case study 1: Human disease network
- Case study 2: Iris flower analysis
- Tableau Desktop versus. Tableau Cloud

Objectives

By the end of this chapter, you will gain hands-on experience with sample data and learn how to create a view, workbook, publish, and manage data on Tableau Cloud. After reading this chapter, the reader, as a developer or site administrator, will learn how to develop highly intuitive Tableau reports. Readers will get a clear, distinct experience in report development and functionality between Tableau Cloud and Tableau Desktop.

Case study 1: Human disease network

Objective: The objective of this case study is to understand the associations between human disease and disorder and genes and other sources.

Here is a description of the dataset: Use the sample data set, i.e., **Edge+Nodes(Disease) .xlsx** file available.

Problem statement: Based on the Excel file provided, we will analyze and develop a dashboard to understand the disease, disorder, and gene associations from the sample data. Use filters, objects, and highlighters wherever possible to make the dashboard interactive.

1. Create a disease network chart to display the associations between disorders, genes, and diseases.

Hint: Use dual axis

2. Create a dynamic tooltip on the network chart to show the total genes and associations related to that disease and disorder.

Hint: On hovering on the disease, respective details will be displayed on the tooltip.

3. Show the total distinct number of disorders, genes, and diseases in the sample data.
4. Add a description text box defining the meaning of the *human disease network*.
5. Make the dashboard interactive using highlighter based on the type of disorder.
6. Schedule the extract refresh on weekly updates.

Lab environment: Use Tableau Cloud (Web version) **Domain:** Healthcare Before we start creating an interactive and intuitive Tableau dashboard, let us first understand the data file. Here, the dataset has 19 variables on which analysis must be done to gather details about disease, genes, and disorder connections. A sample of the dataset reference is shown in [Figure 9.1](#):

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	
D or G	Direction	Disease	Disorder Name	Gene Name	ID	ID Copy	ID Main	Link	Source D/G	Target D/G	Type of disorder	Node	Source	Target	Value	Weight	X	Y	
D	3-methylglu	3-methylglu	3-methylglutacc	OPA3	510	510	510	1 D	G	G	Metabolic		1285	1285	3211	11.61905	1	1214.388	869.1633
D	Aarskog-Sc	Aarskog-Sc	Aarskog-Scott s	FGD1	308	308	308	1 D	G	G	Multiple		468	468	2914	11.61905	1	796.4285	1618.361
D	ABCD synd	ABCD synd	ABCD syndrome	EDNRB	299	299	299	1 D	G	G	Multiple		416	416	3825	11.61905	1	-900.733	-1305.66
D	Abetalipop	Abetalipop	Abetalipoprotein	APOB	154	154	154	1 D	G	G	Metabolic		126	126	2566	12.18703	1	-1645.94	-790.642
D	Abetalipop	Abetalipop	Abetalipoprotein	MTP	154	154	154	1 D	G	G	Metabolic		126	126	1329	12.18703	1	-1645.94	-790.642
D	Achondrog	Achondrog	Achondrogenes	COL2A1	266	266	266	1 D	G	G	Bone		288	288	3831	11.61905	1	-1385.16	673.8155
D	Achondrog	Achondrog	Achondrogenes	SLC26A2	296	296	296	1 D	G	G	Bone		407	407	2203	11.61905	1	-1459.58	1063.805
D	Achondrop	Achondrop	Achondroplasia	FGFR3	309	309	309	1 D	G	G	Skeletal		473	473	3720	11.61905	1	1141.336	-836.87
D	Achromato	Achromato	Achromatopsia	CNGA3	129	129	129	1 D	G	G	Ophthalmologica		282	282	1338	13.1815	1	-1552.89	-184.895
D	Achromato	Achromato	Achromatopsia	GNAT2	129	129	129	1 D	G	G	Ophthalmologica		282	282	1339	13.1815	1	-1552.89	-184.895
D	Achromato	Achromato	Achromatopsia	CNGB3	129	129	129	1 D	G	G	Ophthalmologica		282	282	2851	13.1815	1	-1552.89	-184.895
D	Acquired lc	Acquired lc	Acquired long Q	KCNH2	358	358	358	1 D	G	G	Cardiovascular		686	686	2814	11.61905	1	-540.223	1454.451
D	Acromegalt	Acromegalt	Acromegaly	GNAS	198	198	198	1 D	G	G	Endocrine		575	575	3609	12.18703	1	164.6341	1884.525
D	Acromegalt	Acromegalt	Acromegaly	SSTR5	198	198	198	1 D	G	G	Endocrine		575	575	1348	12.18703	1	164.6341	1884.525
D	Adenocarc	Adenocarc	Adenocarcinoma	BRAF	88	88	88	1 D	G	G	Cancer		199	199	3160	14.42692	1	557.937	-892.551
D	Adenocarc	Adenocarc	Adenocarcinoma	EGFR	88	88	88	1 D	G	G	Cancer		199	199	3161	14.42692	1	557.937	-892.551
D	Adenocarc	Adenocarc	Adenocarcinoma	ERBB2	88	88	88	1 D	G	G	Cancer		199	199	3248	14.42692	1	557.937	-892.551
D	Adenocarc	Adenocarc	Adenocarcinoma	PARK2	88	88	88	1 D	G	G	Cancer		199	199	3292	14.42692	1	557.937	-892.551
D	Adenoma,	Adenoma,	Adenoma, peria	APC	238	238	238	1 D	G	G	Cancer		112	112	3780	11.61905	1	896.659	-1087.94
D	Adenomas	Adenomas	Adenomas	APC	118	118	118	1 D	G	G	Cancer		113	113	3780	13.1815	1	956.7198	-1144.28
D	Adenomas	Adenomas	Adenomas	MUTYH	118	118	118	1 D	G	G	Cancer		113	113	2309	13.1815	1	956.7198	-1144.28
D	Adenomas	Adenomas	Adenomas	PLAG1	118	118	118	1 D	G	G	Cancer		113	113	1361	13.1815	1	956.7198	-1144.28
D	Adenosine	Adenosine	Adenosine deam	ADA	233	233	233	1 D	G	G	Immunological		62	62	3571	11.61905	1	-410.868	-1788.15
>	Sheet1	+																	

Figure 9.1: Human disease network dataset Before you start creating a workbook, you need to connect the data directly to Tableau Cloud. Follow these steps to connect the data directly on Tableau Cloud:

- Navigate to the Tableau Cloud site, then on the homepage, select **New | Workbook**.
- Click on the **Data Source** tab. Now, **Connect to Data | Files**.
- In the next step, upload the data file from the local device.
- Now, click **On This Site** tab and then, select the file published from the list as shown in [Figure 9.2](#).
- Click on the **Sheet1** tab to start creating views.

Connect to Data

Add a new connection to create cross-database joins. [Learn more](#)

On This SiteFilesConnectors

Q

Search for data

TypeAll

Certified

More Filters

Popular Data

Data Sources

See All

	Name	Workbooks	Owner	Location	Connects To	Live / Last extract
	Mental Disorder (MM27MentalHealth) - live	1		Project1	MM27MentalHealth.xlsx	Jan 23, 2024, 9:46 PM
	Human Disease Network	1		Project1	Edge+Nodes (Disease).xlsx	Live
	Superstore Datasource	0		Samples	Sample - Superstore.xls	Live
	GME_stock	0		Project1	GME_stock.csv	Jan 24, 2024, 2:50 PM

Connect

Figure 9.2: Connect to Data (human disease network) Now, we will begin solving the problem statement and build the HUMAN DISEASE NETWORK dashboard, as shown in [Figure 9.3](#):

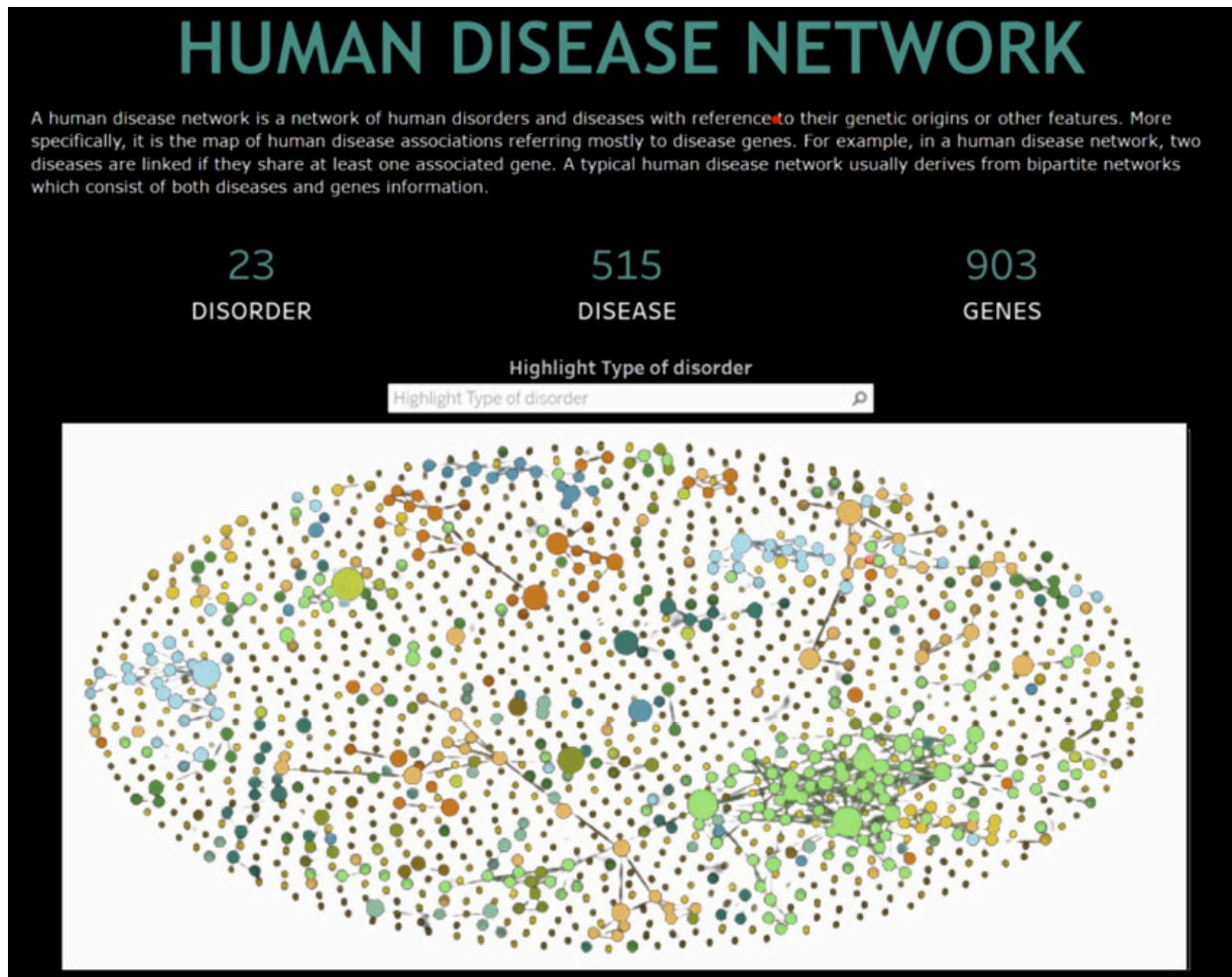


Figure 9.3: Human Disease Network dashboard

1. Create a disease network chart to display the associations between disorders, genes, and diseases.

Tip: Use dual axis

Follow these steps to create a disease network chart:

- a. Drag the measured fields **X** on columns, and **Y** on rows.
- b. Duplicate **Y** rows and put next to **Y** rows.
- c. Right-click the second Y, then click **More Options | Dual Axis**.
- d. On the toolbar, select **Analysis | Uncheck Aggregate Measures**.
- e. Drag the dimension field **Disease** onto the Filters card. Then right-click on Disease and select **More Options | Edit Filter**.

- f. Now, from the **List** menu, select **Use all**. Then **Ok**.
- g. Again, right-click the **Disease** field, then select **Show Filter**.
- h. Drag the **Type of Disorder** on the **Color card** on the **All Marks** card.
- i. Now, click on **Color** card and then **Edit color** | assign **Green-Orange-Teal color palette** and configure **Opacity** to 35%.
- j. Change **Y Marks** to a **Line Chart** and **Y(2) Marks** to a **Shape a Chart**.
- k. On the **Y Marks card**, drag **Direction** on the **Details card** and adjust the **Size card** to 0.
 - i. On **Y(2) Marks** card, drag **Disease** and **Type of Disorder** on **Detail** card. Drag **the Value on the Size** card and configure it to 25%.
 - ii. Now, change the **Shape** card to **filled circles**.
 - iii. Right-click anywhere on Pane, then under **Shading**, assign **Black** color to Worksheet, and under **Grid Lines** and **Zero Lines**, select **All | Off**.
 - iv. Double-click on **Title** and configure the **font color** to **White** and font style to **Tableau Light**.
 - v. Rename the **Sheet1** tab as **Network**.
 - vi. On **Y Marks** and **Y(2) Marks** card, click on **Tooltip** and enhance the tooltip as shown in the figure:

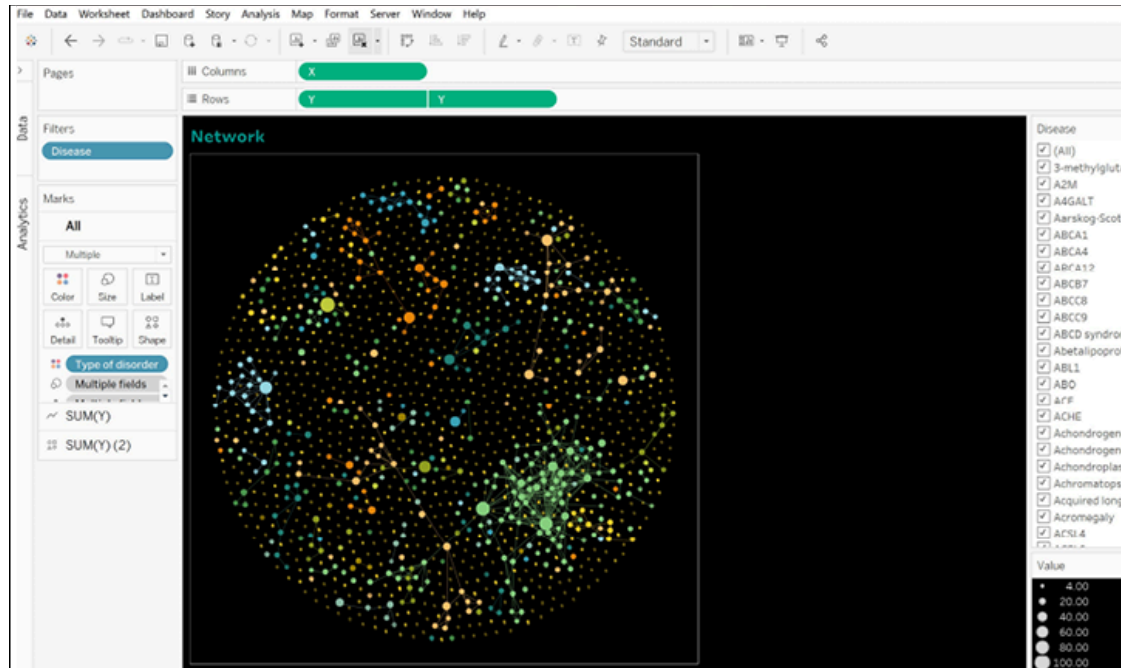


Figure 9.4: Network worksheet

2. Create a dynamic tooltip on the network chart to show the total genes and associations related to the disease and disorder.

Hint: On hovering on the disease, respective details will be displayed on the tooltip.

To create a dynamic tooltip on the **Network** sheet, follow these steps:

- a. First, we need to create a sheet to insert as a tooltip in **Network** worksheet as a tooltip. Hence, let us create a tooltip sheet named **Statistics**.
- b. Click on **New Worksheet** (*Ctrl + Alt + T*).
- c. Right-click on **Sheet2**, then click on **Rename Sheet2** as **Statistics**.
- d. On the toolbar, select **Analysis | Create Calculated Field**.
- e. Rename the field as **#Genes Linked**.
- f. In the Calculation Editor, enter the formula as shown in the following figure:

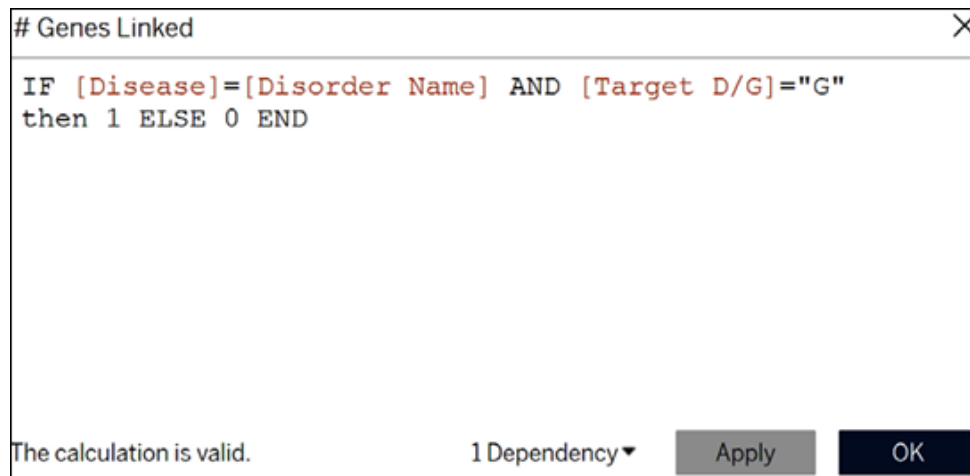


Figure 9.5: Statistics field—#Genes Linked

- g. Create another new calculated field, select **Analysis | Create Calculated Field**.
- h. Rename the field as **#Associated Disease**, as shown in [Figure 9.5](#).
- i. In the Calculation Editor, enter the formula, as shown in the following figure:

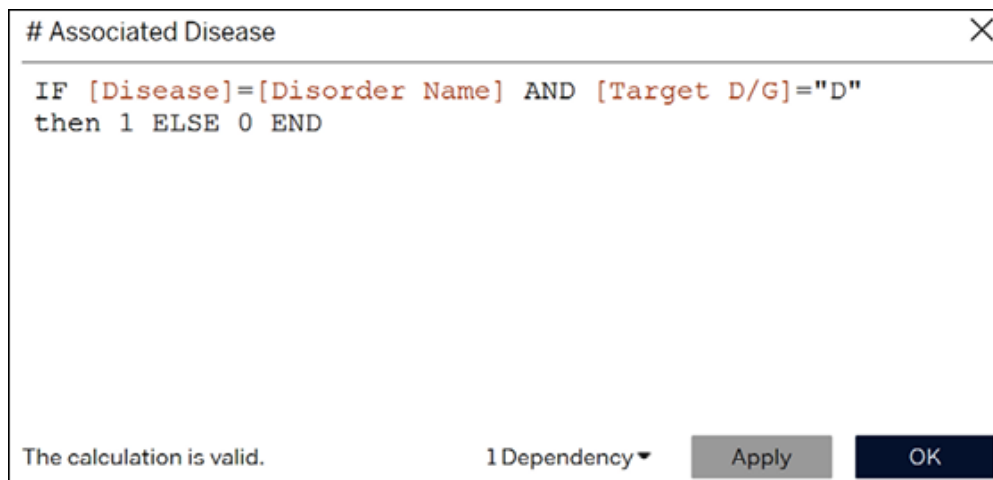


Figure 9.6: Statistics field—#Associated Disease

- j. Now, drag the measured field **Link** to the **Filters** card. Set the **Range of Values** to **1** (Min:1 andMax:1). Then **OK**.
- k. Drag fields **#Genes Linked** and **#Associated Disease** to **Text** Card. Then, edit the text as shown in [Figure 9.7](#).
- l. Click on **Tooltip** and uncheck the **Show Tooltips**.

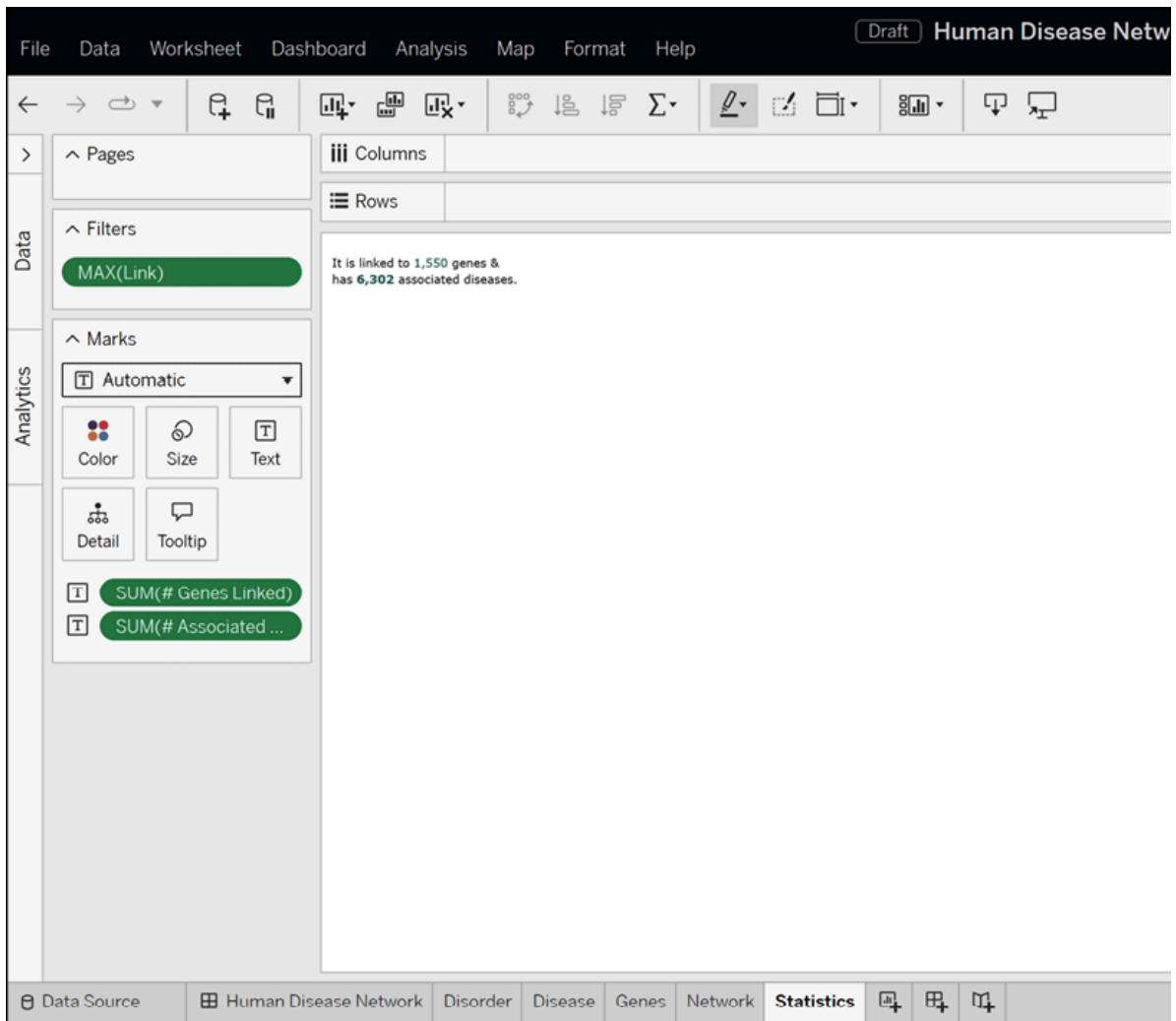


Figure 9.7: Statistics worksheet Now, it is time to insert the tooltip sheet into the Network worksheet.

- Click on the **Network** worksheet.
- On **Y(2) Tooltip** marks card, click on **Insert | Sheets**. Then select the **Statistics** sheet.
- Configure the details on the tooltip, as shown in [Figure 9.8](#):

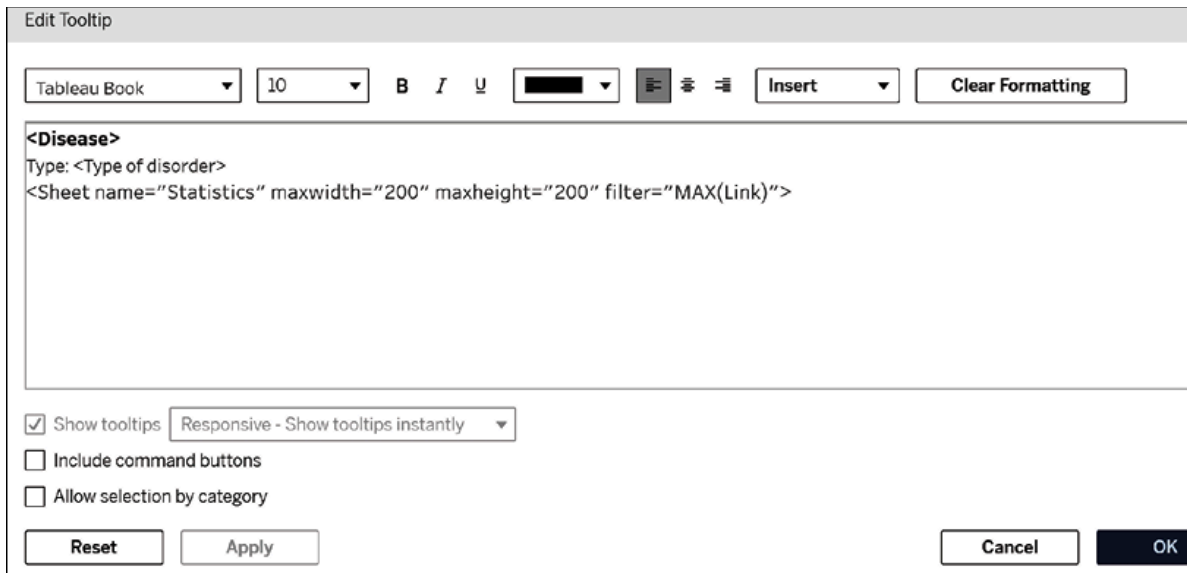


Figure 9.8: Dynamic tooltip on network sheet

3. Show the total distinct number of disorders, genes, and diseases in the sample data.

Follow these steps to do the same:

- Click on **New Worksheet** and rename Sheet 3 as **Disorder**.
- On the toolbar, select **Analysis | Create Calculated Field.**, named as **Total Disorder**.
- In the Calculation Editor, enter the formula as shown in the following figure:

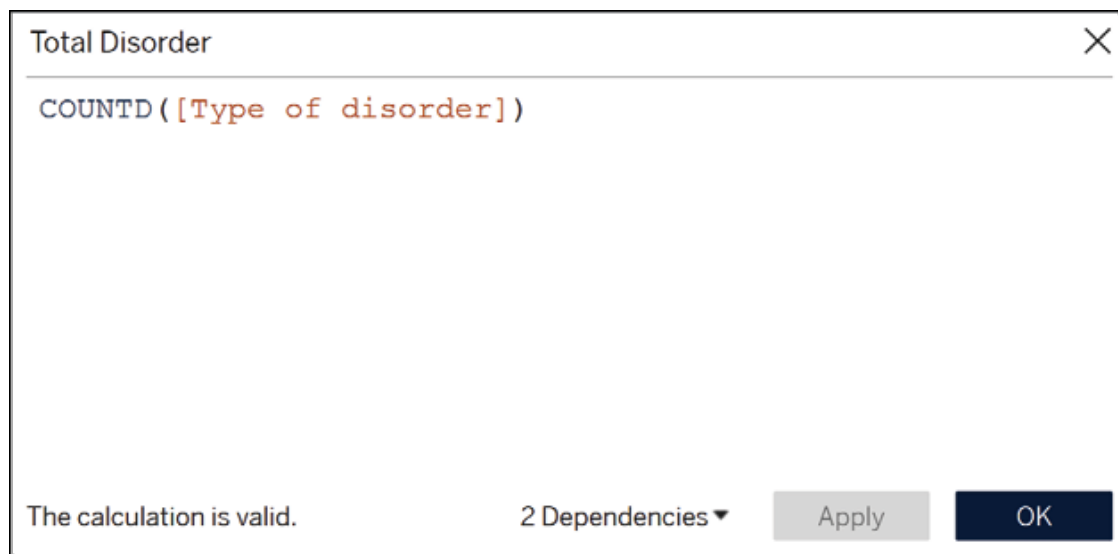


Figure 9.9: Total Disorder—calculated field

- d. Drag the **Total Disorder** field to the **Text Card**. Then **Edit Text**, as shown in the following figure:

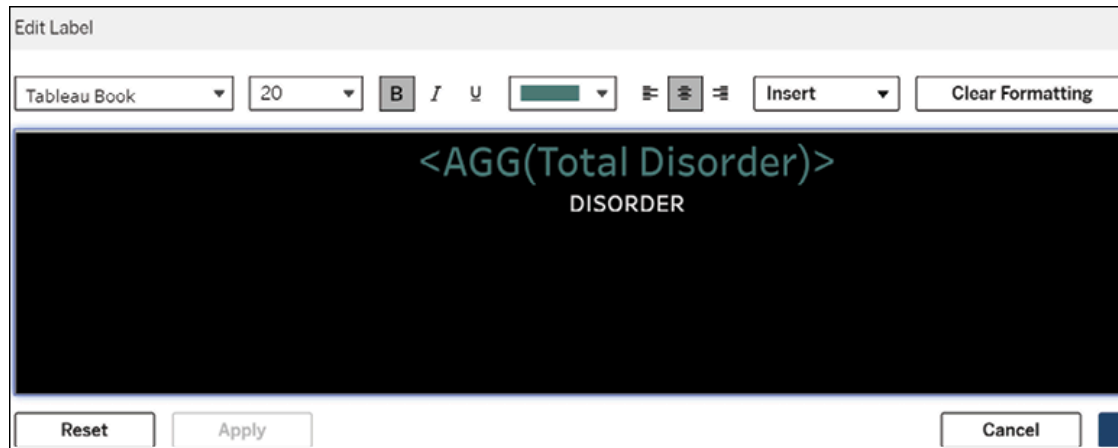


Figure 9.10: Total Disorder text

- e. Right-click anywhere on pane, then under Shading, assign Black color to **Worksheet**, and under **Grid Lines** and **Zero Lines**, select **All | Off** as shown in figure:

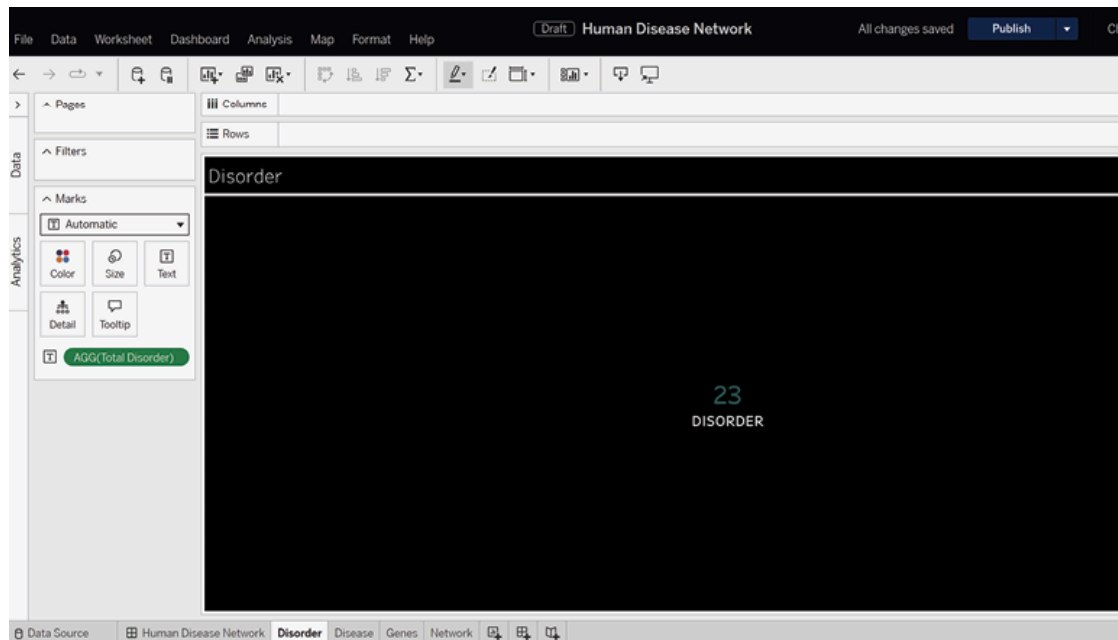


Figure 9.11: Total Disorder sheet

- f. Click on **New Worksheet** and rename Sheet 4 as **Disease**.

- g. On the toolbar, select **Analysis | Create Calculated Field** and name as **Total Disease**.
- h. In the **Calculation Editor**, enter the formula as shown in the following figure:

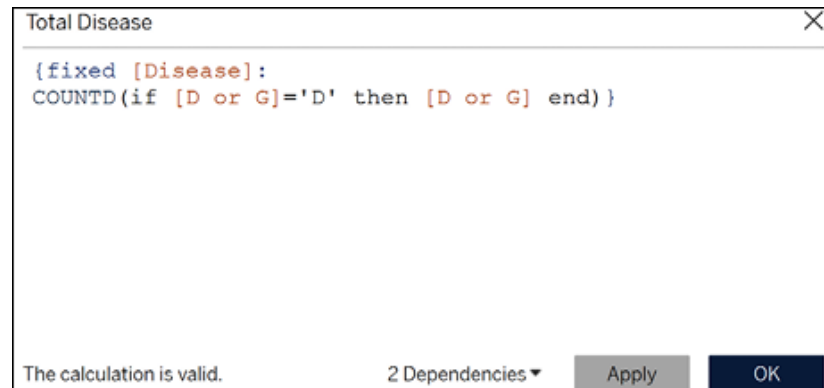


Figure 9.12: Total Disease—Calculated Field

- i. Drag **Total Disease** field to **Text Card**. Then, edit the **Text** and configure it like **Total Disorder**:

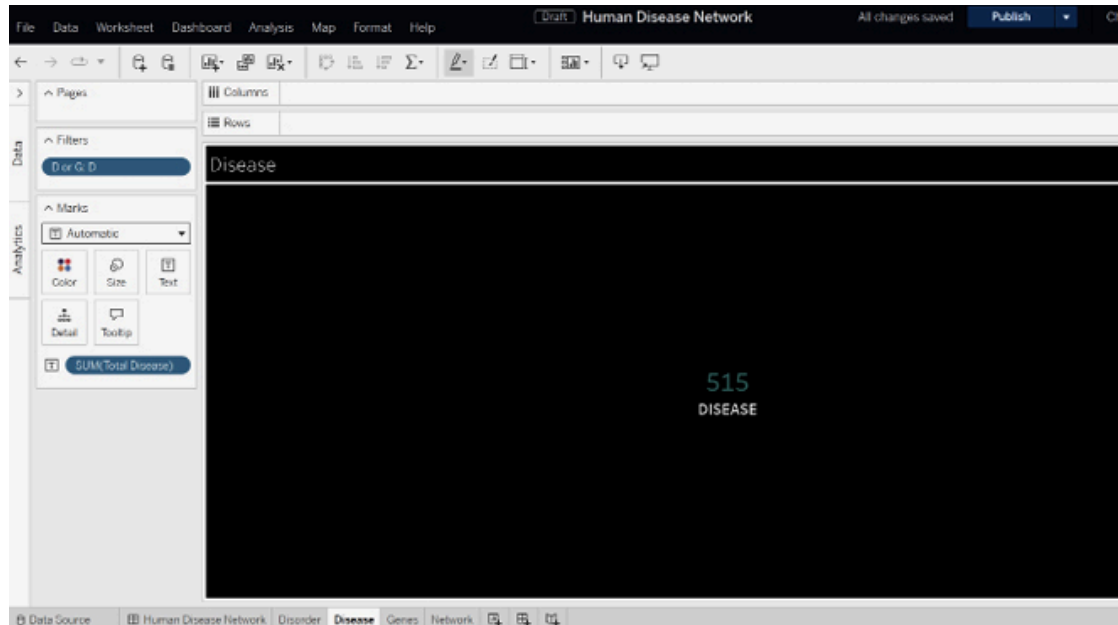


Figure 9.13: Total Disease sheet

- j. Click on **New Worksheet** and rename Sheet5 as **Genes**.

- k. On the toolbar, select **Analysis | Create Calculated Field**, named **Total Genes**.
- l. In the Calculation Editor, enter the formula as shown in the following figure:

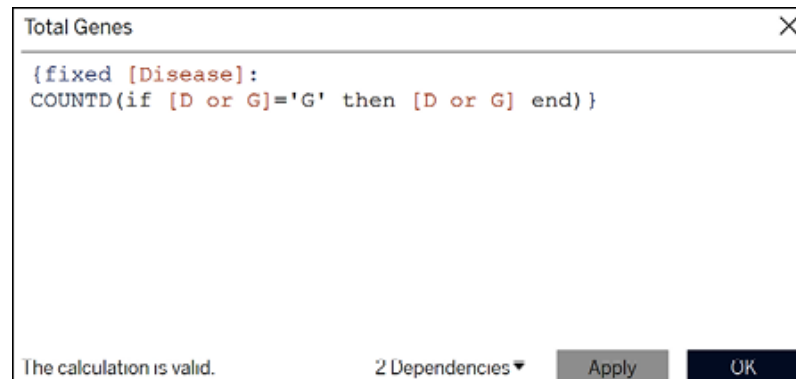


Figure 9.14: Total Genes calculated field

- m. Drag the **Total Genes** field to the **Text Card**. Then, edit the **Text** and configure it like **Total Disorder**:

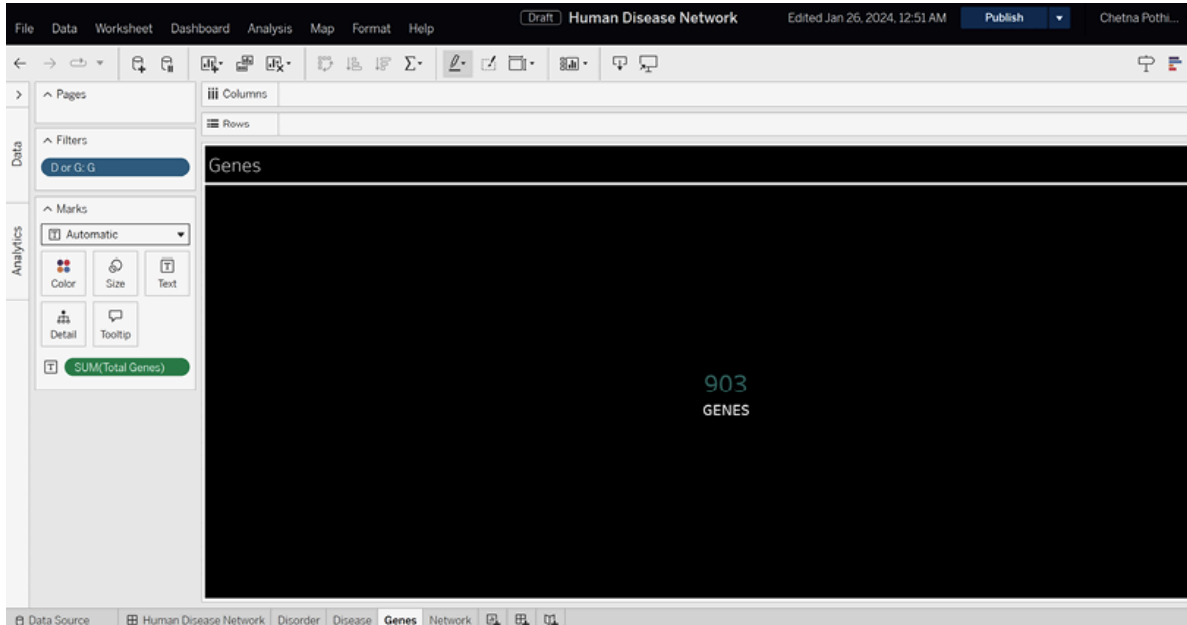


Figure 9.15: Total Genes worksheet

4. Add a description text box defining the meaning of HUMAN DISEASE NETWORK.

- a. Click on **New Dashboard** and rename it as **Human Disease Network**.

- b. Drag the **Vertical Container** on the pane.
- c. Now, drag **Text** under **Objects** and edit text as **HUMAN DISEASE NETWORK**, as shown in [Figure 9.16](#).
- d. Drag another **Text Object** for description and align next to Text Object 1 and edit as shown in [Figure 9.16](#).
- e. Now, drag **the Horizontal Container** and put it next to Text Object 2.
- f. Drag **Blank Object** on top of **Horizontal Container**, then drag **Disorder sheet** just next to Blank Object on Horizontal Container.
- g. Now, drag **Disease sheet** and put next to Disorder sheet. Then drag **Genes sheet** and aligned next to Disease sheet.
- h. Next, drag another **Blank sheet** and put it next to the Genes sheet.
- i. Now, drag the **Network sheet** after Horizontal Container 1.
- j. Remove all the filters.

Hint: Adjust all the sheets, Fit -| Entire View.

5. Make the dashboard interactive using highlighter based on the type of disorder.

Follow these steps to do the same:

- a. Right-click on the **Network Sheet**. Now, select **Highlighter | Type of Disorder**.
- b. Now, to align this Highlighter below **Horizontal Container 1** and above the Network sheet, drag another Horizontal Container and place it next to Horizontal Container 1.
- c. Drag **Blank Object** on put on Horizontal Container 2.
- d. Now, drag this **Highlighter** on and put it next to Blank Object.
- e. Add one more Blank Object next to **Highlighter**. Make sure you adjust all the items equally.

- f. On selecting the **Type of Disorder** on highlighter, the respective associations related to that disorder get highlighted on the network graph.
- g. On hovering on each disease and disorder on Network sheet, the dynamic tooltip will display as shown in [Figure 9.16](#):

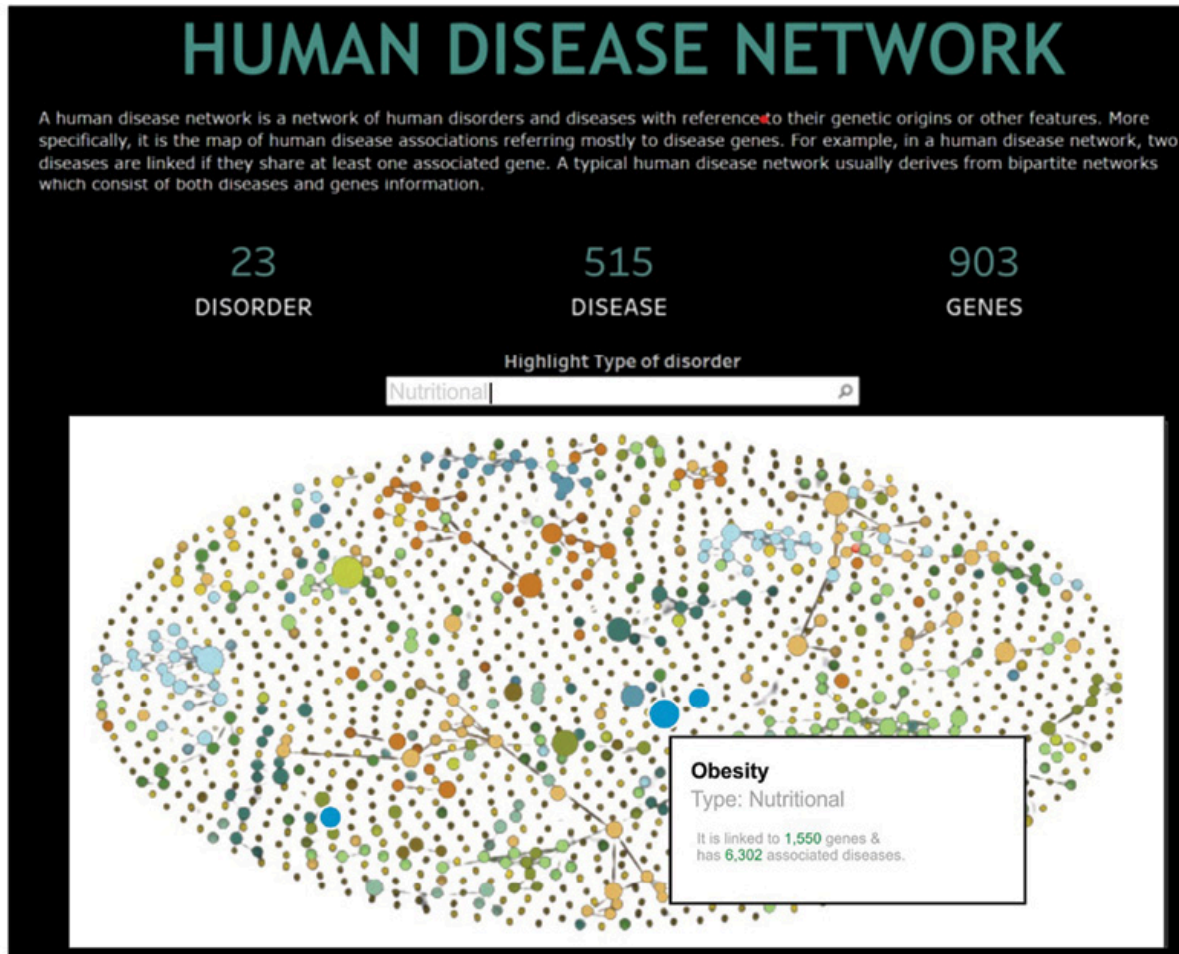
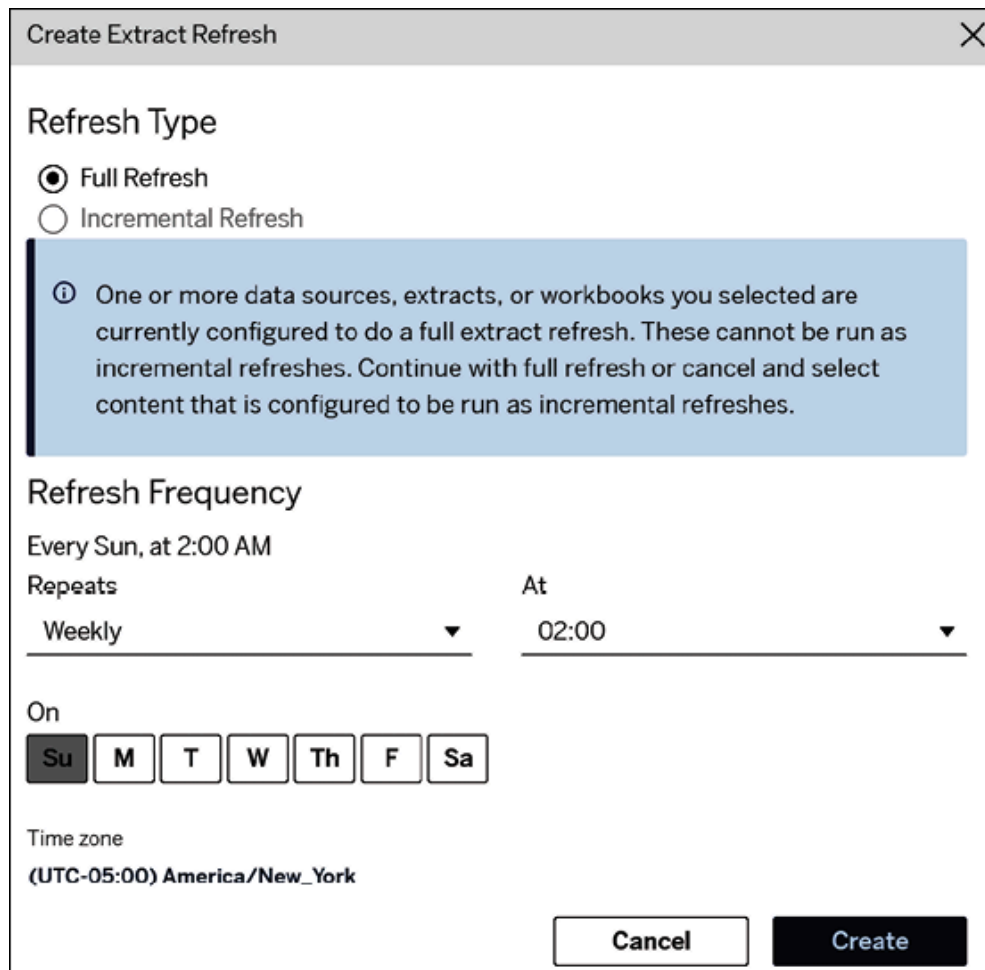


Figure 9.16: Human Disease Network dashboard

6. Schedule the extract refresh on weekly updates.

- a. Navigate to the **Data Source Page** on the Tableau Cloud site.
- b. Select the data source you want to schedule the extract refresh.
- c. Now, select the **Extract Refreshes** tab and then click on the **New Extract Refresh** button.

- d. Now, configure the **Create Extract Refresh** dialog box as shown in the following figure:



The image shows a 'Create Extract Refresh' dialog box with the following configuration:

- Refresh Type:** ☒ Full Refresh, ☐ Incremental Refresh
- Message:** One or more data sources, extracts, or workbooks you selected are currently configured to do a full extract refresh. These cannot be run as incremental refreshes. Continue with full refresh or cancel and select content that is configured to be run as incremental refreshes.
- Refresh Frequency:** Every Sun, at 2:00 AM
- Repeats:** Weekly (dropdown)
- At:** 02:00 (dropdown)
- On:** Su (selected), M, T, W, Th, F, Sa
- Time zone:** (UTC-05:00) America/New_York
- Buttons:** Cancel, Create

Figure 9.17: Human Disease Network Dashboard

- e. Set up the day and time as per your choice to weekly run the refresh.
- f. Lastly, click on **Create**.

Case study 2: Iris flower analysis

Objective: The objective of this case study is to analyze the **IRIS** flower types and classify them based on their features.

Here is the description of the dataset: Use the sample data set i.e. **IRIS.csv** file available.

Problem statement: Based on the comma-separated file (**.CSV**) provided, we will analyze IRIS flower and develop a dashboard segmenting it based on its features and finding the correlation between features from the sample data. Use filters and objects wherever possible to make the dashboard interactive. Follow these steps:

1. Create a scatter plot to display the data distribution.
2. Segment the data based on the features.

Hint: Use Cluster Analysis.

3. Create a dynamic tooltip on a scatter plot chart to display the image of IRIS flowers.

Hint: Use custom shapes.

4. Create Correlation fields and display them.
5. Allow only specific users to view the report.

Hint: Use row-level security.

6. Create the IRIS Flower Analysis Dashboard.
7. Publish the workbook with data sources to the Tableau Cloud site.

Lab environment: Use Tableau Desktop **Domain:** Botany Before we initiate creating an analytical Tableau dashboard, let us first understand the data file. Here, the dataset has five variables that we will use for analysis. A sample of the dataset is given for reference, as shown in [Figure 9.18](#):

A	B	C	D	E
sepal_length	sepal_width	petal_length	petal_width	species
5.1	3.5	1.4	0.2	Iris-setosa
4.9	3	1.4	0.2	Iris-setosa
4.7	3.2	1.3	0.2	Iris-setosa
4.6	3.1	1.5	0.2	Iris-setosa
5	3.6	1.4	0.2	Iris-setosa
5.4	3.9	1.7	0.4	Iris-setosa
4.6	3.4	1.4	0.3	Iris-setosa
5	3.4	1.5	0.2	Iris-setosa
4.4	2.9	1.4	0.2	Iris-setosa
4.9	3.1	1.5	0.1	Iris-setosa
5.4	3.7	1.5	0.2	Iris-setosa
4.8	3.4	1.6	0.2	Iris-setosa
4.8	3	1.4	0.1	Iris-setosa
4.3	3	1.1	0.1	Iris-setosa
5.8	4	1.2	0.2	Iris-setosa
5.7	4.4	1.5	0.4	Iris-setosa
5.4	3.9	1.3	0.4	Iris-setosa
5.1	3.5	1.4	0.3	Iris-setosa
5.7	3.8	1.7	0.3	Iris-setosa
5.1	3.8	1.5	0.3	Iris-setosa
5.4	3.4	1.7	0.2	Iris-setosa
5.1	3.7	1.5	0.4	Iris-setosa
4.6	3.6	1	0.2	Iris-setosa

Figure 9.18: Iris dataset Let us start building the Tableau dashboard. Firstly, you need to connect your data with your workbook. Hence, follow the given instructions:

1. Open your Tableau Desktop app, then click on **Text** file under **Connect to File**. Now, upload the (**.csv**) data file from the local device.
2. Click on the **Sheet1** tab to start creating views.

We will begin solving the problem statement and build the IRIS dashboard, as shown in [Figure 9.19](#):

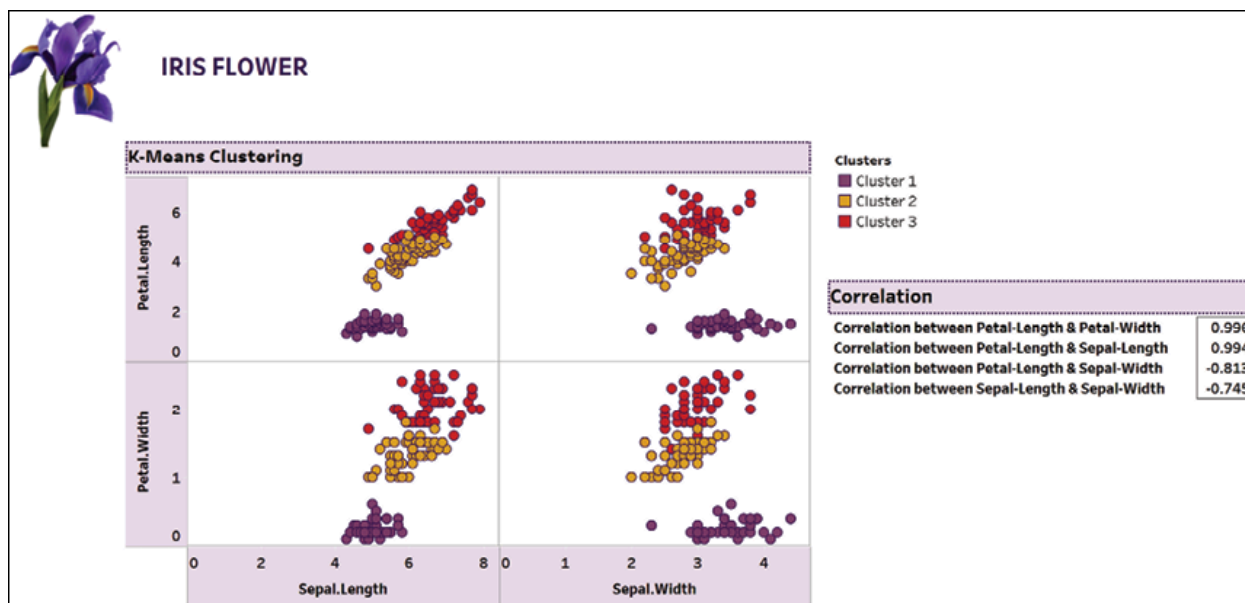


Figure 9.19: Iris dashboard

1. Create a scatter plot to display the data distribution. Here are the steps:

- a. Drag the measured fields—**Sepal.Length** and **Sepal.Width** on **columns** pile.
- b. Drag the measured fields—**Petal.Length** and **Petal.Width** onto the **rows** pile, as shown in [Figure 9.20](#).
- c. Now, rename sheet1 as **scatterplot**.
- d. On the toolbar, select **Analysis** | Uncheck **Aggregate Measures**.
- e. Change **All marks** to a **Circle** chart.
- f. Drag the dimension field **Species** to the **Details** card.
- g. Right-click on the **Worksheet** title and select **Edit Title**. Then, in the dialog box, configure the text to **K-Means Clustering**.
- h. Again, right-click on the **Worksheet** title and select **Format Title** | **Shading**. Then, choose the appropriate color.
- i. Here, select a **light pink** color. Click on **OK**.
- j. Now, right-click on **individual axis**. Then choose **Format** | **Shading**.
- k. Here as well, select a **light pink** color. Click on **OK**.
- l. Right-click anywhere on the worksheet pane and select **Format** | **Lines**. Then click on the **Sheet** tab | **Grid Lines** and make it **None** as shown in [Figure: 9.20](#):

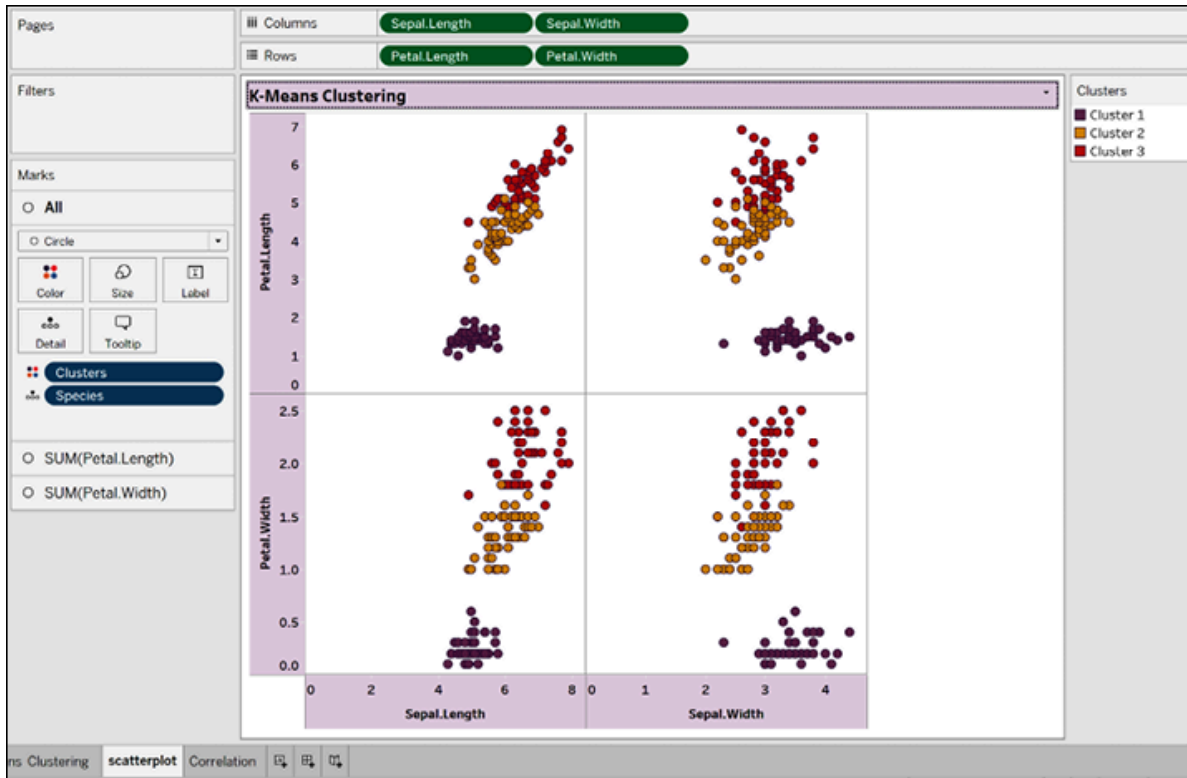


Figure 9.20: IRIS—Scatterplot worksheet

2. Segment the data based on the features.

Hint: Use Cluster Analysis.

Let us apply **Cluster analysis** to group the flower types based on the features on the scatterplot:

- Click on the **Analytics** tab | **Cluster** available on the left pane.
- Drag the cluster and apply it on the **scatterplot** sheet. (Automatic No. of clusters = 3)

Surprisingly, the automatic number of clusters i.e. group of flower types are the exact number of IRIS flowers species types within the dataset as shown in [Figure 9.20](#). This makes the whole analysis highly intuitive and accurate.

Note: In Tableau, cluster analysis is performed based on the k-means Clustering algorithm.

3. Create a dynamic tooltip on a scatter plot chart to display the image of IRIS flowers.

Hint: Use custom shapes.

We must split our tasks into three parts to create a dynamic tooltip with IRIS flower types.

Part1: Download the IRIS flower transparent images:

- a. Browse the IRIS flower and its various types on the Google platform.
- b. Then download three different types of transparent images of IRIS flower species either in **.jpg** or, **.jpeg** or **.png** extension files.
- c. Next, save those images in the following path: **This PC | Documents | My Tableau Repository | Shapes** folder.

Part2: Creating a dynamic Tooltip sheet:

- a. Click on **New Worksheet** (*Ctrl + Alt + T*).
- b. Right-click on **Sheet2**, then click on **Rename** Sheet2 as **iris flower**.
- c. Change the default chart marks card to a **Shapes** chart.
- d. Drag **Species** to **Shape marks** card.
- e. Now, click on **Shape** and then click on the **Reload Images** button from the shape editor.
- f. The new folder and images with the **Shapes** folder will show up as an option, as shown in [Figure 9.21](#):

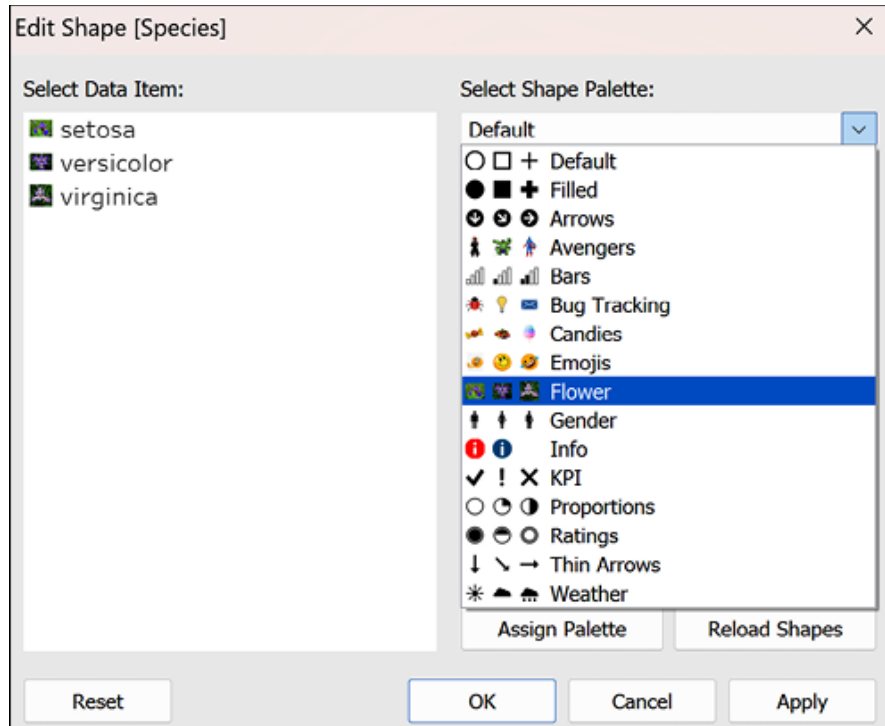


Figure 9.21: IRIS flower—Shape editor dialog box

- g. From **Select Shape Palette**, select the palette, and assign the images respective to **Species** as shown in the following figure:

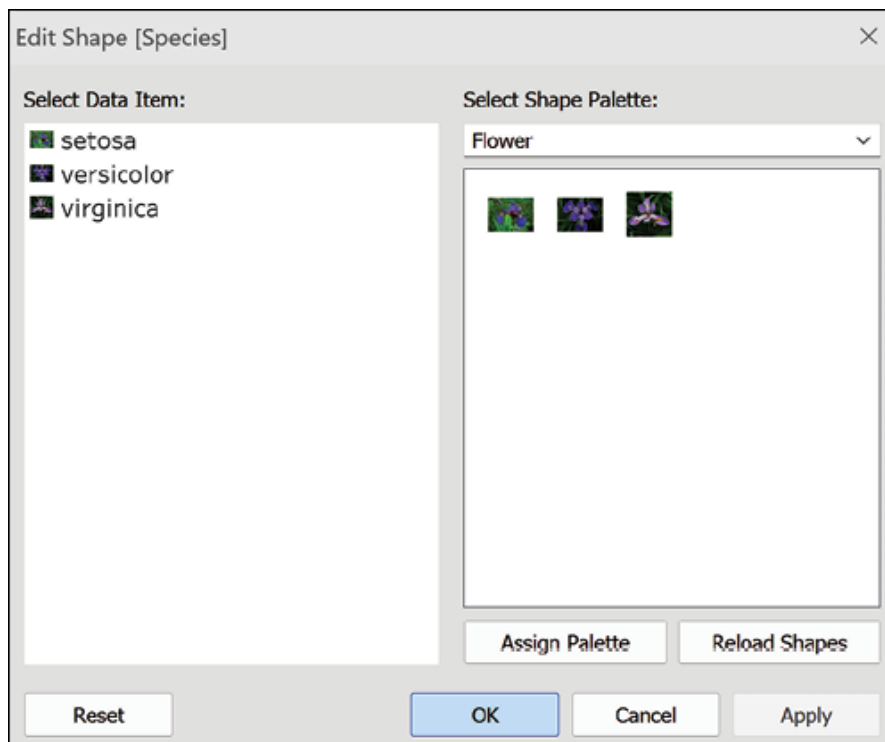


Figure 9.22: IRIS flower—Custom Shape Palette

- h. Drag **Species** to the **Filter card** and then right-click on **Species**.
- i. Select **Edit Filter | Use All** and click on **OK**. Refer to the following figure:

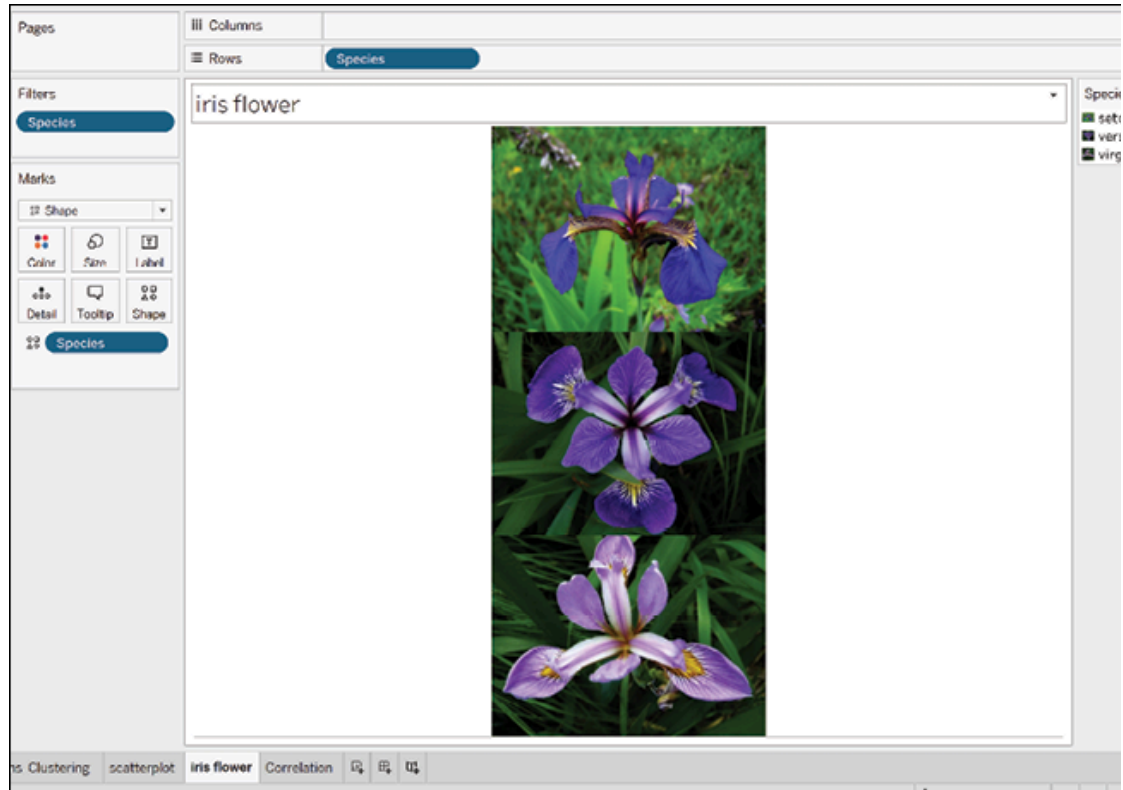


Figure 9.23: IRIS flower tooltip sheet

Part3: Insert the dynamic tooltip sheet in the **scatterplot** worksheet:

- a. Navigate to the **scatterplot** sheet.
- b. Click on **Tooltip** on the scatterplot sheet and then select **Insert -| iris flower sheet** as shown in the following figure:

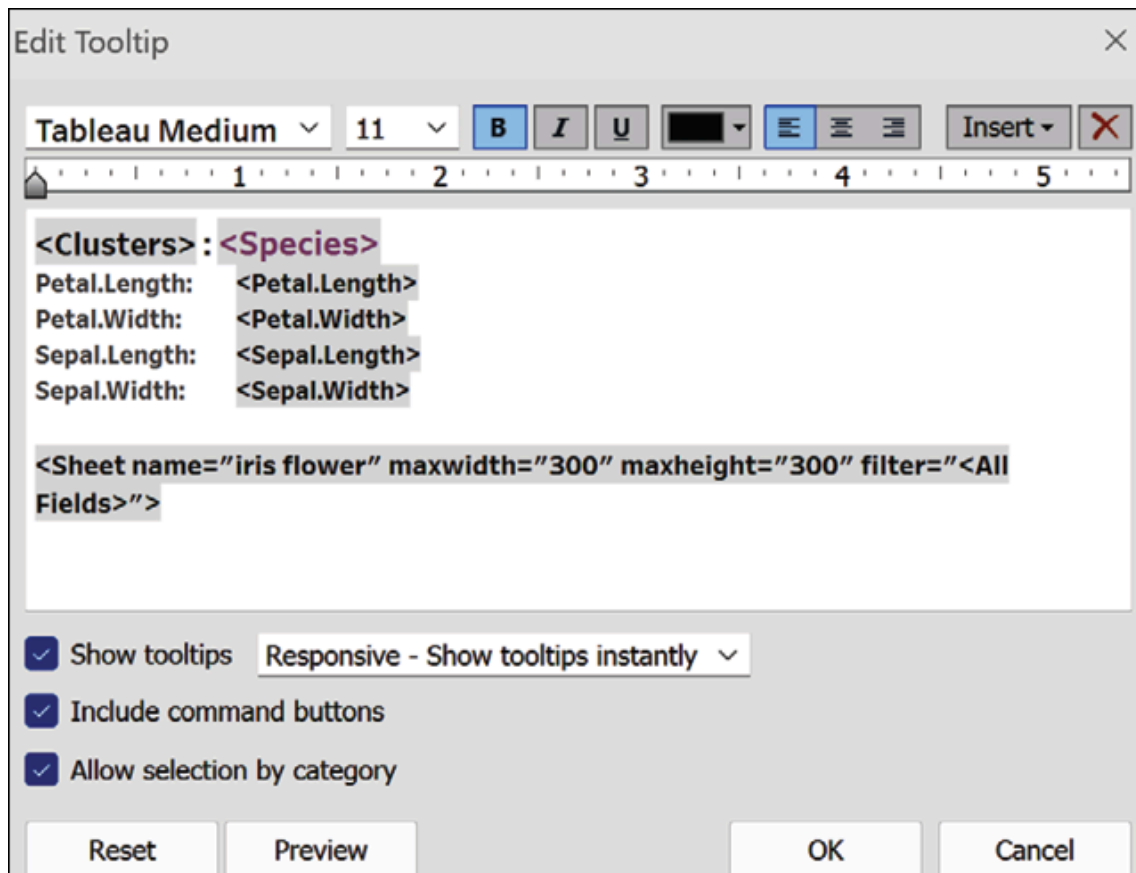


Figure 9.24: Scatterplot sheet Tooltip

4. Create Correlation fields and display:

- a. Click on **New Worksheet** (*Ctrl + Alt + T*).
- b. Right-click on **Sheet 3** and rename it as **Correlation**.
- c. On the toolbar, select **Analysis | Create Calculated Field**, named as **Correlation_petal_l_petal_w**.
- d. In the Calculation Editor, enter the formula as shown in the following figure:



Figure 9.25: Correlation calculated field1

- e. On the toolbar, select **Analysis | Create Calculated Field.**, named as **Correlation_petal_l_sepal_l**.
- f. In the Calculation Editor, enter the formula as shown in the following figure:

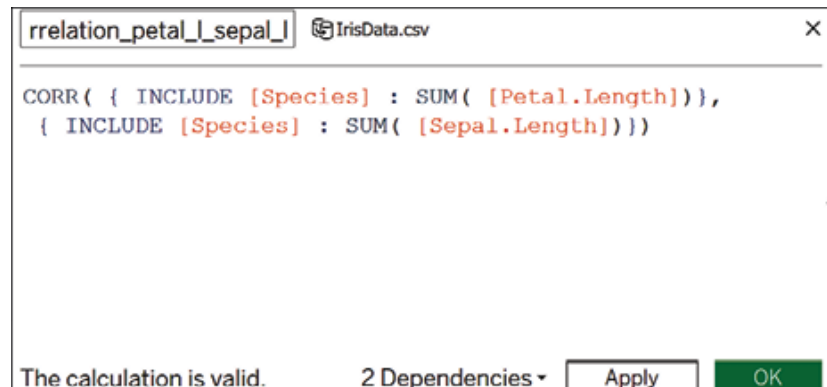


Figure 9.26: Correlation calculated field 2

- g. On the toolbar, select **Analysis | Create Calculated Field** named as **Correlation_petal_l_sepal_w**.
- h. In the **Calculation Editor**, enter the formula as shown in the following figure:

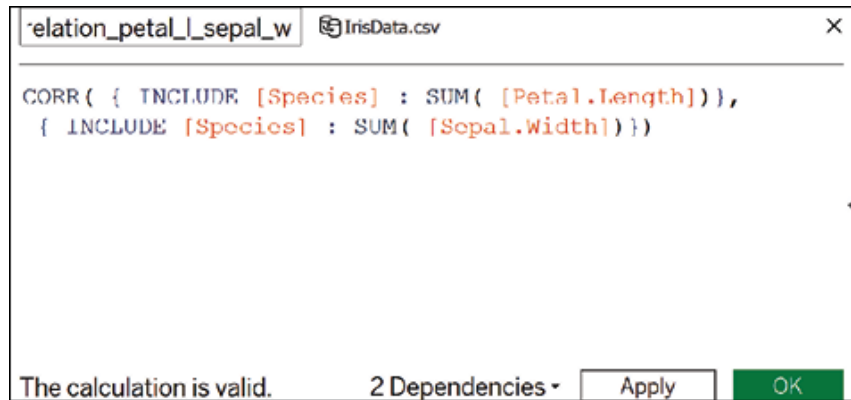


Figure 9.27: Correlation—Calculated field3

- i. On the toolbar, select **Analysis | Create Calculated Field**, named as **Correlation_sepal_l_sepal_w**.
- j. In the Calculation Editor, enter the formula as shown in the following figure:



Figure 9.28: Correlation calculated field4

Follow these steps to create a **Correlation** worksheet:

- a. Drag the **Measure Names** on the **rows** shelf.
- b. Drag the **Measure Values** on **Text marks** card.
- c. Drag **Measure Names** to the **Filter card**, right-click on the field, and select only the **calculated correlated fields** from the options.
- d. Right-click on the **Correlation_petal_l_petal_w** field on the pane and then select **Edit Alias**. Now, rename it as **Correlation between Petal-Length & Petal-Width**.

- e. Right-click on the **Correlation_petal_l_sepal_w** field on the pane and then select **Edit Alias**. Now, rename it as **Correlation between Petal-Length & Sepal-Width**.
- f. Right-click on the **Correlation_petal_l_sepal_l** field on the pane and then select **Edit Alias**. Now, rename it as **Correlation between Petal-Length & Sepal-Length**.
- g. Right-click on the **Correlation_sepal_l_sepal_w** field on the pane and then select **Edit Alias**. Now, rename it as **Correlation between Sepal-Length & Sepal-Width**.

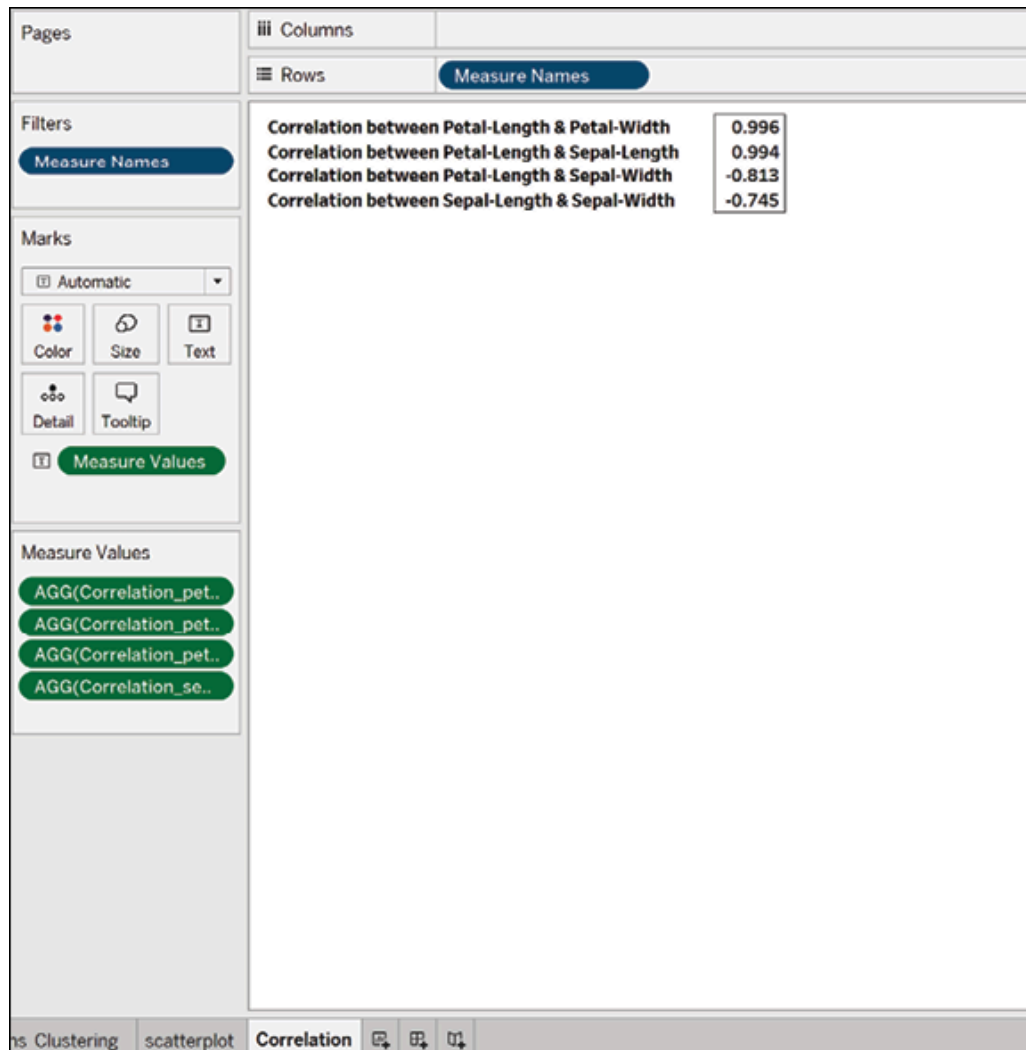


Figure 9.29: Correlation worksheet

5. Allow only specific users to view the report.

Hint: Use row-level security

- a. On the toolbar, select **Analysis | Create Calculated Field**, named **User Filter**.
- b. In the **Calculation Editor**, enter the formula as follows:

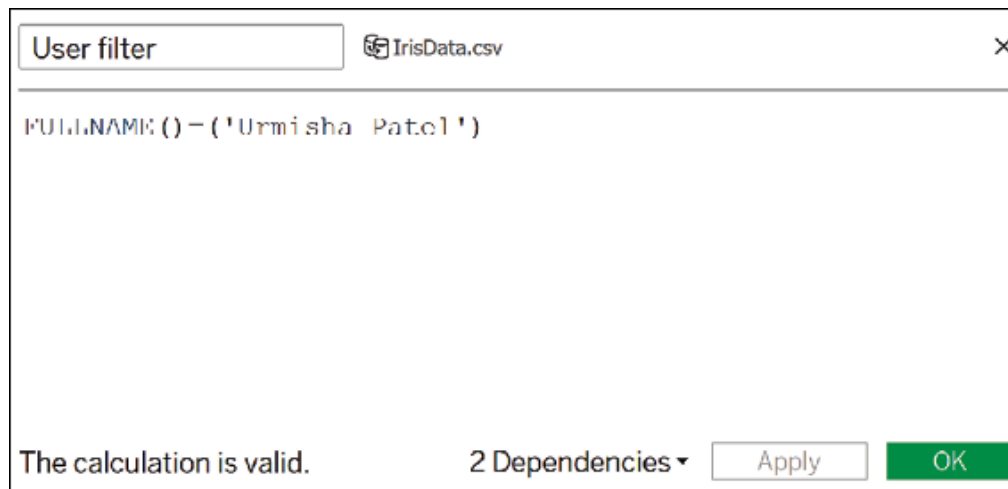


Figure 9.30: Row-level security

- c. Drag this **User Filter** field to the **Filter card** on the **Scatterplot worksheet**.
- d. Right-click the **User Filter** field and then choose **True** from **Select from List** tab.
- e. Click on **OK**.

Note: The usernames defined in the calculated field have the permission to access the views. So, if thumbnail or Tableau Server/Online sign-in does not match with the user filter field then users are not allowed to access the views.

6. Create the IRIS Flower Analysis Dashboard.

- a. Click on the **New dashboard**, which is named **IRIS K-Means Clustering**.
- b. Drag the scatterplot worksheet to the dashboard pane in floating.
- c. Now, drag the correlation worksheet floating next to the scatterplot worksheet.

- d. For Title, drag text object from objects and edit the text as shown in [Figure 9.31](#).
- e. Next, drag the image object. Then, upload the desired image from the local device.
- f. Arrange all the items as shown in [Figure 9.31](#):

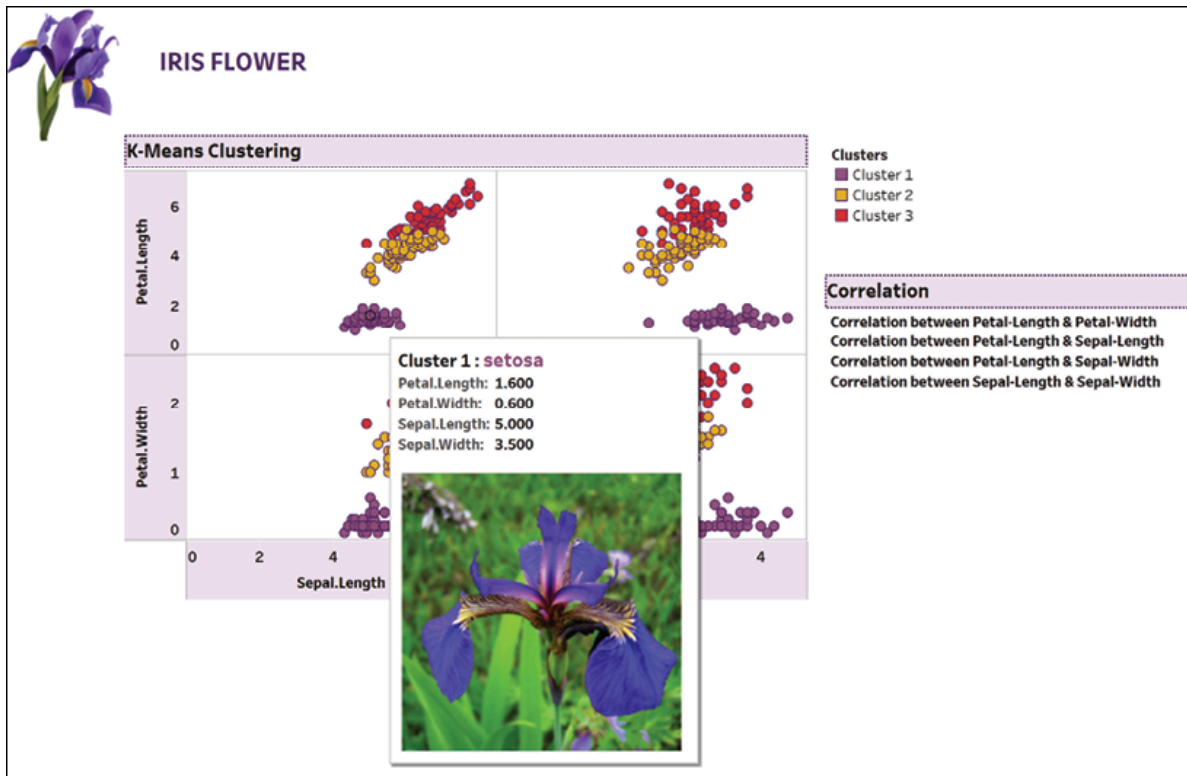


Figure 9.31: IRIS flower analysis dashboard

7. Publish the workbook with data sources to the Tableau Cloud site.

- a. Navigate to **Server | Publish Workbook** to publish the workbook along with the data source as embedded.
- b. Now, configure the settings and details to publish the workbook on the Tableau Online site as shown in [Figure 9.32](#):
- c. Click on **Publish**.

In the defined location, the workbook, along with the data sources, are now published on the site.

×

Publish Workbook to Tableau Online

Project

Project1

Name

IRIS Flower Analysis

Description

Tags

Add

Sheets

All Edit

Permissions

Same as project (**Project1**) Edit

Data Sources

2 embedded in workbook Edit

Generate thumbnails as user:

Chetna Pothiwala

▼

More Options

☐ Show sheets as tabs
☒ Show selections
☐ Include external files

Workbook Optimizer

Publish

Figure 9.32: Publish Workbook IRIS Flower Analysis *Tableau Desktop vs Tableau Cloud*

There are two ways to create/edit workbooks and data sources, either via Tableau Desktop or Tableau Cloud (web version). It is highly recommended to use Tableau Desktop to develop views, as this version provides full functionalities for editing and developing.

The following table helps distinguish between Tableau Desktop and Tableau Cloud:

Tableau Desktop	Tableau Cloud
Tableau Desktop provides FULL functionality for editing and creating workbooks/dashboards and data sources.	Tableau Cloud provides limited functionalities compared to Tableau Desktop for editing and creating workbooks/dashboards but does not allow to create data sources.
Tableau Desktop is used for development.	Tableau Cloud is ideally used for sharing and collaboration.
The best use of Tableau Desktop for developers is to build the intuitive and insightful reports for the decision makers.	The best use of Tableau Cloud is for decision-makers to analyze the report and make better decisions.
Once can share locally the report (.twbx) with others. However, recipients must have Tableau Desktop or Readers in their PC.	Once the workbook is uploaded on Tableau Cloud, recipients do not need to install Tableau Cloud, they just require Login credentials.
By sharing the reports locally, there will not be any restrictions on end users. So, eventually end up with multiple versions of the same reports.	The site administrator can set permissions for each user on Tableau Cloud. Depending on the permissions given, they are allowed to edit, share, download the workbooks, etc.

Table 9.1 : Tableau Desktop vs. Tableau Cloud Conclusion

This chapter provided hands-on experience for readers to create and edit workbooks and data sources on Tableau Cloud and Tableau Desktop. This includes managing content and users on online sites. These exercises offer a glimpse into the real world of Tableau Cloud and Tableau Desktop usage.

Points to remember

Here are some key takeaways from this chapter:

- Tableau Desktop provides full functionalities for editing and creating workbooks and data sources. It is highly recommended to use Tableau Desktop to develop views instead Tableau Cloud.
- Tableau Cloud is ideally used for sharing and collaboration. While Tableau Desktop is used for development.
- If you want to apply cluster analysis, forecasting, or enhanced formatting on data, then use Tableau Desktop.
- Tableau Cloud must be used if you want to maintain the same version of workbooks among other users and developers.
- User permissions can be set either by Tableau Cloud or Tableau Desktop. However, more flexibility and other permissions can be configured on Tableau Cloud.

- Publishing and scheduling data sources can be configured from both Tableau Desktop and Cloud. However, Tableau Cloud provides multiple options. It is highly recommended that Tableau Bridge software be used to publish, share, and schedule the workbooks and data sources from the local machine to Tableau Cloud behind the firewall.

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